Service Manual
6002009602
Getinge 86-series
S-8666, S-8668, S-8668T

2023-09-06 EN REV.K

GETINGE

The model name and serial number should always be provided when ordering spare parts or during queries per telephone or written correspondence:

Serial number of the machine:

....

Machine model:

....

Getinge Disinfection AB
Ljungadalsgatan 11
352 46 Växjö
SWEDEN

Internet: www.getinge.com

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1 INTRODUCTION

This section contains information about the purpose and intended use of the machine.

1.1 About this manual

This manual contains important information about preventive and corrective maintenance of the machine. To ensure safe handling and error-free maintenance, read the manual before performing maintenance. Pay special attention to section *2 SAFETY, page 12*. Some sections are not relevant for all machines; options, selections and customizations can vary.

1.2 Symbols in this manual

In this manual, warnings and information are indicated by the following symbols.

	Indicates a hazardous situation that could result in death, personal injury or damage to property if it is not avoided.
	Points out important details to consider for trouble-free and efficient usage of the machine.

1.3 Intended use statement

The intended use of the 86-series washer-disinfector is to clean, disinfect at a thermally low or intermediate level, and dry surgical instruments (solid and tubular), bowls, basins, glassware, receivers, suction bottles, baby bottles, anesthesia equipment and surgical shoes. The equipment can be made of stainless steel, aluminum, plastic, rubber, silicone or glass.

The device is not intended to clean and disinfect invasive devices as end point processing.

1.4 Intended reader

The content of this manual is intended to be read by authorized service personnel who are performing maintenance on and installing the washer-disinfector. Authorized service personnel are trained to handle installation, service, preventive maintenance, troubleshooting, end of life handling and recycling. Authorized service personnel have extended rights to the control system and the programs.

Authorized service personnel must have completed training through Getinge Academy.

1.5 Potential misuse

- The washer-disinfector is not intended for use with household detergents.
- The washer-disinfector is not intended for cleaning and disinfection of bedpans or urine bottles (ISO 15883-3).
- The washer-disinfector is not intended for cleaning and chemical disinfection of thermolabile endoscopes (ISO 15883-4).

1.6 References

Documentation	Contents	Paper	Electronic media
User manual	Instructions for daily usage.	x	x
Installation manual	Instructions for assembly, installation and commissioning.	x	x
Quick guide	Instructions for frequently performed tasks.	x	x
Declaration of Conformity	Collection of ISO certificates.	x	-
Electrical diagrams	Collection of circuit diagrams.	-	x
Installation drawing	Installation measurements.	x	x
P&I diagrams	Collection of P&I diagrams.	x	x
Service manual	Instructions for service and preventive maintenance.	-	-
Spare parts	List of available spare parts.	-	-
Program sheets	Description of available programs.	-	x
Smart loading trolley fixed height 2.0 for washer-disinfectors.	User manual for fixed height loading trolley (FHT Disinfection).	-	-
User manual Getinge loading equipment.	Instructions for daily use of Getinge loading equipment.	-	-

x = Delivered with the machine.

All documents are available at Getinge Extranet.

1.7 Product liability

1.7.1 Declaration of Conformity

The product compliance is stated in the Declaration of Conformity.

1.7.2 Disclaimer

- The manufacturer cannot guarantee that the product will work as intended if accessories or spare parts other than those approved by the manufacturer are used.
- The product liability only applies when the instructions in this manual have been followed in their entirety.
- The manufacturer accepts no responsibility for personal injuries when the instructions in this manual have not been followed in their entirety.
- The manufacturer reserves the right to change the specification and design of the product without any prior notice.

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1.7.3 Manufacturer

This product was manufactured by:

Getinge Disinfection AB
Ljungadalsgatan 11
352 46 Växjö
SWEDEN

1.7.4 Warranty

The product warranty applies in accordance with the agreement reached between the Getinge sales company and the customer.

1.8 Installation/Service/Repairs

Installation and service must be carried out by service personnel authorized by Getinge, referred to as service personnel in this manual.

Preventive maintenance and cleaning, as described in the user manual, may be performed by the user of the machine, referred to as the operator in this manual.

2 SAFETY

This section highlights important safety aspects that increases safety awareness. The service personnel must follow instructions and be familiar with the safety functions before performing service on the washer-disinfector.

2.1 Safety functions

2.1.1 Door pinch protection

If anything is detected between the door and the door frame when closing, the built-in door pinch protection is activated and the door reopens. The door will not close as long as it is obstructed.

2.1.2 Password protection

The password protected programs can only be started by logged in users. For more information, see section *4.3.6 Manage access rights, page 44*.

2.1.3 Isolator switch

The machine must have a lockable isolator switch for the electric power supply (connected before/outside the machine's electrical connection). The isolator switch is installed by the customer and must be placed on a wall close to the machine. In an emergency, the main power can be switched off using the isolator switch.

2.2 Safety statements

Hazardous operations and situations are highlighted in the text.

Safety statement structure:
• Warning symbol and signal word. • What is the risk? (Short description) • What causes the risk? (More precise information) • How can the risk be avoided?
Example:
WARNING! **Risk of burns** ***The inside surfaces of the chamber and the load may be hot directly after disinfection.*** **- Let the chamber cool down before entering the chamber.** **- Be careful when handling hot loads.** **- Wear protective clothing, such as gloves.**

2.3 General warnings

WARNING!

Risk of burns

The inside of the chamber and the load may be hot.

- Let the chamber and the load cool down before unloading.

- Wear protective gloves.

WARNING!

Risk of electrical shock and burns

There may be remaining voltage behind the cover panels, even when the power is switched off.

- Be careful when removing the cover panels.

- All service behind cover panels must be performed by technicians with documented experience.

- All electrical installations must be performed by an authorized electrician.

WARNING!

Risk of burns

Parts, media and surfaces in the service area can be hot.

- Be careful when working in the service area.

WARNING!

Risk of burns

Hot steam may escape from the chamber when the door is opened after a process is stopped or completed.

- Pay attention when standing next to the machine.

WARNING!

Risk of cut injuries

There are unprotected sharp edges when the cover panels have been removed.

- Installation and service behind the cover panels must only be performed by technicians with documented experience.

- Wear protective gloves.

WARNING!

Risk of burns

There may be hot surfaces when the cover panels have been removed.

- Be careful when removing the cover panels.

- Installation and service behind the cover panels must only be performed by technicians with documented experience.

- Wear protective gloves.

WARNING!

Risk of injuries

The machine contains moving parts.

- Be careful when the doors are open.

- Be careful when cover plates have been removed.

- Installation and service behind the cover panels must only be performed by technicians with documented experience.

WARNING!

Risk of burns

Pressurized steam or hot water may escape from the supply lines if they are not closed.

- All supply lines must be closed when working on the pipe system.

- Installation and service must only be performed by technicians with documented experience.

2.4 Safety labels on the machine

Symbol	Hazard	Action
	Warning or caution	Pay attention.
	Warning: hot surfaces	Pay attention when in contact with the machine's surfaces.
	Warning: dangerous voltage	Pay attention when close to live parts. Do not touch electrical equipment or installations.

2.5 Important

- Installation and servicing must be carried out in accordance with current local rules and regulations.
- The machine must be connected in accordance with the instructions in the Installation manual.
- Check the connections against the machine's product label.
- The power box and electrical cabinet may only be opened by trained personnel.
- Before welding begins on or close to the machine, all wires connected by plugs and sockets must be disconnected from all I/O modules of the control system.

2.6 Personal protection

Working with the machine includes handling heavy loads and contact with hot surfaces and detergents. Wear protective gear, such as, protective footwear, gloves, protective clothing, glasses and other equipment in accordance with local regulations.

2.7 Accident or incident reporting

Accidents or incidents must be reported in accordance with local regulations.

3 PRESENTATION

3.1 Product versions

The machine is available in versions with different configurations:

- Single door or double doors.
- Electrically or steam heated or with a combination of both.
- Different options for incoming and outgoing connections.
- Different options for monitoring.

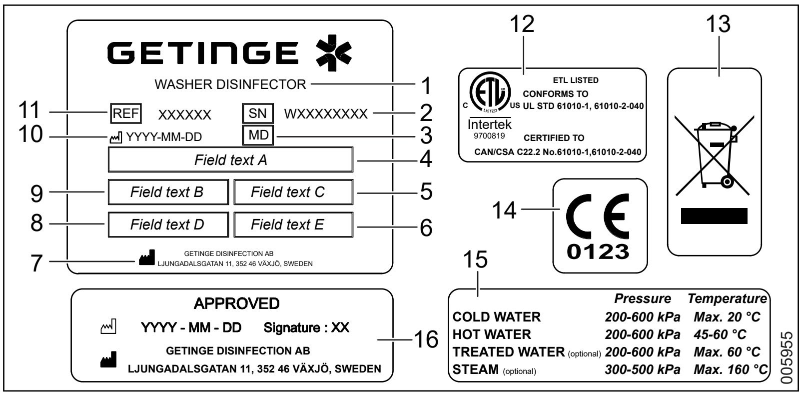
Follow the instructions applicable to the product version in use.

Model	Chamber width (mm)	Chamber length (mm)	Chamber height (mm)	Effective chamber volume (L)	Gross chamber volume (L)	Water temp (°C)	Air temp (°C)
S-8666	650	720	667	312	430	≤95	≤110
S-8668	650	810	667	351	480	≤95	≤110
S-8668T	650	810	667	351	480	≤95	≤110

3.2 Product label

The product label is attached on the inside of the detergent cabinet. The model type and serial number are important when communicating with Getinge.

The product labels are attached on the inside of the detergent cabinet. The product model and serial number are important when communicating with Getinge.



1 Designated purpose	9 Maximum current
2 Serial number	10 Date of manufacture
3 Medical device	11 Machine model
4 Voltage alternative	12 Product listing
5 Frequency	13 Recycling symbol
6 Protection class	14 CE labeling
7 Manufacturer	15 Media connection
8 Fuse	16 Approval marking

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3.3 Recommended detergent

When handling detergents, follow the safety instructions as indicated on the detergent container label and/or in its material safety data sheet.

When selecting detergents, read and follow the instructions and recommendations of the manufacturer of the goods to be cleaned or disinfected.

The following table is a guide.

Always make decisions regarding what detergent to use for cleaning or disinfection of a specific load (material, soil and type of items) together with the detergent supplier.

Make sure the intended use for the detergent corresponds with the program and load.
Contact a local Getinge sales representative for information regarding what detergent to use in a specific program phase.

Area of use	Detergent type
Thermal disinfection of non-invasive, non-critical medical devices, non-blooded waste, aids for the disabled, footwear and wash bowls.	Alkaline Enzymatic
Thermal disinfection of non-invasive, non-critical medical devices, blooded waste, container systems used to transport medical devices, aids for the disabled, footwear and wash bowls.	Alkaline Enzymatic
Thermal disinfection of invasive and critical medical devices, such as invasive equipment, soiled, rigid and non-cannulated surgical instruments.	Alkaline Enzymatic
Chemical disinfection of non-invasive, non-critical medical devices, non-blooded waste, aids for the disabled, footwear and wash bowls.	Alkaline Enzymatic
Chemical disinfection of non-invasive, non-critical medical devices, blooded waste, container systems used to transport medical devices, aids for the disabled, footwear and wash bowls.	Alkaline Enzymatic
For neutralizing alkaline residues.	Acidic
For removal of rust, hard water scale and stains on instruments and in washer-disinfector chamber.	Acidic
For use in the final rinse to speed up the drying of the load.	Rinse aid
For use in the final rinse to lubricate moving parts of surgical instruments and prevent friction corrosion.	Lubricant

3.4 Heating method

The machine can be heated electrically, with steam or with a combination of both (depending on the actual machine configuration).

See the attached P&I diagram for more information about the respective heating method.

Appendix 1 - S-8666/S-8668 - P&I Diagram - Electrically heated

Appendix 2 - S-8666/S-8668 - P&I Diagram - Steam heated

Appendix 3 - S-8666/S-8668 - P&I Diagram - Electrically/Steam heated

Appendix 4 - S-8668T - P&I Diagram - Electrically heated

Appendix 5 - S-8668T - P&I Diagram - Steam heated

Appendix 6 - S-8668T - P&I Diagram - Electrically/Steam heated

Appendix 7 - S-8666/S-8668/S-8668T - Installation drawing

3.5 Accessories

There are several accessories intended for use with this machine, such as wash carts. For a complete list, see the *Getinge loading equipment* manual. For more information about the different loading systems, see the Loading equipment for Getinge 86-series, 88 Turbo & CM320 washer-disinfectors manual.

4 CONTROL SYSTEM

This section contains a description of the control system. This focuses on the more advanced functions that concern administration and settings. The basic functions on the touchscreen and the basic mechanical and general functions are described in the User manual.

The control system consists of hardware and software that interact to control and monitor the machine processes. The system is managed by users from the touchscreen on the operator panel.

Configurations are done from the touchscreen or from an external PC. For more information, see sections *4.3 Touchscreen interface, page 34* and *4.4 Web server interface, page 60*.

The machine's software can be updated. For more information, see section *7.1 Control system software, page 140*.

4.1 Control system overview

4.1.1 Operator panel

The operator panel is a 7" WVGA TFT Power panel with a full-format touchscreen. The whole panel measures 143x198x41 mm (5.6x7.8x1.6 in), (HxWxD). The panel is equipped with integrated processors, memory and other computing hardware.

If the machine is equipped with two doors, there is one panel on each side. The operator panel on the soiled side is called the main panel. The operator panel on the clean side is called the terminal panel.

The control system software is executed from the main panel's computer.

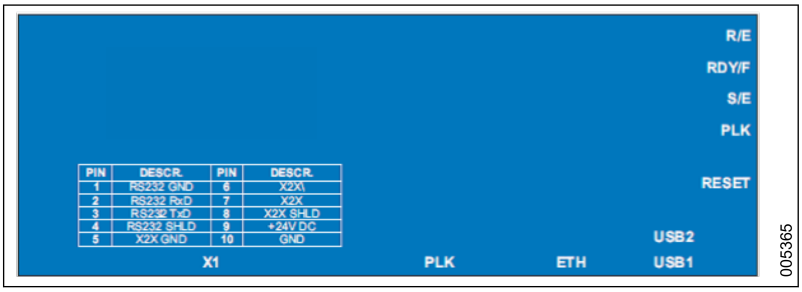
4.1.1.1 Operator panel touchscreen

The touchscreen display resolution is 800x400.

The machine is operated from the touchscreen.

4.1.1.2 Operator panel backside

The operator panel backside is equipped with ports for connections and a **RESET** button. The operator panel can be removed and replaced and the software can be restored from a previous backup.



Status diodes	Physical buttons	Ports
R/E	**RESET**	X1
RDY/F		PLK
S/E		ETH
PLK		USB 1
		USB 2

4.1.1.2.1 Status diodes

The status diodes indicate operator panel status according to the following table:

Diode	Colour	Light	Description
R/E	Green	Steady	Application running.
	Red	Steady	SERVICE mode.
RDY/F	Yellow	Steady	SERVICE or BOOT mode.
	Red	Steady	Temperature too high.
S/E	Green	Steady	The PLK interface is operated purely as an Ethernet TCP/IP interface.
PLK	Green	Steady	Link to network has been set up.
		Flashing	Link to network has been set up. The diode flashes when the network is active.

4.1.1.2.2 Reset button

The system can be set to BOOT mode at any time. BOOT mode is used, for example, when the system is reset from a backup copy. For more information, see sections *4.3.9 Backup and export, page 52* and *7.1.2 Restore control system software from backup, page 141*.

1. Press the button quickly (hold for less than 2 seconds). The display restarts.
2. Press and hold the button for more than 5 seconds. BOOT mode starts.

4.1.1.2.3 Communication ports

The operator panel on the soiled side is equipped with ports as described in the section below.

If the machine is equipped with two doors, there is one panel on each side (clean side and soiled side). The two panels may have different sets of ports.

4.1.1.2.3.1 X1

The X1 contact contains the following:

- 24 V supply for the display
- X2X communication for I/O node
- RS232 communication for printer (option)

4.1.1.2.3.2 X1

The X1 contact contains the following:

- 24 V supply for the display
- X2X communication for I/O node
- RS232 communication for printer (option)

4.1.1.2.3.3 PLK

POWERLINK is used to connect to a network and to communicate with external IT systems, such as Getinge Online and T-doc. A network printer and network storage can also be included in the connected external network.

For more information, see sections *4.3.5.4 Setup printer, page 44*, *4.3.8 Printouts, page 51* and *4.4.9 External Interfaces, page 78*.

4.1.1.2.3.4 ETH

ETH (Ethernet) is used for the following:

- Communication between the panels on the soiled side and the clean side.
- Communication with the supervisor system.
- Connecting an external PC to the machine's Web server. For more information, see section *4.4.1 Connect to the Web server, page 60*.

If using Supervisor, **ETH** is connected to a switch. For more information, see section *4.1.3.4 Double door with Supervisor, page 24*.

4.1.1.2.3.5 USB 1

USB 1 is used for saving reports with data from the wash cycle. The USB function may be installed at delivery or activated later using a specific license key. For more information, see section *4.4.14 Activate options, page 92*.

4.1.1.2.3.6 USB 2

USB 2 is used if the machine is equipped with a bar code scanner.

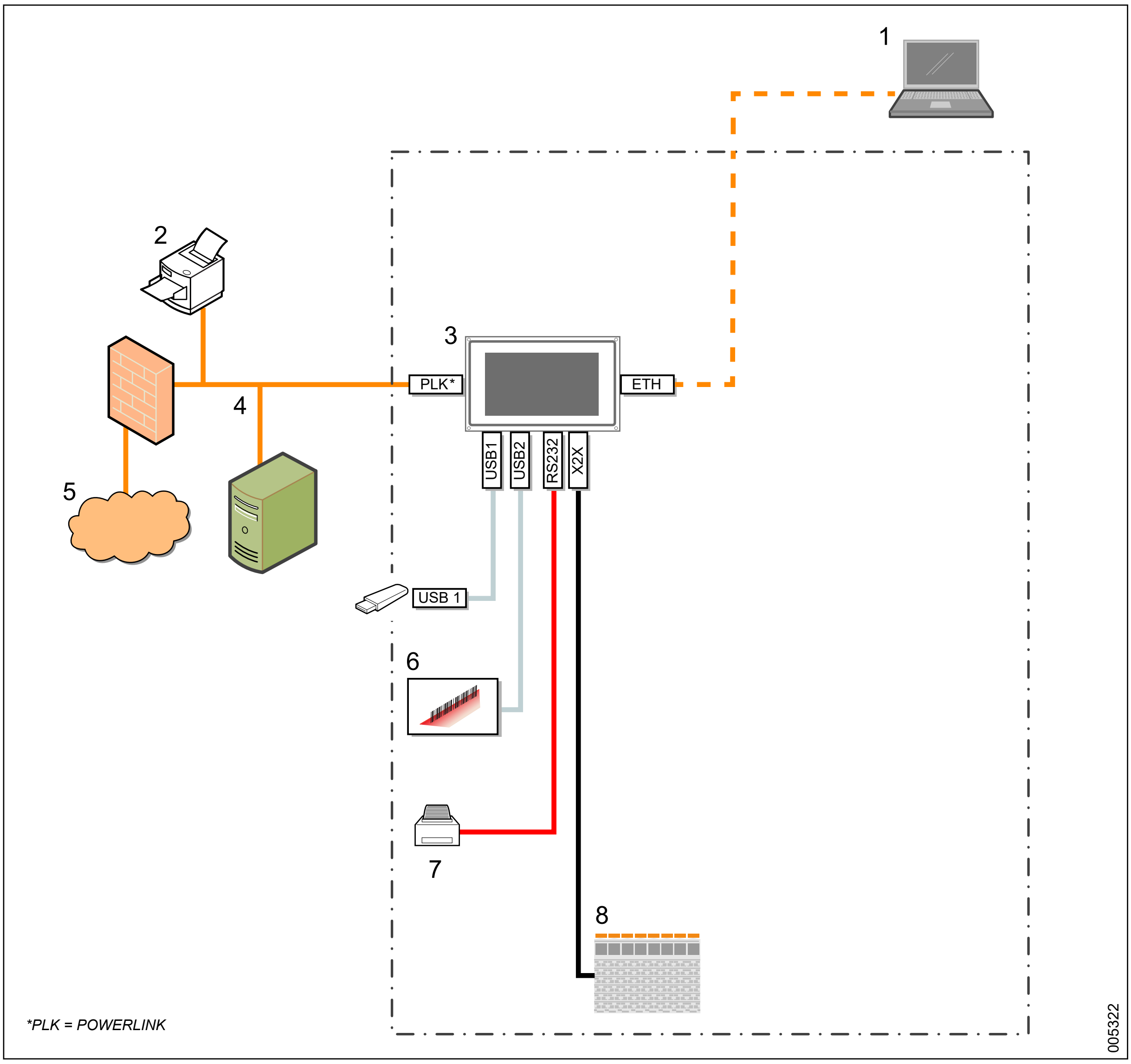
4.1.2 Control System I/O

The control system I/Os are located on a separate I/O-node, connected by an X2X bus link. The I/O node consists of several I/O modules. For more information, see section *7.6.2 Control system modules (I/O and PLC), page 148*.

4.1.3 Communication and peripheral equipment

The machine's internal and external communication (including connection of peripheral equipment) is set up in different ways depending on machine configuration. This is illustrated by the following four versions that show machines with single and double doors; with or without a Supervisor system.

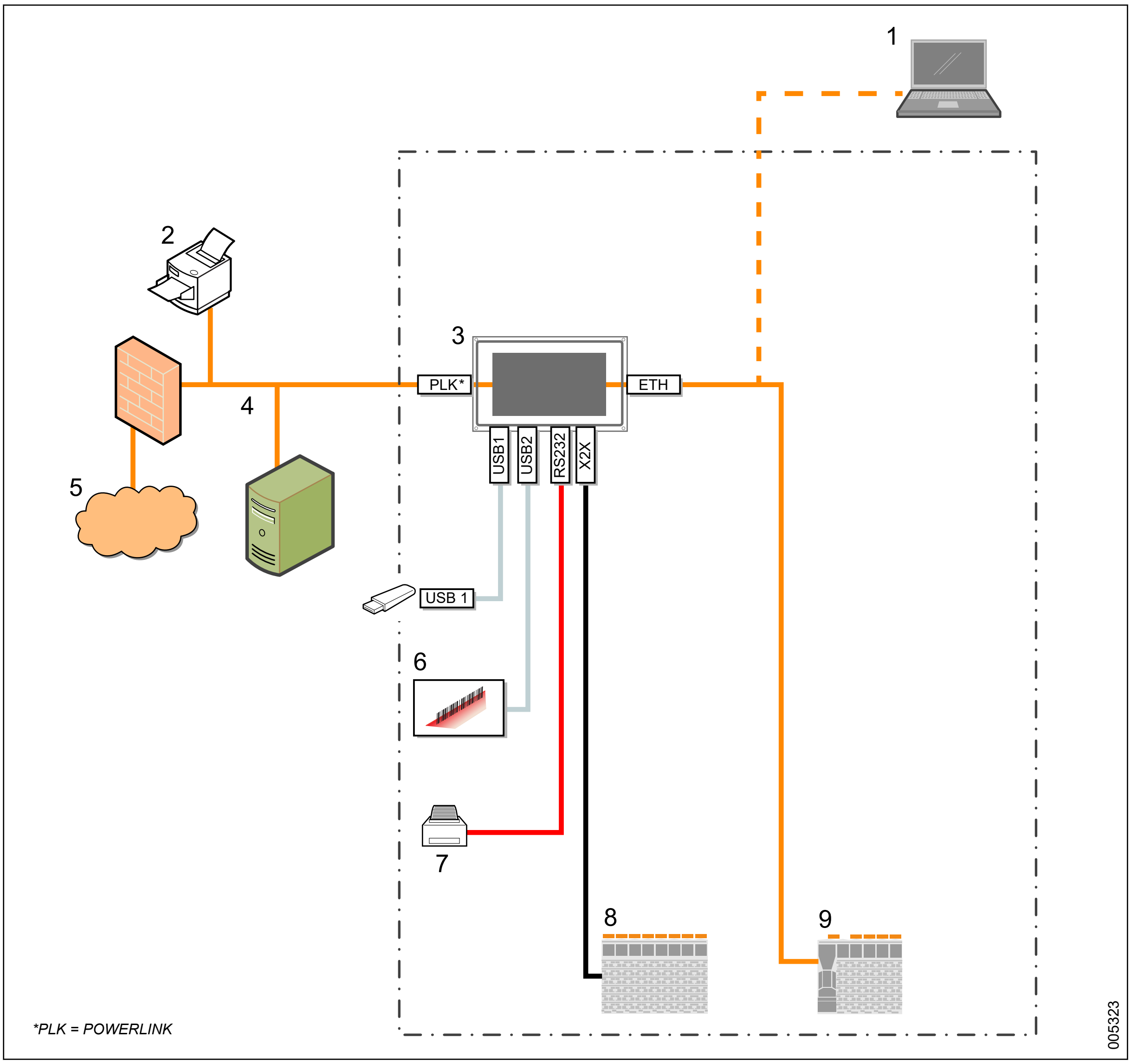
4.1.3.1 Single door without Supervisor



1 Configuration via PC
2 Network printer (option)
3 Soiled side, 7" Power panel
4 Network storage (option)
5 Getinge Online / T-DOC (option)
6 Barcode scanner (option)
7 Local printer (option)
8 Control system and I/O nodes

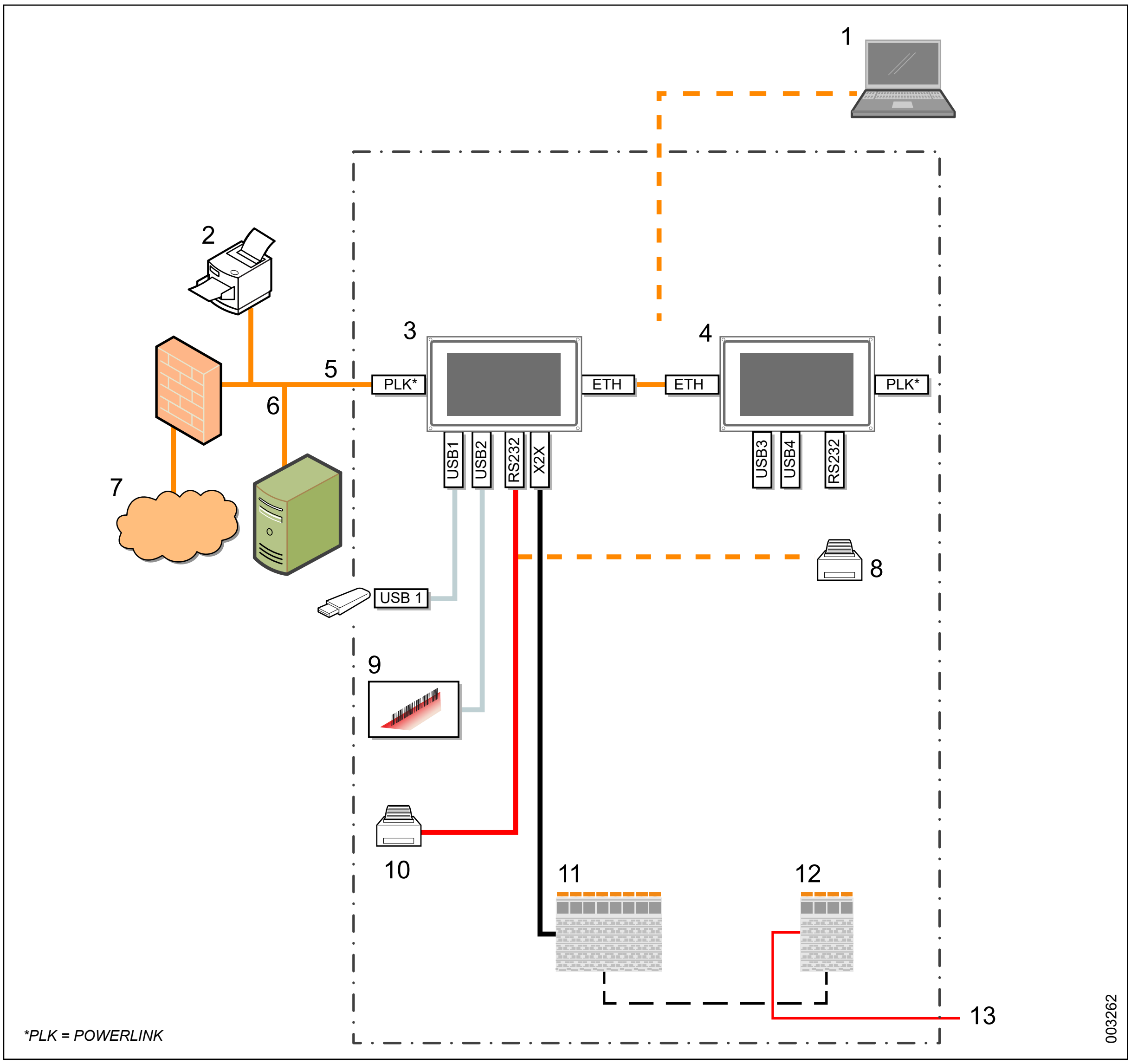
<Doc_SER><Doc_6002009602><Rev.K><Lang_en>

4.1.3.2 Single door with Supervisor



1 Configuration via PC
2 Network printer (option)
3 Soiled side, 7" Power panel
4 Network storage (option)
5 Getinge Online / T-DOC (option)
6 Barcode scanner (option)
7 Local printer (option)
8 Control system and I/O nodes
9 Supervisor

4.1.3.3 Double door without Supervisor



1 Configuration via PC
2 Network printer (option)
3 Soiled side, 7" Power panel
4 Soiled side, 7" Power panel
5 External LAN (option)
6 Network storage (option)
7 Getinge Online / T-DOC (option)
8 Clean side, local printer (option)
9 Barcode scanner (option)
10 Soiled side, local printer (option)
11 Control system and I/O nodes
12 Powerlink module
13 External load system (via Powerlink)

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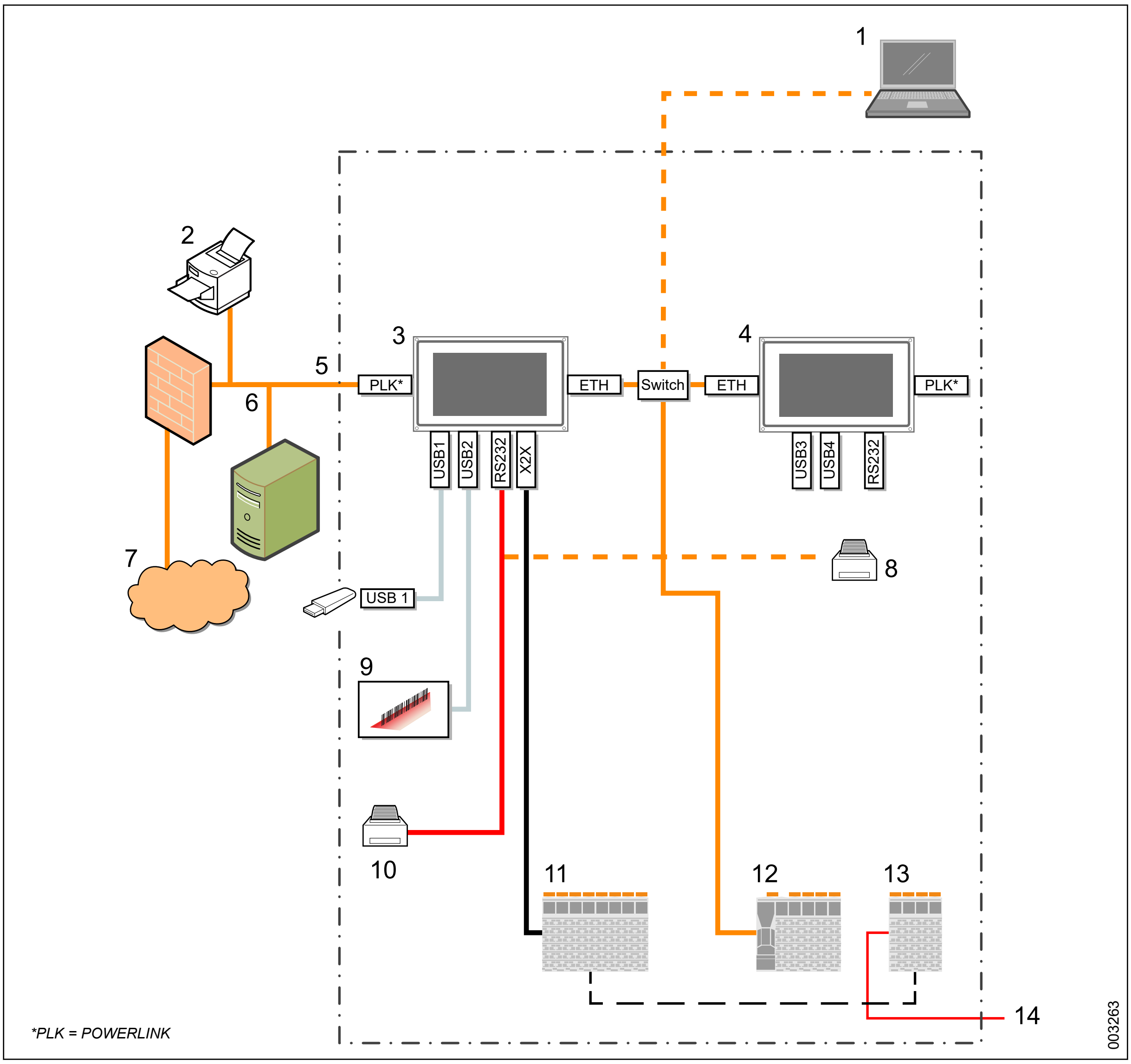
4.1.3.4 Double door with Supervisor



1 Configuration via PC
2 Network printer (option)
3 Soiled side, 7" Power panel
4 Soiled side, 7" Power panel
5 External LAN (option)
6 Network storage (option)
7 Getinge Online / T-DOC (option)
8 Clean side, local printer (option)
9 Barcode scanner (option)
10 Soiled side, local printer (option)
11 Control system and I/O nodes
12 Supervisor
13 Powerlink
14 External load system (via Powerlink)

4.1.3.5 Network printer

It is possible to connect a network printer to the machine. Getinge recommends to use only Lexmark's network printer. For more information, see section *4.3.8 Printouts, page 51*.

4.1.3.6 Network storage

It is possible to save process reports on a network server (FTP or CIFS). For more information, see section *4.4.8 Printer and Report, page 73*.

4.1.3.7 Encryption information

The interface for T-DOC, Getinge Online and ST-AGS uses openSSL version 1.0.1c [14] for communications between the control system G1 and external clients and servers. The signature on the certificate which is used for authenticating clients uses the AES256-SHA signature algorithm with a length of 1028 bits.

Via openSSL, the G1 control system supports the client by establishing a TLS/SSL connection with SSLv3, TLSv1, TLSv1.1 and TLSv1.2 protocols and will also use these certificates and protocols during communication with servers such as Getinge Online.

4.1.4 Monitoring

The system is monitored by using different wash and disinfection parameters. The parameters used depend on the current machine configuration. In addition to the standard monitoring system installed on all machines, there are a number of options measuring different parts of the process.

4.1.4.1 Temperature monitoring

The machine has sensors that monitor temperatures in the following areas:

- Chamber

  Two sensors in the chamber measure the temperature independently of each other. The systems warns if the temperature deviates too much from the set values.
- Drying air

  The sensor ensures that the drying air temperature is optimal.
- Booster tank

  The sensor ensures that the water from the booster tank is hot enough before it is released into the chamber.

4.1.4.2 Detergent flow monitoring

Flow monitoring is used to measure flows of detergents in real time and warns if values deviate from those preset.

For more information, see sections *6.7 Calibration of dose monitors, page 136* and *5.10 Dosing system, page 116*.

4.1.4.3 Conductivity monitoring (option)

The conductivity meter monitors the quality of the water in the final rinse. If the conductivity in the final rinse is higher than the preset value, the machine is emptied and the final rinse is repeated automatically.

If, after a certain number of repetitions, the conductivity is still higher than the preset value, the process is stopped and an error message appears on the touchscreen. The number of repetitions and permitted conductivity can be set.

For adjustment and configuration, see section *7.8 Conductivity meter (option), page 171*.

4.1.4.4 Humidity sensor, drying system (option)

The humidity sensor is located in the machine's drying system, next to the heat exchanger. For more information, see section *7.18 Drying system, page 192*. It measures the humidity level in the outgoing drying air to determine if the goods in the chamber have achieved a sufficient level of dryness.

4.1.4.5 Pressure monitoring, drying system

A differential pressure sensor is placed in the drying system, behind the electrical cabinet. The sensor triggers an alarm if the differential pressure deviates too much from the preset value. A high or low differential pressure can indicate problems with the HEPA filter.

4.1.4.6 Pressure monitoring, water circulation

The pressure monitoring, water circulation monitors the water pressure from the rotating spray arms.

4.1.4.7 Spray arm monitoring SWRSS (option)

Getinge's system for spray arm monitoring (SWRSS) monitors the spray arm rotation. The SWRSS generates a warning if any loading mistake has been made. The system checks that all installed spray wings rotate at least at 5 revolutions per minute.

If the problem is not solved within 15 minutes after the warning, an alarm is triggered.

When SWRSS is installed, there is special equipment available for the washer-disinfector:

- SWRSS evaluation unit
- Antennas
- Tag inside the spray nozzles

When SWRSS is installed, there is special equipment available for the wash cart:

- Trolley indication tag
- Tag inside the spray nozzles
- Clamps for level grids

A spray nozzle with a tag inside is always black.

4.1.4.8 Supervisor (option)

The Supervisor system is a freestanding monitoring system with sensors that warn if the measured values deviate from those that were preset.

The supervisor system is operated from the touchscreen. Alarms from the supervisor system display the prefix **S** in the error message. For more information, see section *4.3.10 View alarms and messages, page 53*.

Depending on the configuration, the Supervisor system monitors the following wash parameters:

- Temperature in the chamber
- Conductivity in the rinse water

- Pressure in the circulation circuit
- Detergent flow between the container and the chamber.

4.2 Program introduction

This section includes general information about programs. Settings and changes are done from the touchscreen and the Web server interface. For more information, see section *4.3.11 Manage programs, page 55* and *4.4.13 Programs, page 84*.

Each washer-disinfector is equipped with a configuration of factory installed programs.

New programs can be created by copying factory installed programs.

Access rights

- Authorized service technicians with Z1 rights can create adapted programs and set settings ranges for parameters, values and rights, as per the limitations that are determined by program type.

4.2.1 Terms and abbreviations

Name	Description
Base program	The base program contains all phases in their correct order. Each base program is product-specific and can therefore not be used for products that it is not intended for.
Adapted program	New program adaptation is made from a copy of the factory-installed program. It can be more restrictive, but cannot have more permissions than its original.
Parameter	Settings/values for different phases in a program (pressure, temperature, time, etc.).
Performance qualification (PQ)	Test to ensure that all equipment functions in accordance with preset values and given product specifications.
Phase	Step in program process.
Program configuration file	The file containing all programs and settings (including button configuration) on the touchscreen. Can be used as a backup on an existing machine or to move a program from one machine to another. Note that the machine type and program version must be identical.
Program Type	Determines program sequences, parameters and maximum setting ranges for the programs that are built on the program type.
Program configuration's control figure	The control figure is unique to the program's configuration and its parameter settings. The control figure must match the safety copy's figure after resetting the program.

4.2.2 Safety/Responsibility

WARNING!

Risk of contaminated load

The disinfection result can be compromised if there is a failure to achieve and validate effective disinfection cycles.

- Follow instructions in the Service manual.

When adapting a program and before using the machine in production, a new Performance Qualification must be done.

Before changing a program parameter, a reference file should be created. The reference file is used as a restore file if the new parameters create problems that are difficult to resolve.

After adapting a program, the machine's functions should be re-validated. The scope of the validation depends on the type of adaptation and the local rules that apply. If full validation is not done, then the reason for this must be described and attached to the validation report.

At least one program is tested as per EN15883, but several factors affect the cleaning/disinfection result. These include type of goods, level of soiling, type of detergent, water quality, water temperature, etc. Parameters must be adapted according to the environment the machine is to be used in.

Responsibility of customer/manufacturer

- The customer has the responsibility to keep themselves updated on applicable application rules that concern testing and validation. If a program is adapted by the customer, then a new validation must be carried out in consultation with the customer's disinfection manager. If there are local rules and regulations or specific customer processes that require more documentation, these must also be carried out by the customer.
- The manufacturer is responsible for the machine fulfilling the criteria in the applicable directive.

For more information about creating and using a reference file, see section *4.3.11.4 Revert a program to a previous version, page 59*.

4.2.3 Program parameters

A number of standard programs are preset on the washer-disinfector upon delivery. See the program sheets for more detailed information about each program. As a standard, the preset values for the programs are used, but it may be necessary to adjust parameters to match a specific wash process.

For editing a program parameter, see section *4.3.11.1 Edit program parameters, page 56* or *4.4.13.3 Parameters, page 88*.

4.2.4 Available program parameters and ranges

Parameter	Program type/Setting range			
	Thermal disinfect	**Chem disinfection**	**BGA disinfect**	**Glassware**
Pre Rinse[1]	0/1/2/3	0/1/2/3	0/1/2/3	0/1/2/3
Pre Rinse time	0-1800 s	0-1800 s	0-1800 s	0-1800 s
Pre Rinse Water [2]	1/2/3/4	1/2/3/4	1/2/3/4	1/2/3/4
Wash 1[3]	0/1	0/1	1	0/1
Wash 1 temp	0-93 °C	0-93 °C	0-93 °C	0-93 °C
Wash 1 time	0-1800 s	0-1800 s	0-3600 s	0-1800 s
Wash 1 Water [2]	1/2/3/4	1/2/3/4	1/2/3/4	1/2/3/4
Wash 1 Dos pump	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4
Wash 1 Dos amount	0-700 ml	0-700 ml	0-700 ml	0-700 ml
Wash 1 Dos temp	0-85 °C	0-85 °C	0-85 °C	0-85 °C
Wash 2[3]	0/1	0/1	0/1	0/1
Wash 2 temp	0-93 °C	0-93 °C	0-93 °C	0-93 °C
Wash 2 time	0-1800 s	0-1800 s	0-1800 s	0-1800 s
Wash 2 Water [2]	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4
Wash 2 Dos pump [4]	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4
Wash 2 Dos amount	0-700 ml	0-700 ml	0-700 ml	0-700 ml
Wash 2 Dos temp	0-85 °C	0-85 °C	0-85 °C	0-85 °C
Wash 3[3]	0/1	0/1	0/1	0/1
Wash 3 temp	0-93 °C	0-93 °C	0-93 °C	0-93 °C
Wash 3 time	0-1800 s	0-1800 s	0-1800 s	0-1800 s
Wash 3 Water [2]	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4
Wash 3 Dos pump [4]	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4
Wash 3 Dos amount	0-700 ml	0-700 ml	0-700 ml	0-700 ml
Wash 3 Dos temp	0-85 °C	0-85 °C	0-85 °C	0-85 °C
Neutralizing [3]	0/1	-	0/1	0/1
Neutr temp	0-93 °C	-	0-93 °C	0-93 °C
Neutr time	0-1800 s	-	0-1800 s	0-1800 s
Neutr Water [2]	0/1/2/3/4	-	0/1/2/3/4	0/1/2/3/4
Neutr Dos pump [4]	0/1/2/3/4	-	0/1/2/3/4	0/1/2/3/4
Neutr Dos amount	0-700 ml	-	0-700 ml	0-700 ml
Neutr Dos temp	0-85 °C	-	0-85 °C	0-85 °C
Chemical Disinf.[3]	-	0/1	-	-
Chem Disinf temp	-	0-93 °C	-	-

Parameter	Program type/Setting range			
	Thermal disinfect	**Chem disinfection**	**BGA disinfect**	**Glassware**
Chem Disinf time	-	0-1800 s	-	-
Chem Disinf Water[2]	-	1/2/3/4	-	-
Chem Disinf Dos pump [4]	-	0/1/2/3/4	-	-
Ch Disinf Dos amount	-	0-700 ml	-	-
Chem Disinf Dos temp	-	0-85 °C	-	-
Rinse[1]	0/1/2/3	0/1/2/3	0/1/2/3	0/1/2/3
Rinse time	0-1800 s	0-1800 s	0-1800 s	0-1800 s
Rinse Water[2,5]	1/2/3/4/5/6/11-16	1/2/3/4/5/6/11-16	1/2/3/4/5/6/11-16	1/2/3/4/5/6/11-16
Final Rinse [3]	1	0/1	0/1	0/1
Final Rinse temp	0-93 °C	0-93 °C	0-93 °C	0-93 °C
Final Rinse time	0-3600 s	0-3600 s	0-3600 s	0-3600 s
Final Rinse Water [2, 5]	1/2/3/4/5/6/11-16	1/2/3/4/5/6/11-16	1/2/3/4/5/6/11-16	1/2/3/4/5/6/11-16
Final Rinse Dos pump [4]	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4	0/1/2/3/4
F. Rinse Dos amount	0-700 ml	0-700 ml	0-700 ml	0-700 ml
Final Rinse Dos temp	0-85 °C	0-85 °C	0-85 °C	0-85 °C
Drying [3]	0/1	0/1	0/1	0/1
Drying Time	0-3600 s	0-3600 s	0-3600 s	0-3600 s
Max Temp Chamber	60-90 °C	60-90 °C	60-90 °C	50-90 °C
Temperature Dryer[6]	60-110 °C	60-110 °C	60-110 °C	60-110 °C
Humidity level [7]	0-50 %	0-50 %	0-50 %	0-50 %
Humidity delay time[8]	0-3600 s	0-3600 s	0-3600 s	0-3600 s
Cooling time [9]	0-1800 s	0-1800 s	0-1800 s	0-1800 s
Cooling temp [10]	60-90 °C	60-90 °C	60-90 °C	60-90 °C
F Rinse Conductivity [11]	0/1/2/3	0/1/2/3	0/1/2/3	0/1/2/3
F Rinse Cond level [12]	0-200 µs	0-200 µs	0-200 µs	0-200 µs
Disinfection type [13]	1/2/3/4	-	3/4	0/1/2
A0-print	0/1	-	0/1	0/1
A0-level [15]	600-6000	600-6000	600-6000	600-6000
Fill delay time [16]	0-60 s	0-60 s	0-60 s	0-60 s
Start Drain time	0-60 s	0-60 s	0-60 s	0-60 s

Parameter	Program type/Setting range			
	Thermal disinfect	**Chem disinfection**	**BGA disinfect**	**Glassware**
Drain time	0-120 s	0-120 s	0-120 s	0-120 s
Pipette [17]	0/1	0/1	0/1	0/1
Pulse on [18]	0-120 s	0-120 s	0-120 s	0-120 s
Pulse off [19]	0-120 s	0-120 s	0-120 s	0-120 s
Open SS Door [20]	0/1	0/1	0/1	0/1
SWRSS [21]	0/1	0/1	0/1	0/1

Parameter	Program type/Setting range				
	Drying	**Deliming**	**Complete drain**	**Door test**	**Machine test**
Wash 1	-	0/1	-	-	-
Wash 1 temp	-	0-93 °C	-	-	0-93 °C
Wash 1 time	-	0-1800 s	-	-	0-1800 s
Wash 1 Water	-	1/2/3/4	-	-	-
Wash 1 Dos pump [4]	-	0/1/2/3/4	-	-	-
Wash 1 Dos amount	-	0-700 ml	-	-	0-700 ml
Wash 1 Dos temp	-	0-85 °C	-	-	0-85 °C
Rinse2	-	0/1/2/3	-	-	0/1/2/3
Rinse time	-	0-1800 s	-	-	0-1800 s
Rinse Water [2, 5]	-	1/2/3/4/5/6/11-16	-	-	1/2/3/4/5/6/11-16
Final Rinse	-	0/1	-	-	0/1
Final Rinse temp	-	0-93 °C	-	-	0-93 °C
Final Rinse time	-	0-3600 s	-	-	0-3600 s
Final Rinse Water [2, 5]	-	1/2/3/4/5/6/11-16	-	-	1/2/3/4/5/6/11-16
Final Rinse Dos pump [4]	-	0/1/2/3/4	-	-	0/1/2/3/4
F. Rinse Dos amount	-	0-700 ml	-	-	0-700 ml
Final Rinse Dos temp	-	0-85 °C	-	-	0-85 °C
Drying [3]	-	0/1	-	-	0/1
Drying Time	0-3600 s	0-3600 s	-	-	0-3600 s
Max Temp Chamber	60-90 °C	60-90 °C	-	-	60-90 °C
Temperature Dryer [6]	60-110 °C	60-110 °C	-	-	60-110 °C
Humidity level [7]	0-50 %	0-50 %	-	-	0-50 %
Humidity delay time [8]	0-3600 s	0-3600 s	-	-	0-3600 s

Parameter	Program type/Setting range				
	Drying	**Deliming**	**Complete drain**	**Door test**	**Machine test**
Cooling time [9]	0-1800 s	0-1800 s	-	-	0-1800 s
Cooling temp [10]	60-90 ºC	60-90 ºC	-	-	60-90 ºC
F Rinse Conductivity [11]	-	0/1/2/3	-	-	0/1/2/3
F Rinse Cond level [12]	-	0-200 µs	-	-	0-200 µs
Disinfection type [13]	-	-	-	-	0/1/2
A0-print[14]	-	-	-	-	0/1
A0-level [15]	-	-	-	-	600-6000
Fill delay time [16]	-	0-60 s	-	-	0-60 s
Start Drain time	0-60 s	0-60 s	-	0-60 s	0-60 s
Drain time	-	0-120 s	0-120 s	-	0-120 s
Pipette [17]	-	0/1	-	-	0/1
Pulse on [18]	-	0-120 s	-	-	0-120 s
Pulse off [19]	-	0-120 s	-	-	0-120 s
Open SS Door [20]	0/1	0/1	-	-	0/1
SWRSS [21]	-	0/1	-	-	0/1

	Description	Comment
1	The number of times the phase is run.	
2	0 = No filling (the water from the previous phase is used) 1 = Cold water 2 = Hot water 3 = Mixed water 4 = Deionized water 5 = Booster tank 6 = Hot water in first (and second) rinse and deionized water in last	
3	0 = Phase deactivated 1= Phase activated	
4	0 = No dosing 1 = Pump 1 2 = Pump 2 3 = Pump 3 4 = Pump 4	

	Description	Comment
5	Normal water level: 1 = Cold water 2 = Hot water 3 = Mixed water 4 = Deionized water 5 = Booster tank 6 = Hot water in first (and second) rinse and deionized water in last. Lower water level and lower pump velocity: 11 = Cold water 12 = Hot water 13 = Mixed water 14 = Deionized water 15 = Booster tank 16 = Hot water in first (and second) rinse and deionized water in last	Only for S-8668T.
6	The set point for the drying unit.	
7	The humidity level that should be equivalent to dry goods. 0 = Not used	
8	The shortest drying time when humidity control is activated	
9	The time the drying fan should run with the heater off.	
10	The chamber temperature that should be reached before stopping the drying fan (with the heater off)	
11	The number of refills in the final rinse phase before the alarm for high conductivity is triggered. 0 = Conductivity check deactivated	
12	Maximum permitted conductivity level after filling in Final Rinse	
13	0 = No disinfection 1 = Time/Temperature-controlled disinfection in Final rinse 2 = A_0 controlled disinfection in Final rinse 3= Time/Temperature controlled disinfection in Wash 1 4 = A_0 controlled disinfection in Wash 1	
14	0 = Deactivated (value not printed out) 1 = Activated (value printed out)	
15	The set A_0 value at A_0 controlled disinfection. This value must also be changed if the Disinfection time/temperature is changed.	
16	Filling time if additional water is needed (time in seconds)	
17	Pipette-/Glassware function. The circulation pump pulses off/on.	
18	Time for each pulse when the circulation pump is on	
19	Time for each pulse when the circulation pump is off	
20	State which door side opens after completed process. 0 = clean side opens (normal function) 1 = soiled side opens	
21	Spray Wing Rotation Supervision System (Flushing monitoring). 0 = Deactivated 1 = Activated	

4.2.4.1 Parameter types

Process-related parameters are managed on the touchscreen or via the Web server according to the following table:

Parameter type	Section	Authorization level	Interface
Program parameter A	*4.4.13.3 Parameters, page 88*	Super Users	Touchscreen **Web server**
Program parameter H	*4.4.13.3 Parameters, page 88*	Authorized service technicians (level 1)	Touchscreen **Web server**
Program parameter I	*4.4.13.3 Parameters, page 88*	Authorized service technicians (level 1)	Touchscreen **Web server**
DIP switches	*4.4.5.1 Machine Setup - DIP Switches, page 66*	Authorized service technicians (level 1)	**Web server**
General parameters	*4.4.5.2 Set general parameters, page 67*	Authorized service technicians (level 1)	**Web server**

Authorization levels are described in section *4.3.6 Manage access rights, page 44*.

4.3 Touchscreen interface

This section handles settings and changes that can be done from the touchscreen. More settings can be done via the **Web server** interface. For more information, see section *4.4 Web server interface, page 60*.

The machine is operated from the operator panel touchscreen. For more information about the touchscreen, see the User manual.

For program-related settings and parameters, see the separate section *4.2 Program introduction, page 27*.

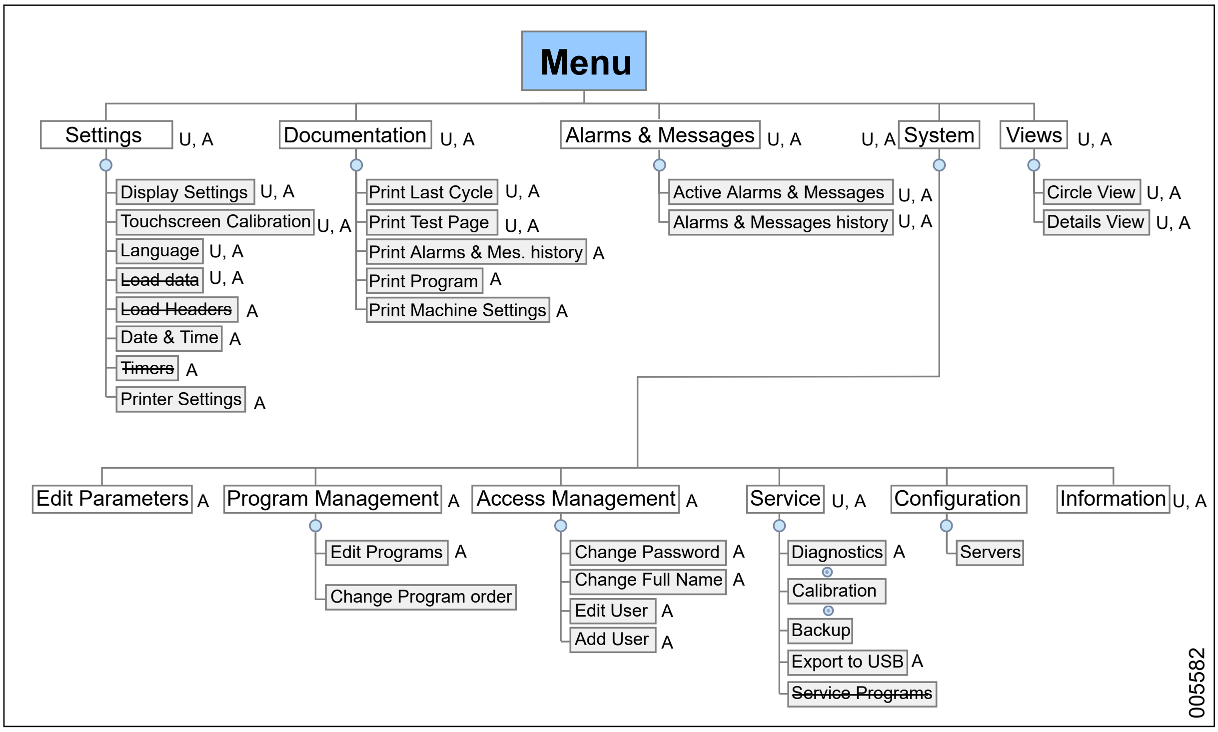
4.3.1 Menu tree

The diagram below shows the structure of the menu tree on the touchscreen.

References to a certain location within the menu structure begin with Path: followed by menu levels stated in bold text, where the different menu levels are differentiated using ">".

Path: **Menu > System > Service > Diagnostics**

Access to certain menu options depends on the authorization level (see section *4.3.6 Manage access rights, page 44*) used when logging on.



The menus that are crossed out are not in use in this configuration.

Access to menu:

U, A = User and Admin have access

A = Admin has access

When logged in as a Getinge Authorized Service Technician, you have access to all menus.

4.3.2 Edit data fields

General information about using the touchscreen, such as navigation and character entry, is described in the User manual. This section describes the data fields used to edit more advanced settings.

To save after editing a data field, tap **Save** to confirm the change.

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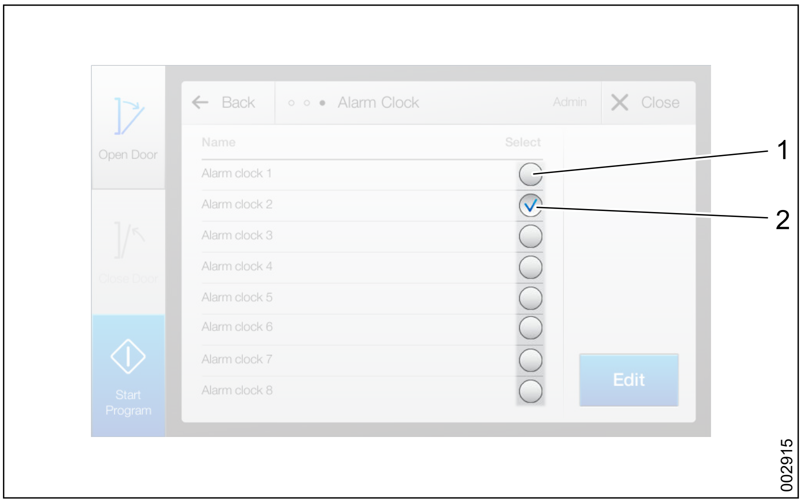
4.3.2.1 Checkboxes

Checkboxes are used to select parameters.



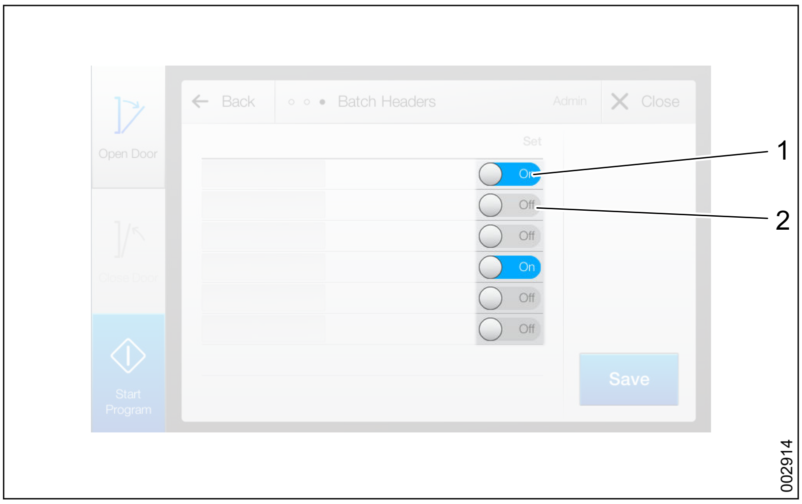
1 Empty checkbox
2 Ticked checkbox

Tick a checkbox by tapping it.

The value you enter or the setting that you change in the following steps will apply for all previously ticked parameters.

4.3.2.2 Off and on buttons

On and **Off** buttons are used to deactivate or activate a specific function.



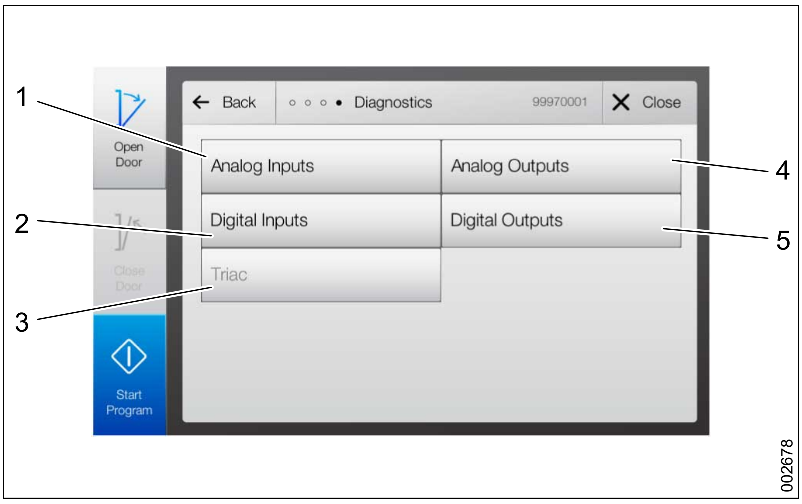
1 **On**
2 **Off**

1. Tap the button to switch between activated and deactivated modes.

4.3.3 Manage inputs and outputs

Check the status of the inputs/outputs or force outputs to troubleshoot the machine.

Path: **Menu** > **System** > **Service** > **Diagnostics**

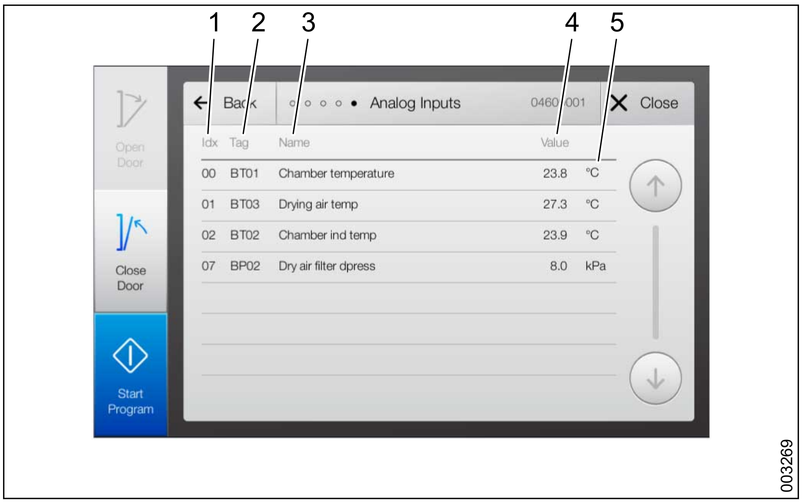


1 **Analog Inputs** (read-only)
2 **Digital Inputs** (read-only)
3 **Triac** This function is not implemented.
4 **Analog Outputs**
5 **Digital Outputs**

4.3.3.1 Analog inputs

Check the values of the analog inputs.

Path: **Menu** > **System** > **Service** > **Diagnostics** > **Analog Inputs**

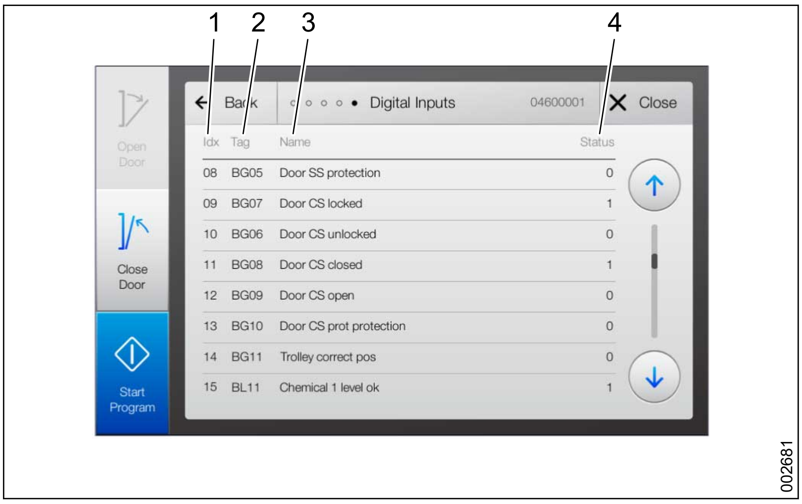


1 **Idx**
2 **Tag**
3 **Name**
4 **Value**
5 **Unit**

4.3.3.2 Digital inputs

Check the status of the digital inputs.

Path: **Menu** > **System** > **Service** > **Diagnostics** > **Digital Inputs**

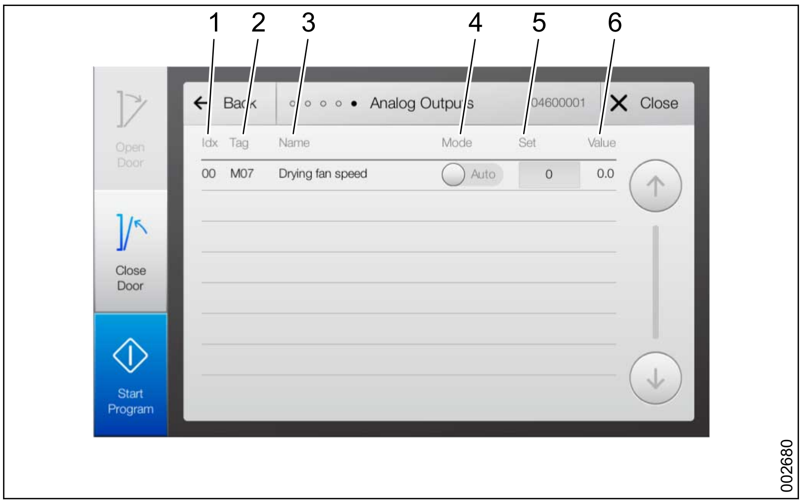


1 **Idx**
2 **Tag**
3 **Name**
4 **Status**

4.3.3.3 Analog outputs

Check the status and force the analog outputs.

Path: **Menu** > **System** > **Service** > **Diagnostics** > **Analog Outputs**



1 **Idx**
2 **Tag**
3 **Name**
4 **Mode**
5 **Set**
6 **Value**

Name	Description
Idx	Index.
Mode	Sets output mode. **Man.** In manual mode, the program's preset values are overrun by the specified value. **Auto**. In automatic mode, the machine uses the program's preset value.
Set	If the output is run in manual mode, a value must be set. The value displays the percentage of the set maximum.
Value	Views the current output value.

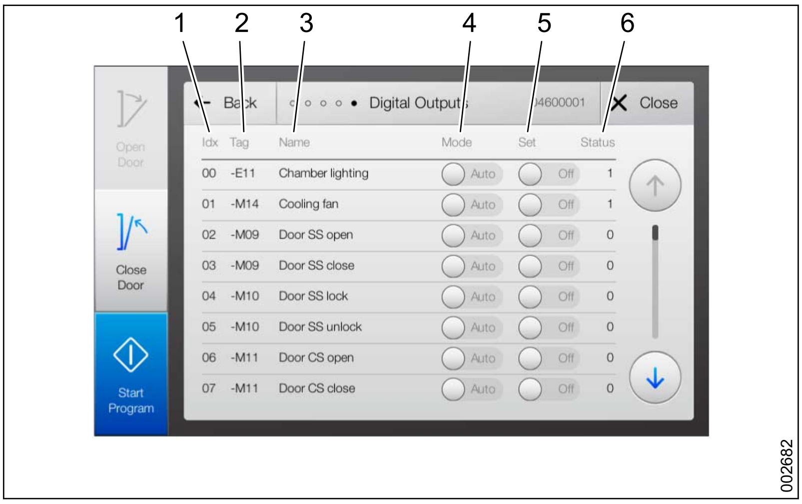
The new values are valid directly after tapping OK.

Programs cannot be started if an output is in manual mode.

4.3.3.4 Digital outputs

Check the status and force the digital outputs.

Path: **Menu** > **System** > **Service** > **Diagnostics** > **Digital Outputs**



1 **Idx**
2 **Tag**
3 **Name**
4 **Mode**
5 **Set**
6 **Status**

Name	Description
Idx	Index.
Mode	Sets output mode. **Man.** In manual mode, the program's preset values are overrun by the specified value. **Auto**. In automatic mode, the machine uses the program's preset value.
Set	If the analog output is run in manual mode, the signal can be toggled to **On** or **Off**.
Status	Indicates the status of the digital output.

Programs cannot be started if an output is in manual mode.

The new values are valid directly after tapping OK.

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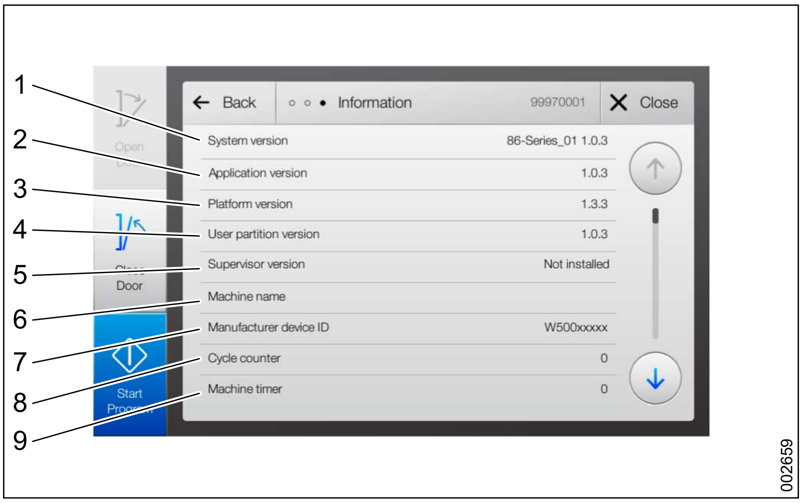
4.3.4 Display machine information

Path: **Menu** > **System** > **Information**.



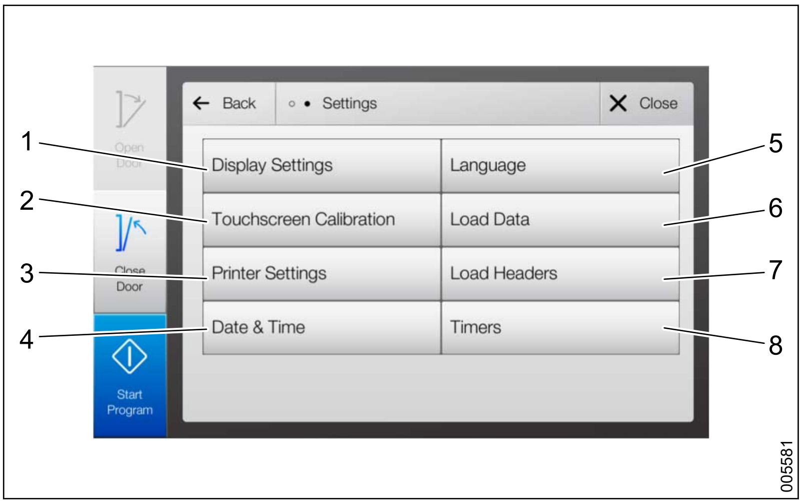
1 **System version**
2 **Application version**
3 **Platform version**
4 **User partition version**
5 **Supervisor version**
6 **Machine Name**
7 **Manufacturer device ID**
8 **Cycle counter**
9 **Machine timer**
10 **Doors closed**
11 **Doors sealed**
12 **External IP address**
13 **Recovery code**
14 **Date & Time**
15 **Panel serial number**

Name	Description
System version	Displays the machine model and software version.
Application version	Displays the application version.
Platform version	Displays the platform software version (Getinge G1 control system).
User partition version	Displays the version.
Supervisor version	Displays the Supervisor system version (if connected).

Name	Description
Machine Name	Displays the machine name or designation. For more information, see section *4.4 Web server interface, page 60*.
Manufacturer device ID	Displays the machine number.
Cycle counter	Displays the total number of completed cycles.
Machine timer	Displays the total number of hours the machine has been running.
Doors closed	This function is not implemented.
Doors sealed	This function is not implemented.
External IP address	Displays the IP address for external connection. For more information, see section *4.4 Web server interface, page 60*
Recovery code	Displays a code for recovering dates or times.
Date & Time	Displays date and time.
Panel serial number	Displays the operator panel serial number.

4.3.5 Settings

Path: **Menu** > **Settings**



1 **Display Settings**
2 **Touchscreen Calibration**
3 **Printer Settings**
4 **Date & Time**
5 **Language**
6 **Load data**
7 **Load Headers**
8 **Timers**

Basic settings are described in the User manual.

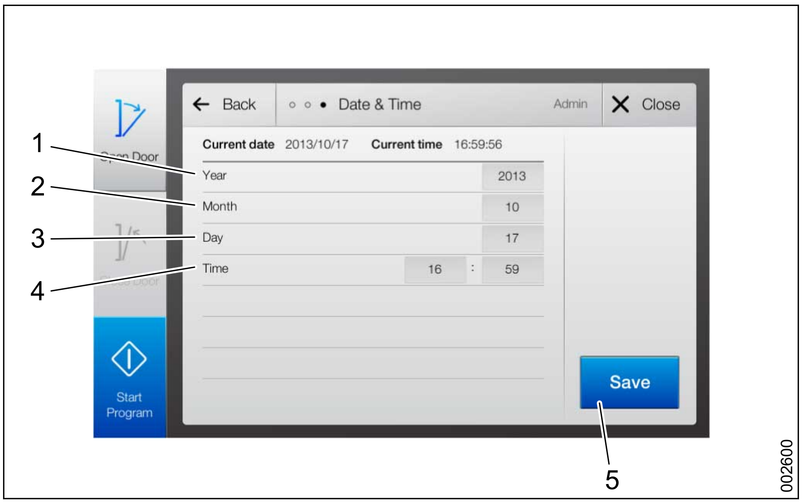
4.3.5.1 Set batch headers

This function is not implemented.

4.3.5.2 Set date and time

Set the date and time for the machine.

Path: **Menu** > **Settings** > **Date & Time**



1 **Year**
2 **Month**
3 **Day**
4 **Time**
5 **Save**

1. Tap a field and the keyboard appears.
2. Set the desired value using the keyboard.
3. Tap **Save** to confirm the change.

4.3.5.3 Set timers

This function is not implemented.

4.3.5.4 Setup printer

Sets how often log data from the wash cycle should be collected.

Depending on how the program is designed, slow or fast log intervals are used for monitoring different phases of the program. Different values can be set for slow and fast log intervals.

1 **Fast log interval**
2 **Slow log interval**

1. Tap a field to edit the value.
2. Tap **Save** to confirm the change.

4.3.6 Manage access rights

4.3.6.1 User names and passwords

A valid user name and password is required to log on to the system.

Logging in as a Getinge Authorized service technician requires the correct Date & Time setting. If the setting is incorrect, log in as 'Admin' with the password '243160' and set the correct date and time.

For more information, see section *4.3.5.2 Set date and time, page 43*.

Without logging in, the user has the authorizations for the user type of Default User. The machine can be used for standard operations, but the user cannot confirm alarms or run programs without a disinfection phase.

Factory settings:
User name = User
Password = 1234
The factory settings may be changed during commissioning.

To add or manage users, log in as Admin. For more information, see section *4.3.6.3 Setup access management, page 47*.

User name: **Admin** (belongs to user group Administrators and Super Users)
Password: **243160**

See section *4.3.6.2 User groups and authorization, page 45* for more information about users, user groups and user types.

4.3.6.2 User groups and authorization

A user can belong to one or more user groups (Users, Super Users, etc.) with different authorization levels (A-Z). The level gives access to different functions.

AST1 = Authorized service technicians level 1, AST2 = Authorized service technicians level 2						
Level/Name	**Access to functions**	**Users**	**Super Users**	**Admin**	**AST1**	**AST2**
Basic process control	Confirm alarms Start password-protected programs	A	A		A	A
Extended process control	Not used		B		B	B
Basic settings	Display settings: - Language - Touchscreen calibration	C	C	C	C	C
Extended settings	Printer settings: - Slow/fast log intervals Machine settings: - Date and time			D	D	D
Basic program	Edit A parameters		E		E	E
Extended program	Set program as Performance Qualified (PQ)		F		F	F
Basic service	Diagnostics: - Analog inputs - Digital inputs		G		G	G
Extended service	Diagnostics: - Analog outputs - Digital outputs - Backup				H	H
Basic access management	Change own: - Full name - Password		I	I	I	I

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AST1 = Authorized service technicians level 1, AST2 = Authorized service technicians level 2						
Level/Name	**Access to functions**	**Users**	**Super Users**	**Admin**	**AST1**	**AST2**
Extended access management	Add user Edit user Delete user			J	J	J
Basic documentation	Print report for last cycle. Print test page	K	K	K	K	K
Extended documentation	Print history for alarms and messages. Print programs. Print Machine Settings.			L	L	L
Calibration	Calibration: - Analog inputs - Dose monitors				M	M
Password 1	Run password-protected (1) programs.	N	N	N	N	N
Password 2	Run password-protected (2) programs.				O	O
Application control	Not used.				P	P
Program control	Modify program names. Modify program order.				Q	Q
Getinge limited (Service)					Z0	Z0
Getinge limited (Program adaptation)					Z1	Z1

There are three standard users that cannot be modified and that do not belong to any user group: Default User, Authorized service technicians and Restricted Services.

Default User:

- used if no login has taken place.
- only has the authorization to start programs.

Authorized service technicians:

- requires special login details.
- may only be used by personnel who are approved by the machine manufacturer.
- has access to all functions and settings on the touchscreen.
- can activate and access the Web server (see section *4.4 Web server interface, page 60*).
- can access program adaptation (see section *4.4 Web server interface, page 60*).

Restricted Services:

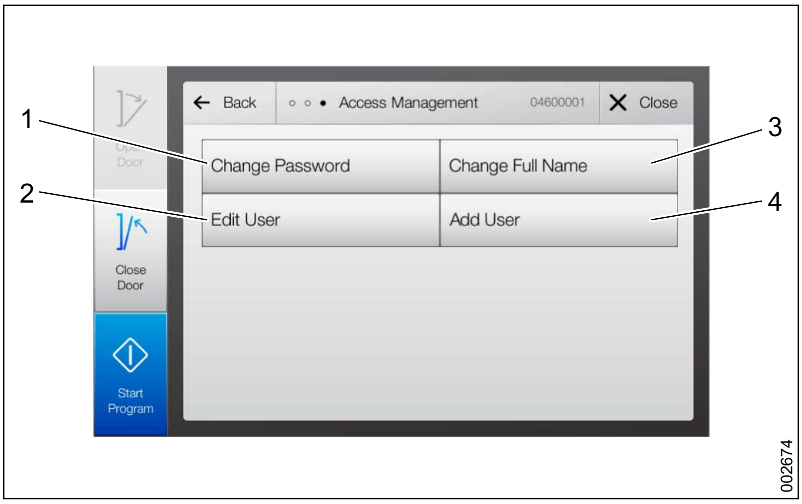
The special user group Restricted Services has the same access rights as the group Authorized service technicians, except for Z1 (A-Q + Z0).

To enable this group, a software license key must be ordered and entered.

4.3.6.3 Setup access management

Add or change a user and change the access rights for a user.

Path: **Menu** > **System** > **Access Management**



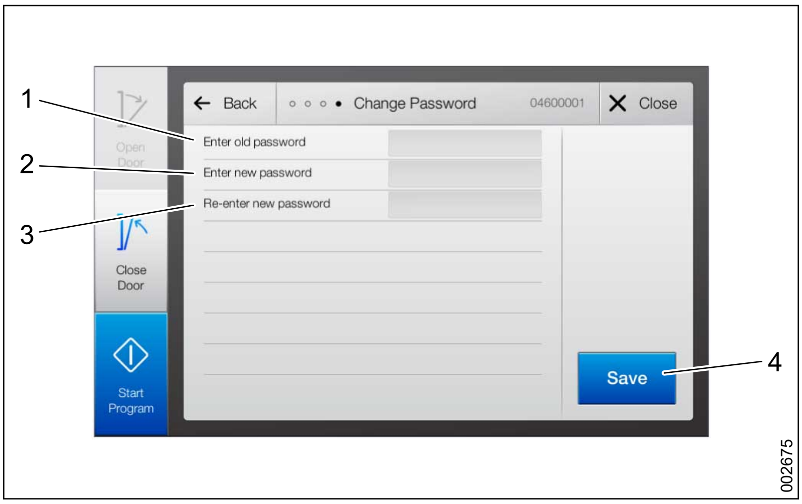
1 **Change Password**
2 **Edit User**
3 **Change Full Name**
4 **Add User**

Additional settings are accessible via the Web server. For more information, see section *4.4 Web server interface, page 60*.

4.3.6.3.1 Change password

All logged-in users (regardless of authorization) can change their password.

Path: **Menu** > **System** > **Access Management** > **Change Password**



1 **Enter Old Password**
2 **Enter new password**
3 **Re-enter new password**
4 **Save**

1. Enter the old password in the upper field.
2. Enter the new password in the next field.
3. Re-enter the password in the last field.
4. Tap **Save** to confirm the change.

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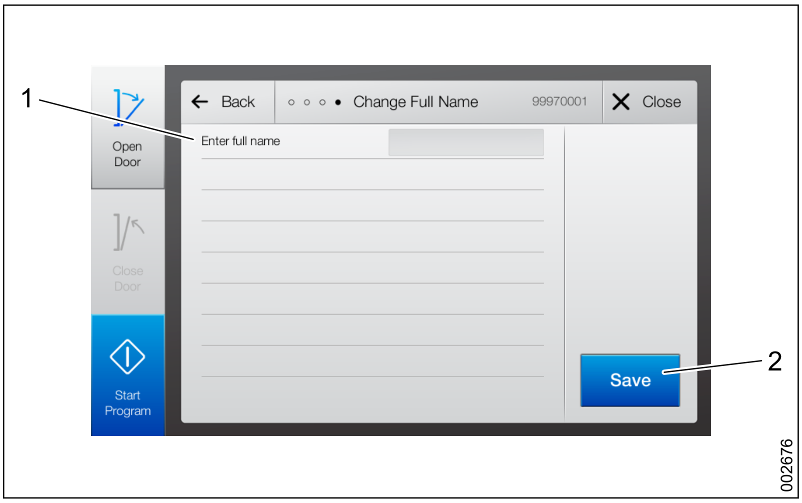
4.3.6.3.2 Change name

Path: **Menu** > **System** > **Access Management** > **Change Full Name**

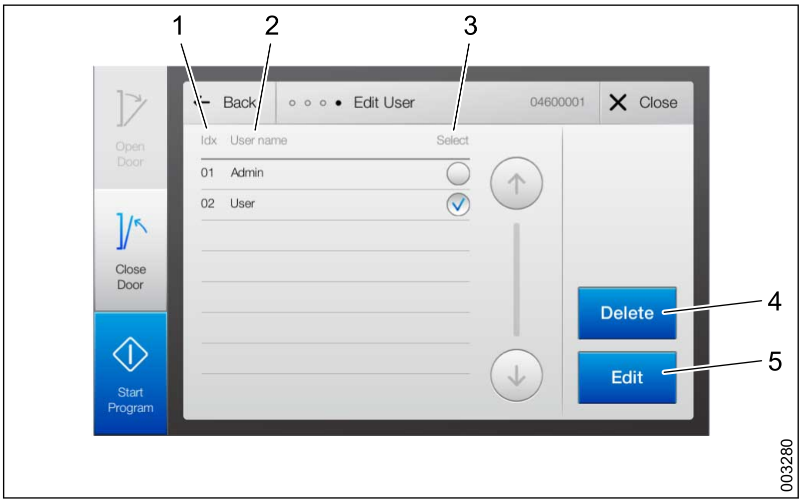


1 **Enter full name**
2 **Save**

1. Tap the field and enter full name.
2. Tap **Save** to confirm the change.

4.3.6.3.3 Edit user account

Path: **Menu** > **System** > **Access Management** > **Edit User**



1 **Idx**
2 **User name**
3 **Select**
4 **Delete**
5 **Edit**

1. Tap **Edit** to proceed to the next step.
2. Tap a field and make changes. See the image below.
3. Tap **Save** to confirm the change.

1 **User name**
2 **New password**
3 **Confirm password**
4 **Full name**
5 **Locked**
6 **Disabled**
7 **Select Groups**
8 **Save**

Name	**Description**
User name	Changes the name of the current user. The user name is used (with password) to log into the system.
New password	Changes the password for the current user.
Confirm password	Re-enter to confirm the new password.
Full name	Changes the full name of the current user.
Locked	Locks or unlocks the current user by tapping the button. A user account is locked after too many incorrect login attempts.
Disabled	Activates/disables a user account by tapping the button. A user cannot log in if deactivated. The user account may be activated at any time with previous user name and password.

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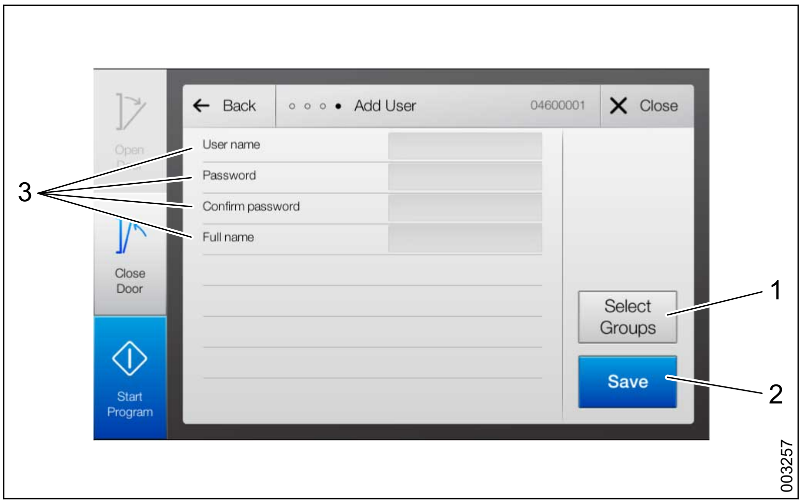
4.3.6.3.4 Add new user

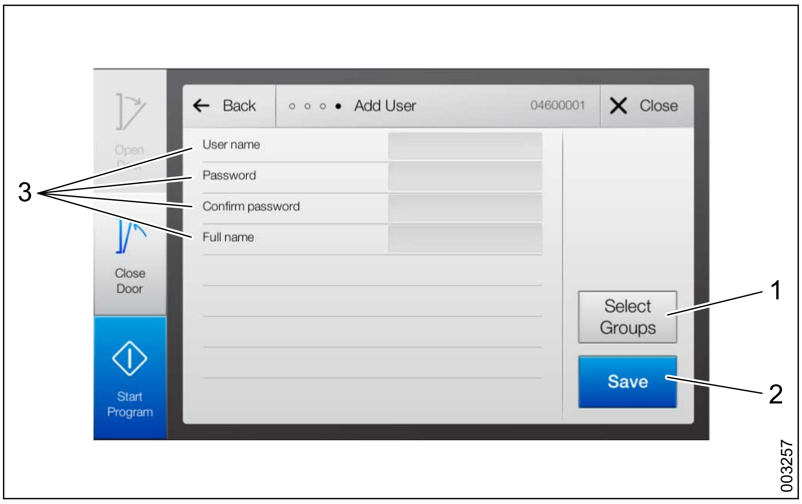
Path: **Menu** > **System** > **Access Management** > **Add User**

Enter the user's information and select the user group to assign the correct authorization to the user.



1 **Select Groups**	4 **Confirm password**
2 **Save**	5 **Password**
3 **Full name**	6 **User name**

1. Tap **Select Groups** to select a user group.



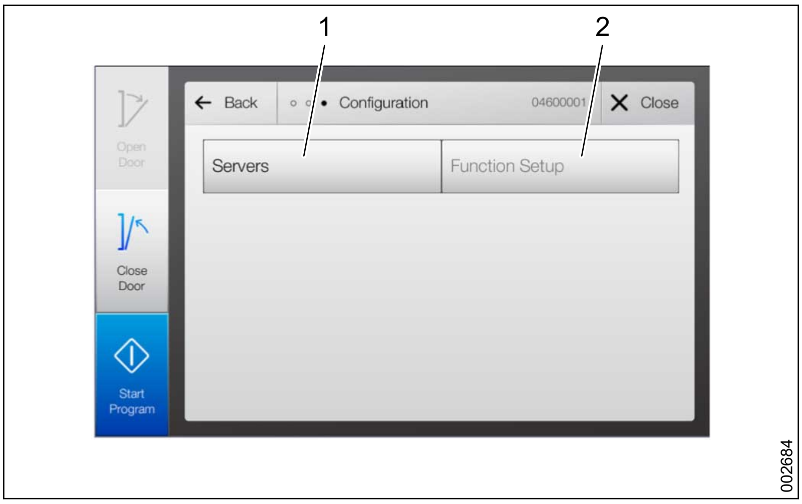
1 **Group**
2 **Select**

2. Tap **Close** to close the window and return to the user's details.
3. Tap **Save** to create the user.

4.3.7 Configure server

Activate the Web server.

Path: **Menu > System > Configuration**



1 **Servers**
2 **Function Setup**

Name	Description
Servers	Activates the Web servers for access via an external PC (special authorization required).

4.3.8 Printouts

4.3.8.1 Print emergency print-outs (login required)

A PDF of the report can be printed or stored on a USB-memory stick. The last cycle, a test page, Alarms and Warnings history, Programs and Machine Settings are included in the report.

Path: **Menu > Documentation> Print Last Cycle**.

1. Tap **Print**.

4.3.8.2 Print test page (login required)

Path: **Menu > Documentation > Print Test Page**.

1. Tap **Print**.

4.3.8.3 Print alarms and messages history (login required)

A list of previously triggered alarms and displayed messages may be printed.

Path: **Menu** > **Documentation** > **Print Alarms & Mes. History**

1. Tick a box to select the number of alarms and messages to print.
2. Tick the box for confirmation if confirmations are to be included in the list.
3. Tap **Continue**.
4. Tap **Print**.

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4.3.8.4 Print program (login required)

A list of all program parameters may be printed to use as a program reference.

Path: **Menu** > **Documentation** > **Print Programs**

1. Select a program for printing.
2. Tap **Print**.
3. Tap **Print** again.

4.3.8.5 Print machine settings

A list of all machine settings may be printed.

Path: Menu > **Documentation** > **Print Machine Settings**

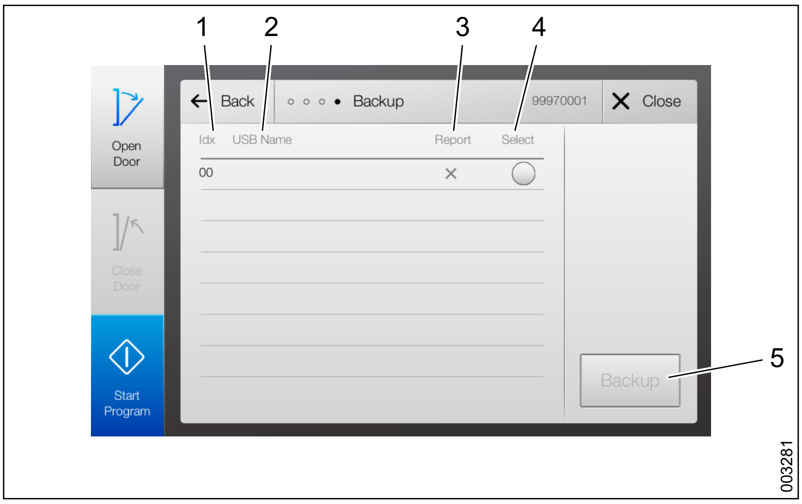
1. Tap **Print**.

4.3.9 Backup and export

4.3.9.1 Backup

Path: **Menu** > **System** > **Service** > **Backup**

A backup copy of the system's configuration files can be used to re-load the machine settings, if the operator panel needs to be replaced. It is possible to save backup files on the USB memory stick regardless of whether it is used for reports or not.



1 **Idx**
2 **USB Name**
3 **Report**
4 **Backup**

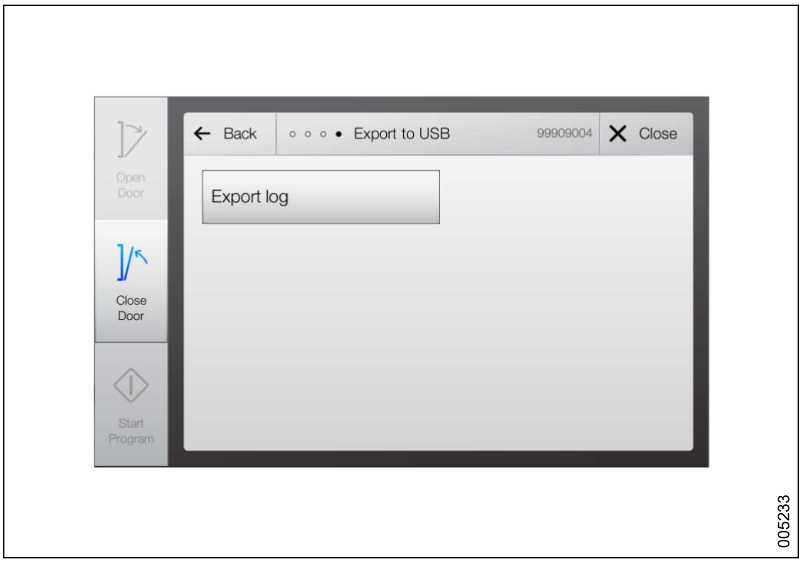
Name	Description
Report	x = The backup is used for reports
Select	Tick to select for backup

1. Select a USB memory stick.
2. Tap **Backup** to save the file to the selected location.

4.3.9.2 Export log

Path: **Menu** > **System** > **Service** > **Export to USB**

An exported system log file can be used for troubleshooting the control system.



4.3.10 View alarms and messages

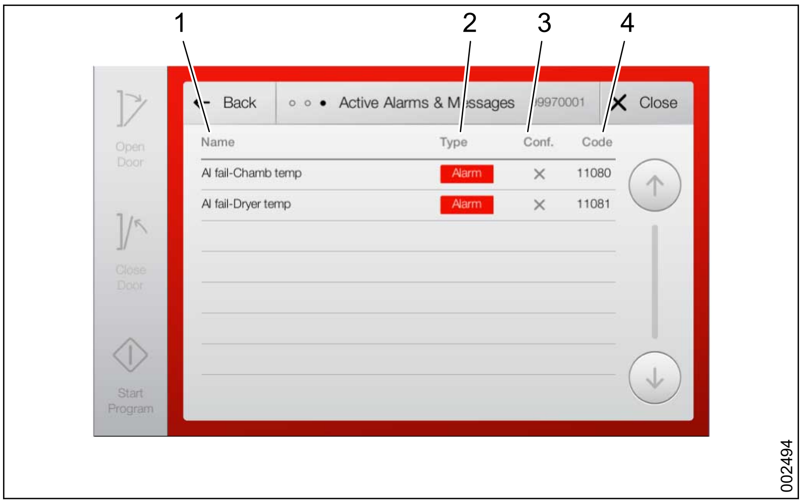
Confirmation of alarms and messages is described in section *9 TROUBLESHOOTING, page 215*.

4.3.10.1 Active Alarms & Messages

Path: **Menu**> **Alarms & Messages** > **Active Alarms & Messages**

This view displays all active alarms (red) and warnings (yellow).

Both confirmed and unconfirmed alarms and warnings are displayed. An alarm or warning must be confirmed and corrected to be removed from the list.



1 **Name**
2 **Type**
3 **Conf.**
4 **Code**

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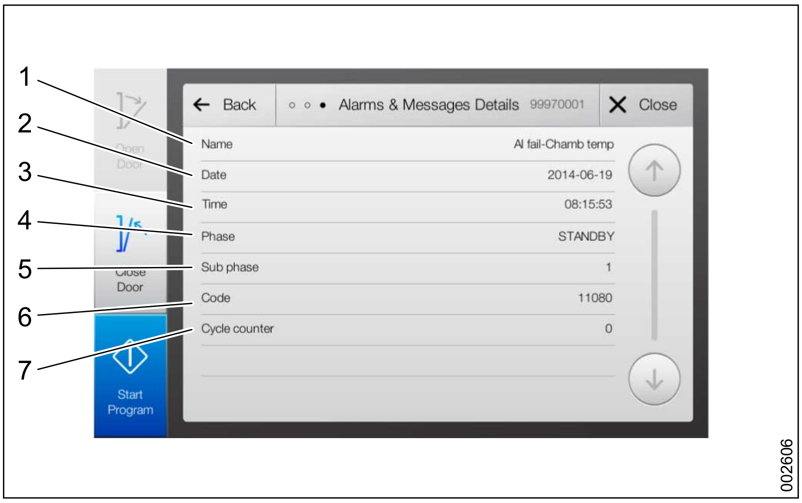
4.3.10.2 Alarms & Messages History

Path: **Menu** > **Alarms & Messages** > **Alarms & Messages History**

The view displays the last 100 alarms and warnings.

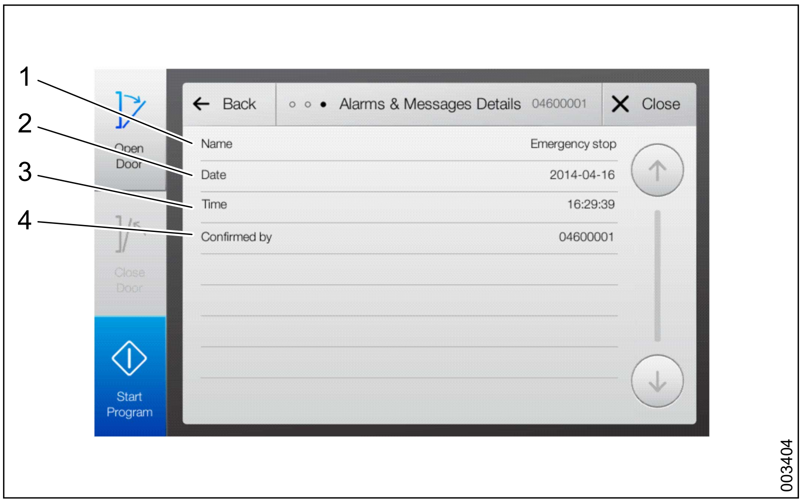
1 **Name**	4 **Time**
2 **Type**	5 **More**
3 **Date**	

If an exception is unconfirmed, the following screen is displayed:



1 **Name**	5 **Sub phase**
2 **Date**	6 **Code**
3 **Time**	7 **Cycle counter**
4 **Phase**	

If an exception is confirmed, the following screen is displayed:



1 **Name**
2 **Date**
3 **Time**
4 **Confirmed by**

Name	Description
Type	Displays the type of exception: Warning (yellow) or Alarm (red).
More	Tap **More** to see detailed information about the exception.
Name	Displays the name of the exception.
Date	Displays the date of the occurrence.
Time	Displays the time of the occurrence.
Phase	Displays the phase in which the exception occurred
Sub phase	Displays the sub phase in which the exception occurred
Code	Displays the exception code.
Cycle counter	Displays the cycle during which the exception occurred.
Confirmed by	Displays which user (user ID) confirmed the exception.

4.3.11 Manage programs

This section handles settings and changes that can be done in relation to a program. For more general information about programs, such as parameters and program ranges, see section *4.2 Program introduction, page 27*.

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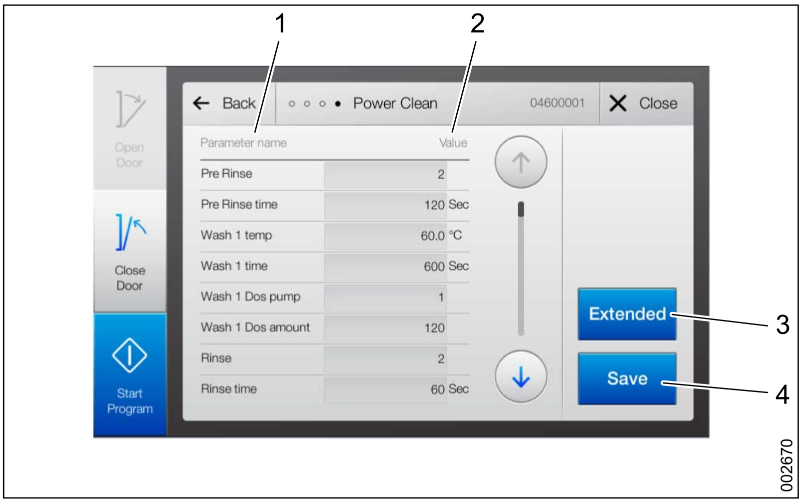
4.3.11.1 Edit program parameters

Path: **Menu** > **System** > **Edit Parameters**

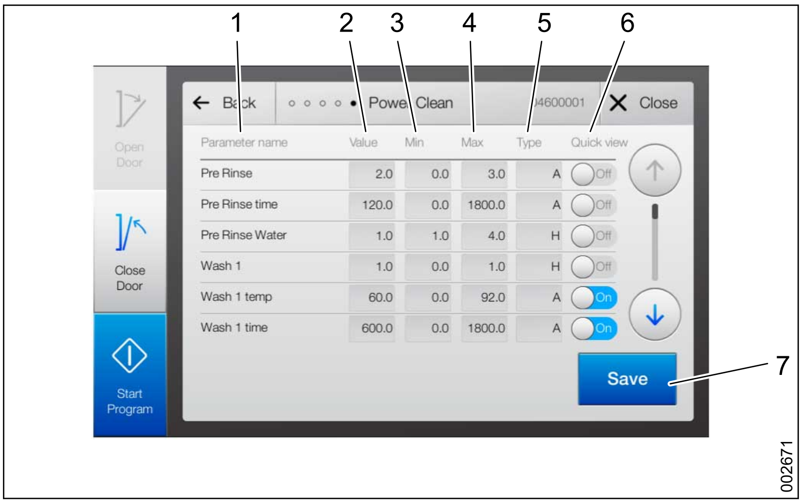


1 **Parameter Name**
2 **Value**
3 **Extended**
4 **Save**

See section *4.2.4 Available program parameters and ranges, page 29* for a list of available parameters.

1. Tap a field in the **Value** column.
2. To reach additional alternatives, tap Extended. This requires special authorization. For more information, see section *4.3.6 Manage access rights, page 44*).
3. Tap **Save** to confirm the change.

Path: **Menu** > **System** > **Edit Parameters** > **Extended**



1 **Parameter Name**
2 **Value**
3 **Min**
4 **Max**
5 **Type**
6 **Quick view**
7 **Save**

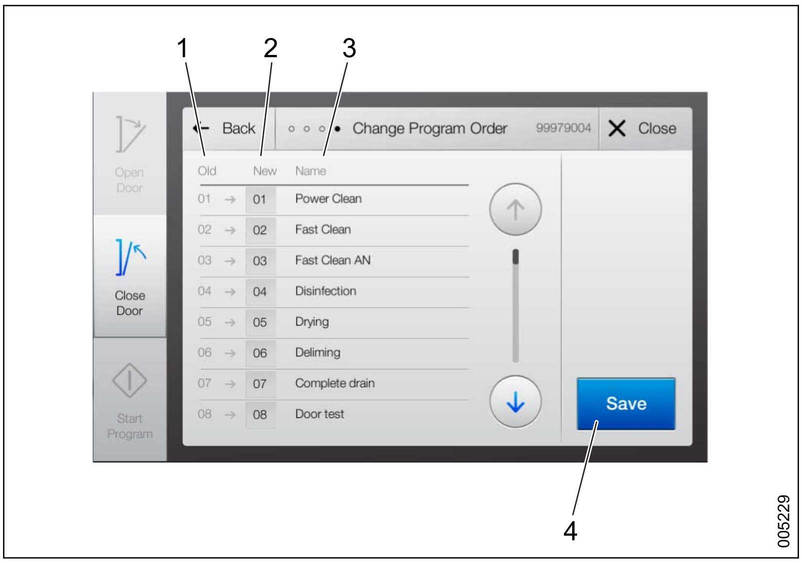
Name	**Description**
Parameter Name	Lists the washing parameters.
Value	Sets the value.

Name	Description
Min/Max	Sets the permitted values (parameter intervals) for editing a certain parameter.
Type	Sets the parameter type: A = Adjustable I = Information only H = Hidden from the user
Quick view	Set to display or not display the value in the **Select Program** view (**Quick view**): **On** = The value is displayed when **Quick view** is activated. For more information, see section *4.4.12 HMI, page 83.* A max of 3 parameters can be activated at the same time. **Off** = The value is not displayed.
Save	Saves the changes made.

4.3.11.2 Change program order

Changing the order of the program buttons in the **Select Program** view requires Authorized Service technicians access rights.

Path: **Menu** > **System** > **Program Management** > **Change Program Order**



1 **Old**	3 **Name**
2 **New**	4 **Save**

1. In the **New** column, tap the program to re-arrange.
2. Enter a digit for a new list placement.
3. Tap **Save** to confirm the change.

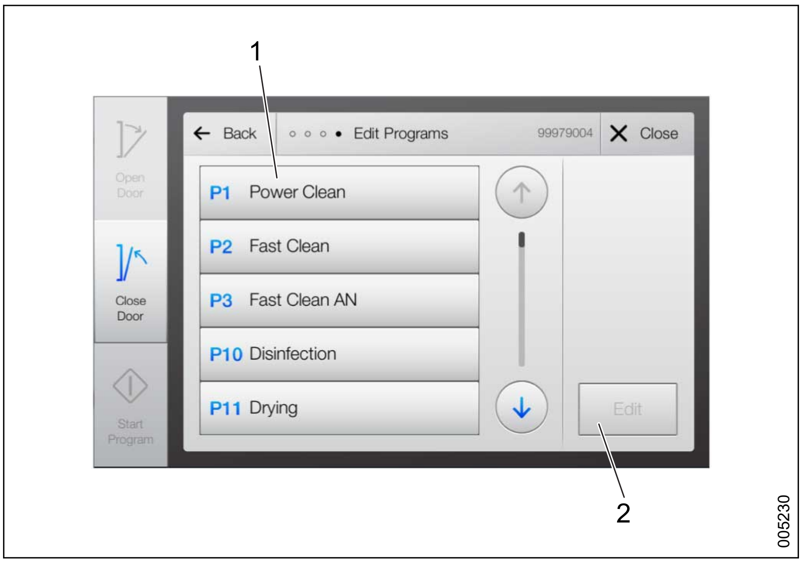
When the placement for one program is changed, the list is automatically re-arranged accordingly.

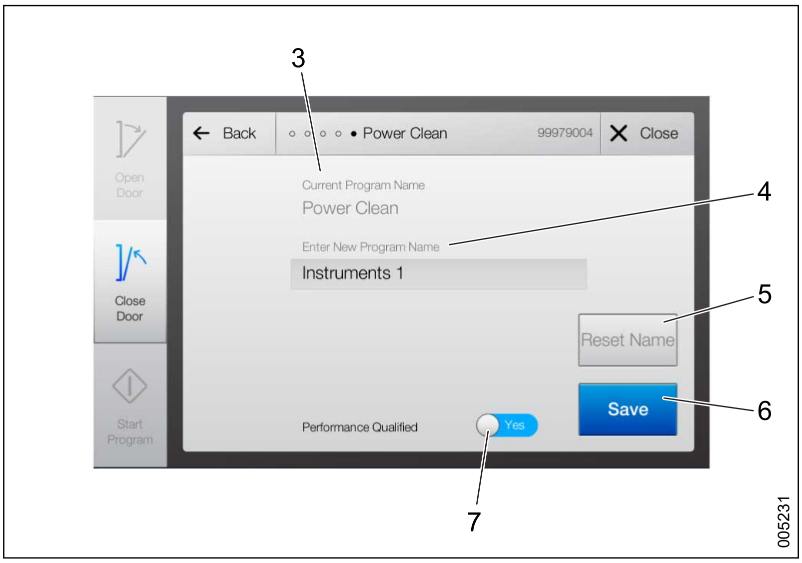
4.3.11.3 Change program name or set to PQ

Path: **Menu** > **System** > **Program Management** > **Edit Programs** > **Edit**

- Renaming a program requires Authorized Service technicians access rights.
- Setting a programs to **Performance Qualified** (PQ) requires Super User access rights.

Before setting a program to Performance Qualified (PQ), a complete PQ validation must be done in accordance with the national rules that apply in the country where the machine is used.





1 Programs list	5 **Reset Name**
2 **Edit**	6 **Save**
3 **Current Program Name**	7 **Yes**
4 **Enter New Program Name**	8 **No**

1. Select the desired program in the programs list.
2. Tap **Edit**.
3. Tap the text box to enter a new program name.
4. Indicate if the program is **Performance Qualified** by selecting **Yes** or **No**.
5. Tap **Save** to confirm the changes.

4.3.11.4 Revert a program to a previous version

4.3.11.4.1 Create and load a program reference file

1. Connect the service PC to the machine and the web interface.
2. Put the machine in standby mode.
3. Export the program reference file according to the instructions in section *4.4.13.5 Import/Export, page 91*. For uploading, follow instructions in the same section.

4.3.11.4.2 Print all the programs

An alternative to creating a program reference file is to print all the programs. This prints or stores all parameter settings in the same way as the process reports. The machine must be installed with a fascia printer, a network printer, network storage or USB storage.

1. On the touchscreen, go to **Menu** > **Documentation** and select **Print Program** (G1).
2. Select one program at a time and tap **Print**. Do this for all programs used.
3. On the touchscreen, go to **Menu** > **Documentation** and select **Print Machine Settings**.
4. Tap **Print**.

4.4 Web server interface

This section handles settings and changes that can be done from the Web server interface. More settings can be done from the touchscreen. For more information, see section *4.3 Touchscreen interface, page 34*.

The control system has a built-in Web server with an associated interface. It is accessed from an external PC with a web browser, through the network port (ETH). Some machine settings that are not accessible from the touchscreen are managed from the Web server.

The language used in the Web interface is English (cannot be altered).

The Web server closes automatically when it is not in use.

The following applies for all changes:

Name	Description
Main menu	Click the Getinge logo to return to the **Main menu**.
Save changes	Click to confirm and save the change.
Reset changes	Click to reset changes to the last saved.
Restart	After saving, a restart must be done for the changes to apply.

For more information about terms and abbreviations displayed on the screen, see section *4.2.1 Terms and abbreviations, page 27*.

4.4.1 Connect to the Web server

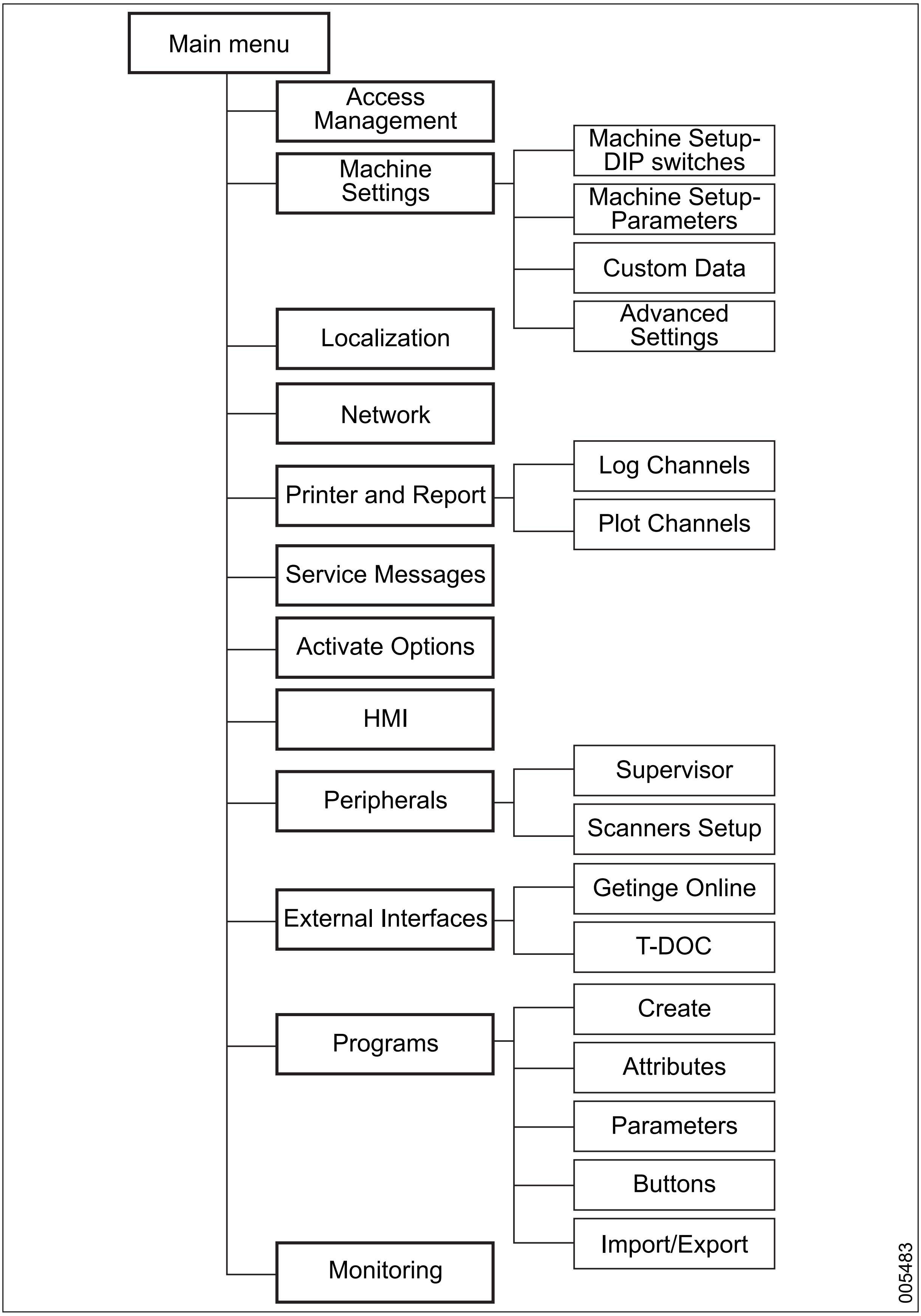
Performing changes from the Web server interface requires logon as an Authorized service technicians. For more information, see section *4.3.6 Manage access rights, page 44*.

1. On the touchscreen, go to **Menu** > **System** > **Configuration** > **Web server**.
2. Shut down any Wi-Fi (wireless network) access on the connected PC.

3. Set the connected PC to use a static IP address (on Windows PCs, usually under network settings). Change settings under TCP/Ipv4 to a relevant LAN connection. Use the following settings, for example:
4. - IP address: **192.168.0.100**
   - Subnet mask: **255.255.255.0**
   - Default gateway: **Leave empty**
5. Connect the PC to the port marked **ETH** (machine's local network).
6. Open the web browser (with JavaScript activated) and go to the address **192.168.0.32**

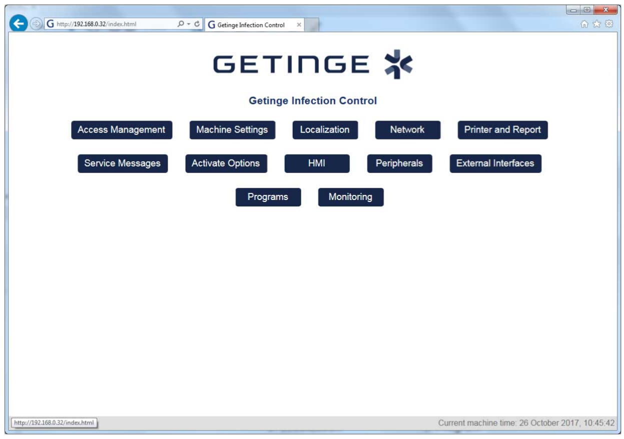
If using a preconfigured Getinge PC, changing network settings may be limited and the Getinge Functions program must be used. Contact your local IT administrator for more information.

4.4.2 Menu tree



4.4.3 Main menu

When first connected to the Web server, the Main menu is displayed. Each button gives access to different configuration categories for the machine.



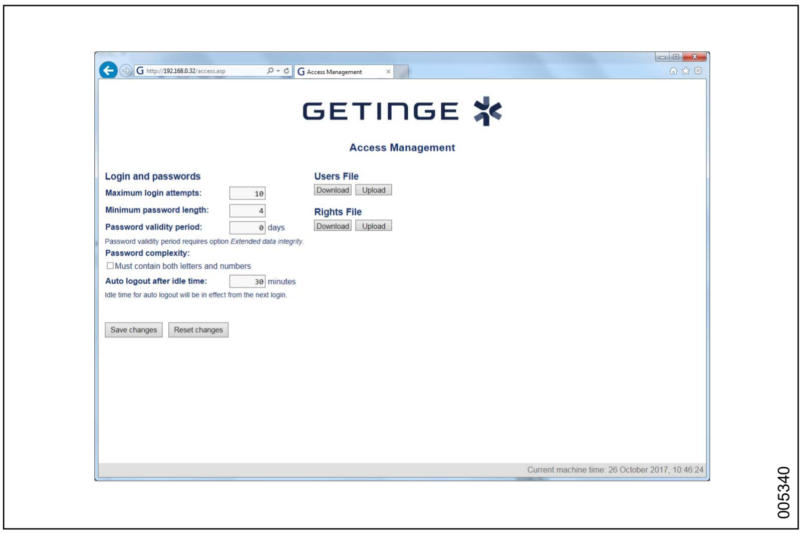
005191

Name	Description
Access Management	Login, password, user, rights, file import/export.
Machine Settings	Manufacturer/end user data. DIP switches (Machine DIP switches). Parameters (Machine parameters). Adaptable data field (Adapted data). Advanced settings (Detailed operating history).
Localization	Date, time, units, language.
Network	IP address, proxy.
Printer and Report	Printer, printed reports. Log/plot channels (select channels).
Service Messages	Change service time, reset message.
Activate Options	Activate options, create license file.
HMI	Changes to the touchscreen.
Peripherals	Supervisor settings. Barcode and RFID settings (not used for this machine).
External Interfaces	Settings for Getinge Online and T-DOC (activate/ deactivate, reset).

Name	Description
Programs	Overview. Delete/create program. Create program (adapt and create). Program attribute. Program parameters (settings). Buttons (changes in the **Select Program** view. Import/Export (import/export of program, button configuration).
Monitoring	Status of the inputs and outputs of the control system.

4.4.4 Access Management

Path: **Main menu** > **Access Management**



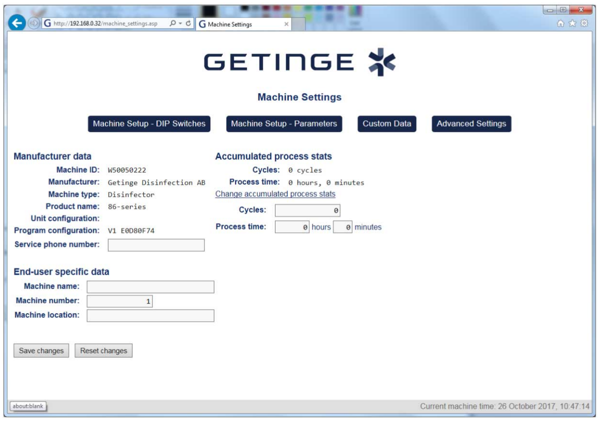
Name	Description
Maximum login attempts	Enter the maximum number of login attempts accepted for a user before the account is locked.
Minimum password length	Enter the minimum number of characters used in the password.
Password validity period	Set the number of days the password is valid. 0 = period of validity is indefinite. To change the period of validity, the function must be activated with a license key.
Password complexity/Must contain both letters and numbers	Tick the box if the password must contain both letters and numbers.
Auto logout after idle time	Set the number of minutes the user may be logged in without being idle.
Users File	The file collects all user's access rights in the system.

Name	Description
Rights File	The file contains a list of all users and user groups added to the system.
Download/Upload	Click to import/export Users'/Rights files to/from the system. This is useful when configuring several machines with the same settings. If the if the upload is not successful, an error message is generated

More settings can be made from the touchscreen. For more information, see section *4.3 Touchscreen interface, page 34*.

4.4.5 Machine settings

Path: **Main menu** > **Machine Settings**



005193

Name	Description
Machine Setup - DIP Switches	Click for DIP switches submenu.
Machine Setup - Parameters	Click for parameters submenu.
Custom Data	Click for custom data submenu.
Advanced Settings	Click for advanced settings submenu.
Machine ID	Displays the unique ID, set by the manufacturer.
Manufacturer	Displays the name of the manufacturer.
Machine type	Displays the machine type.
Product name	Displays the product series name.
Program configuration	Displays the program configuration control figure.
Service phone number	Add a phone number to appear when an alarms occurs.
End-user specific data/Machine Name	Add an end user-specific machine name.
End-user specific data/Machine number	Add an end user-specific machine number.
End-user specific data/Machine location	Add an end user-specific machine location.
Accumulated process stats/Cycles	Displays the total number of wash cycles that the machine has performed.
Accumulated process stats/Process time	Displays the total process time that the machine has been in operation.

Name	Description
Change accumulated process stats/Cycles	Click to change the cycle counter.
Change accumulated process stats/Process time	Click to change the process timer.

The Cycle counter and Accumulated process time fields are mandatory and must include a valid value to be able to save the settings. The details are updated dynamically by the system and should only be edited to correct an incorrect operational history caused by a system error.

Details from this page are displayed on the touchscreen. For more information, see section *4.3.4 Display machine information, page 41*).

4.4.5.1 Machine Setup - DIP Switches

Path: **Main menu** > **Machine Settings** > **Machine Setup - DIP Switches**

Activate/deactivate functions by activating/deactivating the corresponding DIP switch. Available DIP switches are described in section *4.4.5.1.1 List of DIP switches, page 66*. The DIP switch names are product-specific and cannot be changed via the Web server.

1. Tick a DIP switch to activate.
2. Click **Save changes** to confirm and save the change.

4.4.5.1.1 List of DIP switches

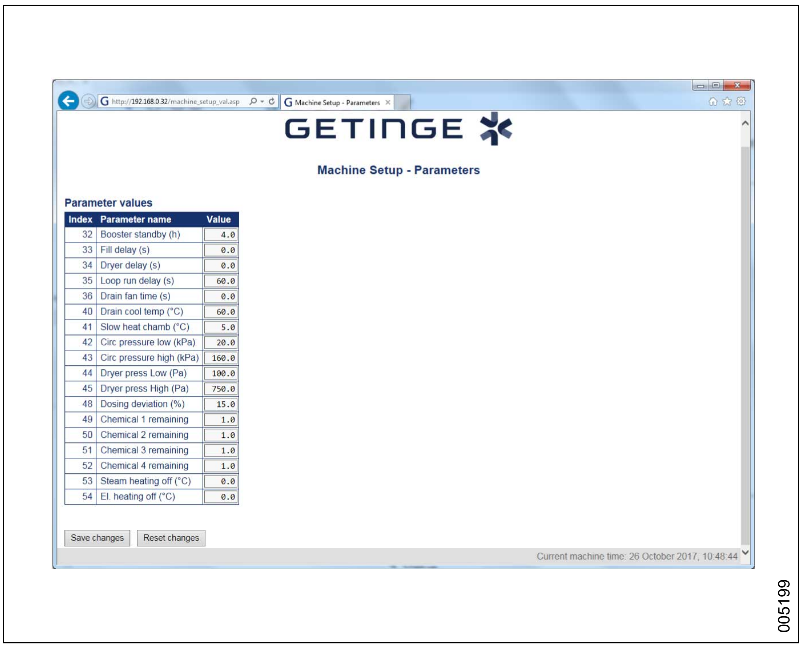
Index	Name*	Description
0	**Single door**	Machine has only one door.
1	**Booster tank**	Machine has booster tank.
2	**Chemical 3**	Machine has dosing pump 3.
3	**Chemical 4**	Machine has dosing pump 4.
4	**Deionized water**	Machine has valve for deionized water.
5	**Supervisor**	Machine has Supervisor system.
6	**Flow control 1**	Flow monitoring for dosing pump 1 activated.
7	**Flow control 2**	Flow monitoring for dosing pump 2 activated.
8	**Flow control 3**	Flow monitoring for dosing pump 3 activated.
9	**Flow control 4**	Flow monitoring for dosing pump 4 activated.
10	**Humidity sensor**	Machine has humidity sensor.
11	**Conductivity check**	Machine has conductivity meter.
12	**SWRSS**	Machine has spray arm monitoring.
13	**Pump pressure**	Machine has pressure monitoring sensor.
15	**Drain cooling**	Machine has drainage cooling valve.
16	**Steam cooling**	Activate drainage cooling when the steam valves are open.
17	**Scanner**	Machine has barcode scanner.
18	**AGS Infeeder**	Machine is connected to AGS or loader.

Index	Name*	Description
19	**AGS Outfeeder**	Machine is connected to AGS or unloader.
27	**A0 Print**	Activate A0 printer function (A0 value calculated and printed in the report).
28	**A0 Disinfection**	Activate A0 disinfection (disinfection time controlled with calculated A0 value).
29	**Door remain closed**	Door(s) do(es) not open automatically.
30	**Validate disinf**	Activate *validation mode*. The machine skips all introductory phases and starts the final rinse/disinfection directly. Used only for HTM2030 testing.
31	**Loop run**	Activate *loop function*. The machine automatically restarts the selected program. Used only for testing. Only used in S-8666/S-8668.

**DIP switch names not translated.*

4.4.5.2 Set general parameters

Path: **Main menu** >**Machine Settings** > **Machine Setup - Parameters**



Name	Description
Index	Displays the index number of the machine parameter.
Parameter name	Displays the name of the machine parameter.
Value	Displays the value of the machine parameter

4.4.5.2.1 General parameters

Index	Name*	Description	Interval	Default selection
32	**Booster standby (h)**	After a wash cycle, the booster tank is set to standby mode to keep the water heated. Decide how long the standby period will be before turning off the booster tank entirely.	0-12	4
33	**Fill delay (s)**	Refill the chamber for the stated number of seconds (used to increase water volume).	0-60	0
34	**Dryer delay (s)**	Delay before the drying fans are run (offers more time to start the external air extractor).	0-300	0
35	**Loop run delay (s)**	Delay before the program starts (applies when *Loop run* DIP switch is activated).	0-3600	60
40	**Drain cool temp (°C)**	If the chamber temperature exceeds the stated maximum value the drainage cooling valve opens before draining.	40-95	60
41	**Slow heat chamb (°C)**	The chamber temperature has to increase by the stated value every 5 minutes (alarm limit).	2-10	5
42	**Circ pressure low (kPa)**	Lowest permissible water pressure in the circulation system (alarm limit).	5-40	20
43	**Circ pressure high (kPa)**	Highest permissible water pressure in circulation system (alarm limit).	100-250	160
44	**Dryer press Low (Pa)**	Lowest permissible differential pressure in the drying system (alarm limit).	50-200	100
45	**Dryer press High (Pa)**	Highest permissible differential pressure in the drying system (alarm limit).	400-900	750
47	**Drying full speed (%)**	The speed of the drying fan when it is running at full speed.	75-100	100
48	**Dosing deviation (%)**	Highest differential dosing amount deviation (± in %)	5-25	15
49	**Chemical 1 remaining**	Number of processes that can be run after the level sensor is deactivated.	0-10	1
50	**Chemical 2 remaining**	Number of processes that can be run after the level sensor is deactivated.	0-10	1
51	**Chemical 3 remaining**	Number of processes that can be run after the level sensor is deactivated.	0-10	1
52	**Chemical 4 remaining**	Number of processes that can be run after the level sensor is deactivated.	0-10	1
53	**Steam heating off (°C)**	Number of degrees below set value at which the heating is switched off.	0-3	0

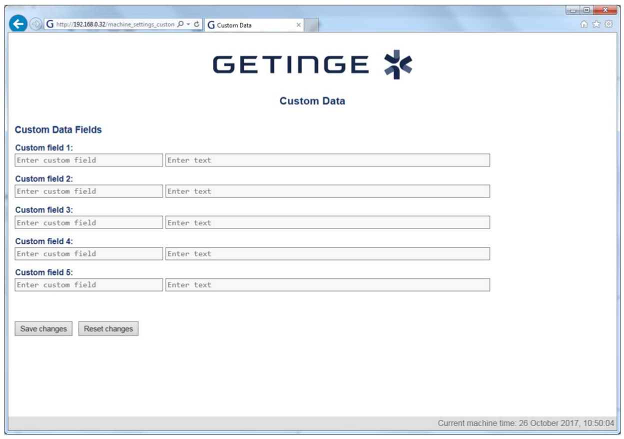
Index	Name*	Description	Interval	Default selection
54	**El. heating off (°C)**	Number of degrees below set value at which the heating is switched off.	0-1	0
55	**Steam cool time (s)**	The pulse time of the drain cooling valve when the steam condensate cooling function is activated.	2-15	5

**Machine parameter names are not translated.*

4.4.5.3 Custom Data

This function is not implemented.

Path: **Main menu** > **Machine Settings** > **Custom Data**



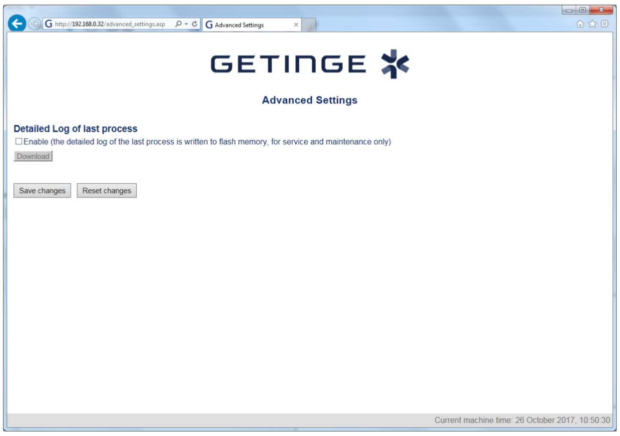
005203

Name	Description
Custom field 1, 2, 3 and 4	Edit the four available fields for adapted data.

4.4.5.4 Advanced settings

Activation of the detailed log.

Path: **Main menu** > **Machine Settings** > **Advanced Settings**



005204

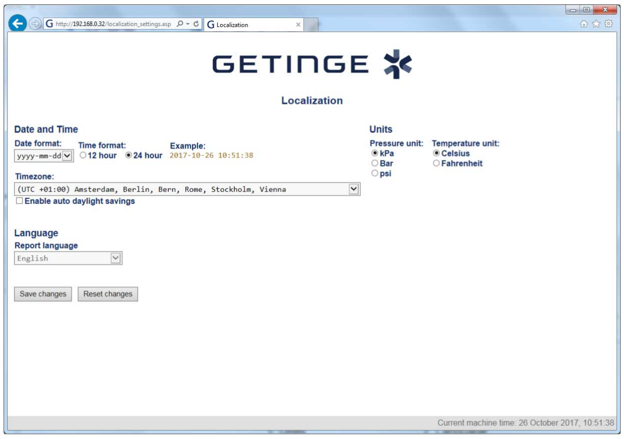
Name	Description
Detailed Log of last process	Tick to activate saving of a detailed log for the last process. This function may be used for test runs and should not be activated for normal use.
Download	Click to create a detailed log file.

To export logs from the machine, see section *4.4.13.5 Import/ Export, page 91*.

4.4.6 Localization

Setup of the display values on the touchscreen and in reports.

Path: **Main menu** > **Localization**

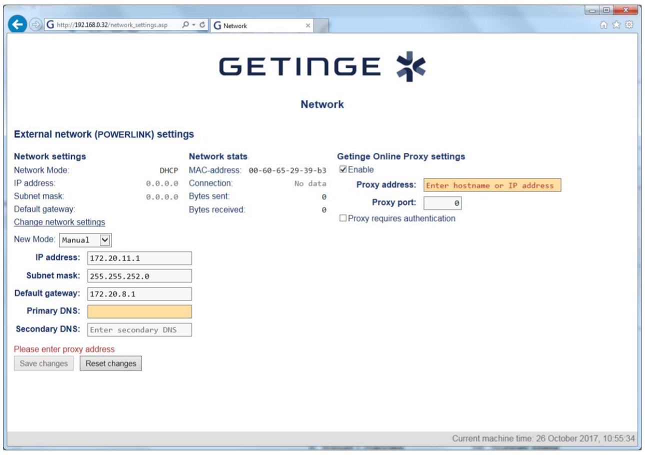


005207

Name	**Description**
Date format	Enter a date format.
Time format	Enter a time format.
Example	Displays the format as an example.
Time zone	Set the time zone to the geographical location of the machine or according to special requirements
Enable auto daylight saving	Tick to enable daylight saving.
Report language	Select the language for the report.
Pressure unit	Tick the unit to select.
Temperature unit	Tick the unit to select.

4.4.7 Network

Path: **Main menu** > **Network**



005208

Name	Description
Network mode	Displays the network mode.
IP address	Displays the IP address.
Subnet mask	Displays the subnet mask.
Default gateway	Displays the default gateway.
Change network settings/Network mode	Select IP type, DHCP (automatic or manual).
Change network settings/IP address	Set IP Address.
Change network settings/Subnet mask	Set subnet mask.
Change network settings/Default gateway	Set standard gateway. A restart is required for the new setting to apply.
Change network settings/Primary DNS	Set the primary DNS address.
Change network settings/Secondary DNS	Set a secondary DNS address.
MAC-address	Displays the MAC-address.
Connection	Displays the status of the network connection.
Bytes sent	Displays the bytes sent.
Bytes received	Displays the bytes received.
Getinge Online Proxy settings/Enable	Tick if a proxy server is used to connect to an external interface.
Getinge Online Proxy settings/Proxy address	Enter the address of the proxy server.
Getinge Online Proxy settings/Proxy port	Set the port for the proxy server.
Getinge Online Proxy settings/Proxy requires authentication	Tick if a user name and password are required to logon to the proxy server.

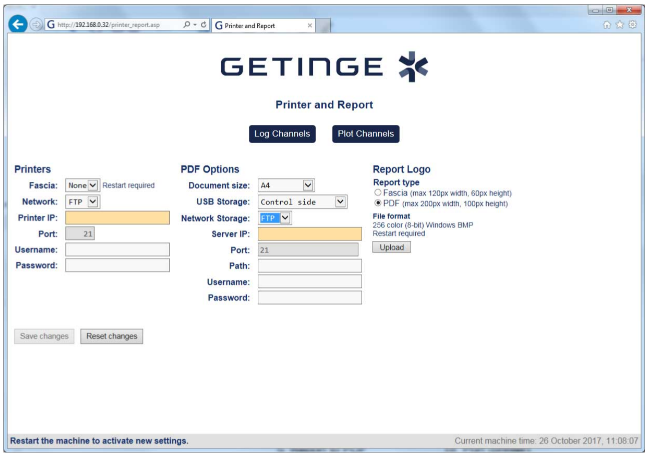
The settings refer to the network port that is used to connect to the external interface, see section *4.1.1.2.3.3 PLK, page 20*.

Adapt settings according to the relevant network and any external IT systems connected to the machine, see section *4.4.9 External Interfaces, page 78*.

4.4.8 Printer and Report

Setup of the locations for saving and printing process reports. Also setup of the process graph.

Path: **Main menu** > **Printer and Report**



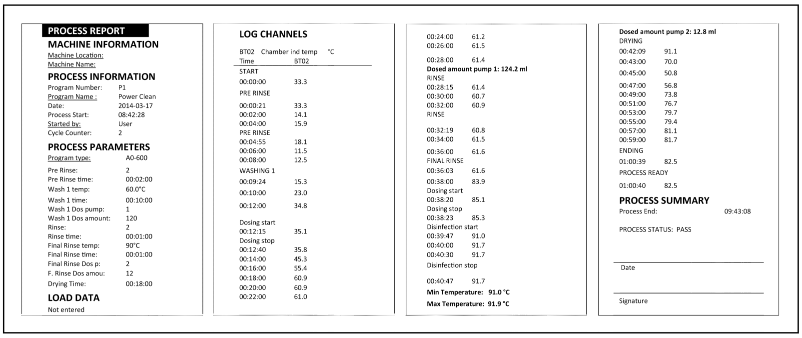
005210

Name	Description
Log Channels	Click for log channels submenu.
Plot Channels	Click for plot channels submenu.
Printers/Fascia	The fascia printer is the machine's local printer (control panel). Select none or 2” according to the current machine configuration. A restart is required for the new settings to take effect.
Printers/Network	FTP = Full size printer connected via network port (FTP). The **Network** printer is an option and must be activated with a license key. For more information, see section *4.4.14 Activate options, page 92*.
Printers/Printer IP	Enter the network printer's IP address.
Printers/Port	Enter the port used for connecting to the network printer.
Printers/Username	Enter a user name for printer access.
Printers/Password	Enter a password for printer access.
PDF Options/Document size	Set the PDF document size.
PDF Options/USB Storage	Set USB storage to "Control Side" if saving the reports on a USB memory stick. **USB Storage** is an option and must be activated with a license key. For more information, see section *4.4.14 Activate options, page 92*.
PDF Options/Network Storage	Set when saving the reports on the network (FTP or CIFS). **Network Storage** is an option and must be activated with a license key. For more information, see section *4.4.14 Activate options, page 92*.
PDF Options/Server IP	Enter the servers' IP address for reports.

Name	Description
PDF Options/Port	Enter the port to use for the reports.
PDF Options/Path	Enter the search path for the server report location.
PDF Options/Username	Enter the user name for server access.
PDF Options/Password	Enter a password for the PDF option.
Report type/Fascia (max 120px width, 60px height)	Tick to change the fascia printer logo.
Report type/PDF (max 200px width, 100px height)	Tick to change the PDF report logo.
Report type/File format	Displays the format of the logo file.
Report type/Upload	Click to upload a new report logo.

The options for printing (locally/network) or saving reports (locally on USB/network) depend on the current machine configuration.

A report of the important process data from the last wash cycle can be printed or saved. For more information, see section *4.3.8 Printouts, page 51*.



PROCESS REPORT
MACHINE INFORMATION
Machine Location:
Machine Name:
PROCESS INFORMATION
Program Number: P1
Program Name : Power Clean
Date: 2014-03-17
Process Start: 08:42:28
Started by: User
Cycle Counter: 2
PROCESS PARAMETERS
Program type: A0-600
Pre Rinse: 2
Pre Rinse time: 00:02:00
Wash 1 temp: 60.0°C
Wash 1 time: 00:10:00
Wash 1 Dos pump: 1
Wash 1 Dos amount: 120
Rinse: 2
Rinse time: 00:01:00
Final Rinse temp: 90°C
Final Rinse time: 00:01:00
Final Rinse Dos p: 2
F. Rinse Dos amou: 12
Drying Time: 00:18:00
LOAD DATA
Not entered

LOG CHANNELS
BT02 Chamber ind temp °C
Time BT02
START
00:00:00 33.3
PRE RINSE
00:00:21 33.3
00:02:00 14.1
00:04:00 15.9
PRE RINSE
00:04:55 18.1
00:06:00 11.5
00:08:00 12.5
WASHING 1
00:09:24 15.3
00:10:00 23.0
00:12:00 34.8
Dosing start
00:12:15 35.1
Dosing stop
00:12:40 35.8
00:14:00 45.3
00:16:00 55.4
00:18:00 60.9
00:20:00 60.9
00:22:00 61.0

00:24:00 61.2
00:26:00 61.5
00:28:00 61.4
Dosed amount pump 1: 124.2 ml
RINSE
00:28:15 61.4
00:30:00 60.7
00:32:00 60.9
RINSE
00:32:19 60.8
00:34:00 61.5
00:36:00 61.6
FINAL RINSE
00:36:03 61.6
00:38:00 83.9
Dosing start
00:38:20 85.1
Dosing stop
00:38:23 85.3
Disinfection start
00:39:47 91.0
00:40:00 91.7
00:40:30 91.7
Disinfection stop
00:40:47 91.7
Min Temperature: 91.0 °C
Max Temperature: 91.9 °C

Dosed amount pump 2: 12.8 ml
DRYING
00:42:09 91.1
00:43:00 70.0
00:45:00 50.8
00:47:00 56.8
00:49:00 73.8
00:51:00 76.7
00:53:00 79.7
00:55:00 79.4
00:57:00 81.1
00:59:00 81.7
ENDING
01:00:39 82.5
PROCESS READY
01:00:40 82.5
PROCESS SUMMARY
Process End: 09:43:08
PROCESS STATUS: PASS
Date
Signature

If an FTP server is used, it must be set to accept passive transfers.

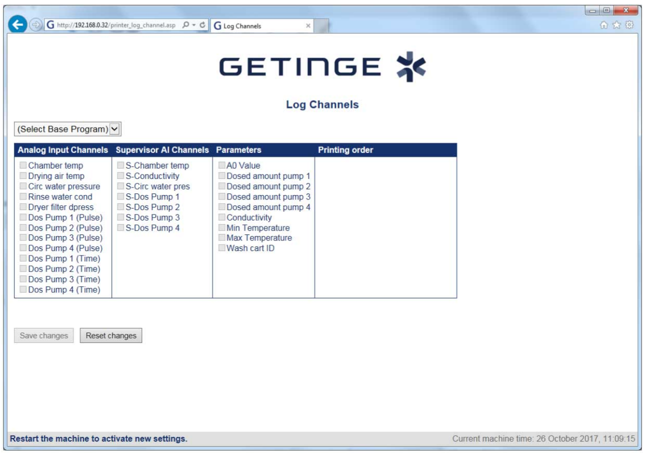
If a network printer is used, a Lexmark printer is recommended. Compatibility with other printers cannot be guaranteed.

1. Select printer and storage type for reports.
2. Set which parameters are to be included in the report (proceed to Log channels and Plot channels).

4.4.8.1 Log channels

Path: **Main menu** > **Printer and Report** > **Log Channels**

The page is used to set the parameters included in the reports for different programs. Log channel values are numeric in reports.



005212

Name	Description
Select Base Program	Drop-down menu for selecting program.
Analog Input Channels	Tick to select channels to include in the report. The column contains all available channels, depending on the current program. Mandatory channels are preselected and cannot be edited.
Supervisor AI Channels	Tick to select supervisor channels to include in the report. The column contains all available channels, depending on the current program. Mandatory channels are preselected and cannot be edited. AI = Analog Input.
Parameters	Tick to select parameters to include in the report. The column contains all available channels, depending on the current program. Mandatory channels are preselected and cannot be edited.
Printing order	Set the printing order for the channels, in the report. The order cannot be changed for all channels. The first channel on the list is printed as the first to the left in the report. Drag and drop here to select the print order.

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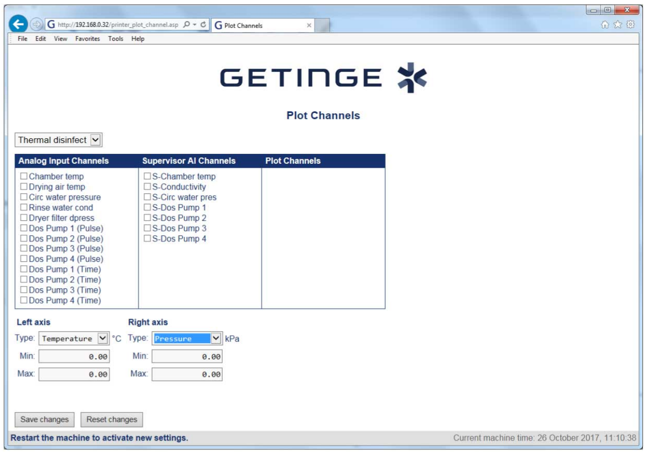
4.4.8.2 Plot channels

This view is used to set the plot channels used in the process reports for different programs.

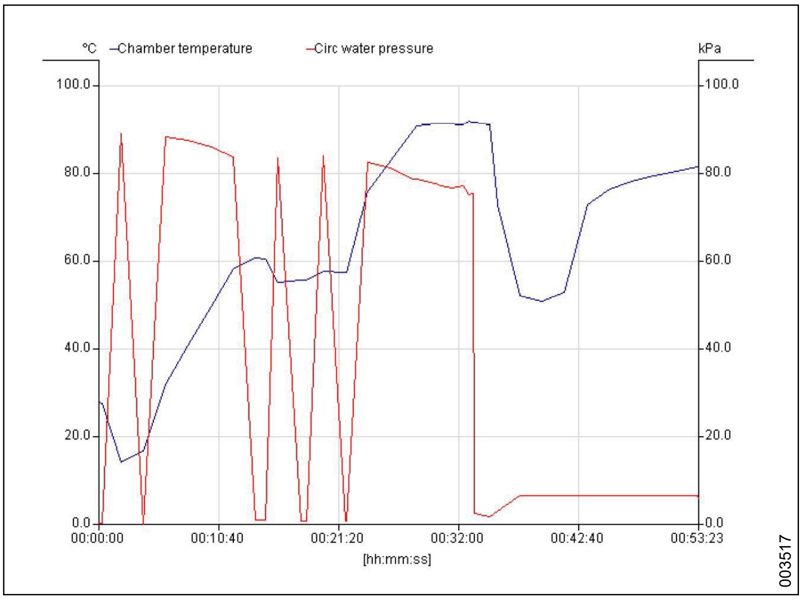
Path: **Main menu** > **Printer and Report** > **Plot Channels**



005213

Name	Description
Select Base Program	Drop-down menu for selecting program type.
Analog Input Channels	Tick to select channels to include in the report. The column contains all available channels, depending on the current program. Mandatory channels are preselected and cannot be edited. A maximum of 6 can be selected.
Supervisor AI Channels	Tick to select supervisor channels to include in the report. The column contains all available channels, depending on the current program. Mandatory channels are preselected and cannot be edited. AI = Analog Input
Left axis/Type	Enter the type displayed on the left axis.
Left axis/Max	Enter the minimum value for the left axis.
Left axis/Min	Enter the maximum value for the left axis.
Right axis/Type	Enter the type displayed on the right axis.
Right axis/Min	Enter the minimum value for the right axis.
Right axis/Max	Enter the maximum value for the right axis.

Plot channels work best with network channels and when reports are saved as PDFs.



Set plot channels:

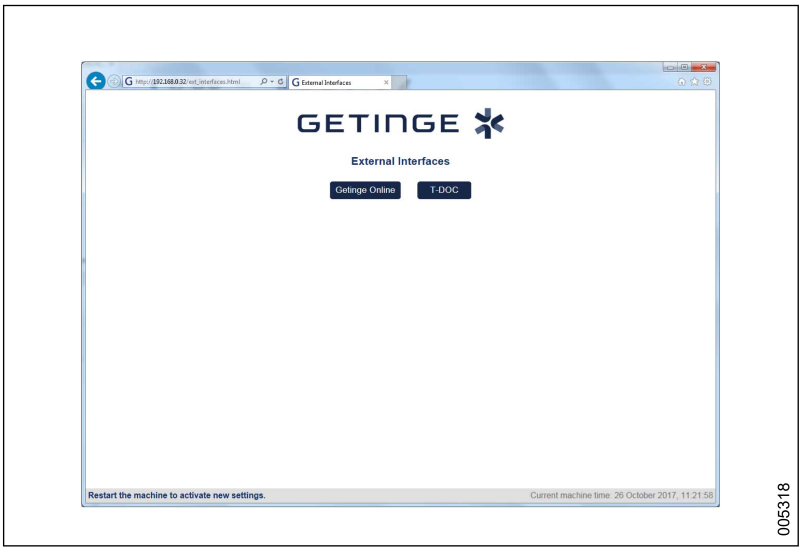
1. In the drop-down menu for **Select Base Program**, select a program type.
2. Set the display mode for the **Left axis** and **Right axis** by selecting a **Type** and add its minimum and maximum value.
3. Tick the channels to include in the report. When a channel is selected, a display option for this channel appears in the **Plot Channels** column.
4. In the **Plot Channels** column, set the axis to use for each channel. The device using the channel must correspond with the display option for right and left axes: **Type**, **Min** and **Max**
5. Click **Save changes**.
6. Repeat the procedure for each program type.

4.4.9 External Interfaces

Path: **Main menu** > **External Interfaces**

For general network settings, see section *4.4.7 Network, page 72.*

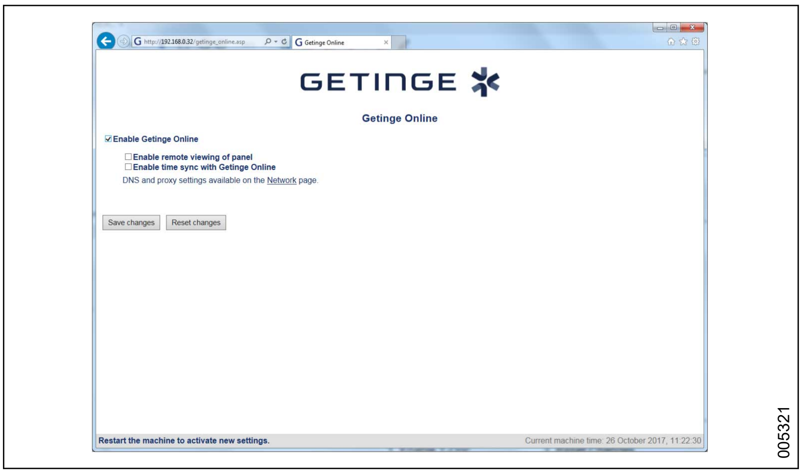
Refer to the documentation that accompanies each system for information not provided in this manual.



Name	Description
Getinge Online	Click for Getinge Online submenu.
T-DOC	Click for T-DOC submenu.

4.4.9.1 Getinge Online

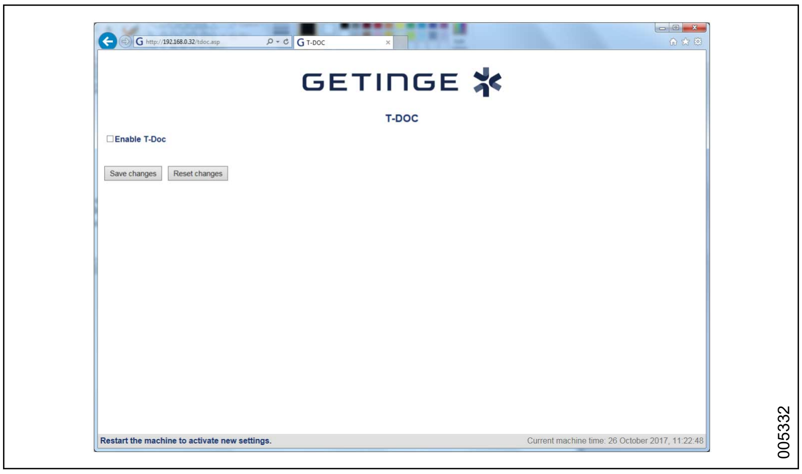
Path: **Main menu** > **External Interfaces** > **Getinge Online**



Name	Description
Enable Getinge Online	Tick to enable Getinge Online.
Enable remote viewing of panel	Tick to activate remote access.
Enable time sync with Getinge Online	Tick to activate time synchronization with Getinge Online.

4.4.9.2 T-Doc

Path: **Main menu** > **External Interfaces** > **T-DOC**



Name	Description
Enable T-Doc	Tick to enable T-Doc

4.4.10 Peripherals

Path: **Main menu** > **Peripherals**

Name	Description
Supervisor	Click to set values for system alerts.
Scanners Setup	Click for barcode reader settings. This function is not implemented.

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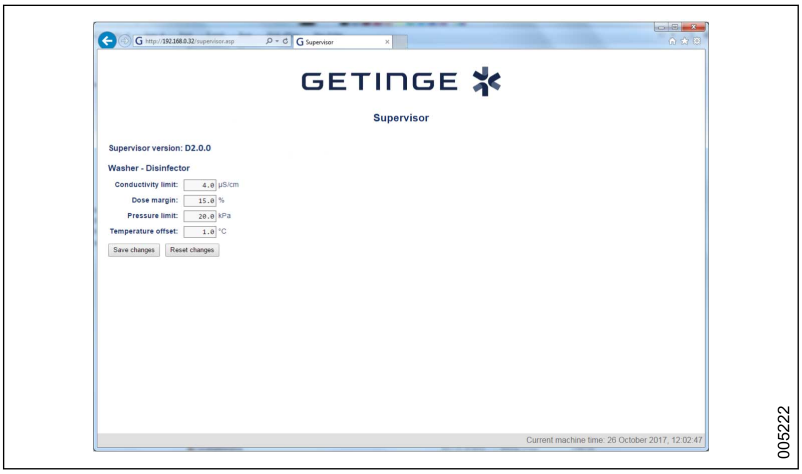
4.4.10.1 Supervisor

Path: **Main menu** > **Peripherals** > **Supervisor**

The deviation from a set value decides the trigger point for the Supervisor system. If a value exceeds the allowed deviation, the system triggers a warning. A suitable DIP switch must be activated to use the Supervisor system. For more information, see section *4.4.5.1 Machine Setup - DIP Switches, page 66*.

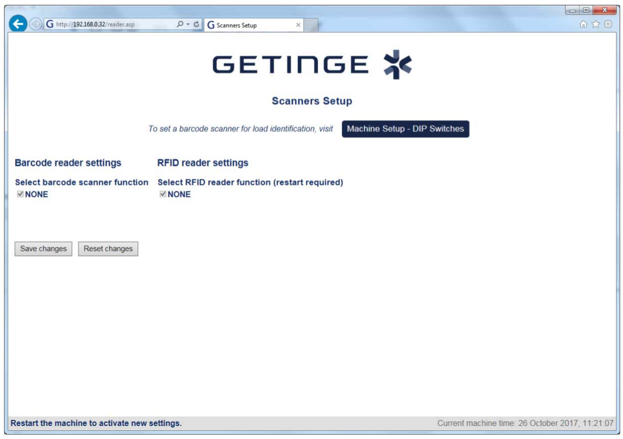


Name	Description	Default values
Supervisor version	Displays the version of the Supervisor system.	-
Conductivity limit	Enter the allowed deviation for the conductivity.	10 uS/cm
Dose margin	Enter the allowed deviation for the dosage system.	15 %
Pressure limit	Not used for this machine.	N/A
Temperature offset	Enter the allowed deviation for the temperature. The value should usually not be changed.	1.0 °C
Pressure deviation limit	Enter the allowed deviation for the pressure.	10 kPa

4.4.10.2 Scanners Setup

Path: **Main menu** > **Peripherals** > **Scanners Setup**

This function is not implemented. These settings are not relevant for a washer-disinfector. To activate the program selection barcode scanner, see section *4.4.5.1 Machine Setup - DIP Switches, page 66*.

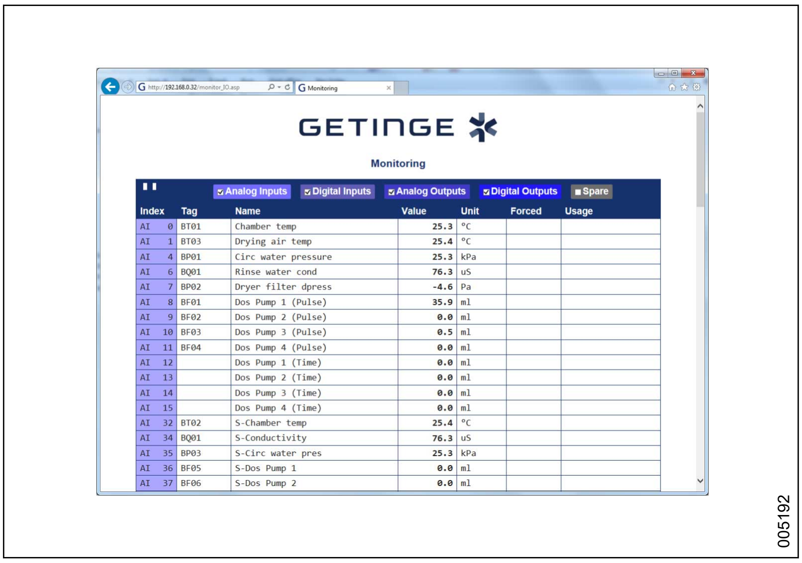


005307

4.4.11 Monitoring

Path: **Main menu** > **Monitoring**

The view is used to monitor in- and outputs. Use drag and drop to change the order in the list.

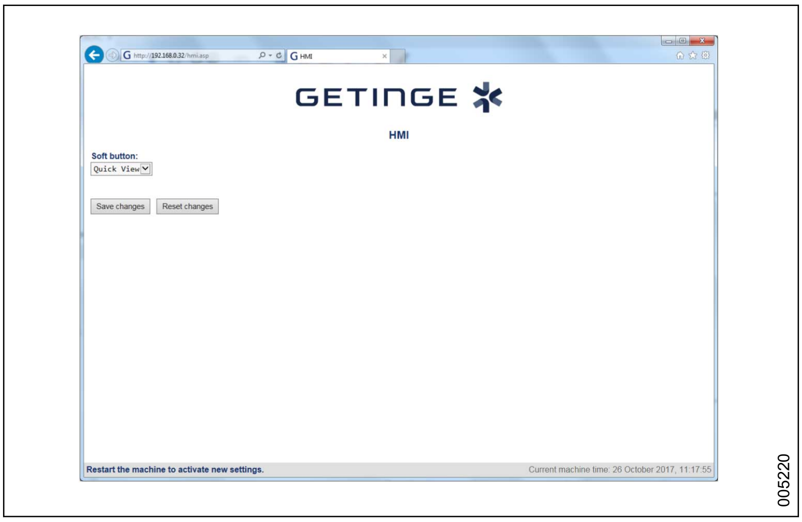


Name	Description
"	Click on the symbol to freeze the in- and outputs values momentarily.
Analog Inputs	Tick to display the analog input values.
Digital Inputs	Tick to display the digital input status.
Analog Outputs	Tick to display the analog output values.
Digital Outputs	Tick to display the digital outputs status.
Spare	Tick to display the spare inputs/outputs.
Index	Displays the I/O index number.
Tag	Displays the I/O tag.
Name	Displays the I/O name.
Value	Displays the actual I/O value/status.
Unit	Displays the value unit.
Forced	**Forced** is displayed when the output is forced.
Usage	**Spare** is displayed when the I/O is used as a spare.

4.4.12 HMI

Adaption of the **Select Program** view.

Path: **Main menu** > **HMI**



Name	Description
Soft button/None	Standard setting for the user interface. Only the program selection buttons are displayed.
Soft button/Quick view	Displays extra information: the name of the program and the three wash parameters.

Soft button - None

Soft button - Quick view

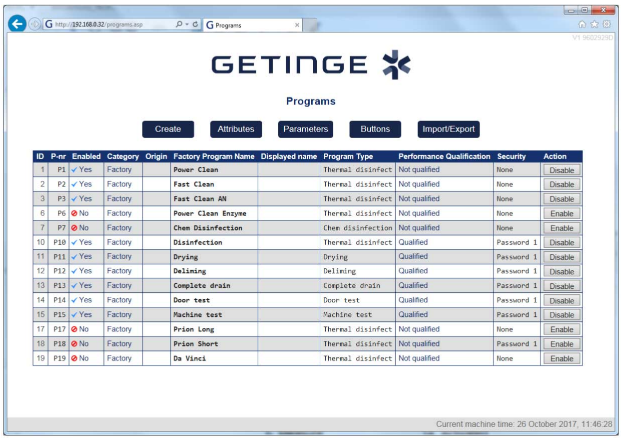
002656

The displayed parameters are set from the touchscreen. For more information, see section *4.3.11.1 Edit program parameters, page 56*.

4.4.13 Programs

Create new programs and edit existing programs. Select which program should be visible. Before changing a program, see section *4.2 Program introduction, page 27*.

Path: **Main menu** > **Programs**



005333

Name	Description
Create	Click to create program.
Attributes	Click to set up program attributes.
Parameters	Click to configure program parameters.
Buttons	Click to edit buttons in **Select Program** view.
Import/Export	Click to Import and export program configurations.
ID	Displays the program ID.

Name	Description
P-nr	Displays the program number shown in the **Select Program** view.
Enabled	Displays if the program is active (**Yes**) or inactive (**No**). An inactive program cannot be started or added to the **Select Program** view.
Category	Displays whether the program is factory installed or customized.
Origin	Displays the P-nr for the base program. It also displays whether the program is within or outside of the base program's setting area. In-range = all parameter values are within the base program's setting range. Out-range = at least one parameter has a value that lies outside of the base program's setting range.
Factory Program Name	Displays the original name of the factory installed program.
Displayed Name	
Program Type	Displays the function of the program type.
Performance Qualification	Displays if a **Performance Qualification** has been performed.
Security	Displays the program security level; the access rights required to start the program.
Action	Buttons for deactivating, activating and deleting programs. Deactivate = changes to **No** in the column **Enabled**. Activate = changes to **Yes** in the column **Enabled**. Delete = only deactivated standard programs can be deleted. A deleted program is completely removed and the action cannot be undone.

WARNING!

Risk of contaminated load

Changes in program parameters or incorrect use of a program can affect the cleaning result.

- Follow instructions in the Service manual.

When adapting a program and before using the machine in production, a new Performance Qualification must be done.

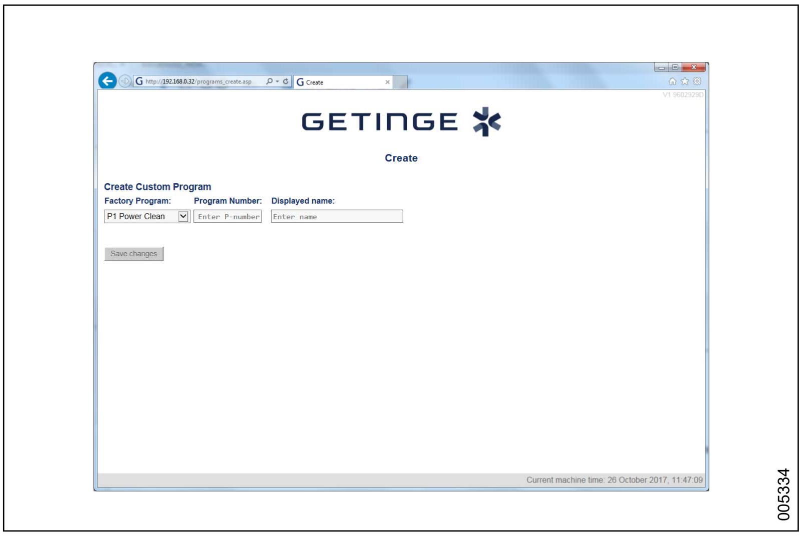
Adaptations of programs can only be carried out by authorized service technicians with Z1 rights.

Before using an adapted program:

- Check that the program is created for the intended load.
- Check the Intended use and other statements in the manual to ensure that nothing prevents the program from functioning properly.

4.4.13.1 Create

Path: **Main menu** > **Programs** > **Create**



Name	Description
Factory Program	Select a program to copy from the drop-down list. The program must be intended for the same type of application as the base program.
Program Number	Enter a unique P number for the new program.
Displayed name	Enter a unique name for the new program. The name is displayed in the **Select Program** view. Choose a name that contains a description of the function.

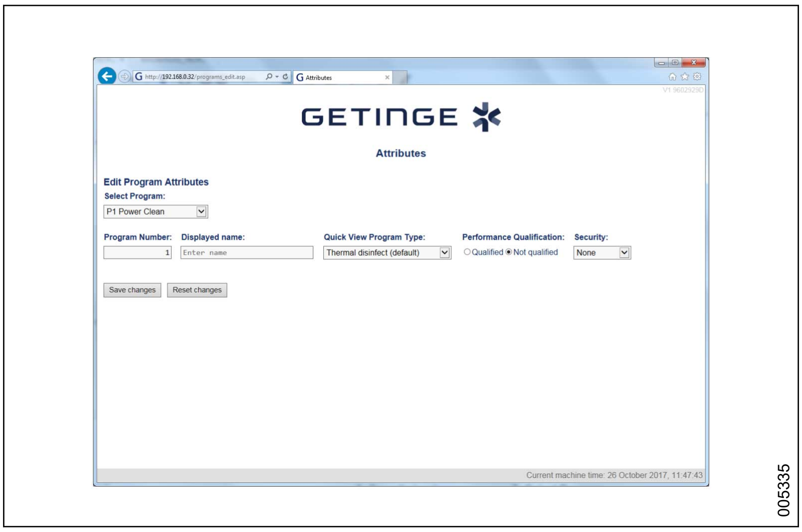
Create a new program by copying a factory installed program. The new program inherits the copied program's parameters.

When adapting a program and before using the machine in production, a new Performance Qualification must be done.

4.4.13.2 Attribute

Path: **Main menu** > **Programs** > **Attributes**

This view is used to change a program attribute.



Name	Description
Select Program	Select program from the drop-down list.
Program Number	Enter a P number for the program. Ensure that the number is not used by another program. If the field is left blank, the program gets the same number as the base program.
Quick View Program Type	Choose the name displayed in the **Select Program** view on the touchscreen when **Quick view** is activated. Program type or base program is shown.
Performance Qualification	Tick **Qualified** if a PQ has been done.
Security	Select a security level. This must be done before the program is run: **None** = The program can be started by anyone. **Confirm** = The program can be started by anyone after confirmation at the start. **Login 1** = Authorized personnel must be logged in to start the program. **Login 2** = Authorized personnel with Super User rights must be logged in to start the program.

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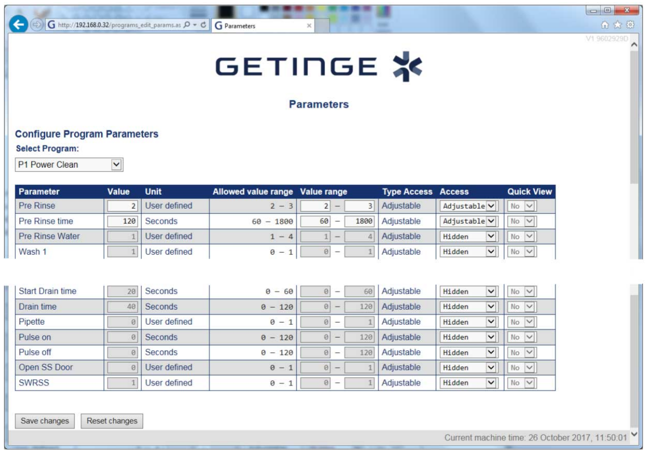
4.4.13.3 Parameters

Path: **Main menu** > **Programs** > **Parameters**



005336

Name	Description
Configure Program Parameters	The program type determines which parameters can be changed. The allowed value range is also determined by the program type and can be less restrictive than the factory installed. Changed program parameters can affect the program's process, performance and security. Access to adaptation of parameters is limited by: - Technician's rights in the system - Parameter's access type - Parameter's setting range - Base program's settings
Select Program	Select program from the drop-down list.
Parameter	Displays the **Parameter** name.
Value	Displays the **Parameter** value. The setting range is specified in **Value range**. If **Type Access** is set to **Adjustable**, the value can only be changed by an authorized service technician.
Unit	Displays the unit for the selected parameter. (Cannot be changed.)
Allowed value range	Displays the maximum setting range for the parameter. This may depend on the value set in **Value range**. If this is the case, then the corresponding value is color marked in the list.
Value range	Enter the setting range for the parameter. The value may be more restrictive or the same as the value in **Allowed value range**. If **Type Access** is set to **Adjustable**, the value may only be changed by an authorized service technician.

Name	Description
Type Access	The program type's access settings define if the parameter is adaptable. If set to **Adjustable**, the value may only be changed by an authorized service technician. The following parameters can only be changed by an authorized service technician: - **Value** - **Value range** - **Access**
Access	The parameter access settings can be changed within the range specified in **Allowed value range**. - **Hidden** = The value is visible and can only be changed by an authorized service technician. - **Info** = The value is visible and can only be changed by an authorized service technician. The parameter is printed in the process report. - **Adjustable** = The value is visible and can only be changed by an authorized service technician, and from the touchscreen by users belonging to the Super Users group. The parameter is printed in the process report.
Quick view	Click to change between **Yes** or **No**.

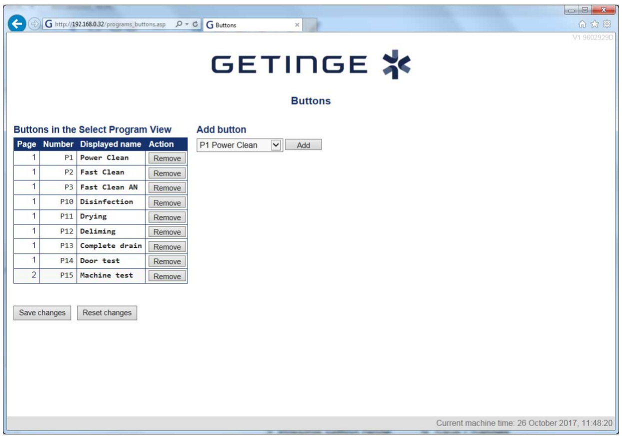
Original program

States what factory installed programs have been copied and have formed the basis for adaptations.

4.4.13.4 Buttons

Path: **Main menu** > **Programs** > **Buttons**

Configures the program buttons on the **Select Program** view.



005337

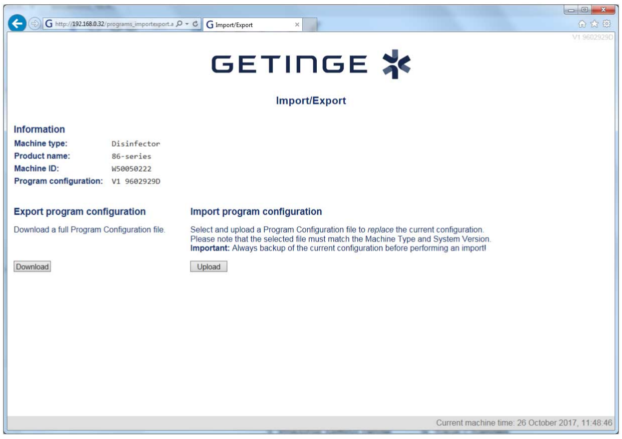
Name	Description
Page	Displays the page the program is displayed on. To rearrange the list, drag and drop items.
Number	The program's P number is changed by changing attributes.
Displayed name	The name is displayed in the **Select Program** view. The name is changed by changing attributes.
Action	Click **Remove** to delete the program in the **Select Program** view. Deleted programs can be added in **Add button**.
Add button drop-down list	Select a program to add to the program list in the **Select Program** view. The list contains all available programs. More than one button can be added for the same program.
Add	Click to add a program to the program list.

4.4.13.5 Import/Export

The program configuration may be exported from one machine and imported into another machine.

It is only possible to import files from a machine type range. The program configuration in the main version/top version must match. Subversions do not need to match.

Path: **Main menu** > **Programs** > **Import/Export**



005338

Name	Description
Machine type	Displays the machine type.
Product name	Displays the product series name.
Machine ID	Displays the machine ID.
Program configuration	Displays the program configuration control figure.
Export program configuration/Download	Click to export a zip file. Select a file location and save.
Import program configuration/Upload	Click to import a new configuration file.

Import (**Upload**) file:

1. Create backup copies of the existing configuration.
2. Ensure that the new program configuration file matches the machine type and the program configuration version.
3. Ensure that the new program configuration file is complete and not modified.
4. Note the current value in the **Program configuration** field.
5. Click **Upload** and select the program configuration file for import.

After upload, restart the machine and check that the control figure (from step 4) is correct.

When the file is uploaded, the following are checked:

- Standard syntax for naming
- Machine type
- Program configuration's compatibility

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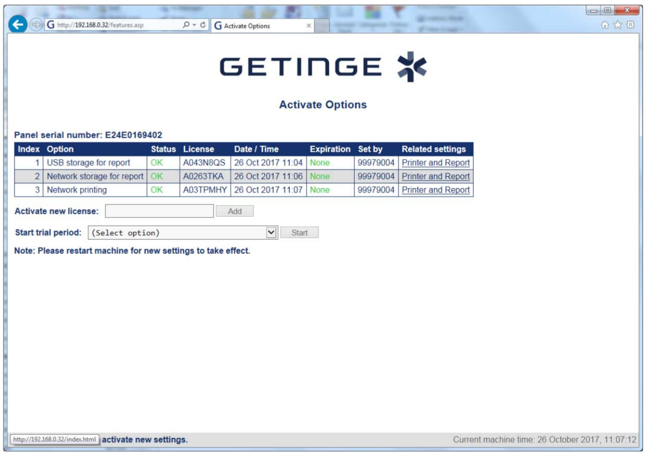
4.4.14 Activate options

Path: **Main menu** > **Activate Options**



005215

Name	Description
Panel serial number	Displays the serial number of the operator panel.
Index	Displays the option index number.
Option	Displays the names of the installed options.
Status	A function can have the status OK, Expired or Trial period.
License	Displays the license key of the installed options.
Date / Time	Displays the date and time when the option was installed.
Expiration	Displays the expiration time of the trial version of the option.
Set by	Displays the ID of the technician who installed the option.
Related settings	Displays related settings.
Activate new license/Add	Enter a valid license key for a function and click **Add** to activate the new license.
Start trial period/Start	Drop-down menu for selecting the function for which the trial period is valid. A trial period can only be used for certain functions. Click **Start** to activate the period.
Note: Please restart machine for new settings to take effect.	The machine needs to be restarted for the new changes to take effect.

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The machine needs to be restarted for the new changes to take effect.

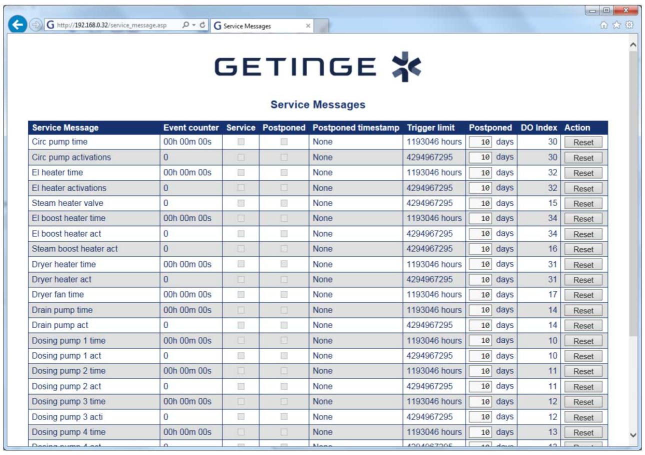
Certain software functions require a special license key to be used. The choice of available languages depends on the current machine configuration.

For creating a new license file, see section *7.1.3 Restore serial number after replacing operator panel , page 143*.

4.4.15 Service Messages

Path: **Main menu** > **Service Messages**

This function is not implemented. All values are set to the maximum to avoid the service message being displayed on the touchscreen. Triggered service messages can be reset as needed.



005214

Name	Description
Service Messages	Displays the text in the service message on the touchscreen.
Event counter	Displays the time passed or number of runs since the last reset.
Service	Indicates whether service has been carried out.
Postponed	Indicates whether service has been postponed.
Postponed timestamp	Indicates when service is postponed.
Trigger limit	Displays the time or number of runs that will trigger the message.
Postponed	Displays the number of days until the message is shown again.
DO index	Displays which output is monitored.
Action/Reset	Click to reset (confirm) that a triggered service message has been dealt with.

5 PREVENTIVE MAINTENANCE

The amount of maintenance required depends largely on the quality of the incoming water and how often the machine is used. The maintenance interval will have to be determined in each individual case. Getinge Disinfection AB recommends that the maintenance be performed at the intervals indicated in section *5.1 Service table, page 94*.

Preventive maintenance and system testing must be carried out to maintain normal functioning, and to ensure safety and the machine's proper operation.

In addition to system testing, we recommend that appropriate cleaning tests and temperature validation be performed according to EN ISO 15883.

Document performed service measures in accordance with local regulations.

The required maintenance interval will depend largely on the quality of the incoming water, how often the machine is used and the nature of the goods to be cleaned. The maintenance interval will have to be determined in each individual case. Getinge recommends that the stated maintenance operations be done at least according to the specified intervals.

Maintenance operations are sorted into categories related to location or function of the component that is to be maintained.

Maintenance instructions for loading equipment can be found in the respective documentation. For more information, see section *1.6 References, page 10*.

Before starting servicing work:

- a disinfection program must be run. If disinfection cannot be achieved, disinfect the machine with disinfectant.
- drain all the tanks.
- switch off power to the machine.
- shut off steam and water supplies.

5.1 Service table

Activity/Component Inspection/Replacement	**Interval**	**Time required***
	Each year /4000 cycles	
General		
5.3.1 Checking electrical cables and connection points, page 97	•	10 min
5.3.2 Checking ventilation fans in the machine, page 97	•	5 min
5.3.3 Checking that the machine is level, page 97	•	5 min
5.3.4 Checking Quick guide, page 97	•	1 min
5.3.5 Checking alarms and message history, page 98	•	2 min

Activity/Component Inspection/Replacement	Interval	Time required*
	Each year /4000 cycles	
5.3.6 Checking docking, page 98	•	5 min
5.3.7 Checking the touchscreen, page 98	•	5 min
5.3.8 Checking the printer (option), page 98	•	5 min
The chamber		
5.4.1 Cleaning the chamber, page 99	•	20 min
5.4.2 Checking spray arms, page 99	•	20 min
5.4.3 Checking spray arm manifolds, page 100	•	10 min
5.4.4 Checking the strainer in the bottom of the chamber, page 100	•	5 min
5.4.5 Checking guide rails, page 100	•	5 min
5.4.6 Checking lighting , page 100	•	5 min
5.4.7 Checking chamber overflow, page 101	•	5 min
Door		
5.5.1 Checking and adjusting door, page 101	•	120 min
5.5.2 Checking door seal, page 110	•	5 min
5.5.3 Checking wires, page 111	•	5 min
Process/Booster tank		
5.6.1 Checking check valve, page 111	•	5 min
Water connections		
5.7.1 Checking inlet connections, water, page 113	•	5 min
5.7.2 Checking inlet filters, water, page 113	•	5 min
5.7.3 Checking hose and pipe connections, water, page 114	•	5 min
Drain system		
5.8.1 Checking hose and pipe connections, drain, page 114	•	10 min
5.8.2 Checking check valve, drain (S-8666/S-8668), page 114	•	10 min
5.8.3 Drain tank (S-8668T), page 115	•	10 min
Steam system (option)		
5.9.1 Checking steam filter, page 115	•	10 min
5.9.2 Checking hose and pipe connections, steam, page 116	•	20 min
Dosing system		
5.10.2 Replacing transport hoses, page 118	•	10 min
5.10.3 Replacing the pump hose, page 118	•	10 min

Activity/Component Inspection/Replacement	Interval	Time required*
	Each year /4000 cycles	
5.10.4 Checking flow meters, page 119	•	10 min
5.10.5 Checking the dosage amount, page 120	•	30 min
5.10.6 Checking level sensors and suction lances, page 120	•	5 min
5.10.7 Labelling suction lances, page 121	•	5 min
Drying system		
5.11.2 Replacing a HEPA filter, page 123	•	10 min
5.11.3 Checking seals and hoses, page 123	•	5 min
5.11.4 Checking humidity sensor, page 123	•	5 min
5.11.5 Checking dryer fans, page 124	•	10 min
5.11.6 Checking the check valves, page 124	•	5 min
Test run (performed at each service)		
5.12.1 Function and leakage inspection, page 125		60 min
5.12.2 Temperature check, page 126		

**The amount of time required is estimated with the machine pulled out and the cover plates removed. The time can also vary depending on the installation settings, operating conditions and equipment level.*

5.2 Consumables

The following consumables are necessary for preventive maintenance and are not covered by the product warranty. See the spare parts list for the article numbers.

Consumables	Maintenance section
Bearing, bearing rings and screws	*5.4.2 Checking spray arms, page 99*
Spare parts kits, solenoid valve	*5.7 Water connections, page 113*
Transport hose	*5.10.2 Replacing transport hoses, page 118*
Pump hose	*5.10.3 Replacing the pump hose, page 118*
HEPA filter	*5.11.2 Replacing a HEPA filter, page 123*

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5.3 General

5.3.1 Checking electrical cables and connection points

Checking of electrical cables and connection points is performed to prevent both personal injuries and damage to the machine.

The following checks are performed on all cables found in or connected to the power box.

- Check that the isolator switch is working.
- Check that the power cord to the grid power is intact and undamaged and that it is installed as per the instructions in the instruction manual.
- Check that all electrical cables for electricity supply are correctly connected to the installation terminal blocks on the machine.
- Check that all the power cables are firmly secured in the connections. If necessary, tighten the screw connectors.
- Perform a visual inspection of all cables. Check that there are no damaged, non-insulated or unconnected cables.

5.3.2 Checking ventilation fans in the machine

Ventilation fans are used to increase air turnover and reduce the temperature in enclosed spaces. A faulty or poorly working fan can negativley affect the operation and lifespan of the component.

Ventilation fans are found at the following locations:

- Control cabinet (1 M15)
- Power box (1 UH02 - M16)
- Machine top (1 M14, 1 M17 and 1 M18)

Check the ventilation fans as follows:

- Check that the fan works and that it provides sufficient air flow.
- Clean the fan if needed.

5.3.3 Checking that the machine is level

This check is performed during installation and does not usually need to be repeated if the machine is not moved. A machine that is no longer level may make the loading and unloading of wash carts more difficult and may result in reduced washing capability.

Check and adjust the level:

1. Use a spirit level to check levels using the machine's guide rails inside the chamber.
2. Adjust where necessary, using the feet adjustment screws. The machine must be level with an error margin of ±2 mm.

5.3.4 Checking Quick guide

Check that the following are posted close to the machine.

- Quick guide
- Goods positioning sign

5.3.5 Checking alarms and message history

Check alarms and messages history and total number of completed cycles and enter the required details in a suitable log book. Note if an abnormally high number of warnings or alarms have been triggered in a specific area.

Necessary information can be displayed on the touchscreen:

- To see alarms and messages, go to **Main menu > Alarms and messages > Alarms and messages history** (see section *5.3.5 Checking alarms and message history, page 98*).
- To see the total number of completed cycles, go to **Main menu > System > Information** (see section *4.3.4 Display machine information, page 41*).

5.3.6 Checking docking

Dockings with reduced function may affect the water flow in the circulation system and reduce wash capability.

Check dockings as follows:

- Check that there is no visible damage to any parts.
- Check that the spring-loaded mechanism works and that the docking does not get stuck in the compressed position.
- Replace parts that are damaged or have impaired function with new parts.

5.3.7 Checking the touchscreen

Check the touchscreen as follows:

- Check that the touchscreen is undamaged and can be easily read.
- Check that the touchscreen function works.
- Clean the touchscreen when necessary. Use an alcohol-based cleaning agent or medicinal alcohol.

5.3.8 Checking the printer (option)

Check the printer as follows:

- Replace the paper roll if needed. If so, perform the actions in the following step by step list.
- Check that the printer lid is closed.

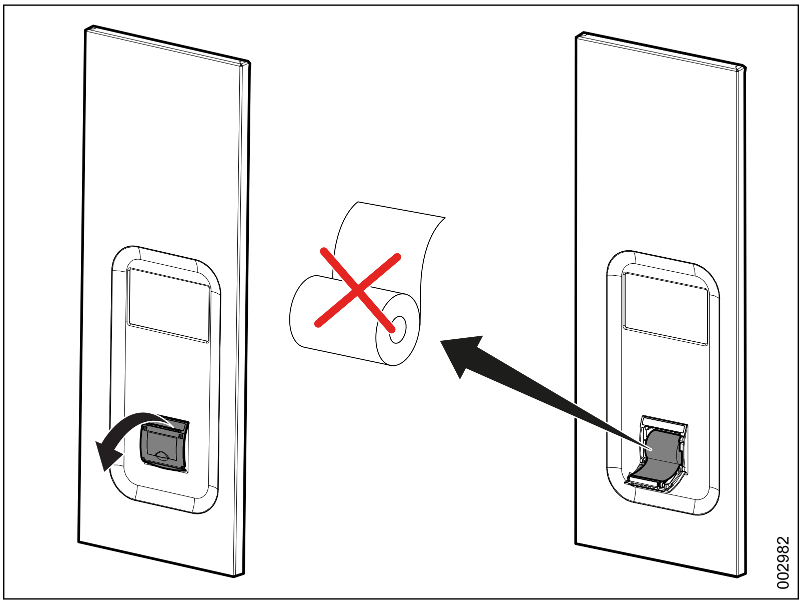
The built-in thermal printer (option) does not have a colored ink cartridge that needs to be replaced, but instead uses heat-sensitive paper for printing.

Note! Never use the printer without paper. Do not pull on the paper feed when the printer is on.

Replace the paper roll on the built-in printer as follows:

1. Open the printer lid by gently pulling out and then down.
2. Remove the existing roll and replace it with a new one. Make sure to position the paper roll in the correct orientation.

3. Close the lid on the printer.
4. Print a test page. See section *4.3.8.2 Print test page (login required), page 51*.



5.4 The chamber

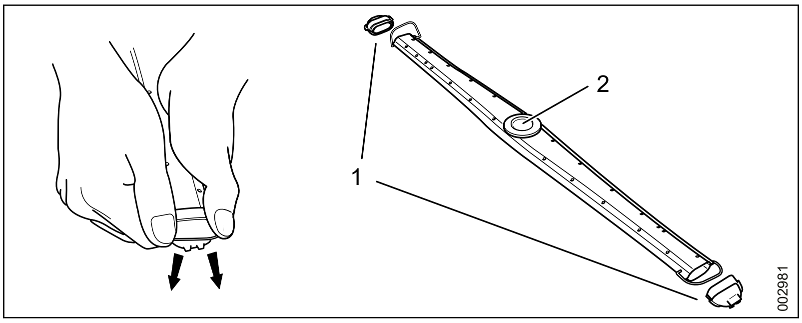
5.4.1 Cleaning the chamber

Clean the chamber by running the **Deliming** program together with an acidic detergent that removes rust, stains and hard water scale from washer-disinfector chambers. Contact a local Getinge sales representative for information regarding what detergent to use.

Cleaning of the chamber is best performed in conjunction with test runs.

5.4.2 Checking spray arms

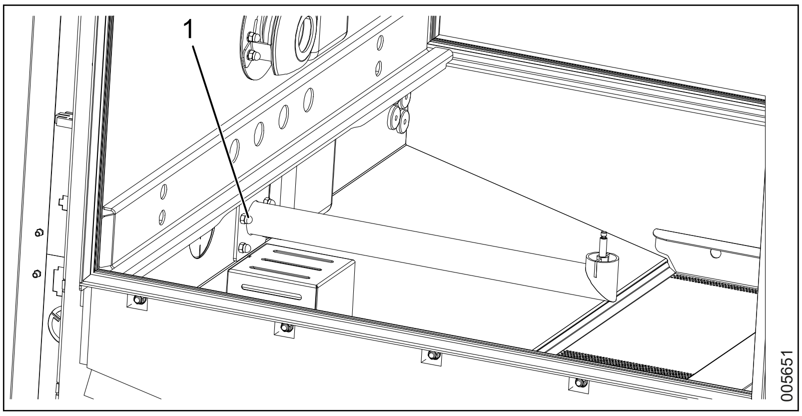
- Check that all spray arms are correctly fitted.
- Check that the spray arms can rotate freely.
- Check the spray arm bearings, and replace damaged or worn parts.
- Check that the spray arm holes are not blocked.
- Clean if necessary. The spray nozzles can be removed to facilitate cleaning.



1 Spray nozzle

2 Spray arm bearings

5.4.3 Checking spray arm manifolds



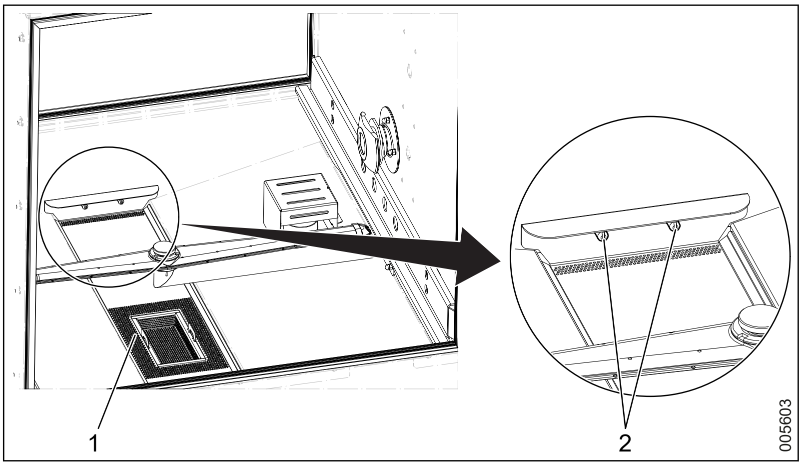
1 Dome nuts

Remove and clean the spray arm manifold (the part that connects to the spray arm) by unscrewing the four dome nuts in the lower section of the chamber.

5.4.4 Checking the strainer in the bottom of the chamber

Clean where necessary and inform the user that daily maintenance must be performed.

Check that the strainer in the bottom of the chamber is correctly positioned and that the strainer steering holes are connected to the steering pins before pressing the strainer firmly in place.



1 Strainer

2 Steering pins

5.4.5 Checking guide rails

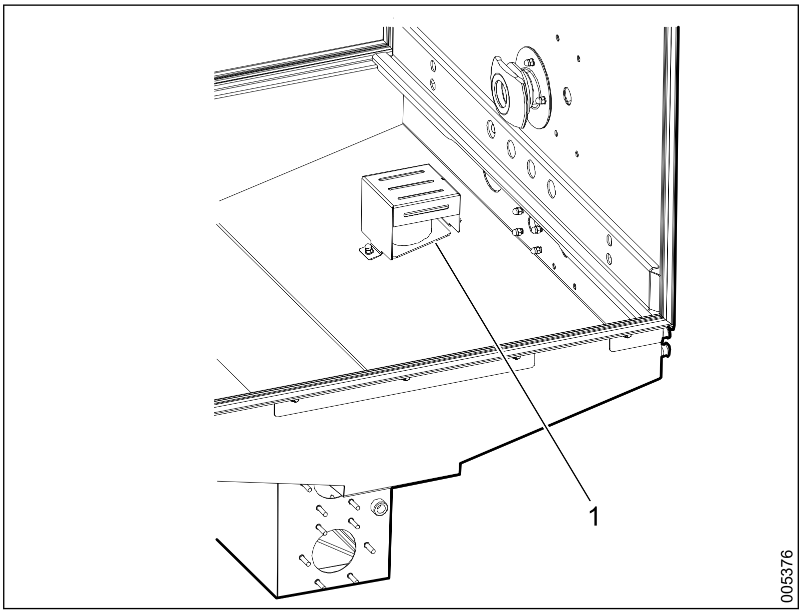
Check that the guide rails work correctly and are not damaged. It must be possible to move the wash carts easily backwards and forwards.

5.4.6 Checking lighting

Check that the lamp that lights up the chamber space works and replace it if necessary. To access the lamp during replacement, the protective plate above the door on the soiled side must be removed.

5.4.7 Checking chamber overflow

Check that the overflow float (plastic part) is free to move up and down.



1 Overflow float

5.5 Door

5.5.1 Checking and adjusting door

Checking and adjustment of doors must be carried out on both the soiled side and the clean side (when appropriate).

5.5.1.1 Operating door in manual mode

To operate the door manually, go to **Menu > System > Service > Diagnostics > Digital outputs**.

See section *4.3.3.4 Digital outputs, page 40* for more information.

5.5.1.2 Checking signals

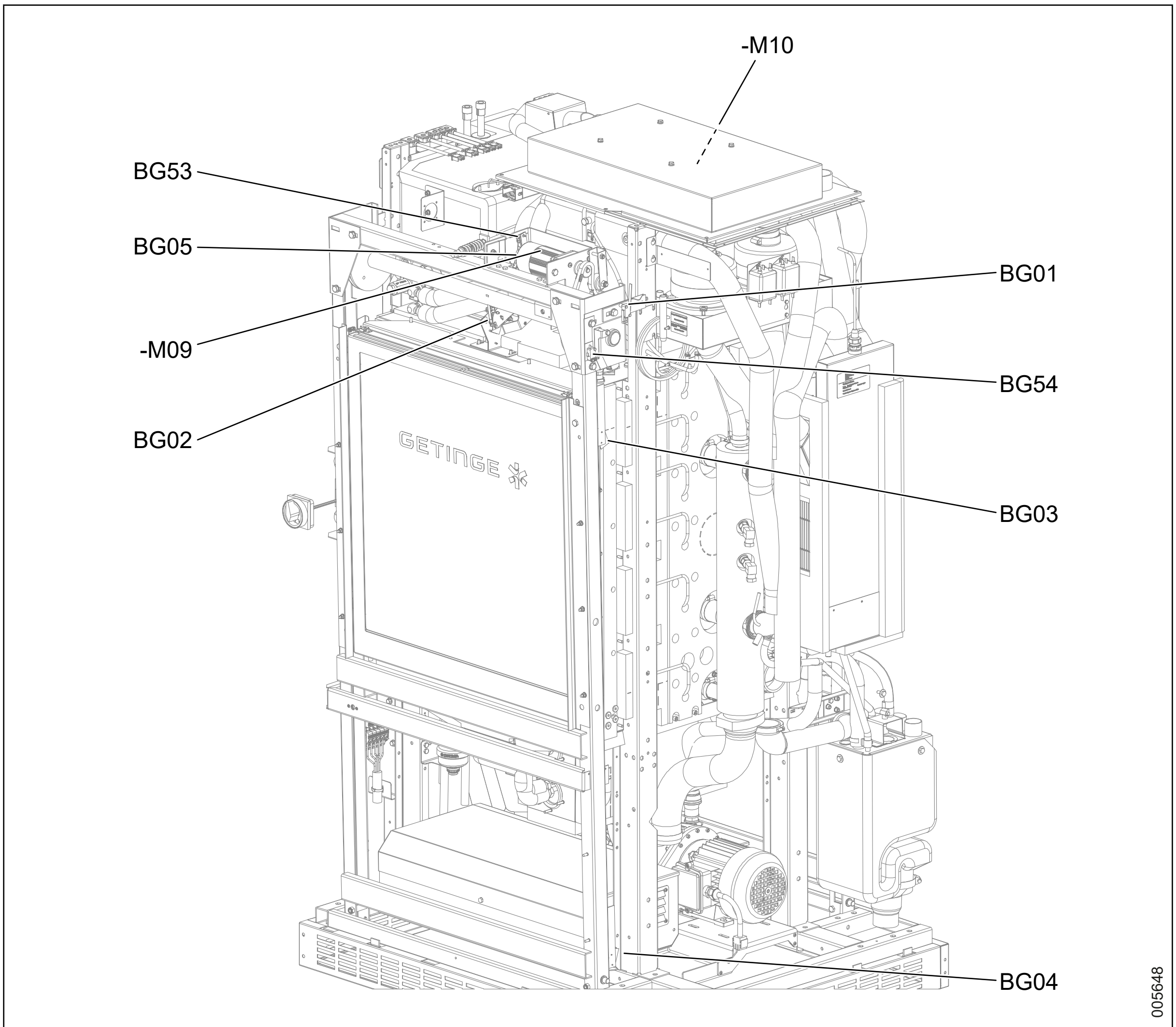
Check the signal from the switch to the control system by reading the relevant diodes in the I/O module. See section *7.6.2.1 Checking module status, page 148* and *4.3.3.2 Digital inputs, page 38* for more information.

5.5.1.3 Door equipment - Overview

The following section covers door switches and door motors for the soiled and clean sides.

Positions are indicated with codes for each piece of equipment.

5.5.1.3.1 Door equipment, soiled side



BG54	Door soiled side opening obstruction	**-M10**	Door locking motor soiled side (lock/ unlock)
BG05	Door soiled side crush protection closing, software	**-M09**	Door motor soiled side (open/close)
BG52	Door soiled side locked hardware protection	**BG01**	Door soiled side unlocked
BG53	Door soiled side crush protection, hardware	**BG03**	Door soiled side closed
BG51	Door soiled side unlocked hardware protection	**BG04**	Door soiled side open

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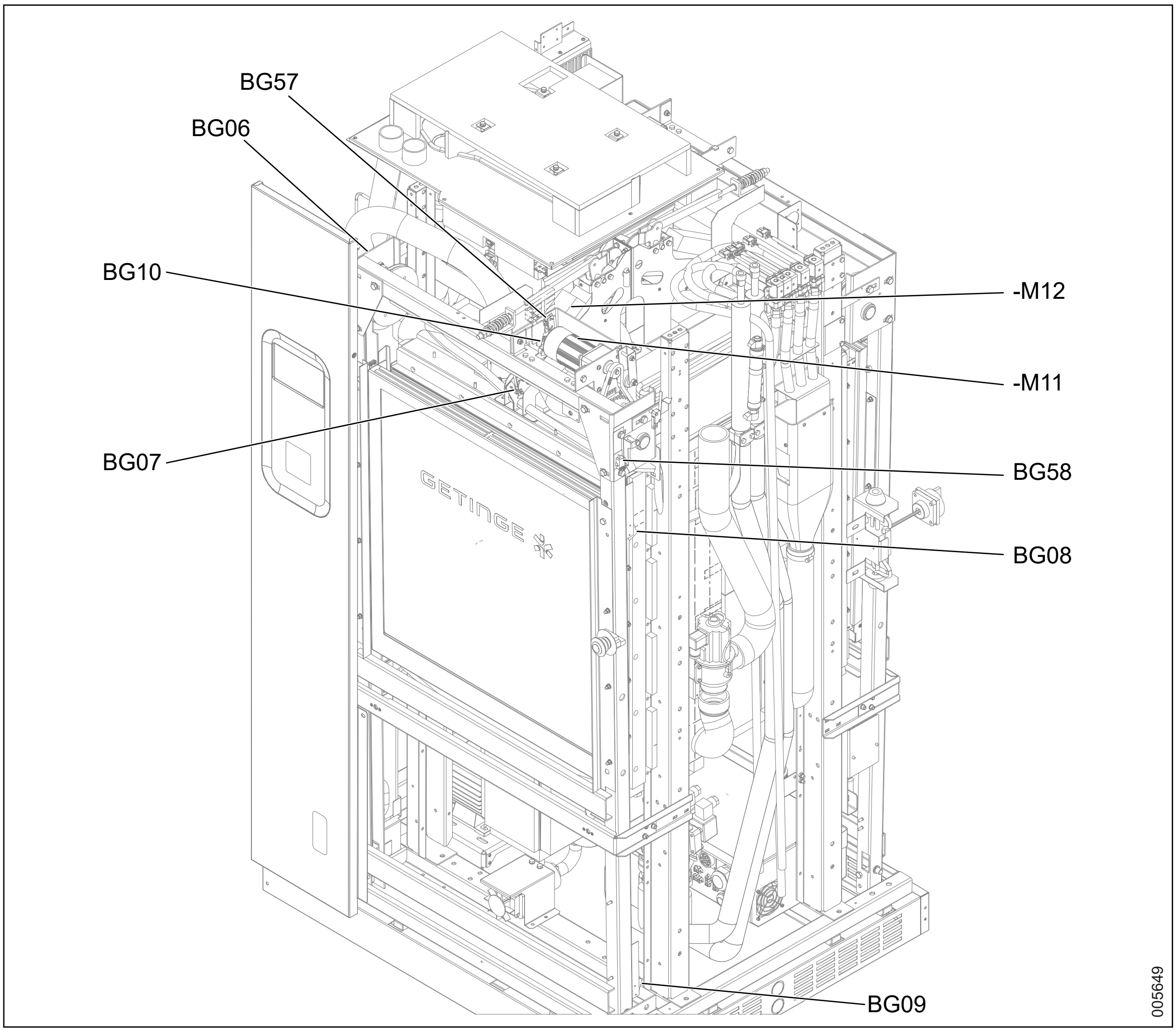
5.5.1.3.2 Door equipment, clean side



BG10	Door clean side return switch upward	**BG55**	Door clean side unlocked
BG57	Door clean side crush protection upward	**-M11**	Door motor clean side (open/close)
-M12	Door motor clean side (lock/unlock)	**BG58**	Door clean side cord slack downward
BG06	Door clean side unlocked	**BG08**	Door clean side closed
BG56	Door clean side locked hardware protection	**BG09**	Door clean side open

5.5.1.4 Adjustment - door open (door lowered)

5.5.1.4.1 Adjusting door open (door lowered) - vertically

Affected switch

Pos	Description	I/O module	Voltage
BG04	Soiled side - Door open	KF06, channel 4	24 VDC
BG09	Clean side - Door open	KF06, channel 12	24 VDC

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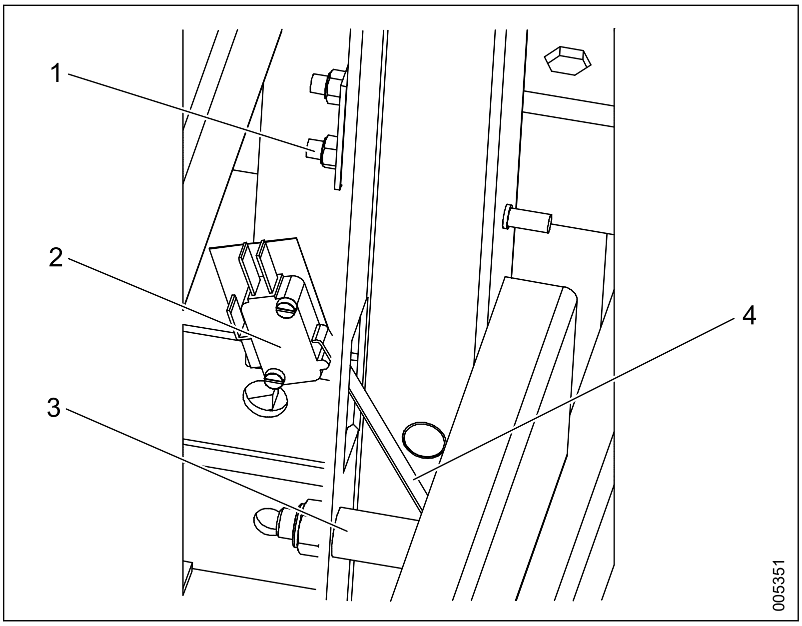
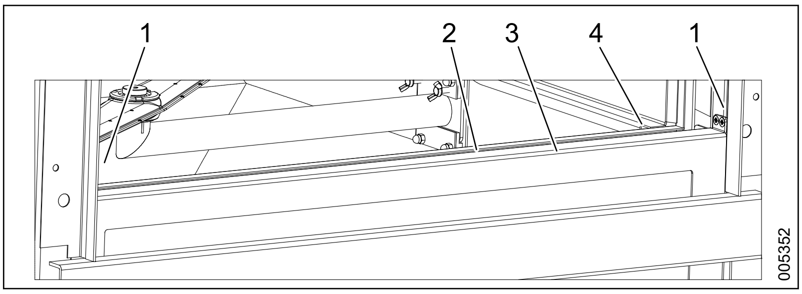
Adjust the switch as follows:

- The switch arm must be triggered before it reaches the stop.
- The switch (BG04/BG09) is adjusted to ensure the upper door edge is at the same height as the track.
- The cords must be taut when the door is open.

The switch must not reach the stop.



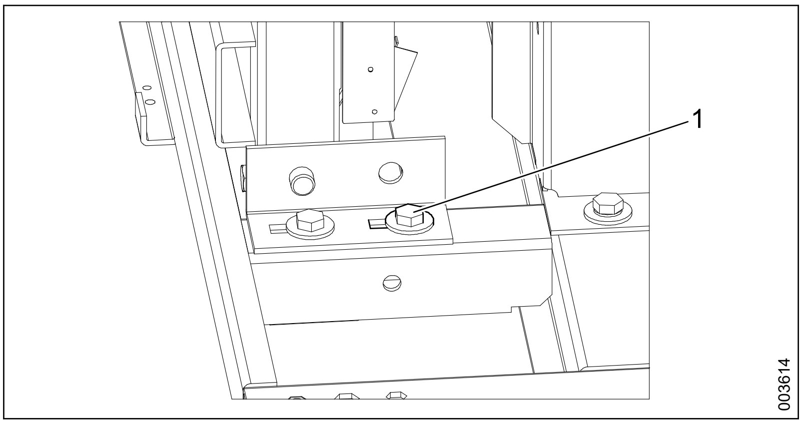
1 Bolts (adjusting switch
2 BG04
3 Stop
4 Switch arm



1 Wire
2 Gap
3 Upper door edge
4 Guide rails

5.5.1.4.2 Adjusting door open (door lowered) - horizontally

1. Adjust horizontally by sliding the left and right brackets inward and outward.



1 Bracket

2. Adjust to create a 2-3 mm gap between the door and chamber, when the door is pressed against the chamber.
3. Adjust so that the gap between the door and chamber is as parallel as possible.

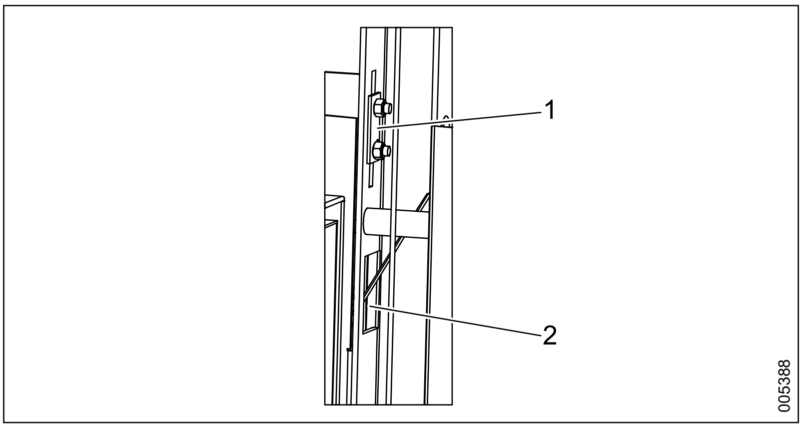
5.5.1.5 Adjustment - door closed (door up)

Affected switch

Pos	Description	I/O module	Voltage
BG03	Soiled side - Door closed	KF06, channel 3	24 VDC
BG08	Clean side - Door closed	KF06, channel 8	24VDC

The door closed switches (BG03 and BG08) usually never require readjustment.

The switch must be adjusted to the center of the adjustment slot.



1 Adjustment, switch door closed (door up)
2 BG03

5.5.1.6 Adjustment - cord slack protection

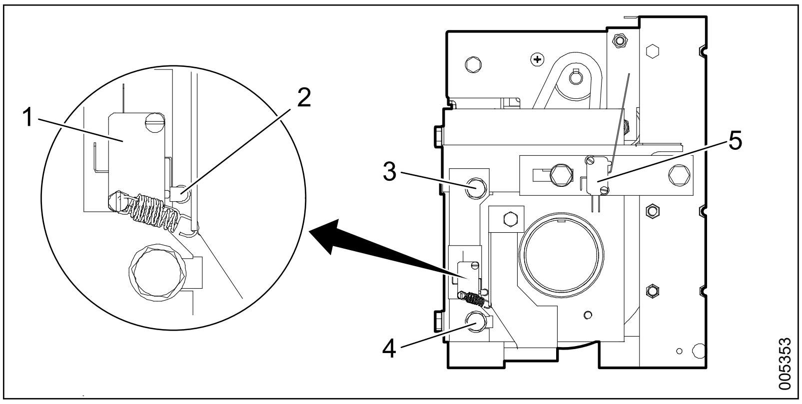
The switch triggers if the door stops on the way downward (the cords slacken).

Affected switch

Pos	Description	I/O module	Voltage
BG54	Soiled side - Cord slack	N/A	230 VAC
BG58	Clean side - Cord slack	N/A	230 VAC

Adjust as follows:

1. Position the door halfway (about 100 mm) from being closed.
2. Unscrew screws: adjustment screw, upper and adjustment screw; lower to adjust the position of the switch.
3. Adjust the switch so that the switch contact point is almost touched. Leave a space of 1 mm.



1 BG54
2 Switch contact point
3 Adjustment screw, upper
4 Adjustment screw, lower
5 BG01

5.5.1.7 Crush protection

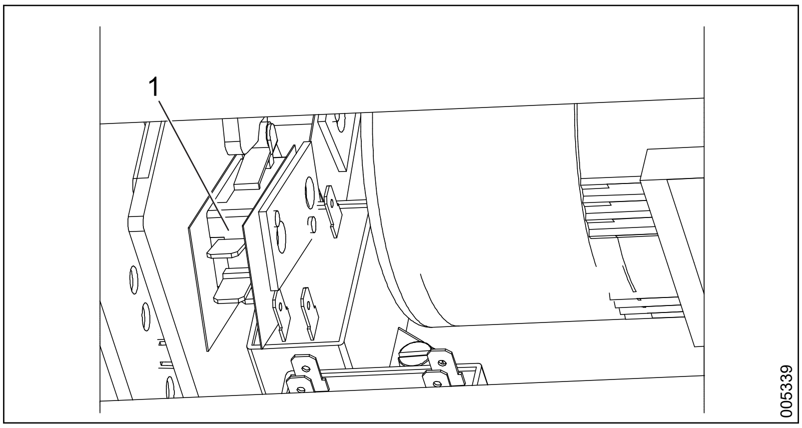
The crush protection consists of two different switches on each door (soiled and clean side):

The return switch is triggered when the door is on its way upward if the opposing force is greater than the set value and it then returns to the open position.

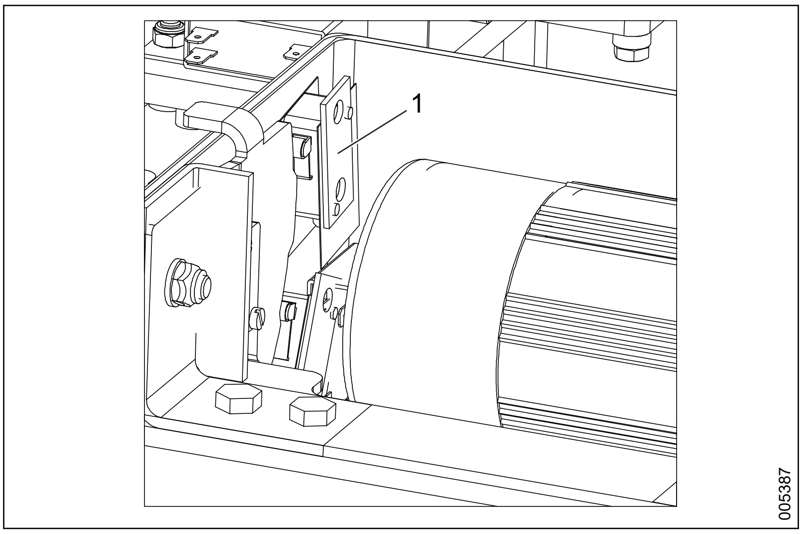
The hardware protection stops the door by cutting the power if the return switch does not work properly.

Affected switch

Pos	Description	I/O module	Voltage
BG05	Soiled side - Crush protection, return switch	KF06, channel 5	24 VDC
BG10	Clean side - Crush protection, return switch	KF06, channel 10	24 VDC
BG53	Soiled side - Crush protection, hardware protection	N/A	230 VAC
BG57	Clean side - Crush protection, hardware protection	N/A	230 VAC



1 BG05



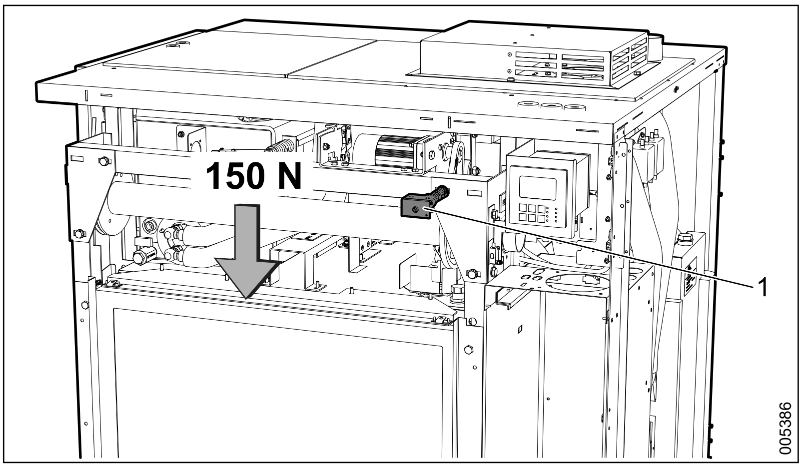
1 BG53

Check the power setting and adjust if required according to the following:

5.5.1.7.1 Check crush protection

Check the crush protection as follows:

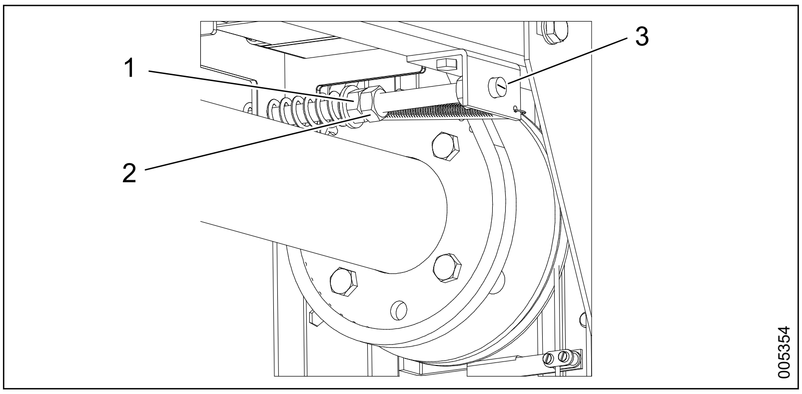
- Use a force meter to check that the crush protection is triggered when subjected to an opposing force of 150 N (permitted range: 130-150 N).
- Remember that if the adjustment spring is too loose, there is a risk that the door will move erratically or cannot be fully closed.



1 Adjusting spring, crush protection

5.5.1.7.2 Adjust crush protection

1. Loosen lock nut.
2. Hold a flat screwdriver at the screwdriver slot and adjust the crush protection using the adjustment nut: Turn clockwise to increase the force.
3. Check setting according to section *5.5.1.7.1 Check crush protection, page 107*.



1 Adjustment nut
2 Lock nut
3 Slot for screwdriver

5.5.1.8 Adjustment - door unlocked

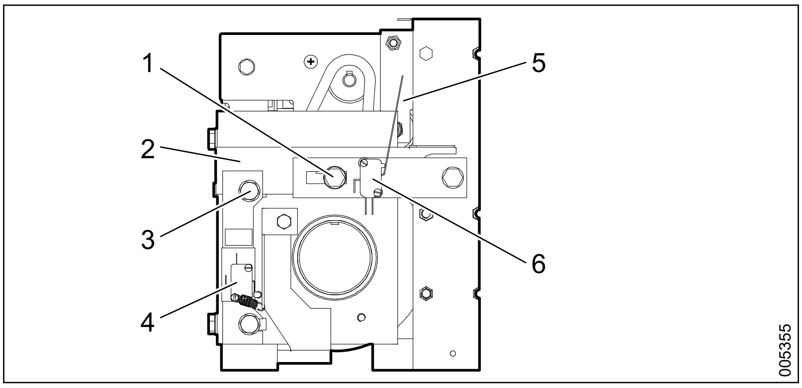
Hardware protection is triggered if the signal switch does not work.

Affected switch

Pos	Description	I/O module	Voltage
BG01	Soiled side - Door unlocked, signal switch	KF06, channel 2	24 VDC
BG06	Clean side - Door unlocked, signal switch	KF06, channel 7	24 VDC

Adjust as follows:

1. Loosen screw, adjust switch to adjust the position of the signal switch.
2. Adjust the signal switch's (BG01/BG06) position by sliding the glide plate in or out: The switch arm must be completely straight (when the door is in the unlocked position).
3. Check that it slides easily between the unlocked and locked positions. Otherwise, adjust the glide force with the screw to adjust glide force.



1 Screw to adjust glide force	4 BG54
2 Glide plate	5 Switch arm
3 Screw to adjust switch	6 BG01

5.5.1.9 Adjustment - door locked

The switches are used to indicate that the door is in the locked position.

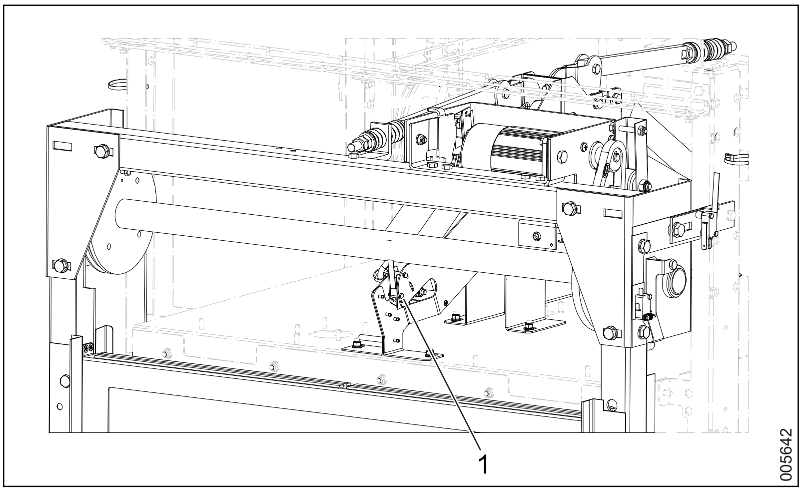
Hardware protection is triggered if the signal switch does not work.

Affected switch

Pos	Description	I/O module	Voltage
BG02	Soiled side - Door locked	KF06 channel 1	24 VDC
BG07	Clean side - Door locked	KF06 channel 6	24 VDC

Adjust as follows:

1. Unscrew the screw to adjust the signal switch.



1 Screw

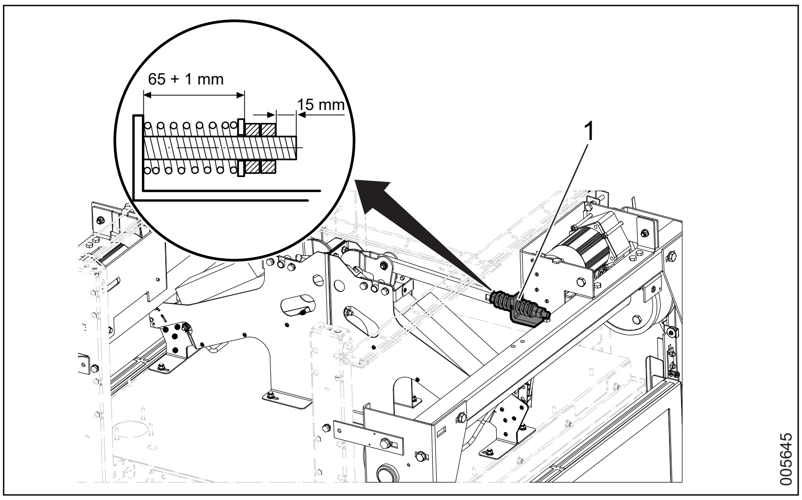
2. Adjust the signal switch's (BG02/BG07) position to fully trigger its arm (when the door is in the locked position).
3. Adjust the locking force according to section *5.5.1.9.1 Adjusting locking force, page 110*

5.5.1.9.1 Adjusting locking force

When the door reaches its top position, the door locking motor pulls the door in from the outer to the inner position.

Adjust the locking force according to the following illustration.

The spring must never be fully compressed.



1 Adjustment spring, locking force

5.5.2 Checking door seal

The door seal consists of a rubber strip that lines the chamber opening.

The purpose of the check is to investigate the seal's ability to form a tight seal between the door and the chamber.

Failure to perform this inspection can result in unwanted leakage of water and steam.

- Check that the seals are not damaged and that they are securely in place around the chamber opening.
- Check that the elasticity of the seals is sufficient and that they have not become too hard or brittle.
- Clean or replace the seal if necessary.
- Perform a functional check according to section *5.12.1 Function and leakage inspection, page 125*.

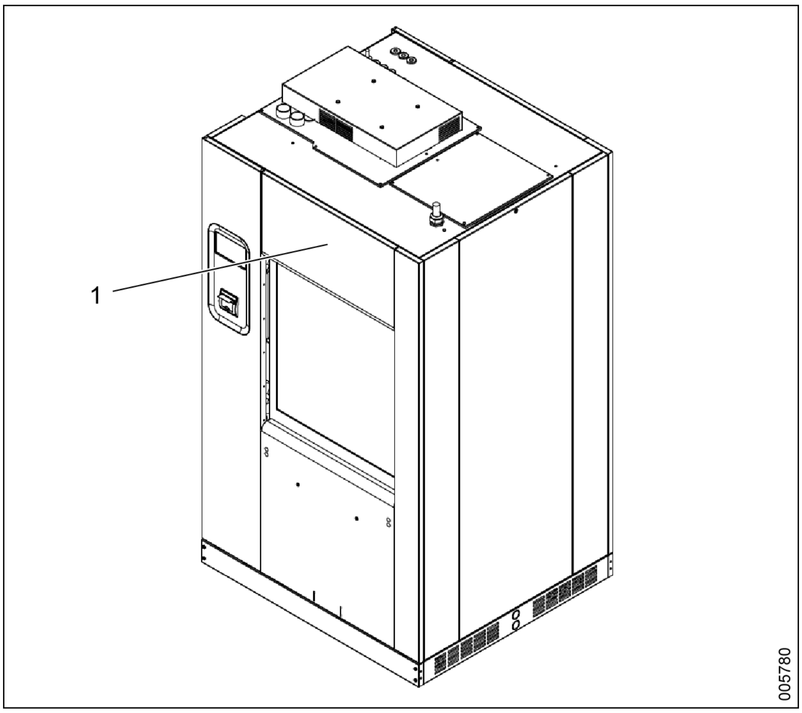
5.5.3 Checking wires

Check the condition of the wires to the sensors controlling the door; remove and replace if necessary.

5.6 Process/Booster tank

5.6.1 Checking check valve

1. Turn off the power to the washer-disinfector.
2. Remove the upper cover plate on the clean side of themachine.



1 Upper cover plate

3. Loosen the hose clamp and the overflow draining hose from the booster/process tank.

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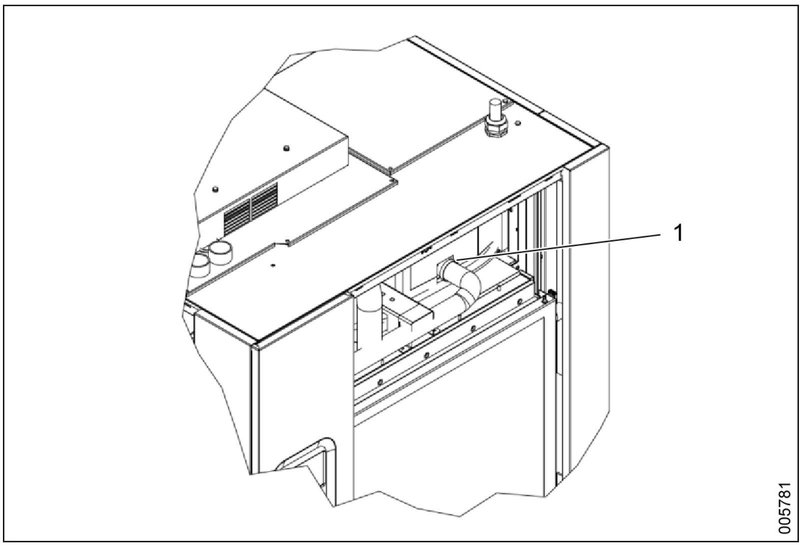
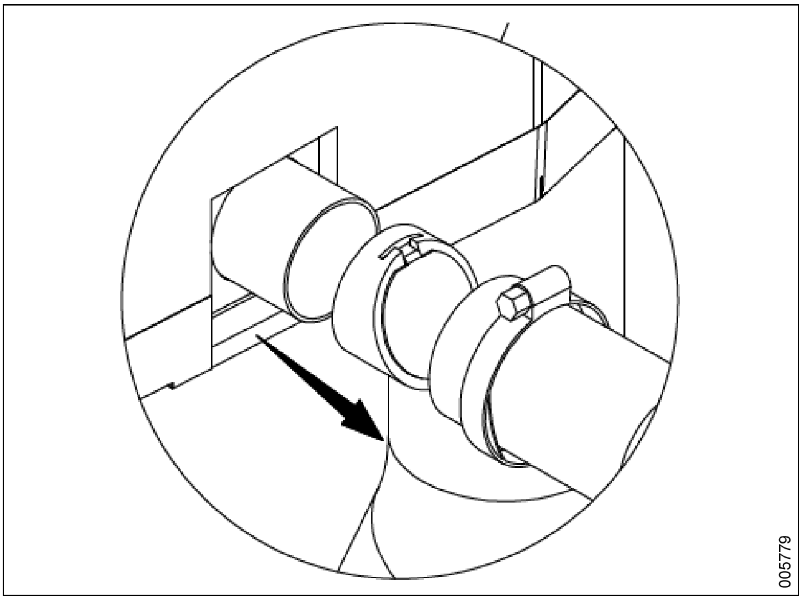


1 Hose clamp and the overflow draining hose

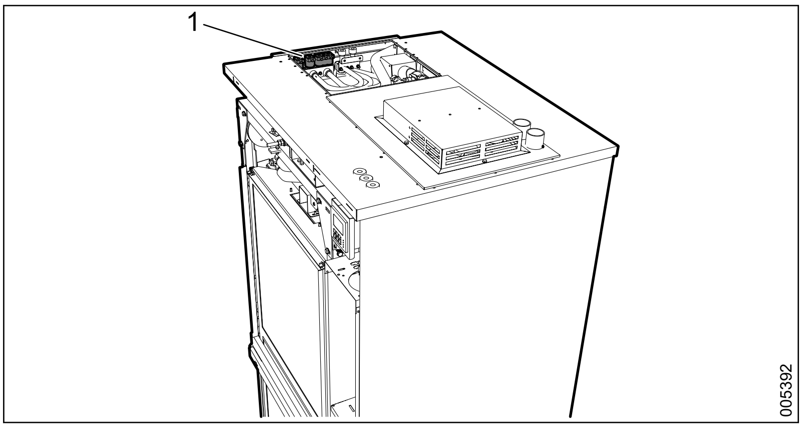


4. Check that the ckeck valve can be opened and closed without obstructions.
5. Install the upper cover plate on the clean side of the machine.
6. Turn the power off.

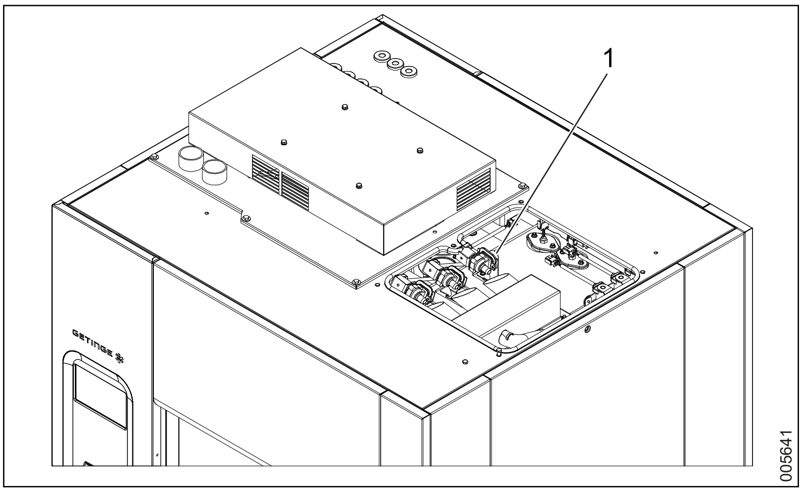
5.7 Water connections

5.7.1 Checking inlet connections, water

Check that the connections to and from the solenoid valves are correctly installed and are not leaking.



1 Solenoid valves, S-8666/S-8668



1 Solenoid valves, S-8668T

5.7.2 Checking inlet filters, water

Checks and cleaning of the solenoid valves are done to prevent clogging that could result in leaks and slow filling.

Check the valves on the water connection by dismantling them according to the illustration below. Clean the flow limiters and filters if necessary.

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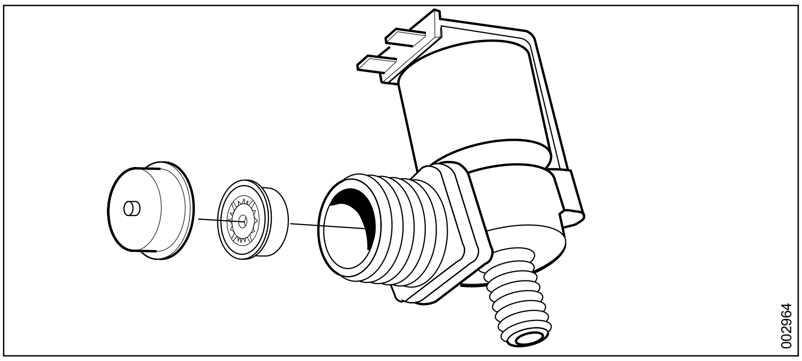


5.7.3 Checking hose and pipe connections, water

Check all hose and pipe connections between the inlet and the chamber for leaks.

5.8 Drain system

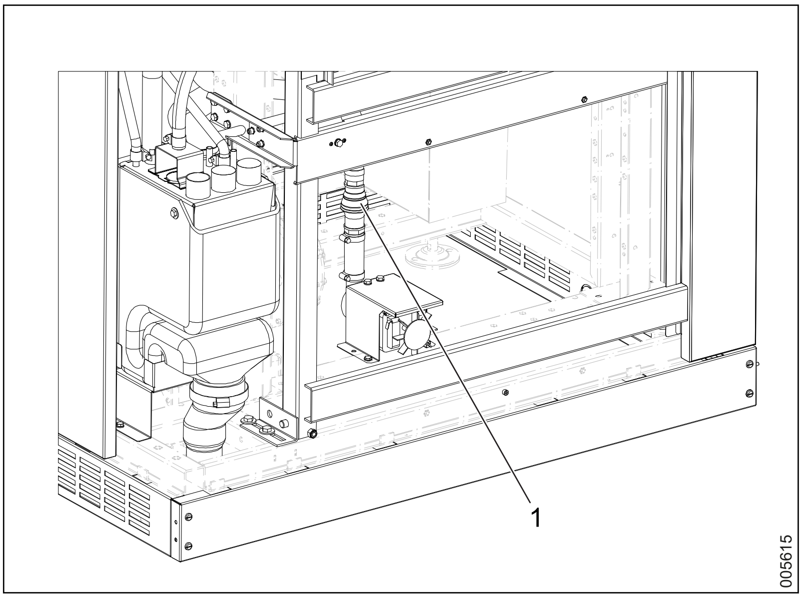
5.8.1 Checking hose and pipe connections, drain

Check that all connections in the drainage system are correctly installed and that no leaks are present.

5.8.2 Checking check valve, drain (S-8666/S-8668)

The drain check valve is located after the drain pump outlet.

A faulty check valve can allow waste water to leak back into the chamber.



1 Drain check valve

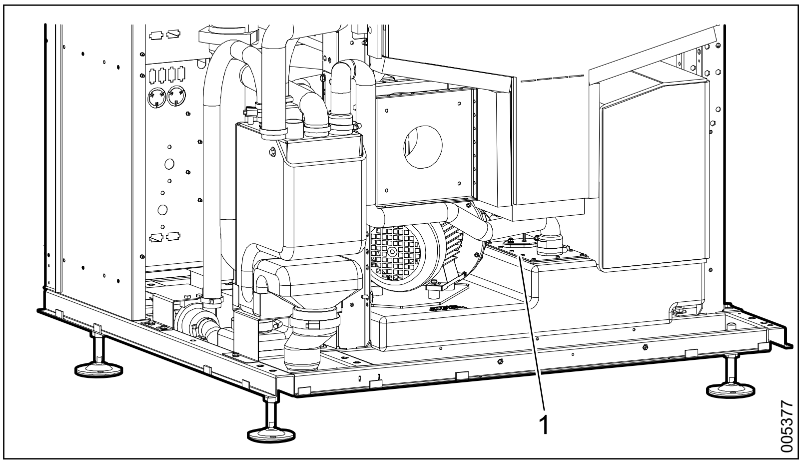
Remove the check valve and test for proper operation. Hold the check valve in a vertical position when testing it.

Water should be allowed to flow from only one direction - away from the chamber. If defective, replace the check valve.

Install the check valve so that the direction of flow is towards the drain.

5.8.3 Drain tank (S-8668T)

For manual cleaning, follow the instructions in section *7.12.5 Drain tank (S-8668T), page 180*.



1 Access to the drain tank

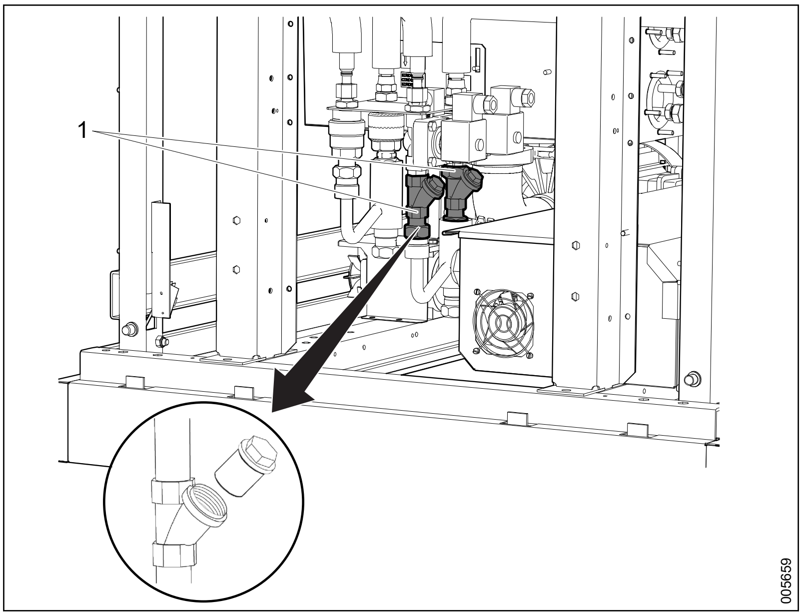
5.9 Steam system (option)

Checks of the steam system are only performed on machines that use steam heating.

5.9.1 Checking steam filter

Unscrew and clean the steam filter next to the valve for incoming steam.

S-8666 and S-8668



1 Steam filter S-8666 and S-8668

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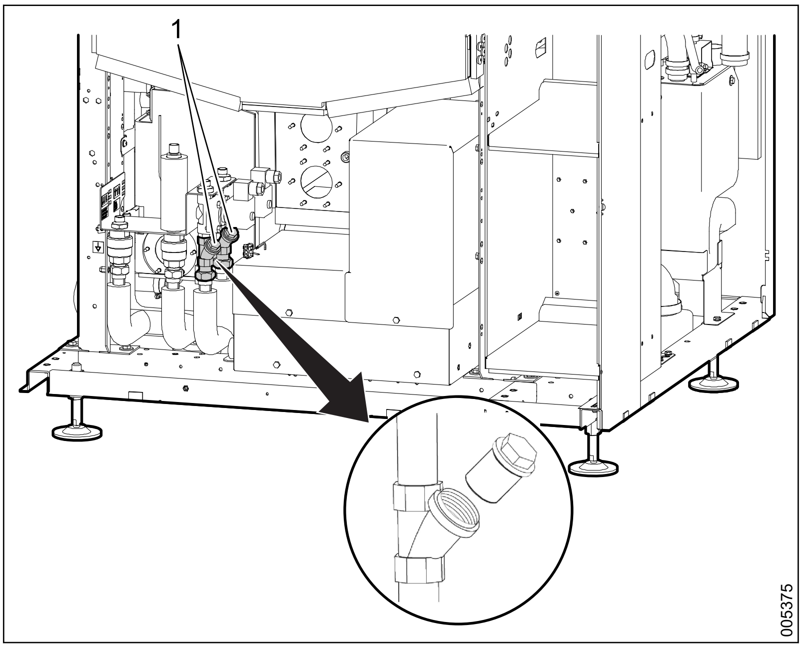
S-8668T



1 Steam filter S-8668T

For inspecting and repairing the steam valve and condensate trap, see section *7.17.1 Steam valves and condensate traps, page 190*.

5.9.2 Checking hose and pipe connections, steam

Check all hoses and pipes in the steam system for leaks.

Replace damaged steam hoses according to the instructions:

1. Unscrew the screw-on connector and remove the existing hose.
2. Lay and install the replacement hose in the same way as the old hose.
3. Check that no leaks occurred.

5.10 Dosing system

When handling detergents, follow the safety instructions as indicated on the detergent container label and/or in its material safety data sheet.

The following inspection measures are to ensure the proper operation of the dosing system. Not following the inspection measures may result in inadequate/no cleaning and/or lime deposits.

The dosing system hoses can harden and become brittle after long periods of use. Their service life is highly dependent on which chemicals are used.

5.10.1 Dosing hoses

There are two types of hoses in the dosing system:

- Transport hoses
- Pump hoses (hoses inside pumps)

Transport hoses are fitted between the dosing pump and the detergent container and between the dosing pump and the machine. There are different grades of transport hose, where the grade used is determined by the detergent used, see section *5.10.1.1 Dosing hoses from factory, page 117*.

Pump hoses are fitted inside the dosing pumps. They are exposed to greater mechanical force and therefore must comply with stringent strength demands. The pump hose material should be adapted to cope with the detergent they are used for in accordance with section *5.10.1.3 Recommended dosing hoses, page 118*.

5.10.1.1 Dosing hoses from factory

Dosage No.	Transport hose	Pump hose
Dosing 1	503645300, Tygon F-4040-A	6027521101, Tygon A-60-G
Dosing 2	503645300, Tygon F-4040-A	6027521104, Pharmed BPT
Dosing 3	503645300, Tygon F-4040-A	6027521101, Tygon A-60-G
Dosing 4	503645300, Tygon F-4040-A	6027521101, Tygon A-60-G

The hose kit delivered with the machine consists of; one 6027521101, Tygon A-60-G one 6027521104, Pharmed BPT and one 6027521105, Versilon F-5500-A. Replace the hoses as recommended on the detergents used, see section. *5.10.1.3 Recommended dosing hoses, page 118*. The hoses in the detergent pumps are exposed both to chemical wear and mechanical wear. To achieve safe operation and satisfactory service life, it is important that the correct hose material is selected with regard to the chosen detergent. The hoses in the detergent pumps are consumables parts with an expected service life of 1 year.

5.10.1.2 Dosing hoses available for ordering

Part No.	Type of hose	Design
6027521101, Tygon A-60-G	Pump hose	Black, label text on bag: (N-3/16-2-WT6 SET)
6027521104, Pharmed BPT	Pump hose	Beige, label text on bag: (P-3/16-2-WT6 SET)
6027521105, Versilon F-5500-A	Pump hose	Black, label text on bag: (F-3/16-2-WT6 SET)
503645300, Tygon F-4040-A	Transport hose	Transparent, yellow
6001662701, Tygon A-60-G	Transport hose	Black (white text), for Heavy Soil

5.10.1.3 Recommended dosing hoses

5.10.1.3.1 Dosing hoses tested and recommended for Getinge Clean

When handling detergents, follow the safety instructions as indicated on the detergent container label and/or in its material safety data sheet.

Detergent	Recommended pump hose
Getinge Clean Heavy Soil Detergent	6027521101, Tygon A-60-G
Getinge Clean Universal Detergent	6027521101, Tygon A-60-G
Getinge Clean Enzymatic Detergent	6027521104, Pharmed BPT
Getinge Clean MIS Detergent	6027521101, Tygon A-60-G
Getinge Clean Prion Detergent	6027521101, Tygon A-60-G
Getinge Clean Neutralizer	6027521104, Pharmed BPT
Getinge Clean Neutralizer Plus	6027521101, Tygon A-60-G
Getinge Clean Instrument Lubricant	6027521105, Versilon F-5500-A
Getinge Clean Instrument Lubricant Plus	6027521105, Versilon F-5500-A
Getinge Clean Rinse Aid	6027521104, Pharmed BPT

5.10.1.3.2 Dosing hoses recommended for non-Getinge detergents

Dosing hoses recommended for non-Getinge detergents	
Type of detergent	**Recommended pump hose**
Alkaline detergent	6027521101, Tygon A-60-G
Acidic detergent (dishwasher detergent, fabric softener, neutral)	6027521104, Pharmed BPT
Acidic detergent based on phosphoric acid	6027521105, Versilon F-5500-A
Enzymes	6027521104, Pharmed BPT
Lubricants	6027521105, Versilon F-5500-A

The chemical composition of different detergents on the market can vary considerably. If uncertain when selecting, contact the detergent supplier.

5.10.2 Replacing transport hoses

The replacement interval stated in the service table is the minimum recommended. Check and replace the hoses on a regular basis to prevent leakages from occurring and to prevent the dose amount to deviate.

The dosing must be checked after replacement of the transport hose (see section *6.7 Calibration of dose monitors, page 136*).

5.10.3 Replacing the pump hose

There are different types of pump hoses, and the type used is determined by the detergent used. Make sure to use the right type of hose according to section *5.10.1.3 Recommended dosing hoses, page 118*.

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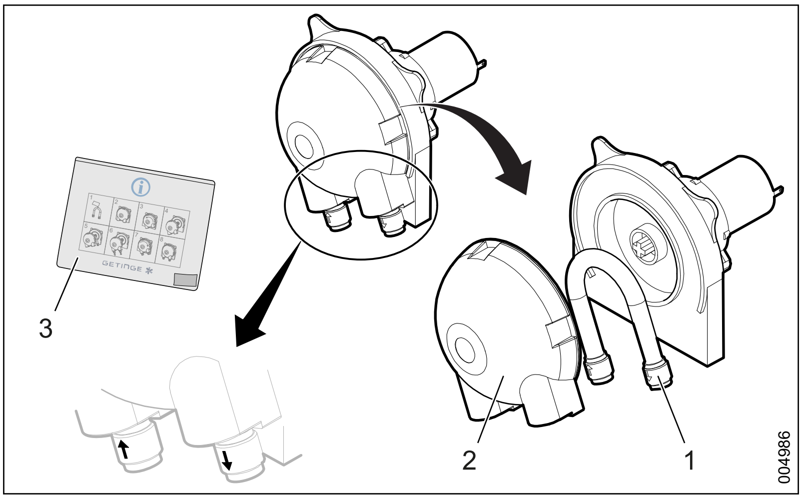
The replacement interval stated in the service table is the minimum recommended. Check and replace the hoses on a regular basis to prevent leakages from occurring and to prevent the dose amount to deviate.

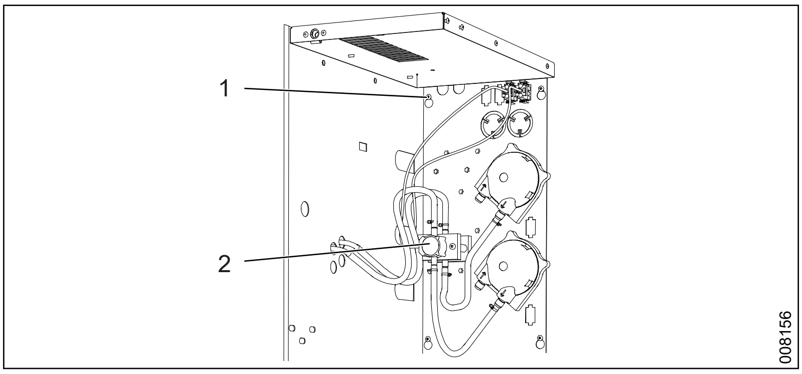


1 Pump hose
2 Cover
3 Plastic bag including grease

1. Remove the cover.
2. Replace the existing pump hose with a new one.
3. Refit the cover.
4. Check the dosing, see section *6.7 Calibration of dose monitors, page 136*.

5.10.4 Checking flow meters

Rinse the flow sensors with warm water (~40°C) regularly. The interval stated in the service table is the minimum recommended. The sensors impeller can get stuck or slower by crystallized or dried chemicals in the sensor house.



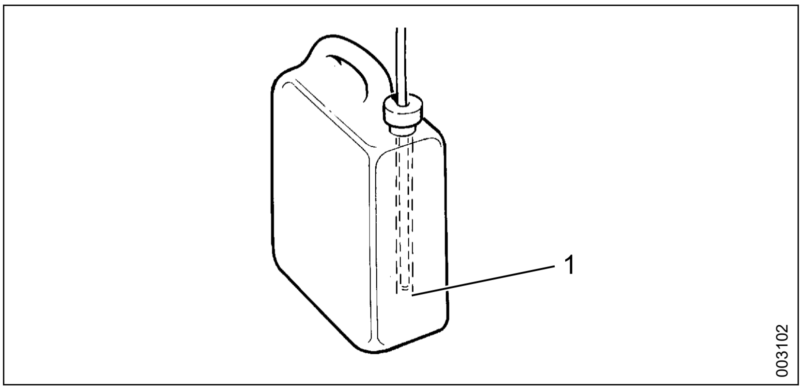
1 Screw (one in each corner).
2 Flow sensor

1. Remove the 4 screws on the panel.
2. Pull the hoses out through the panel and lift off the panel.
3. Remove the flow sensor.
4. Rinse the flow sensor with warm water (~40°C).
5. Install the flow sensor.
6. Install the panel.

5.10.5 Checking the dosage amount

Check that the correct amount of detergent is dosed for each pump according to the instructions in section *6.7 Calibration of dose monitors, page 136*.

5.10.6 Checking level sensors and suction lances



1 Level sensor, suction lance

Handle the suction lances with care. Clean and replace the suction lances if necessary.

A level sensor is fitted to the end of each suction lance. An alarm will be displayed at the start of the program if there is no detergent in the container.

Do the following to check the level sensor:

1. Lift the suction lance out of the container.
2. Check that the signal to the control system is not lost for the level sensor in question according to the following:

Container/Suction lance	Module	Channel
Dosing 1	KF06	12
Dosing 2	KF06	13
Dosing 3	KF07	1
Dosing 4	KF06	2

3. Repeat the procedure for all fitted containers.

See section *7.6.2.1.1 Main mode, page 149* for more information about the modules in question.

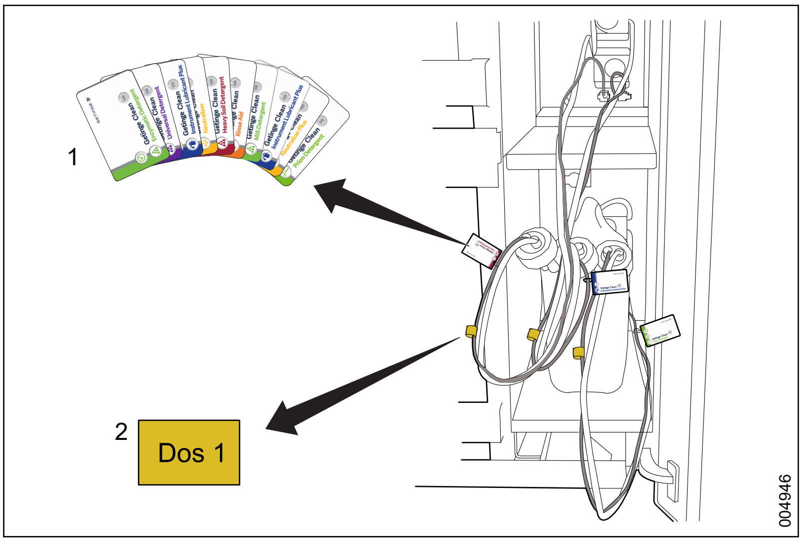
A missed alarm signal for an empty container can result in the process running without any detergent.

5.10.7 Labelling suction lances

When changing from one type of detergent to another, the dosing pump must be recalibrated.

The suction lances must be labelled to ensure that the correct detergent is used.

Getinge range detergent:

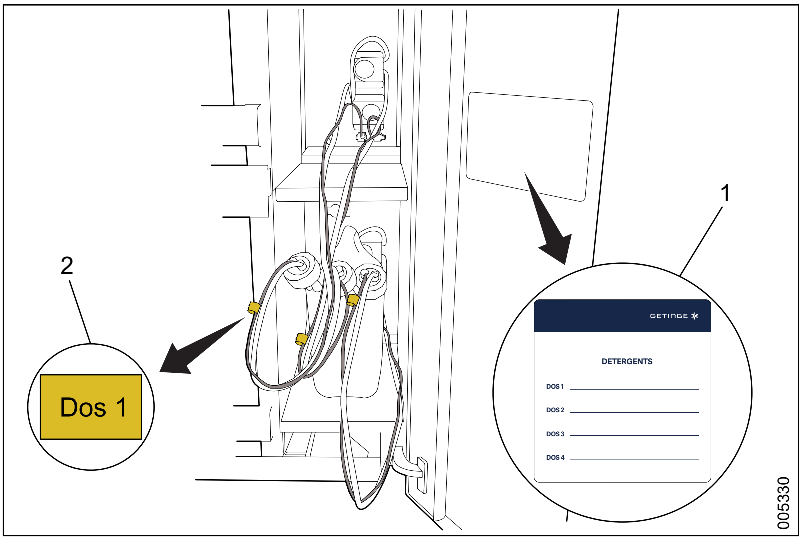


1 Detergent label

2 Dosage tag

1. For each dosing pump, choose the correct Getinge detergent label.
2. Using a cable tie, attach the label to the grey cable between the suction lance and the dosing pump.

Other detergents (non-Getinge range):



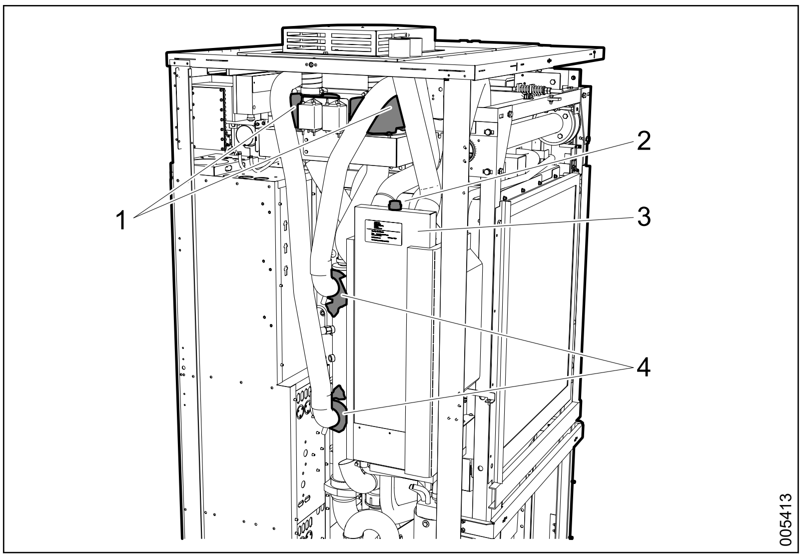
1 List of detergents
2 Dosage tag

1. Attach the list of detergents to the inside of the door.
2. For each dosing pump, mark the cables with dosage tags for Dos 1, Dos 2, Dos 3 and Dos 4.
3. In the list of detergents, lines 1, 2, 3 and 4, write the name of the detergent used for each dosage pump. Ensure that the number corresponds to the right dosage pump and tag.

5.10.8 Dosing hoses

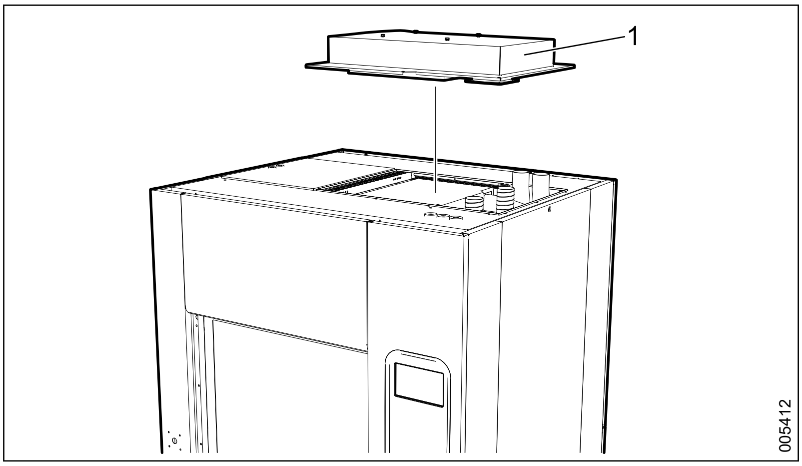
5.11 Drying system

5.11.1 Overview



1 Dryer fans
2 Humidity sensor
3 Heat exchanger
4 Check valves, main pipe

5.11.2 Replacing a HEPA filter



1 Cover

The filter cleans the air before it reaches the chamber. The filter is located at the top of the machine.

The replacement interval stated in the service table is the minimum recommended. Replacement may be required more often, especially in dirty environments.

Do the following to replace sterile filters:

1. Remove the top of the machine.

   The power plugs to the ventilation fans must be disconnected before the cover is removed.
2. Unlock the filter holder by undoing the two hasps: one on each long side.
3. Open the cover to the filter holder by leaning it toward the side where the air connections come from the dryer.
4. Remove the old filter and replace with a new one.

   The side of the filter that has a slightly higher edge should face downwards upon assembly. Make sure that the filter seals tight; otherwise there is a risk of alarm.
5. Test run the machine with a new filter according to section *6.5 Calibration of signals from differential pressure sensor, HEPA filter, page 134*.

   The value of the differential pressure should be between 200 and 400 Pa.

5.11.3 Checking seals and hoses

Check all hoses and seals (air inlet and outlet) in the drying system, including the heat exchanger.

5.11.4 Checking humidity sensor

The drying system's humidity sensors measure humidity in the exiting air and it is mounted adjacent to the heat exchanger.

Check each humidity sensor as follows:

1. Locate the humidity sensor (see section *5.11.1 Overview, page 122*).
2. Unscrew the humidity sensor from the heat exchanger.
3. Carry out a visual inspection of the humidity sensor.
4. Clean the humidity sensor or replace its accompanying filter as necessary.

5.11.5 Checking dryer fans

The machine has two dryer fans that control the circulation of warm air in the chamber and the drying system.

Carry out an operational check of the fans as follows:

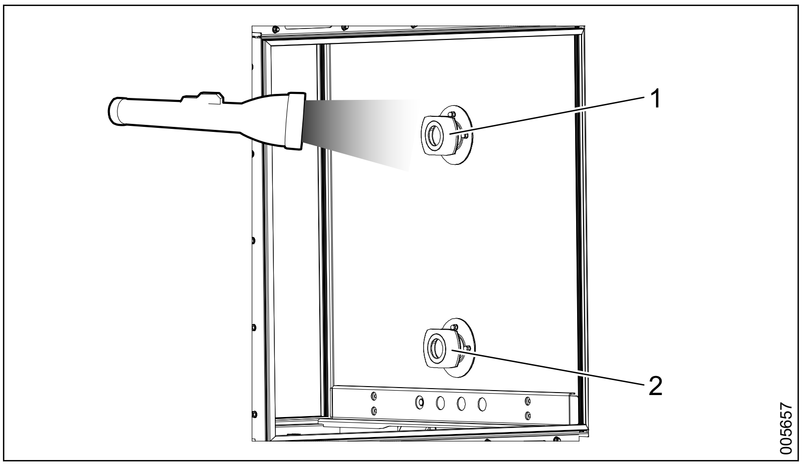
1. Run the fans in manual mode (see section *4.3.3 Manage inputs and outputs, page 37*).
2. Remove the power plug for one fan at a time and check that each fan is operating properly.

5.11.6 Checking the check valves

The drying system has two check valves that are made of stainless steel. They are located adjacent to the machine's main pipe and prevent water from the chamber leaking into the drying system. It is very important that check valves are checked regularly according to the service table to avoid leakage that may cause downtime.

Check the check valves:

1. Use a flashlight to illuminate the docking in the chamber.



1 Upper docking
2 Lower docking

2. Check that the check valve is not damaged.
3. Check that there are no lime deposits.

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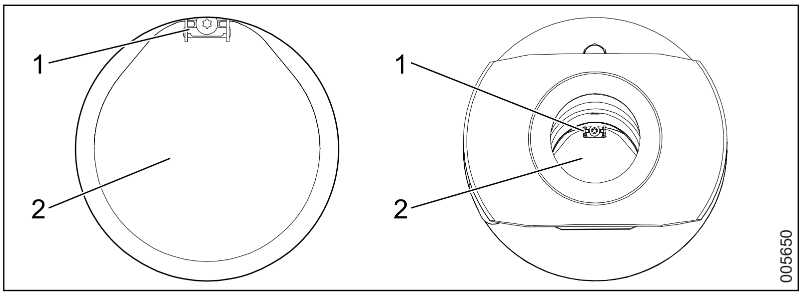
4. Check from inside of the chamber that the check valve hangs correct and is in the closed position.



1 Hinge
2 Check valve plate

5. If the check valve is damaged, replace it according to section *7.18.5 Replacing check valves, page 194*.

5.12 Test runs

5.12.1 Function and leakage inspection

The purpose of test runs is to check the functioning of the various components and to identify any leakage.

Perform a test run as follows:

Inspection before running the program

1. Check that there is water in the circulation pump.

   If necessary, fill the chamber with water by running a water valve in manual mode* (on S-8668T, the process valve must be open).
2. Close and lock the door(s) to the chamber.
3. Run the circulation pump in manual mode and check the direction of rotation.

   The motor must rotate in the proper direction as indicated by the arrow on the pump motor. If it is rotating in the wrong direction, cut the power, change the phase sequence in the electrical connection and restart the machine.
4. Run all dosing pumps in manual mode* and check that they are functioning properly.

*Manual mode is described in section *4.3.3.4 Digital outputs, page 40*.

Check when running the program

1. Lift off the lower panel on the soiled side. This will make it easier to detect any leakage.
2. Load a wash cart with goods into the chamber and close the door.
3. Start a program.
4. Check the following points while the program is in progress:

- Check that the machine fills properly with water.
- Check that there are no abnormal sounds coming from the circulation pump.

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- Carefully check for leaks beneath the chamber. Tighten the hose clamps if there are any leaks.
- Check the water, steam and waste connections. Tighten the hose clamps if there are any leaks.
- Check that the circulation pump connections are tight and no water is leaking from the pump housing.
- Check that the dosing pumps work at the correct time in accordance with the program description and that detergent is being pumped.
- Check that there are no noises coming from the dryer fans during the drying process.

5. Unload the wash cart after the cycle is complete and check that the goods are dry.

5.12.2 Temperature check

A temperature check should not be confused with a temperature validation in OQ/PQ.

Temperature checks are performed by measuring the temperature inside the chamber with external measurement equipment during a running program. The measuring equipment must be capable of continuously recording temperature and time.

Perform temperature control as follows:

1. Attach an external reference temperature sensor to the temperature sensor in the chamber.
2. Start an appropriate program and check the following:

- Check that the temperature in the chamber follows the accompanying temperature curve (to ensure that the heating function is working properly).
- Check the temperature deviation between the reference temperature and the temperature in the control system at 90°C. If the temperature deviation is greater than ±1°C, the sensor needs to be calibrated.

See section *6 CALIBRATION, page 129* for calibration instructions.

5.12.3 Electrical check

The purpose of test running is to check the safe state of the electrical equipment after maintenance.

Inspection before the test run:

- If any fuse is replaced, make sure that the rating and characteristics are correct according to the electrical documentation.
- If any electrical components are replaced, make sure that the connections are well tightened.
- If any electrical components are replaced, make sure that the high voltage insulation still is valid.
- After the cover plates are mounted again, make sure that the earth bonding is still is okay for all the cover plates that have been removed.

Perform a test run as follows:

1. Attach an external temperature measurement on typical goods.
2. Select an appropriate program.
3. Run the program and check for leakages and abnormal sounds.
4. Make sure that the program was completed without any alarms and with a temperature that is approved for the selected program.

5.13 Storing the machine

Perform the steps in this section when the machine will not be used for a short or long period of time (planned operational outage) or if it is being moved.

Before storage:

1. Run a program with detergent and disinfection. This is done to clean and disinfect the chamber.
2. Run the **Complete drain** program.
3. Lift out the containers from the dosing cabinet.

Handle the suction lances with care. Do not hit the suction lance against the container to remove excess detergent.

4. Remove the suction lances from the detergent containers (have a cloth ready to wipe up any excess detergent) and place them in a suitable container with water.
5. Rinse the dosing system with water. This is done by running the dosing pump in manual mode, see section *4.3.3 Manage inputs and outputs, page 37*.
6. Leave the suction lances in the water container and run a program. This is done to flush, clean, drain and dry the chamber.
7. Lift out and clean suction lances and level sensors.
8. Empty the dosing system of water by running the dosing pump in manual mode.
9. Close the door(s) to the chamber.
10. Switch off the main power switch.
11. Drain the circulation pump (see section *7.11.3 Draining the circulation pump, page 178*).

When starting the machine:

Calibration of dose monitors is always carried out after hoses have been replaced in the dosing system.

1. Replace all the hoses in the dosing system (see section *5.10 Dosing system, page 116*). This is recommended for long-term storage.
2. Place the detergent containers in the dosing cabinet.

3. Manually fill up the hoses in the dosing system with detergent. This is done by running the dosing pump in manual mode (see section *4.3.3 Manage inputs and outputs, page 37*).
4. Calibrate dose monitors (see section *6.7 Calibration of dose monitors, page 136*).
5. Run a program without any goods. This is done to clean the chamber and verify that the correct amount of detergent was used.
6. Check and set the time and date on the touchscreen. See section *4.3.5.2 Set date and time, page 43*.

6 CALIBRATION

Placement and removal of the machine's various sensors is presented in section *7 CORRECTIVE MAINTENANCE, page 140* and section *7.10.3 Installation and calibration, page 176*.

See applicable electrical diagrams for more information.

6.1 General

A calibration check or calibration must be done according to the service intervals in the service table. For more information, see section *5 PREVENTIVE MAINTENANCE, page 94*.

A calibration (check) of the temperature measurements and general analog signals (pressure, conductivity) must be done:

- if the sensor has been replaced.
- if the input module has been replaced.

A calibration (check) of the detergent dosing flow must be done:

- if the sensor has been replaced.
- if the input module has been replaced.
- if the pump hose has been replaced.
- if the detergent has been changed.

6.1.1 Calibration methods

6.1.1.1 Liquid bath

Liquid baths are used to achieve a stable measurement temperature during calibration of temperature sensors. The calibration is performed with two different reference temperatures:

- water bath = low. The water bath can hold water at room temperature (about 20 °C).
- oil bath = high. The oil bath must be heated until the temperature is stabilized (about 90 °C).

When measuring the temperature:

- use an external thermometer in the liquid bath.
- check that the reading of the thermometer has stabilized before calibrating.

6.1.1.2 Signal generator

Calibration of the signal for the relevant sensor takes place using an external signal generator (0-5 V, 0-10 V, 0-20 mA, 4-20 mA).

6.1.1.3 External differential pressure gauge

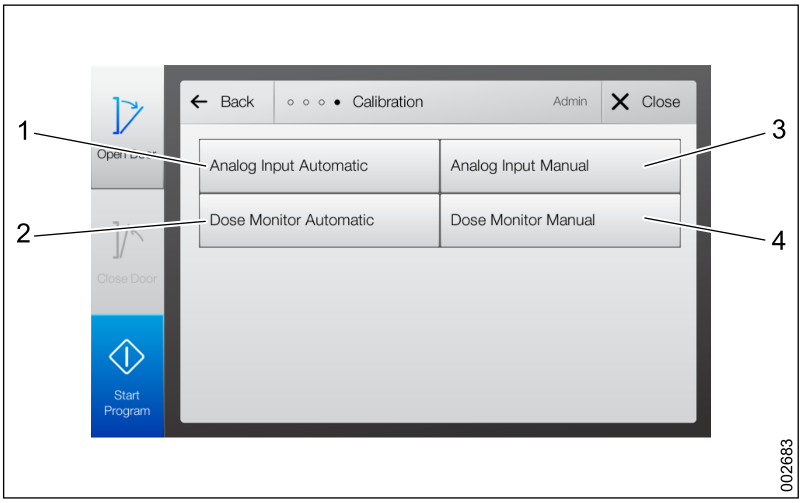
Connect an external differential pressure gauge in parallel with the existing machine gauge.

6.1.1.4 Measuring beaker

Used to measure the dosed amount of detergent during calibration of dose and flow monitors.

6.1.2 Calibration menu

Path: **Menu** > **Service** > **System** > **Calibration.**



1 **Analog Input Automatic**
2 **Dose Monitor Automatic**
3 **Analog Input Manual**
4 **Dose Monitor Manual**

Sensors for analog inputs and dose monitoring can be calibrated. Choose between automatic and manual calibration methods for each category.

Automatic calibration means that the machine's sensors are calibrated in different ways in relation to measured reference values. Liquid bath or connection of resistance in the I/O module can for example be used to obtain reference values for the temperature.

Manual calibration means that one or more values are stated manually to correct deviation in the sensor.

6.1.2.1 Analog inputs - Automatic

Path: **Menu** > **Service** > **System** > **Calibration** > **Analog Input Automatic**

Display images can differ somewhat from those in the manual depending on which sensor is to be calibrated.

1. In the **Select** column, select the sensors to be calibrated by tapping the corresponding radio button.
2. Tap **Calibrate**.

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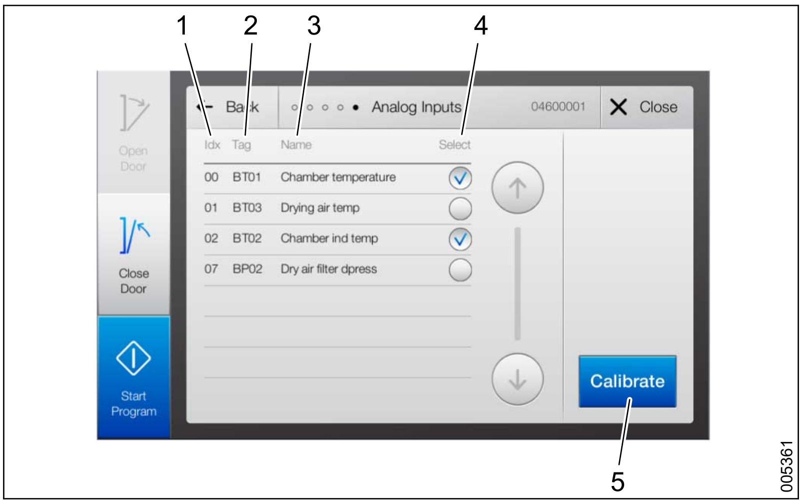
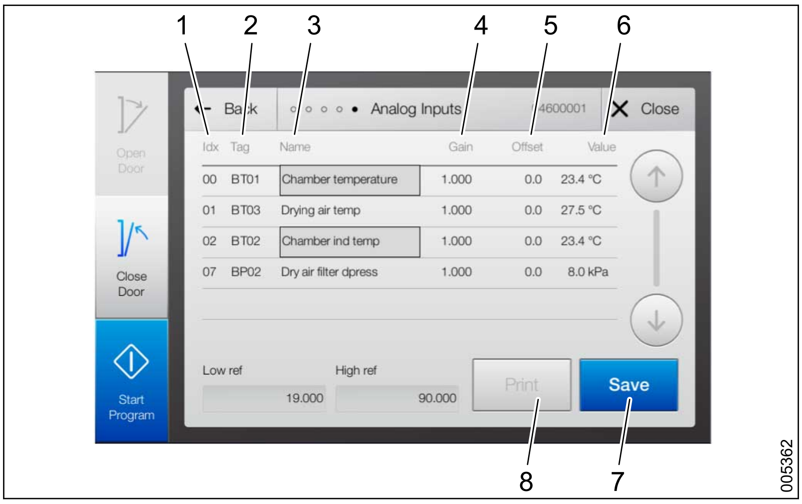


1 **Idx**	4 **Select**
2 **Tag**	5 **Calibrate**
3 **Name**	

3. Tap the **Low ref** field and enter a reference value.
   The measurement **Value** must be updated and must match the entered value.



1 **Idx**	5 **Offset**
2 **Tag**	6 **Value**
3 **Name**	7 **Save**
4 **Gain**	8 **Print**

4. Tap **High ref** and enter a reference value. The measurement **Value** must be updated and must match the entered value.
5. Tap **Save** to save the calibration values.

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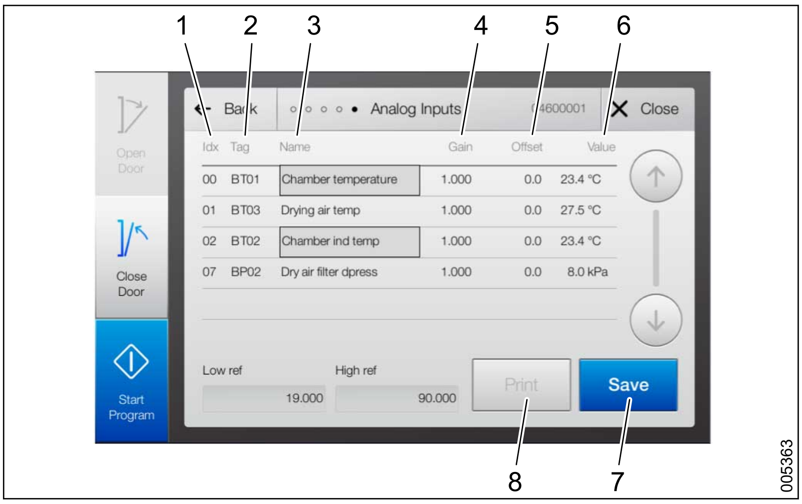
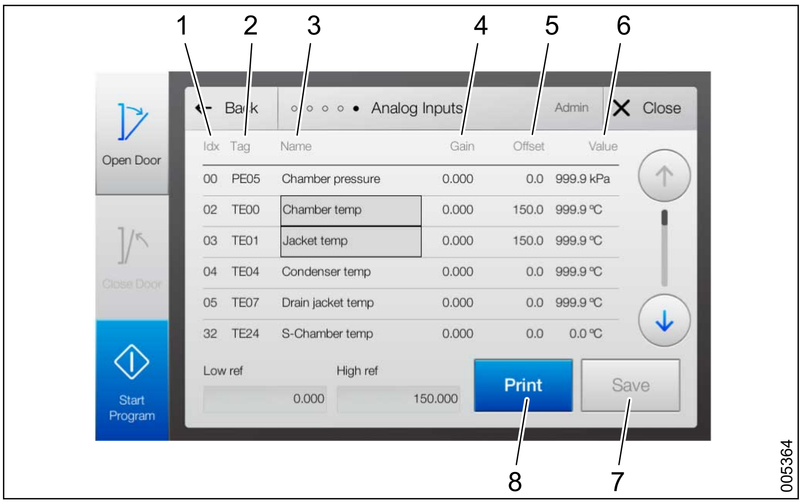


1 **Idx**
2 **Tag**
3 **Name**
4 **Gain**
5 **Offset**
6 **Value**
7 **Save**
8 **Print**

6. Print out a calibration report by tapping **Print** (optional).



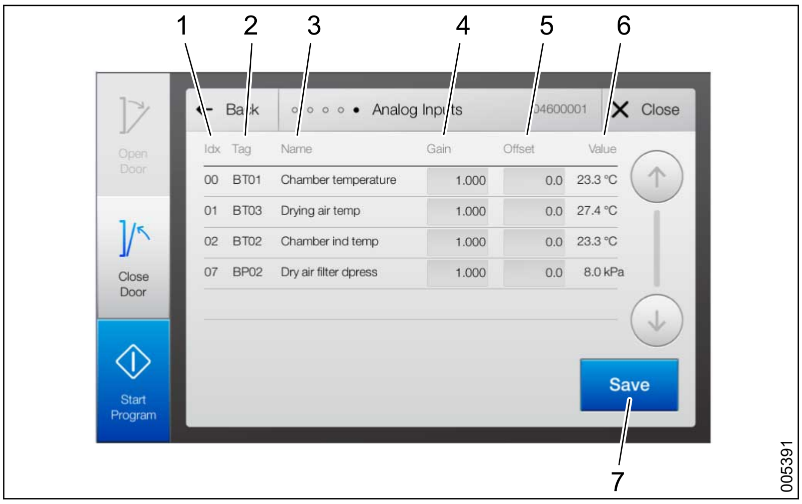
1 **Idx**
2 **Tag**
3 **Name**
4 **Gain**
5 **Offset**
6 **Value**
7 **Save**
8 **Print**

7. Tap **Close** to close the window.

6.1.2.2 Analog inputs - Manual

Path: **Menu** > **Service** > **System** > **Calibration** > **Analog Input Manual**

Follow instructions to set analog inputs manually:



1 **Idx**
2 **Tag**
3 **Name**
4 **Gain**
5 **Offset**
6 **Value**
7 **Save**

1. Tap the **Gain** or **Offset** field to be changed for the desired sensor.
2. Enter new values.
3. Tap **Save** to confirm the change and save the calibration.
4. Tap **Close**.

6.1.2.3 Dose Monitor Automatic

For more information, see section *6.7.1 Automatic calibration of dose monitors, page 136*.

6.1.2.4 Dose Monitor Manual

For more information, see section *6.7.2 Manual dose monitor calibration, page 139*.

6.2 Calibrating temperature sensors

Calibration is performed by entering two different reference temperatures: one low and one high. Use the water bath as a reference fluid for the low temperature and the oil bath as a reference fluid for the high temperature.

Perform the following steps to calibrate a temperature sensor with a liquid bath (same instructions apply for all temperature sensors):

1. Prepare a liquid bath according to section *6.1.1.1 Liquid bath, page 129*
2. Remove the relevant temperature sensor according to the description in section *7 CORRECTIVE MAINTENANCE, page 140*.
3. Follow the instructions in section *6.1.2.1 Analog inputs - Automatic, page 130*.
4. Use the liquid baths to calibrate according to the following:

   Set **Low ref** for the temperature in the water bath.

   Set **High ref** for the temperature in the oil bath.

6.3 Calibration of signals from humidity sensor, drying air

Use the signal generator (0 - 5 V) to calibrate the signal from the sensor.

Follow the general instructions in section *6.1.2.1 Analog inputs - Automatic, page 130*.

1. Connect a signal generator to KF04:
   + at 22
   - at 23
2. Set the signal generator to 0 V.
3. Enter value 0 in the **Low ref** field.
4. Set the signal generator to 5 V.
5. Enter value 100 in the **High ref** field.

6.4 Calibrating signals from pressure sensors, circulation pump

Proceed as follows to calibrate signals from pressure sensors with the signal generator (4-19.8 mA):

1. Follow the instructions in section *6.1.2.1 Analog inputs - Automatic, page 130*.
2. Connect a signal generator to KF04:
   + at 11
   - at 13
3. Set the signal generator to 4 mA.
4. Enter value 0 in the **Low ref** field.
5. Set the signal generator to 19.8 mA.
6. Enter value 250 in the **High ref** field.

6.5 Calibration of signals from differential pressure sensor, HEPA filter

The differential pressure sensor is connected to the drying filter in the machine's upper space. For more information, see section *7.18.1 Overview, page 192*.

The differential pressure sensor is used to monitor the following:

- both fans are running and working as intended
- the HEPA filter is properly mounted
- the HEPA filter is not blocked
- the HEPA filter is not damaged

The differential pressure sensor is calibrated by running the fan in manual mode and entering reference values for max and min speeds.

Perform reference measurements of the differential pressure:

1. Close the machine doors.
2. Connect an external differential pressure sensor parallel to the existing machine sensor.
3. Reset the external differential pressure sensor.

4. Start the machine's fan: Go to **Menu > System > Service > Diagnostics > Analog Inputs** and set the fan to manual (**Man.**).
5. Tap the field in the **Set** column and enter the value 100 to set the fan speed to maximum.
6. Check the reading of the external differential pressure sensor. The reading should be between 200 and 400 Pa.

   If it is higher, the filter may be clogged. If it is lower, there may be a leak.
7. Read the value on the machine's differential pressure sensor (go to **Menu > System > Service > Diagnostics > Analog Inputs**). If the value deviates more than 10% compared to the external differential pressure meter, calibration must be performed.

Perform calibration:

1. Continue to run the fan in manual mode at maximum speed.
2. Stop the fan by removing the power plug to the fan.
3. Go to **Menu > System > Service > Calibration > Analog Input Automatic.**
4. Tick **Dryer air filter differential pressure.**
5. Tap **Calibrate.**
6. Enter value 0 in the **Low ref** field.
7. Start the fan by reinserting the power plug for the fan (the fan must run at maximum speed).
8. Enter the value obtained from the external differential pressure sensor in the **High ref** field. Check that the measurement value on the panel rises to the reading of the external differential pressure sensor.
9. Reset the fan: Go to **Menu > System > Service > Diagnostics > Analog Outputs** and change from

   Man. to **Auto.**
10. Reset the differential pressure gauge.

For more information, see section *6.1.2.1 Analog inputs - Automatic, page 130* and *4.3.3.1 Analog inputs, page 37*.

6.6 Calibration of signals from conductivity meter

The self-contained conductivity meter has a built-in calibration function. Besides the calibration from the machine touchscreen (section *6.1.2.1 Analog inputs - Automatic, page 130*), the calibration of signals from the conductivity meter also requires management on the conductivity meter unit (located in the control cabinet)

Calibrate the conductivity meter:

1. Press E on the conductivity meter.
2. Enter code 22 with + and −.
3. Press E.

   Setup 1 appears.
4. Choose Output with + and −.

5. Press [E].
   Sel. Type appears.
6. Select SIM with [+] and [−]
7. Press [E].
8. Set value with [+] and [−].
9. Enter 4 mA.
10. On the touchscreen, go to **Menu > Service > System > Calibration > Analog Input Automatic**.
11. Select conductivity meter and then Calibrate to proceed.
12. Select **Low ref** and enter the value 0.
    Check that the measurement value for conductivity is updated.
13. On the conductivity meter, use [+] and [−] to set 19.8 mA.
14. On the touchscreen, select **High ref** and enter the value 500.
    Check that the measurement value for conductivity is updated.
15. Tap **Save** to confirm the changes and save the calibration values.
16. Tap **Close** to close the calibration window.

Log out of the conductivity meter:

1. Press [+] and [−] at the same time. Output appears in the display.
2. Press [+]. Sel. Type appears.
3. Choose Lin. with [−].
4. Press [E].
5. Press [+] and [−] twice at the same time to log out of the conductivity meter.

6.7 Calibration of dose monitors

Getinge's washer-disinfectors are type-tested using Getinge's own process chemical range, Getinge Clean. The washer-disinfector, including the dosing equipment, is adapted and tested with these detergents. Flow monitoring only works with detergents with viscosities lower than 10 cSt.

6.7.1 Automatic calibration of dose monitors

Path: **Menu** > **System** > **Service** > **Calibration** > **Dose Monitor Automatic**

If the machine is equipped with flow monitoring, the dosing meters should be calibrated in terms of both time and pulsing.

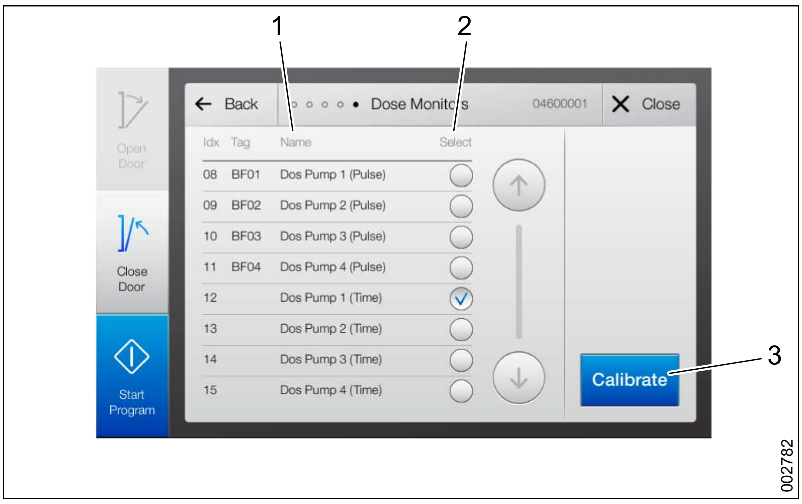
Use the dose amount the pump is set to dose in a process as the desired dose amount in the calibration.

Calibration of dosing meters includes calibration of dosing amounts and flow monitoring.

Calibration of dosing quantities (time-based) is done on all machines. Calibration of dosing quantities (time-based) and flow monitoring (pulse-based) is done on machines equipped with flow monitoring.

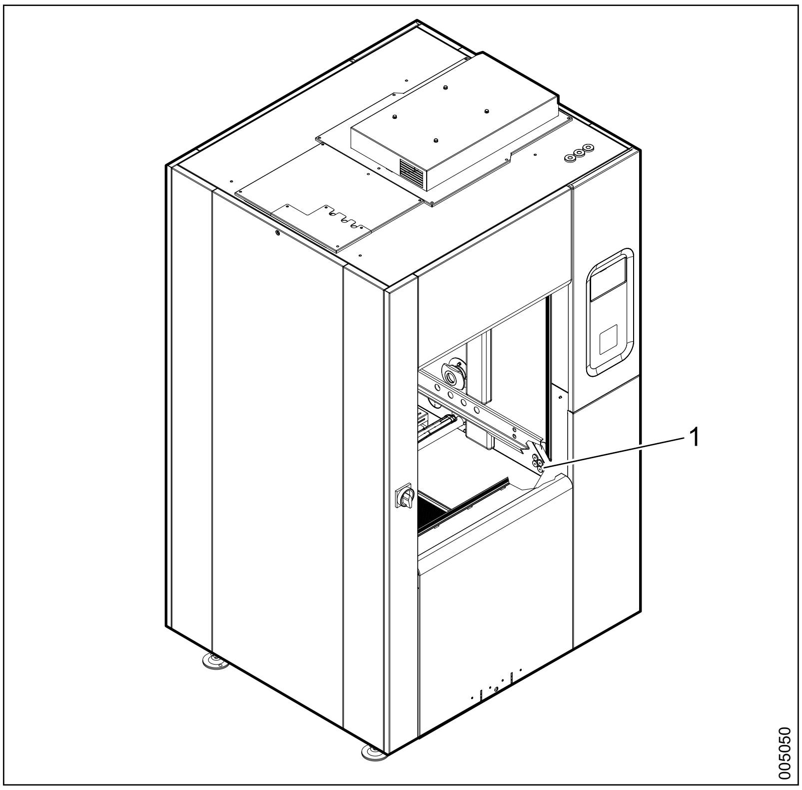
Follow these instructions to calibrate a certain pump; all pumps must be calibrated individually. To calibrate both dosing quantities and flow monitoring, begin by calibrating dosing quantities for all pumps, and then calibrate flow monitoring for all pumps.

1. Tick the pump/calibration method to calibrate in the **Select** column.
2. Tap **Calibrate**.



1 **Name**
2 **Select**
3 **Calibrate**

3. Place a measuring beaker under the outlet in the chamber, as shown in the illustration.



1 Detergent outlet

4. Tap the **Desired** field - a window will open. Enter the desired does amount. The dosing pump starts as soon as you tap **OK**.

1 **Desired**

2 **Measured**

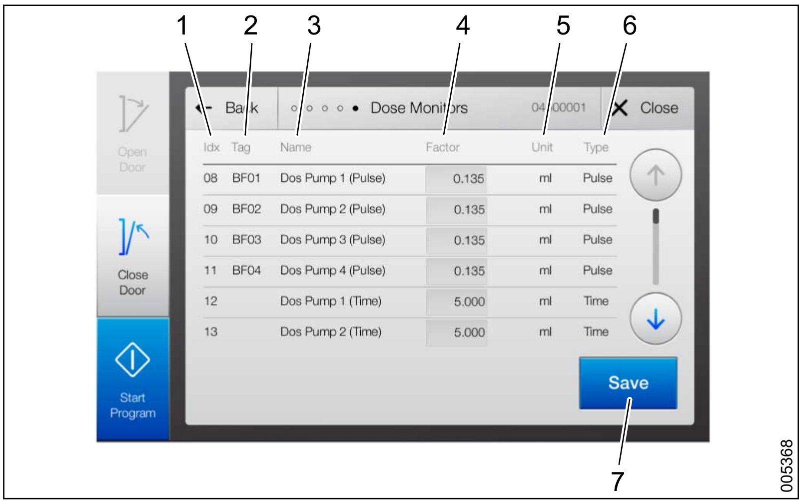
3 **Save**

5. When the dosing pump stops, check the measuring beaker and enter the measured amount of detergent in the **Measured** field.
6. Tap **Save** to confirm and save the calibration.
7. Repeat the procedure (steps 1-7) until the desired dosing amount is the same as the measured dosing amount.

6.7.2 Manual dose monitor calibration

Path: **Menu** > **System** > **Service** >**Calibration** > **Dose Monitor Manual**

A manual calibration of a dosing pump is done to set the dosing deviation value (%) based on the dose amount (ml).



1 **Idx**
2 **Tag**
3 **Name**
4 **Factor**
5 **Unit**
6 **Type**
7 **Save**

1. Tap the **Factor** column field for the sensor that should be calibrated.
2. Enter the desired dose amount value in the window that appears and tap **OK**.
3. Repeat if necessary for more sensors.
4. Tap **Save** to confirm and save the calibration.

If the Supervisor system is installed, calibrate the input from the sensor in the same way as for the standard sensors.

7 CORRECTIVE MAINTENANCE

This section describes corrective maintenance. Preventive maintenance is described in section *5 PREVENTIVE MAINTENANCE, page 94*.

The corrective measures are grouped according to the component's function or its placement in the machine.

Before starting servicing work:

- a disinfection program must be run. If disinfection cannot be achieved, the machine must be disinfected with a disinfectant.
- all the tanks must be drained.
- the power to the machine must be switched off.
- the steam and water supplies must be shut off.

7.1 Control system software

7.1.1 Control system software update

For an update package, contact Getinge Disinfection AB Technical support: ServiceNow.

Complete instructions for the update are included in the software package as well as a verification form that should be filled in and sent back to Getinge Disinfection AB after the update. The software updates the main panel (operator panel on the soiled side), the terminal panel (operator panel on the clean side) and the **Supervisor** system.

Only a Getinge authorized service technician is allowed to update the software on the machine.

Before starting the update:

1. Create a back-up copy of the control system. For more information, see section *4.3.9.1 Backup, page 52*.
2. Check the **System version** , **Cycle counter** and **Machine timer** values. For more information, see section *4.3.4 Display machine information, page 41*.
3. If **Supervisor** is installed, note the calibration data for the **Supervisor** system. For more information, see section *6 CALIBRATION, page 129*.
4. If the update includes new/changed parameters, changed parameter ranges, or new programs, it is not possible to keep the programs. Print or store the program parameters to be able to re-enter them. For more information, see section *4.3.9 Backup and export, page 52*.
5. If the machine is connected to an external network, remove the cable connected to the PLK port. For more information, see section *4.1.3 Communication and peripheral equipment, page 20*.
6. Follow the instructions included in the software package.

After the update is completed:

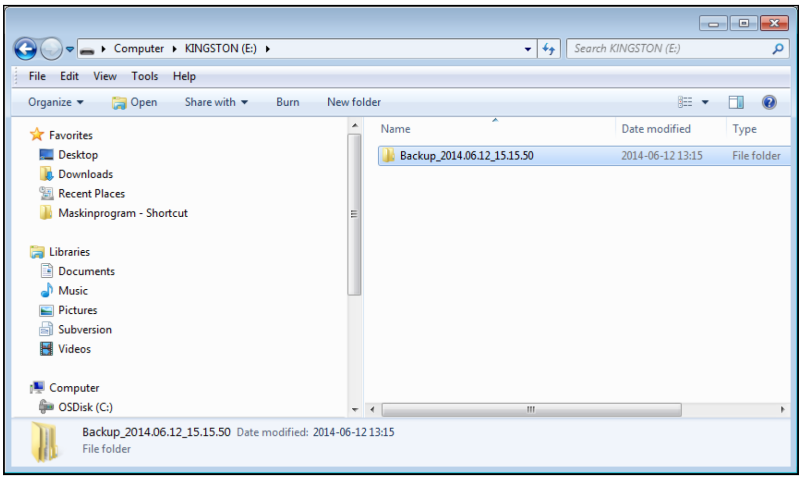
1. Restart the whole system by turning it off and back on.
2. If **Supervisor** is installed, re-enter the calibration data for the **Supervisor** system.
3. Re-enter the values for the **Cycle counter** and the **Machine timer**. For more information, see section *4.4 Web server interface, page 60*.
4. If the machine was connected to an external network, reconnect the cable to the PLK port.

7.1.2 Restore control system software from backup

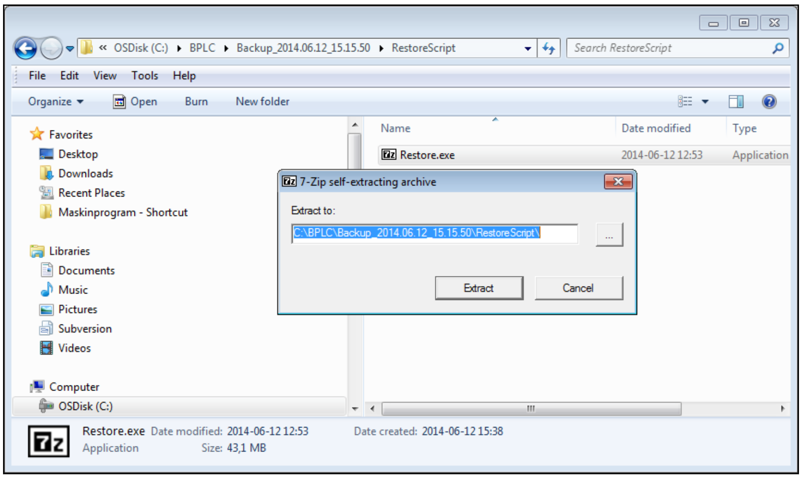
Restoring from a backup copy resets the system to the status prior to making the backup copy. For example, settings, licenses and contents in user files and authorization files can be erased.

Reset the machine's PLC with a backup copy saved on a USB memory stick.

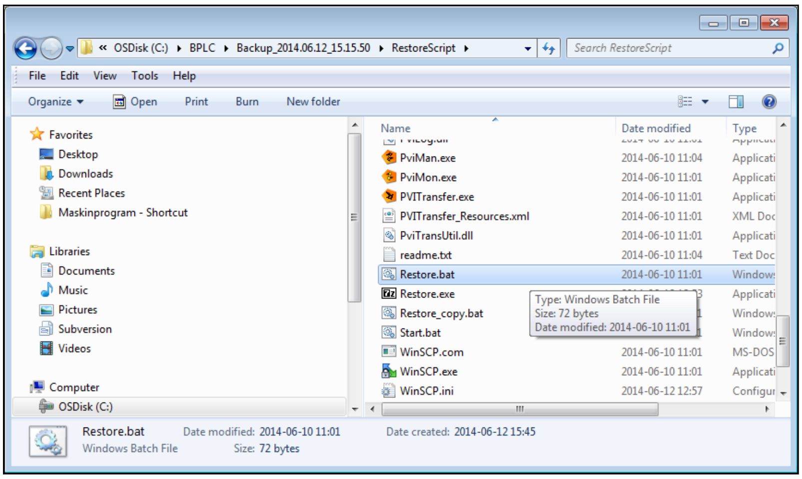
1. Copy the backup folder from the USB memory stick to the PC.



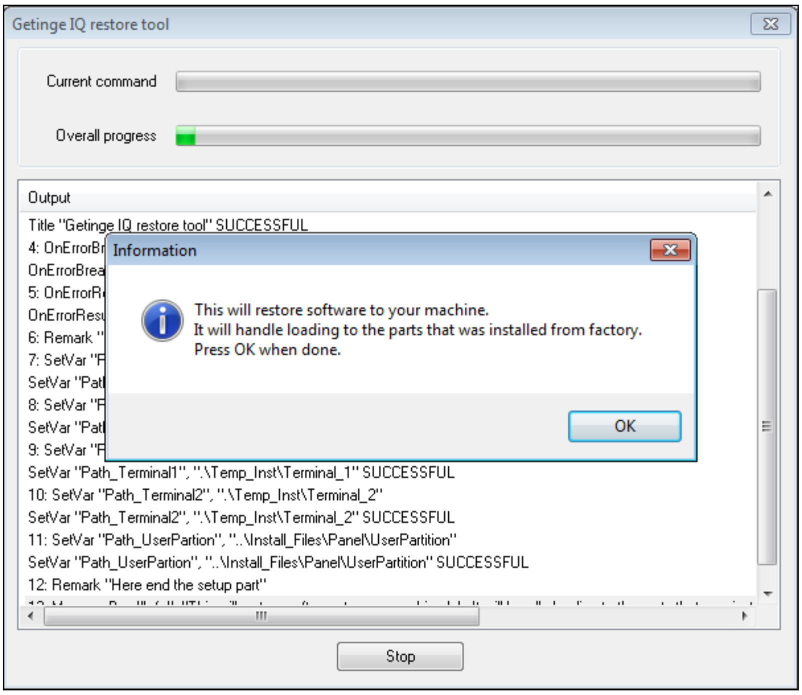
2. In the folder copied to the PC, go to the **RestoreScript** folder and double-click the **Restore.exe** file.
3. To save the restore files to the PC, in the pop-up window, click on **Extract** without making any changes in the **Extract to** field.



4. Connect the PC using the Ethernet cable to the port marked **ETH** on the back of the operator panel. Open the door to the control cabinet to access it. The PC must be configured as described in section *4.4.1 Connect to the Web server, page 60*.
5. In the **RestoreScript** folder, double-click the **Restore.bat** file.



6. Follow the instructions on the screen and place the operator panel in boot mode by pressing the RESET button. For more information, see section *4.1.1 Operator panel, page 18*.



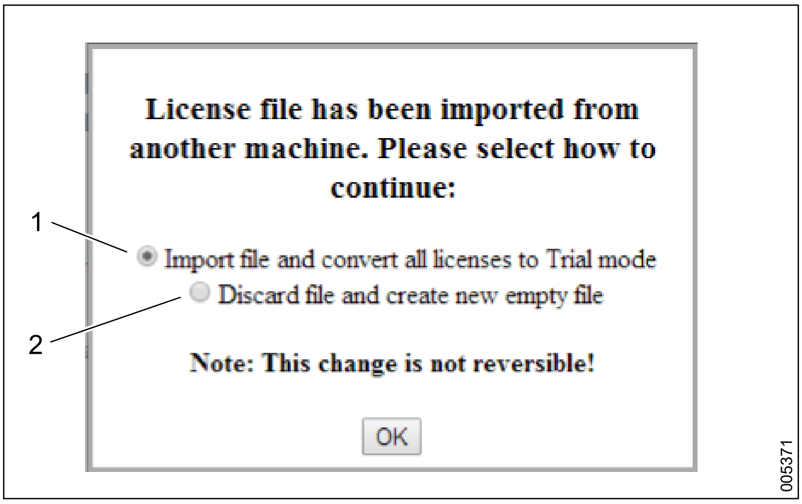
7. When the restore is complete, restart the machine by turning the power off and then back on.

7.1.3 Restore serial number after replacing operator panel

The changes are not reversible.

When the operator panel is replaced and the control system is restored from a backup copy, the serial number for the new operator panel will not correspond with the one in the license file. When the machine starts, a warning message will appear on the touchscreen, saying that a new license file must be configured. When the license key has been changed, there will be no warning message at the next machine start.

1. Connect to the **Web server**. For more information, see section *4.4.1 Connect to the Web server, page 60*.
2. Go to the **Activate Options** page. For more information, see section *4.4.14 Activate options, page 92*. A message is displayed:



3. Tick the box for **Import file and convert all licenses to Trial mode** or **Discard file and create new empty file**.
4. Click **OK** to confirm and save.

Name	Description
Import file and convert all licenses to Trial mode	All selections will be run in the trial mode. A valid license must be entered within two weeks; otherwise the selections will stop working.
Discard file and create new empty file	All selections are turned off.

7.2 Barcode reader

If the barcode reading is malfunctioning, ensure that the barcode tag is readable before replacing the scanner and the USB extension cable.

7.3 Printer

The following configuration must be set on the printer before it is used on the machine.

Menu item	Settings
Path: **Settings** > **Settings** > **GeneralSettings > Timeouts > Print Timeout**	Disabled
Path: **Settings** > **Settings** > **Print Settings** > **Setup Menu** > **Print Language**	PCL Emulation
Path: **Settings** > **Settings** > **Print Settings** > **PCL EmulMenu** > **PCL Emulation Settings** > **Lines Per Page**	70

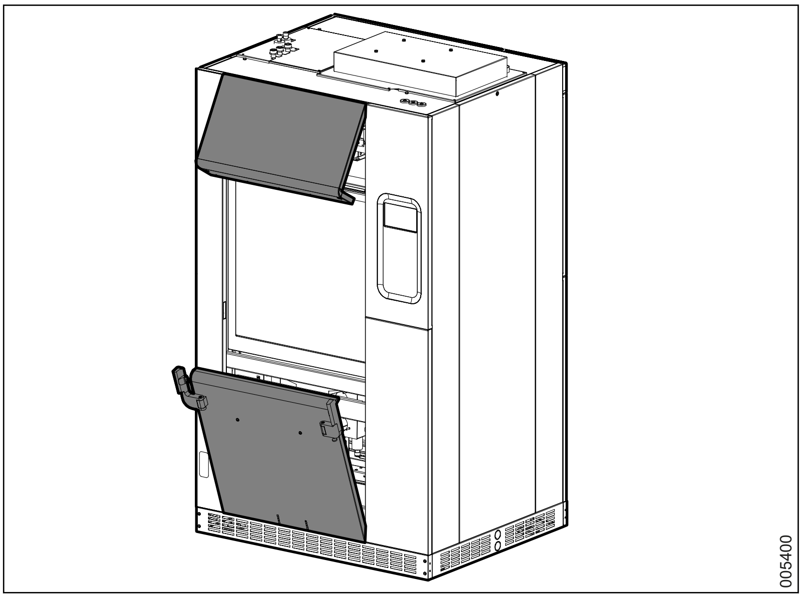
Very short log interval settings can create excessively long process reports, resulting in the printing not being finished within five minutes at the end of a cycle. In this case, the **Print Last Cycle** button is grayed out.

For information on how to set the configuration on the printer, see the printer manual.

7.4 Cover panels

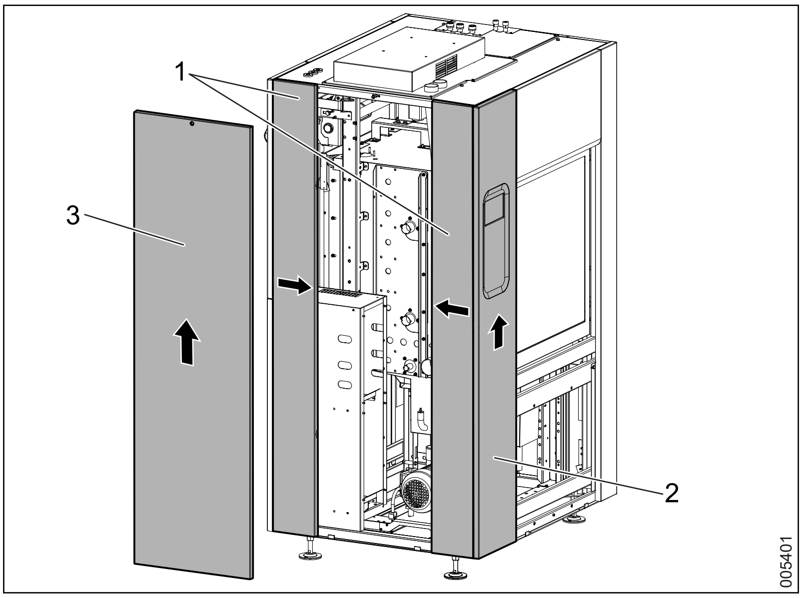
When handling cover panels, ensure that no cables are trapped or damaged.

Remove the cover panels, front:



1. Loosen the screws at the lower edge of the upper trim panel.
2. Tilt outwards to release and lift the panel off.
3. Loosen the two screws on the front of the lower trim panel.
4. Tilt outwards to release and lift the panel off.

Remove cover panels, right side:



1 Outer panels

2 Clean side display panel

3 Central panel

Step 1

1. Loosen the screws holding the clean side display panel in place. The screws are fitted to the right of the panel on the machine's inside.
2. Press the panel upwards to release and remove it.

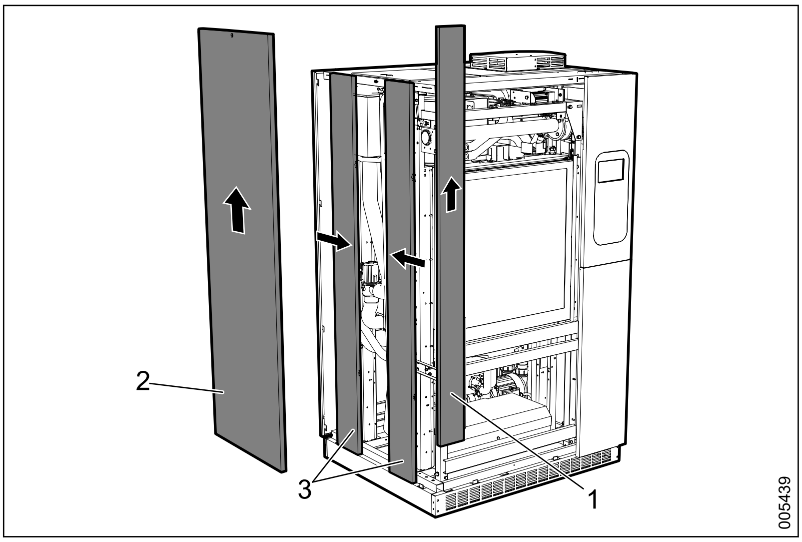
Step 2

3. Loosen the screws on the central panel's top section.
4. Tilt the panel outwards to release and lift the panel off.

Step 3

5. Press the upper part of the outer panels inwards. Tilt the panel outwards to release and lift the panel off.

Remove cover panels, left side



1 Main switch panel
2 Central panel
3 Outer panels

Step 1

1. Make sure that the main switch is turned on. Loosen the screws holding the main switch panel in place. The screws are fitted to the right of the panel on the machine's inside.
2. Press the panel upwards to release and remove it.

Step 2

3. Loosen the screw for the central panel's top section.
4. Tilt the panel outwards to release and lift the panel off.

Step 3

5. Press the upper part of the outer panels inwards. Tilt the panel outwards to release and lift the panel off.

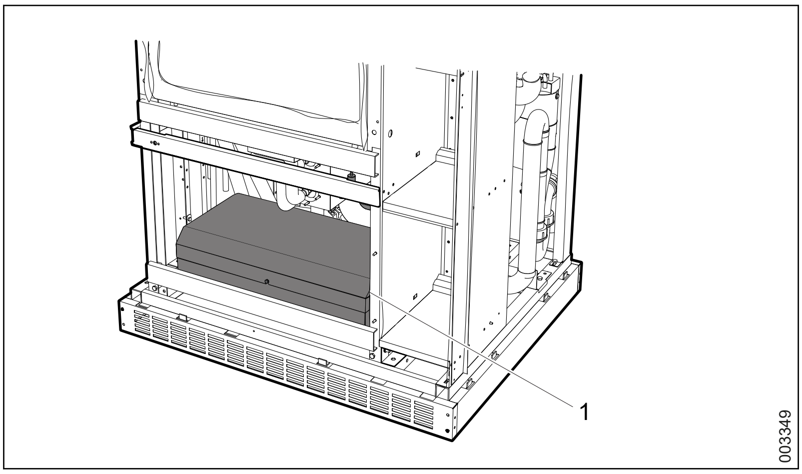
7.5 Power box

7.5.1 Overview

Remove the lower trim panel on the front of the machine, to access the power box. For more information, see section *7.4 Cover panels, page 144*.

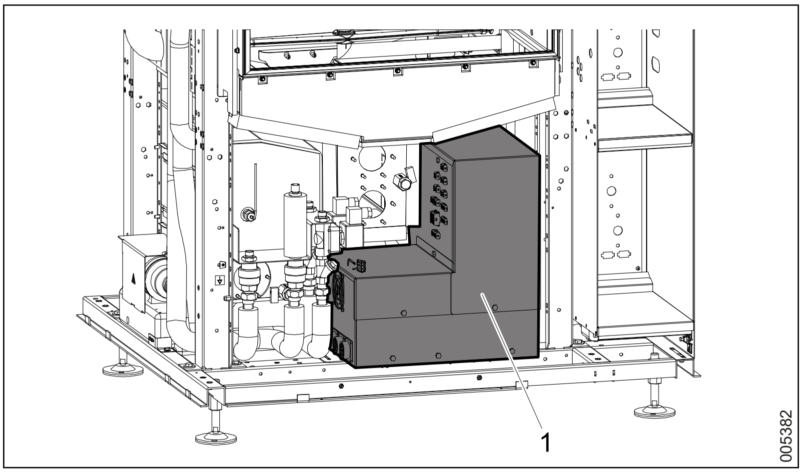
The position of the power box is shown in the following illustrations:

S-8666 and S-8668



1 Power box S-8666 and S-8668

S-8668T



1 Power box S-8668T

7.5.2 Fuses

The following table applies to all machine configurations and to all heating methods (Steam, Electricity, Steam+Electricity, 200-415 V).

Label	Type	Current	Description
F10	D	25 A	Transformer, 230 VAC, T01
F11	C	10 A	Power supply, T02
F12	C	4 A	UH02-UH03, 230 VAC
F13	C	10 A	Electrical outlet 230 VAC (option) RCCD included
F20	C	6 A	UH02-UH03, 24 VDC
F21	B	10 A	Power supply FSLC (option)
F22	B	10 A	Power supply FSUC (option)

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7.6 Control cabinet

7.6.1 Fuses

The following table applies to all machine configurations and to all heating methods (Steam, Electricity, Steam+Electricity, 200-415 V)

Label	Type	Current	Description
F101	T	2 A	Control panel soiled side, A01
F102	T	2 A	Control panel clean side, A02
F103	T	2 A	I/O node KF01
F104	T	2 A	I/O node KF01, internal power supply KF02-KF07 (S-8666 and S-8668) I/O node KF01, internal power supply KF02-KF09 (S-8668T)
F105	T	2 A	I/O node KF01, internal power supply KF09-KF13 (S-8666 and S-8668) I/O node KF01, internal power supply KF09-KF15 (S-8668T)
F106	T	2 A	Printer, USB Hub
F107	T	2 A	Spray arm monitoring (SWRSS)
F108	T	2 A	Humidity sensor, exit air
F109	T	2 A	Supervisor, PLC
F110	T	2 A	Supervisor, internal I/O supply KF93-KF95
F111	T	2 A	I/O node for AGS 2.0, power supply KF21-KF24
F112	T	2 A	Drain valve
F113	T	2 A	Conductivity transmitter

7.6.2 Control system modules (I/O and PLC)

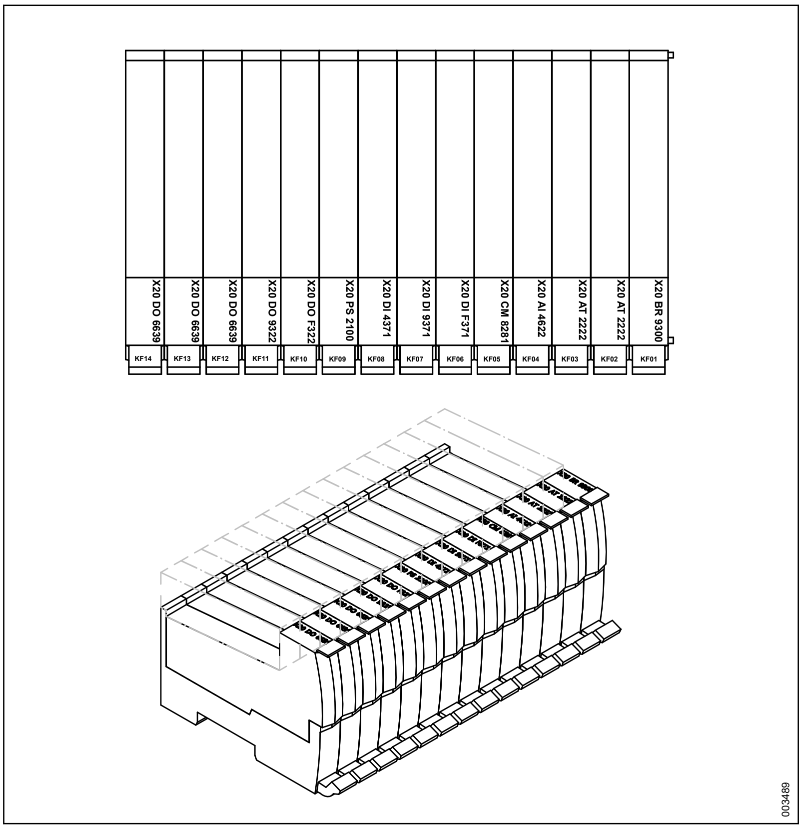
7.6.2.1 Checking module status

Information related to the module status can be read from each module and used for troubleshooting.

The modules are presented based on the nodes in which they are used. For more information about individual modules (in addition to the description in this manual), see www.br-automation.com.

7.6.2.1.1 Main mode

S-8666 and S-8668:



KF01	X20BR9300	Bus receiver X2X
KF02	X20AT2222	Analogue input module RTD
KF03	X20AT2222	Analogue input module RTD
KF04	X20AI4622	Analogue input module
KF05	X20CM8281	Combo module
KF06	X20DIF371	Digital input module
KF07	X20DI9371	Digital input module
KF08	X20DI4371	Digital input module
KF09	X20PS2100	Power supply module
KF10	X20DOF322	Digital output module
KF11	X20DO9322	Digital output module
KF12	X20DO6639	Digital output module
KF13	X20DO6639	Digital output module
KF14	X20DO6639	Digital output module

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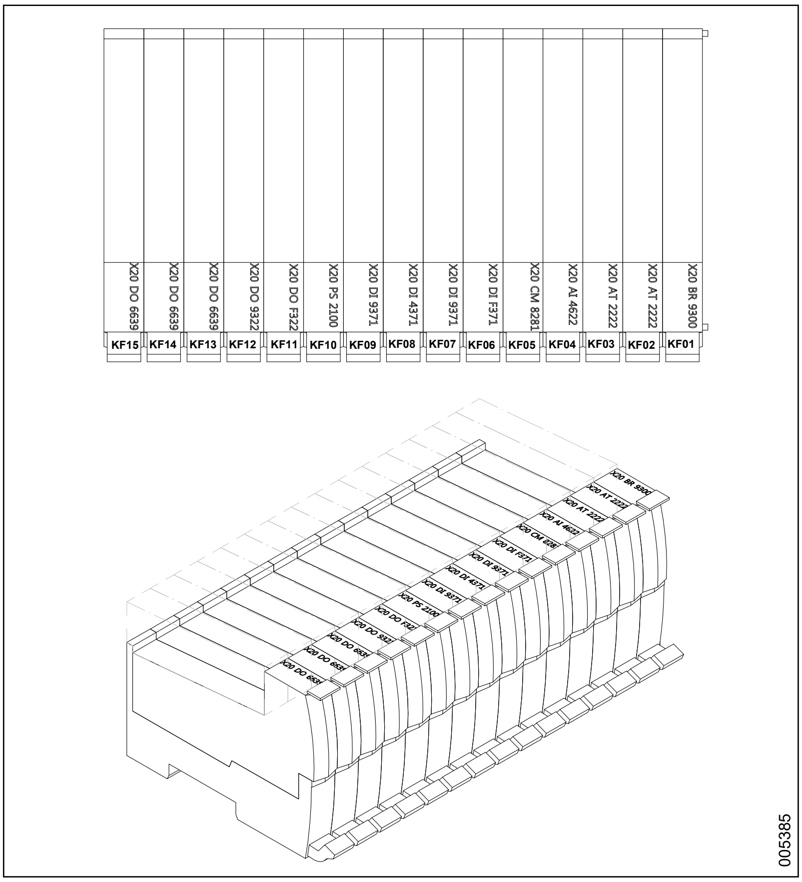
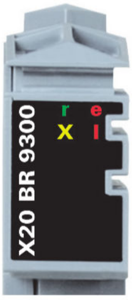
S-8668T:

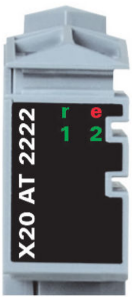


KF01	X20BR9300	Bus receiver X2X
KF02	X20AT2222	Analogue input module RTD
KF03	X20AT2222	Analogue input module RTD
KF04	X20AI4622	Analogue input module
KF05	X20CM8281	Combo module
KF06	X20DIF371	Digital input module
KF07	X20DI9371	Digital input module
KF08	X20DI4371	Digital input module
KF09	X20DI9371	Digital input module
KF10	X20PS2100	Power supply module
KF11	X20DOF322	Digital output module
KF12	X20DO9322	Digital output module
KF13	X20DO6639	Digital output module
KF14	X20DO6639	Digital output module
KF15	X20DO6639	Digital output module

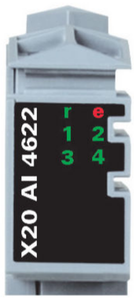
7.6.2.1.1.1 X20BR9300

Figure	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			Double flash	Indicates one of the following conditions: • X2X Link power supply is overloaded • I/O supply too low • Input voltage for X2X Link supply too low
	e + r	Steady red / single green flash		Invalid firmware
	X	Orange	Off	No communication at the X2X Link
			On	X2X Link communication in progress
	l	Red	Off	X2X Link supply in the acceptable range
			On	X2X Link power supply is overloaded Solution: Use an additional feed module PS3300

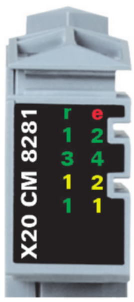
7.6.2.1.1.2 X20AT2222

Figure	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			On	Error or reset state
			Single flash	Warning/error for an I/O channel. Overflow or underflow of the analog inputs.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 2	Green	Off	The input is switched off
			Blinking	Overflow, underflow or broken connection
			On	The analog/digital converter is running, value is OK

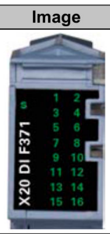
7.6.2.1.1.3 X20AI4622

Figure	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			On	Error or reset state
	e + r	Steady red / single green flash		Invalid firmware
	1 - 4	Green	Off	Open connection or sensor is disconnected
			Blinking	Overflow or underflow of the input signal
			On	The analog/digital converter is running, value is OK

7.6.2.1.1.4 X20CM8281

Figure	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			Single flash	Warning / error for an I/O channel. Level monitoring for digital outputs has been triggered.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 4	Green		Input status of the corresponding digital input
	1 - 2	Orange		Output status of the corresponding digital output
	1	Green	Off	Open connection or sensor is disconnected
			Blinking	Overflow or underflow of the input signal
			On	The analog/digital converter is running, value is OK
	1	Orange	Off	Value = 0
			On	Value $\neq$ 0

7.6.2.1.1.5 X20DIF371

Image	LED	Color	Status	Description
	S	Green	Off	No power to module
			Single flash	Reset mode
			Blinking	PREOPERATIONAL mode
			On	RUN mode
		Red	Off	Module supply not connected or everything OK
		Red on / Green single flash		Invalid firmware
	1 - 16	Green		Input status of the corresponding digital input

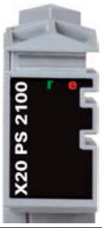
7.6.2.1.1.6 X20DI9371

Image	LED	Color	Status	Description
	r	Green	Off	No power to module
			Single flash	Reset mode
			Blinking	PREOPERATIONAL mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything OK
	e + r	Red on / Green single flash		Invalid firmware
	1 - 12	Green		Input status of the corresponding digital input

7.6.2.1.1.7 X20DI4371

Image	LED	Color	Status	Description
	r	Green	Off	No power to module
			Single flash	Reset mode
			Blinking	PREOPERATIONAL mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything OK
	e + r	Red on / Green single flash		Invalid firmware
	1 - 4	Green		Input status of the corresponding digital input

7.6.2.1.1.8 X20PS2100

Figure	LED	Color	Status	Description
	r	Green	Off	No power to module
			Single flash	RESET mode
			Blinking	PREOPERATIONAL mode
			On	RUN mode
	e	Red	Off	No power to module or everything OK
			Double flash	LED indicates one of the following states: • I/O supply too low • X2X link voltage too low
	e + r	Red on / Green single flash		Invalid firmware

7.6.2.1.1.9 X20DOF322

Figure	LED	Color	Status	Description
	S	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	PREOPERATIONAL mode
			On	RUN mode
		Red	Off	Module supply not connected or everything OK
			Single flash	Warning/Error on an I/O channel. Level monitoring for digital outputs has been triggered.
		Red on / Green single flash		Invalid firmware
	1 - 16	Orange		Output status of the corresponding digital output

7.6.2.1.1.10 X20DO9322

Figure	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	PREOPERATIONAL mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything OK
			Single flash	Warning/Error on an I/O channel. Level monitoring for digital outputs has been triggered.
	e + r	Red on / Green single flash		Invalid firmware
	1 - 12	Orange		Output status of the corresponding digital output

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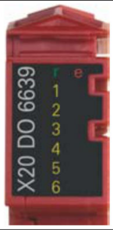
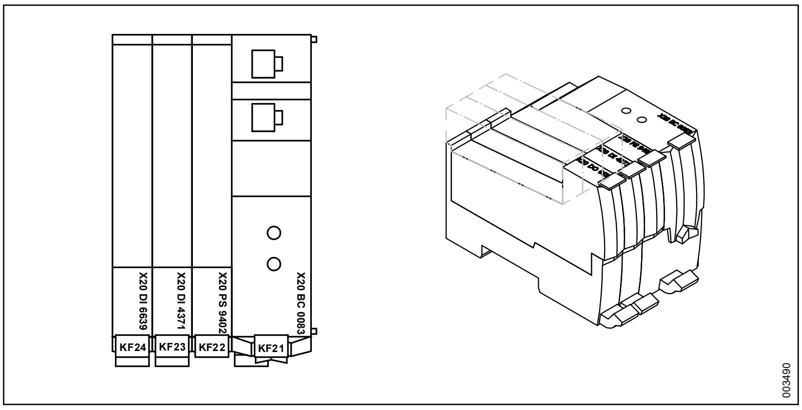
7.6.2.1.1.11 X20DO6639

Figure	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	PREOPERATIONAL mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything OK
			On	Error or reset status
	e + r	Red on / Green single flash		Invalid firmware
	1 - 6	Orange		Output status of the corresponding digital output

7.6.2.1.2 AGS node (option)



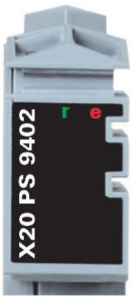
KF21	X20BC0083	Bus controller powerlink
KF22	X20PS9402	Supply module
KF23	X20DI4371	Digital input module
KF24	X20DO6639	Digital output module

How to set the station addresses for the AGS nodes is described in section *7.6.2.2.4 Setting addresses, page 167*.

7.6.2.1.2.1 X20BC0083

Figure	LED	Color	Status	Description
	S/E[1)]	Green	Off	No power supply or mode is NOT_ACTIVE. In this mode, the bus controller waits for about 5 seconds after restarting. No communication with the bus controller is possible. If no POWERLINK communication is detected during these 5 seconds, the bus controller goes into the BASIC_ETHERNET mode. If POWERLINK communication is detected before this time passes, however, the bus controller goes into the PRE_OPERATIONAL_1 mode.
			Flickering	BASIC_ETHERNET mode. The bus controller did not detect any POWERLINK communication. In this mode, direct communication with the bus controller is possible using UDP. If POWERLINK communication is detected while in this mode, the bus controller goes into the PRE_OPERATIONAL_1 mode.
			Single flash	PRE_OPERATIONAL_1 mode. With operation on a POWERLINK V1 master, the bus controller goes directly into PRE_OPERATIONAL_2 mode. With operation on an POWERLINK V2 manager, the CN (Controlled Node) waits for the reception of a SoC frame and then switches over to PRE_OPERATIONAL_2 mode.
			Double flash	PRE_OPERATIONAL_2 mode. In this mode the bus controller is normally configured by the manager. A command (POWERLINK V2) or setting the data valid flag in the output data (POWERLINK V1) then switches the mode to READY_TO_OPERATE.
			Triple flash	READY_TO_OPERATE mode. In a POWERLINK V2 network, the manager then switches via command to OPERATIONAL mode. In a POWERLINK V1 network, the bus controller then switches automatically to OPERATIONAL mode as soon as input data are present.
			On	OPERATIONAL mode
			Blinking	STOPPED mode. No output data sent nor input data received. Only the appropriate command from the manager can enter or leave this mode.
		Red	On	The bus controller has encountered an error (failed Ethernet frames, increased number of collisions on the network, etc.). **Note**: The LED blinks red several times immediately after startup. This is not an error.
	L/A IFx	Green	On	A link to the remote station has been established.
			Blinking	A link to the remote station has been established and there is activity on bus.

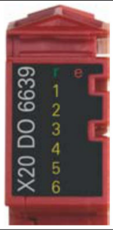
7.6.2.1.2.2 X20PS9402

Figure	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			Double flash	Indicates one of the following conditions: • The bus controller / X2X Link power supply is overloaded • I/O supply too low • Input voltage for bus controller / X2X Link too low
	e + r	Steady red / single green flash		Invalid firmware

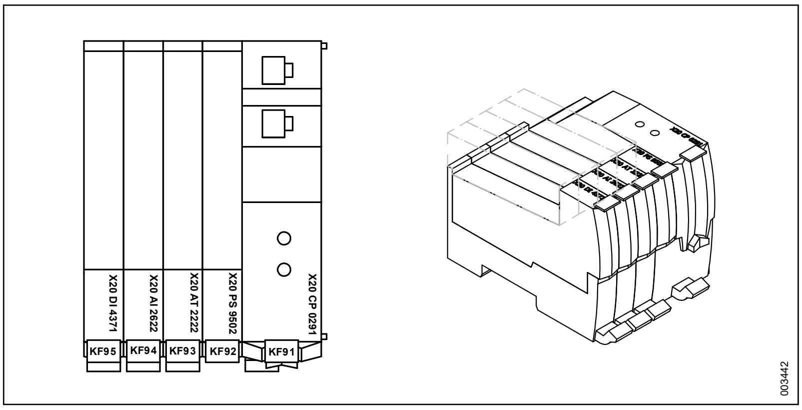
7.6.2.1.2.3 X20DI4371

Image	LED	Color	Status	Description
	r	Green	Off	No power to module
			Single flash	Reset mode
			Blinking	PREOPERATIONAL mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything OK
	e + r	Red on / Green single flash		Invalid firmware
	1 - 4	Green		Input status of the corresponding digital input

7.6.2.1.2.4 X20DO6639

Figure	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	PREOPERATIONAL mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything OK
			On	Error or reset status
	e + r	Red on / Green single flash		Invalid firmware
	1 - 6	Orange		Output status of the corresponding digital output

7.6.2.1.3 Supervisor-PLC (option)



KF91	X20CP0291	Compact plc
KF92	X20PS9502	Supply module
KF93	X20AT2222	Analog input module RTD
KF94	X20AI2622	Analog input module gen 2xAI
KF95	X20DI4371	Digital input module 4xDI

How to set the station addresses for the Supervisor-PLC is described in section *7.6.2.2.4 Setting addresses, page 167*.

7.6.2.1.3.1 X20CP0291

CP0201

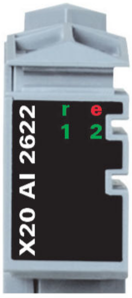
CP0291, CP0292

LED	Color	Status	Description
R/E	Green	On	Application running
	Red	On	SERVICE mode
RDY	Yellow	On	SERVICE or BOOT mode
L/A	Green	On	A link to the Ethernet remote station has been established.
		Blinking	A link to the Ethernet remote station has been established. The LED blinks when Ethernet activity is present on the bus.

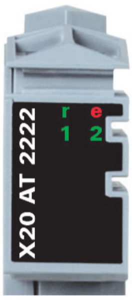
7.6.2.1.3.2 X20PS9502

Figure	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			Double flash	Indicates one of the following conditions: • The CPU / X2X Link power supply is overloaded • I/O supply too low • Input voltage for CPU / X2X Link too low
	e + r	Steady red / single green flash		Invalid firmware
	S	Yellow	Off	The CPU does not send any data via the RS232 interface.
			On	The CPU sends data via the RS232 interface.
	C	Yellow	Off	The CPU does not send data via the CAN bus interface.
			On	The CPU sends data via the CAN bus interface.
	T	Yellow	Off	The terminating resistor integrated in the BB27 or BB37 bus module is turned off.
			On	The terminating resistor integrated in the BB27 or BB37 bus module is turned on.

7.6.2.1.3.3 X20AI2622

Figure	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			On	Error or reset state
	e + r	Steady red / single green flash		Invalid firmware
	1 - 2	Green	Off	Open connection or sensor is disconnected
			Blinking	Overflow or underflow of the input signal
			On	The analog/digital converter is running, value is OK

7.6.2.1.3.3.1 X20AT2222

Figure	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			On	Error or reset state
			Single flash	Warning/error for an I/O channel. Overflow or underflow of the analog inputs.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 2	Green	Off	The input is switched off
			Blinking	Overflow, underflow or broken connection
			On	The analog/digital converter is running, value is OK

7.6.2.1.3.4 X20DI4371

Image	LED	Color	Status	Description
	r	Green	Off	No power to module
			Single flash	Reset mode
			Blinking	PREOPERATIONAL mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything OK
	e + r	Red on / Green single flash		Invalid firmware
	1 - 4	Green		Input status of the corresponding digital input

7.6.2.2 Replacing modules

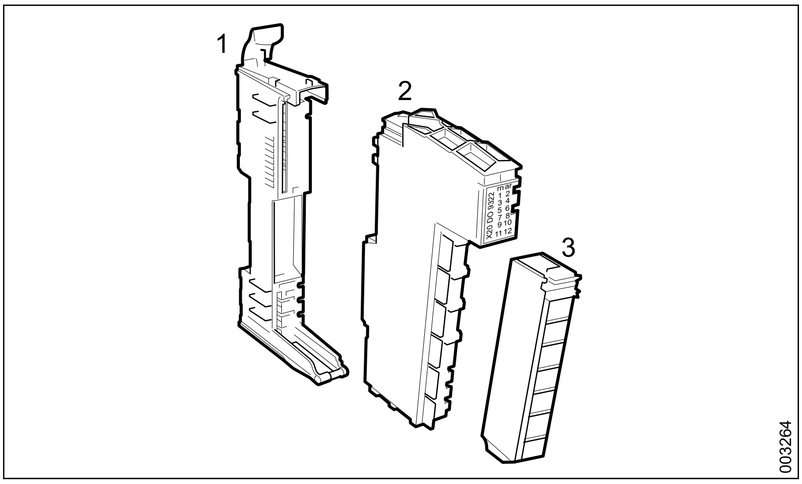
Only use Getinge original parts.

7.6.2.2.1 General

7.6.2.2.1.1 Module management

- Store the constituent parts in the package until just before installation.
- Do not touch the connection points on the parts.
- Take necessary precautions against static electricity.

7.6.2.2.1.2 Constituent parts



1 Bus module
2 Electronic module
3 Terminal block

7.6.2.2.2 Method 1

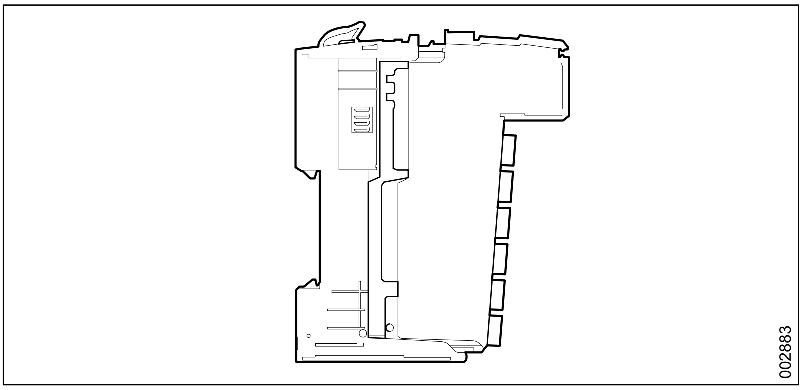
Method 1 describes how to assemble all constituent parts into one unit before fitting it onto the mounting rail on the machine.

7.6.2.2.2.1 Assembling parts

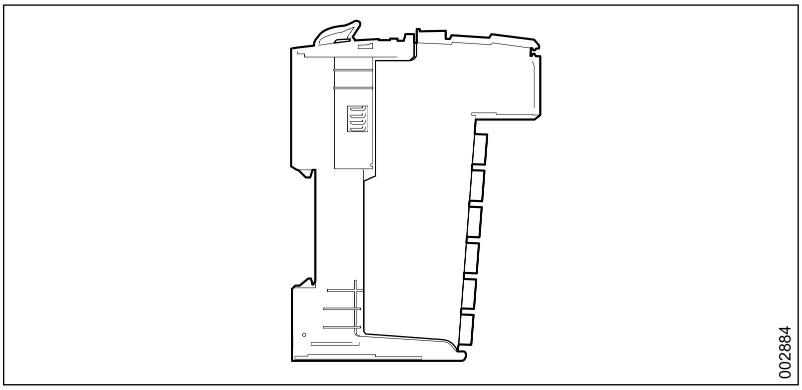
This section describes how to assemble the electronic module, bus module and terminal block into one unit.

The section *7.6.2.2.2.2 Connecting the assembled parts, page 161* then describes how the assembled parts are connected to each other.

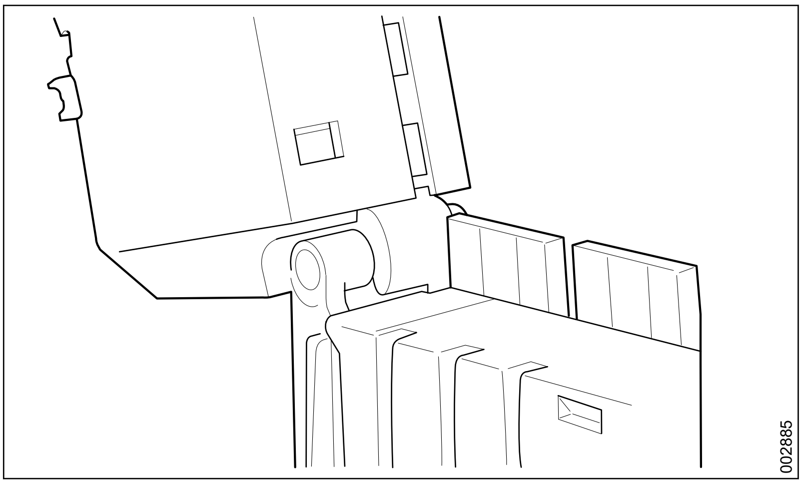
1. Remove the packaging and check that the parts have no visible damage.
2. Put the electronic module in the tracks on the bus module.



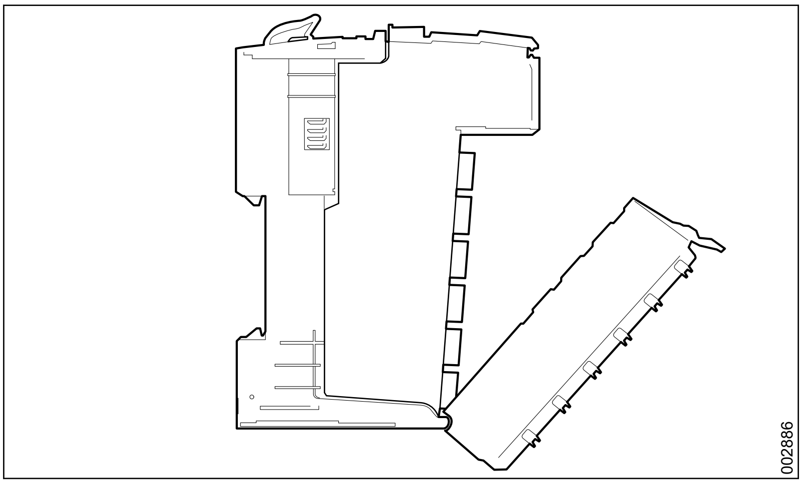
3. Press the electronic module and the bus module together.



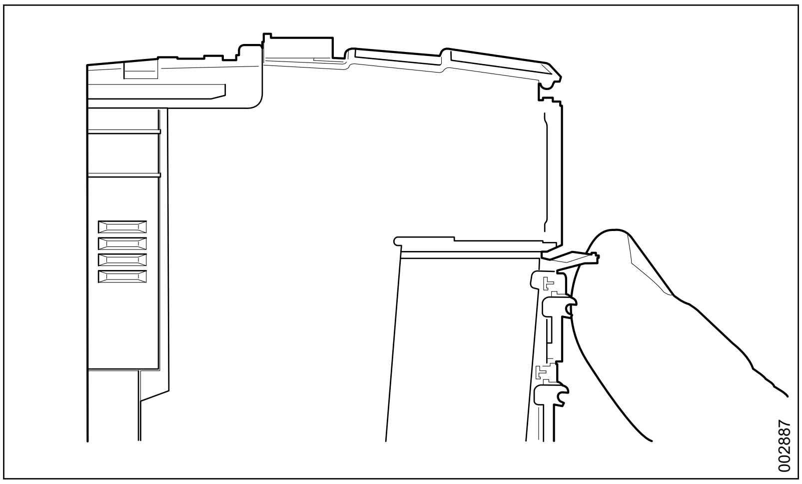
4. Hang the lower part of the terminal block in place on the bus module.



5. Rotate the terminal block upwards and press it in place in the bus module. The terminal block locks into place with an audible click from the locking arm.



6. If the terminal block does not lock into place, the locking arm must be pushed upwards by hand.



7.6.2.2.2.2 Connecting the assembled parts

This section describes how to connect the assembled parts, the electronic module, the bus module and the terminal block into one unit.

The section *7.6.2.2.2.3 Mounting on rail, page 163* describes how to attach the unit to the mounting rail in the machine.

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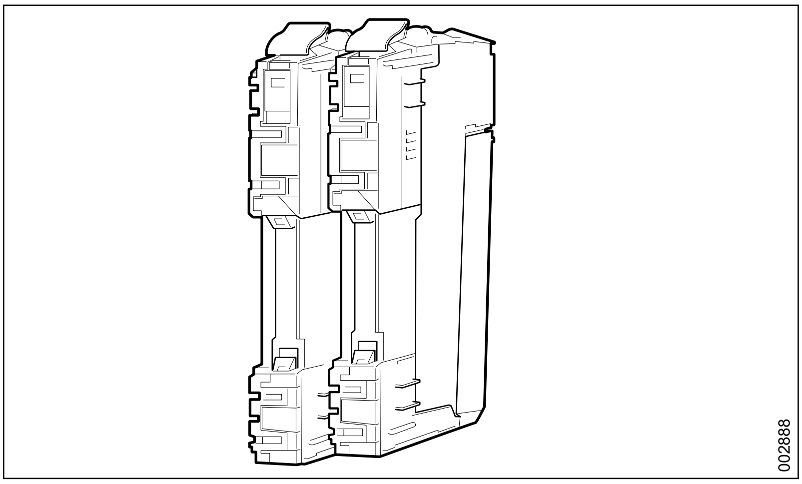
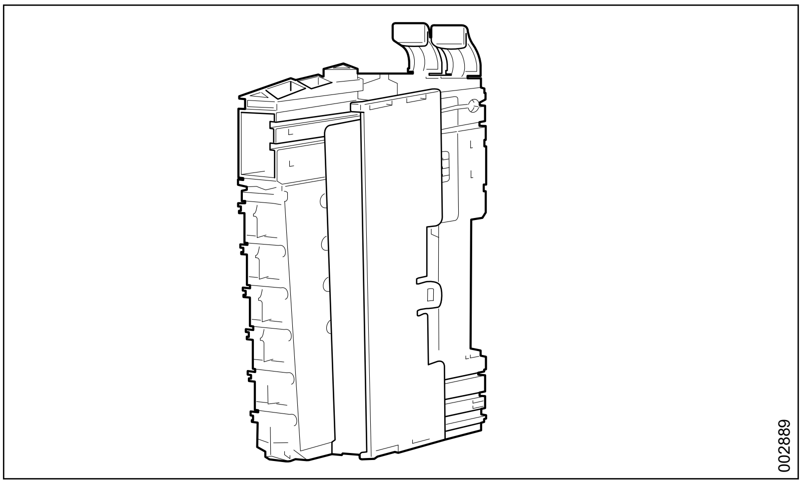
1. Connect the modules to each other from left to right.

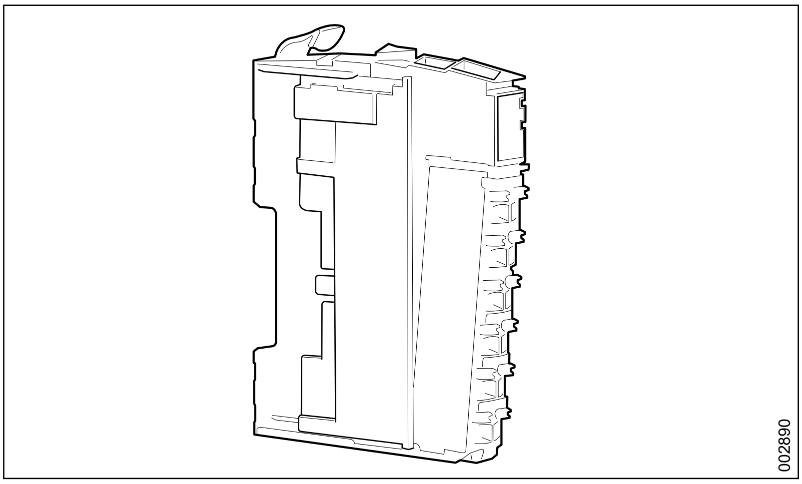


2. Hold one module in the left hand. With the other hand, push the second module from behind onto the first one. Ensure that the tracks and rails on the modules fit.
3. Continue pushing the module forwards until they are in line with each other.
4. Continue in the same way with the rest of the modules.



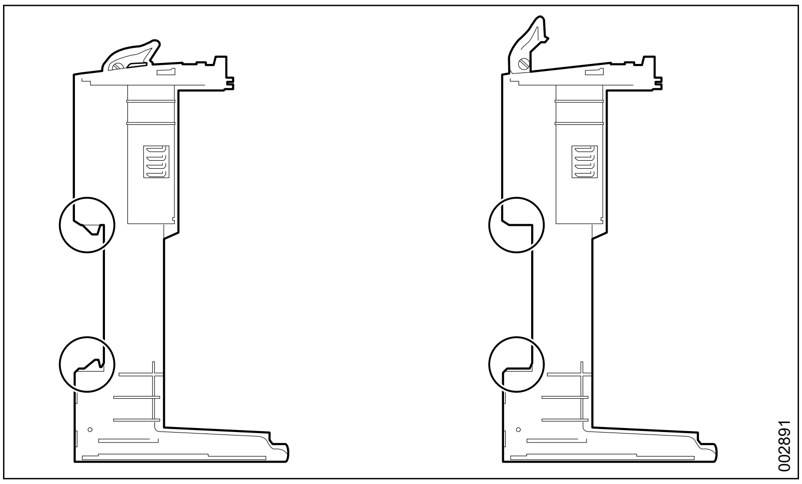
5. Place the electronic module in the bus module and press hard to fit the two modules together.
6. Put the lower part of the terminal block in position on the electronics module and press it upwards. The terminal block locks into place with an audible click from the locking arm.

7. Put the left-hand locking plate on the left-hand module and slide it into the tracks. Finally, move the locking plate forwards.



7.6.2.2.2.3 Mounting on rail

1. Press the locking arm all the way up on all bus modules to open the locking mechanism and enable installation on the mounting rail.



2. Hang the bus modules in the required position on the mounting rail.
3. Release the locking arm and ensure that the locking mechanism closes and attaches the bus modules on the mounting rail.

7.6.2.2.3 Method 2

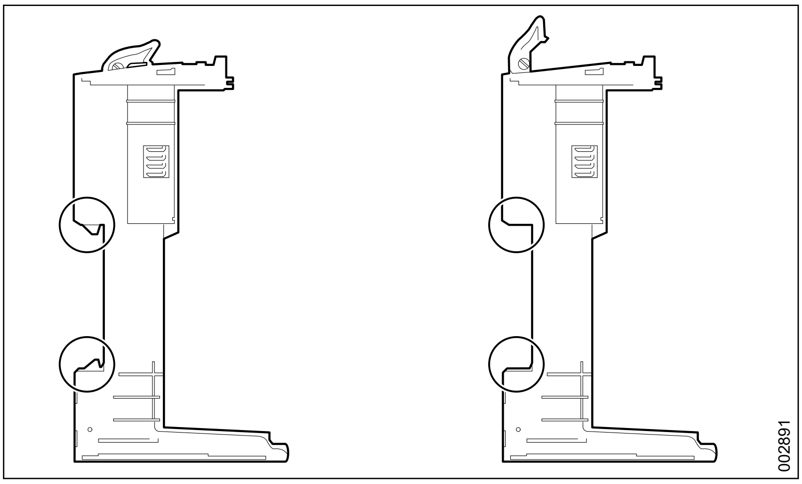
Method 2 describes how to fit the bus module on the mounting rail before installing the electronic module.

7.6.2.2.3.1 Fitting bus module to rail

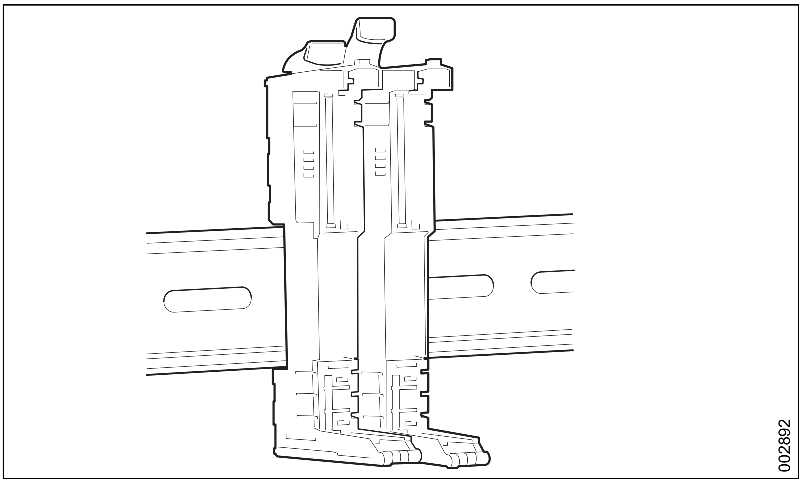
This section describes how the bus modules are attached to the mounting rail.

The section *7.6.2.2.3.2 Fitting electronic module and terminal block, page 164* then describes how to fit the electronic module and accompanying terminal block to the bus modules.

1. Open the packaging and check that the part has no visible damage.
2. Press the locking arm all the way up on the bus module to open the locking mechanism and enable installation on the mounting rail.



3. Hang the bus module in the required position on the mounting rail.
4. Release the locking arm and make sure the locking mechanism closes and attaches the bus module to the mounting rail.
5. Put the next bus module in the tracks of the previously fitted bus module.

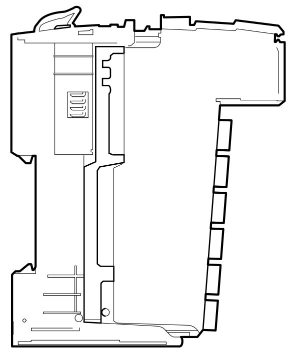


6. Open the locking arm and slide the bus module against the mounting rail.
7. Release the locking arm and make sure the locking mechanism closes and attaches the bus module to the mounting rail.
8. Continue in the same way with the remaining bus modules.

7.6.2.2.3.2 Fitting electronic module and terminal block

In the following instructions, the bus control units are first attached to the mounting rail. Then the electronic modules are connected, and finally the terminal blocks are connected.

1. Put the accompanying electronic module in the tracks on the bus module furthest to the left.



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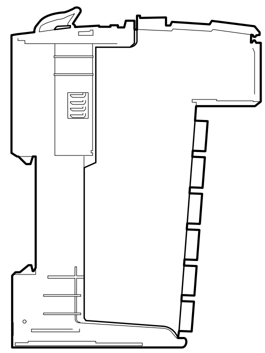
2. Press the electronic module and the bus module together.



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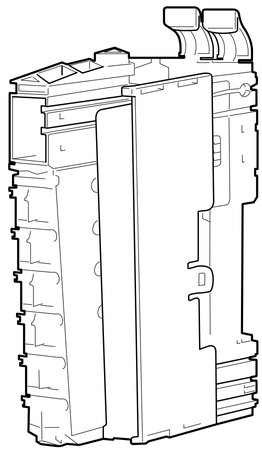
3. Continue in the same way to the second-last module.
4. Put the right-hand locking plate into the tracks on the bus module from the front and push it the whole way in.



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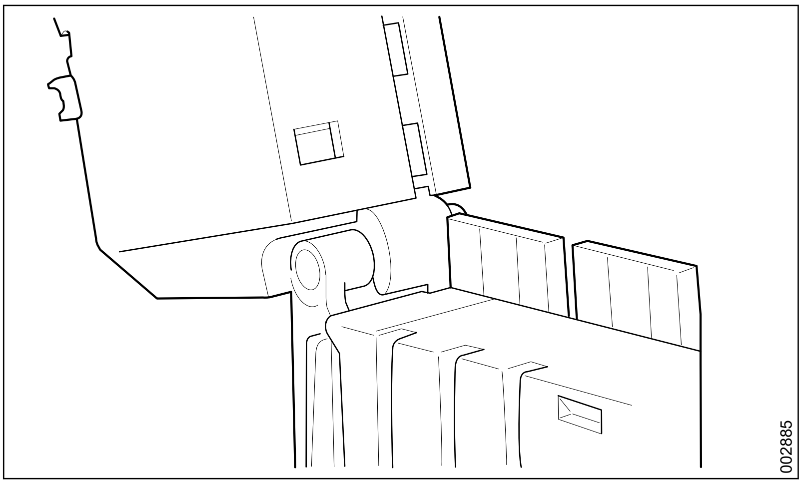
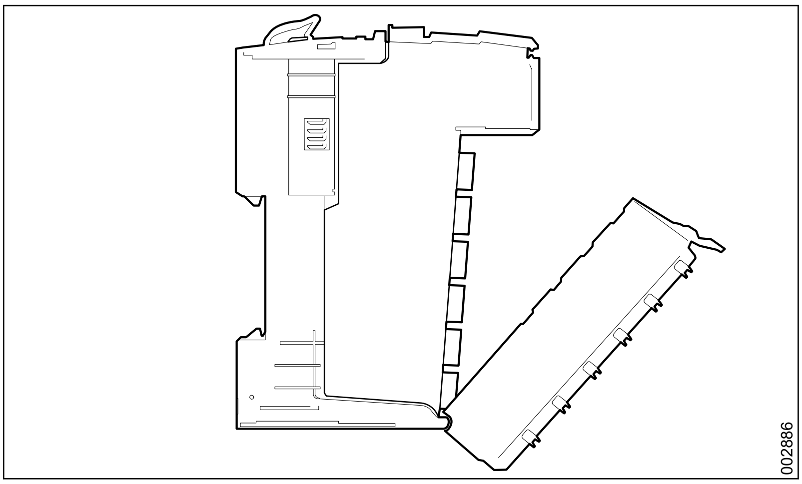
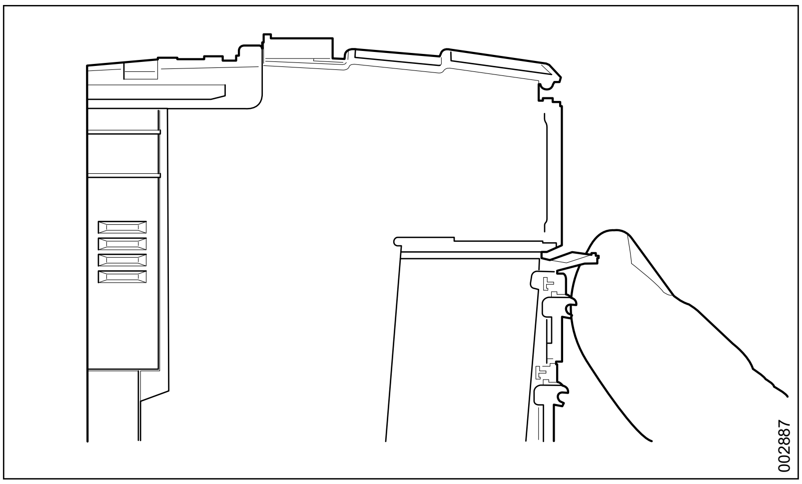
5. Place the electronic module in the bus module and press hard to join the two modules together.
6. Hang the terminal block in place on the bus module at the extreme left.



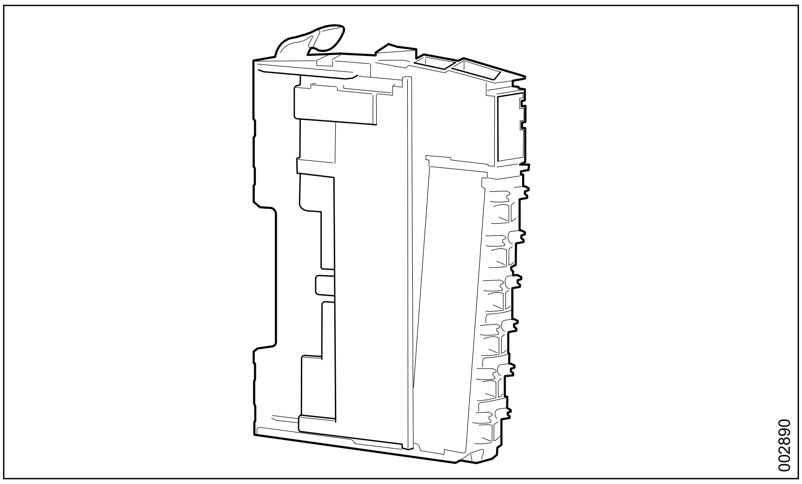
7. Rotate the terminal block upwards and press it in place in the bus module. The terminal block locks into place with an audible click from the locking arm.



8. If the terminal block does not lock into place, the locking arm must be pushed upwards by hand.

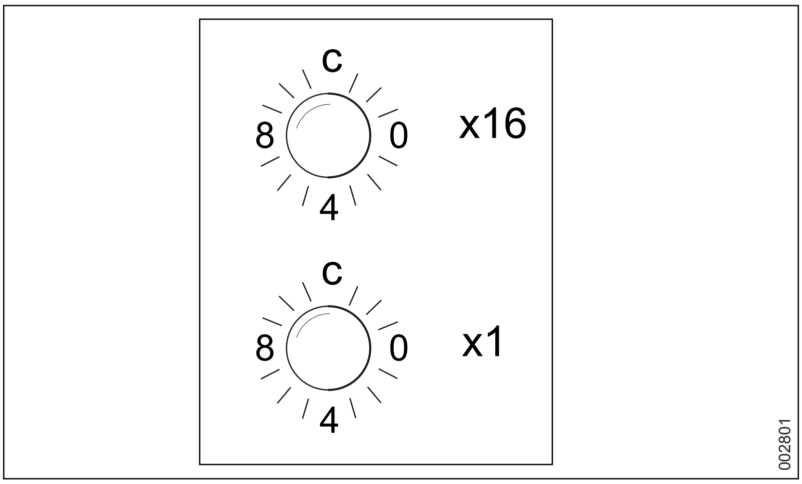


9. Continue in the same way with the rest of the terminal blocks.
10. Put the left-hand locking plate on the left-hand module and slide it into the tracks.
11. Attach the locking plate by sliding it forward.



7.6.2.2.4 Setting addresses

When a node module is replaced, the station address must be set using a manual control. Use the same setting for the new module as for the one replaced.



The station addresses for the different washer-disinfectors during installation in the **AGS** must be set according to section *7.6.2.2.4.1 Station addresses, AGS, page 167*.

The station address for the **Supervisor-PLC** is set according to section *7.6.2.2.4.2 Station address, Supervisor, page 168*.

7.6.2.2.4.1 Station addresses, AGS

Node/name	Station	x16	x1
Washer-disinfector 1	81	5	1
Washer-disinfector 2	82	5	2
Washer-disinfector 3	83	5	3
Washer-disinfector 4	84	5	4
Washer-disinfector 5	85	5	5
Washer-disinfector 6	86	5	6
Washer-disinfector 7	87	5	7
Washer-disinfector 8	88	5	8

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Node/name	Station	x16	x1
Washer-disinfector 9	89	5	9
Washer-disinfector 10	90	5	A

For more information, see separate documentation for the AGS.

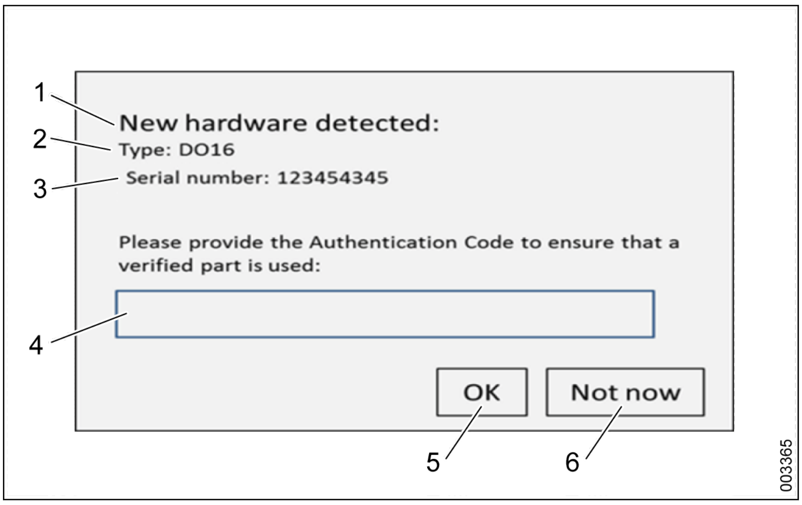
7.6.2.2.4.2 Station address, Supervisor

Node/name	Station	x16	x1
Supervisor PLC	33	2	1

7.6.2.2.5 Approving new hardware

When a new electronic module has been installed, the part must be approved using a specific authentication code. The code is supplied with the original part together with the order from a Getinge sales company.

When the faulty module has been replaced with a new one and the machine is started, a dialog box opens, requesting the authentication code for the new part.



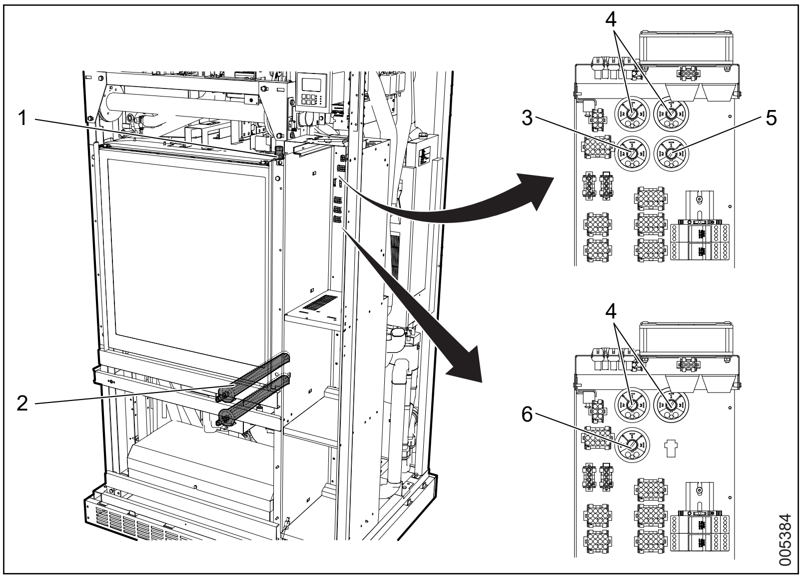
1 Message	4 Enter code
2 **Type**	5 **OK**
3 **Serial number**	6 **Not now**

- Enter a valid authentication code and press **OK** to approve the new electronic module.
- Or press **Not now** to start the system without providing an authentication code.

The dialog box is displayed every time the machine is started until a valid authentication code is provided. If other parts are missing authentication codes, other dialog boxes will open one after another.

If the supplied authentication code is not accepted by the system, contact Getinge for assistance.

7.7 Chamber and main pipe



1 Temperature sensor	4 Pressure switch, main pipe
2 Element	5 Pressure switch, high water level (S-8668T)
3 Pressure switch, low water level (S-8668T)	6 Pressure switch, water level (S-8666 and S-8668)

7.7.1 Replacing a temperature sensor

Sensor: ***BT01/02***

*Connector:***UH03 -X117**

1. Remove the old sensor by pulling it out of the seal.
2. Disconnect the old sensor from the connector.
3. Push the new sensor in through the seal.
4. Connect the new sensor to the connector.

Ensure that the sensor does not penetrate too far into the chamber.

After replacing the sensor in the chamber, the sensor must be calibrated according to section *6 CALIBRATION, page 129*.

7.7.2 Replacing pressure switch

The pressure switches are fitted on the control cabinet's rear plate, behind the other control electronics. Two pressure switches measure the pressure in the drying air pipes and also detect leaks from the chamber to the drying system.

S-8666 and S-8668:

One pressure switch measures the water level in the chamber.

S-8668T:

Two pressure switches measure the water level in the chamber.

7.7.3 Replacing an element

The heating is done by electricity or steam.

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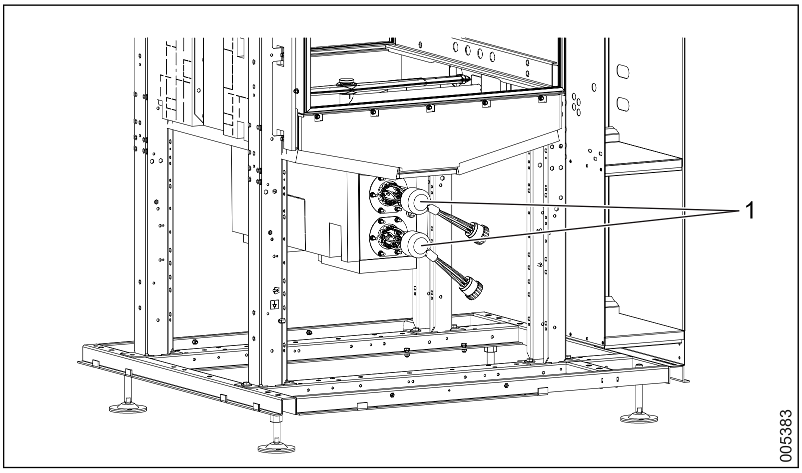
The electrical elements have built-in overheating protection. If the overheating protection has tripped, the whole element must be replaced.



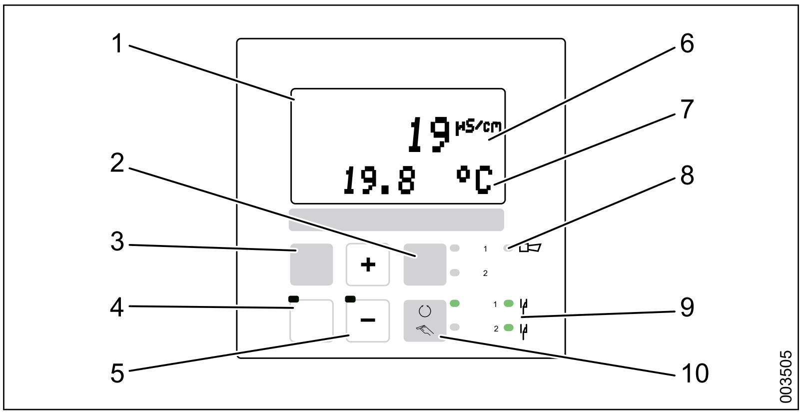
1 Elements

Replacing the heating element:

1. Remove the protective cover on the element.
2. Unscrew the nuts holding the element flange in place and remove the element flange.
3. Disconnect the element from the wires.
4. Remove the element.
5. Fit an O-ring to the element.
6. Fit the new element to the machine in the reverse order.

7.8 Conductivity meter (option)

The conductivity sensor is fitted above the machine's control cabinet.



Name	Description
1. Screen	Displays the current values.
2. Relay button/indicator	* Press to switch the relays in manual mode. Indicator for the active relays.
3. Calibration button	To be used when calibrating the sensor.
4. Enter button E	Press to confirm a function when configuring the sensor.
5. + or −	Press to increase or decrease a value or scroll through functions when configuring the sensor.
6. Measured value	Displays the measured value (conductivity).
7. Secondary measured value	Displays the secondary measured value (temperature).
8. Alarm indicator	Displays the alarm functions from the sensor.
9. Relay indicator	* Displays the status of the relays.
10. Manual/Auto button/indicator	* Press to switch between manual and automatic mode for the relays. Indicates which mode is active.

* The relays are not used by the washer-disinfector.

The normal measuring range is 0-500 µS/cm.

7.8.1 Set the output signal

1. Press E on the conductivity meter.
2. Press + or − to enter code 22.
3. Press E. Setup 1 appears.
4. Press + or − to choose Output.
5. Press E. Lin appears on the upper line and Sel. Type appears on the bottom line.
6. Press E.
7. Press + or − to select the value 4-20 mA.
8. Press E.

9. Press + or − to set 0/4 mA to 0.000 µS/cm.
10. Press E.
11. Press + or − to set 20 mA to 500 µS/cm.
12. Press E.
13. Press + and − at the same time to log out and save the settings.

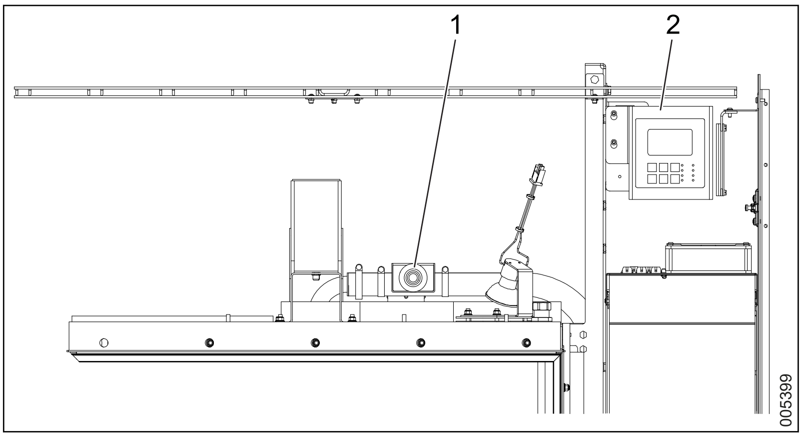
7.8.2 Set the cell constant

1. Press E.
2. Press + or − to enter code 22.
3. Press E. Setup 1 appears.
4. Press E until the display shows Cellconst.
5. Press + or − to set the relevant cell constant; see the calibration certificate.
6. Press E three times. The display shows Setup 1.
7. Press + and − at the same time to log out and save the settings.

7.8.3 Calibration

See section *6.6 Calibration of signals from conductivity meter, page 135*.

7.8.4 Removing a measuring cell



1 Conductivity sensor

2 Conductivity sensor control unit

1. To access the conductivity sensor, remove the top front cover plate of the soiled side.
2. To access the conductivity sensor control unit, open the top cabinet door from the soiled side.

7.9 Door

The door is closed and locked in two steps:

1. To close, the door is raised upwards by the door motor.
2. To lock, the door frame is pulled towards the chamber sealing by the door closing motor.

Switches used in the door function are described in a subsequent section.

7.9.1 Operating door in manual mode

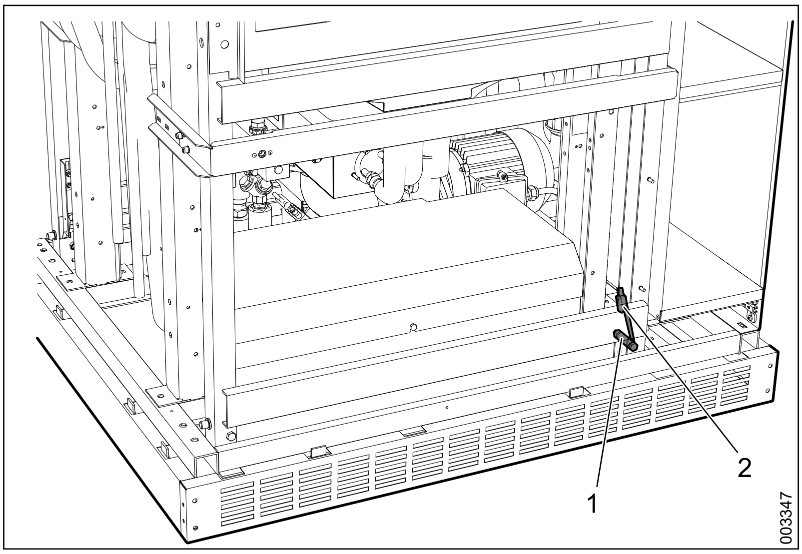
To operate the door manually, go to **Menu** > **System** > **Service** > **Diagnostics** > **Digital Outputs**

For more information, see section *4.3.3.4 Digital outputs, page 40*.

7.9.2 Adjusting door switches

For more information, see section *5.5.1 Checking and adjusting door, page 101*.

7.9.3 Replacing seal



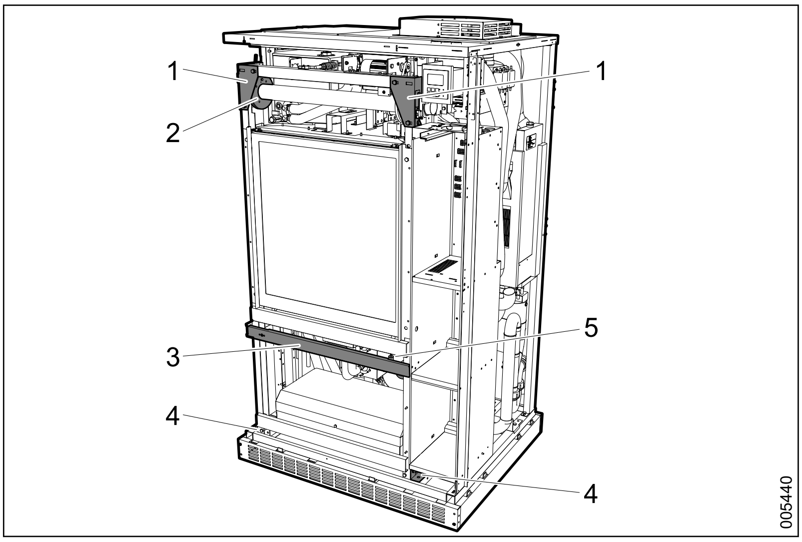
1 Bolt

2 Door switches

Image of S-8666 and S-8668.

1. Remove the door switches and the bolts.
2. Use the touchscreen to operate the door in manual mode and lower the door far enough to make it possible to allow the seal to be removed. Avoid lowering the door all the way down to the floor. For more information, see section *4.3.3.4 Digital outputs, page 40*.
3. Pull off the old seal.
4. Check the silicone and if necessary, apply new silicone.
5. Fit the new seal and press it firmly into place.
6. Use the touchscreen to operate the door in manual mode and raise the door to its upper position.
7. Refit the door switches and bolts.

7.9.4 Removing door



1 Upper fastener
2 Roller, wire
3 Support beam
4 Lower fastener
5 Drain hose

3. Remove the front panels. For more information, see section *7.4 Cover panels, page 144*.
4. Remove the door switch (DOOR OPEN) in the lower section of the frame. For more information, see section *5.5.1.3 Door equipment - Overview, page 101*.
5. Loosen the drainage hose at the bottom of the door.
6. Use the touchscreen to operate the door in manual mode and lower the door until it rests on the bolts. For more information, see section *4.3.3.4 Digital outputs, page 40*.
7. Disconnect the wire from the roller by loosening the locking screw.
8. Separate the door frame and the door closing mechanism (the part that pulls the door frame onto the chamber).
9. Disconnect the remaining switches in the door frame. For more information, see section *5.5.1.3 Door equipment - Overview, page 101*.
10. Secure the frame to ensure that it does not fall forwards.
11. Free the top edge of the door frame by loosening the fasteners located on each side of the door frame.
12. Loosen the support beam.
13. Free the lower edge of the door frame by loosening the fasteners located on each side of the door frame.
14. Lift out the frame and door and place it on a table or on the floor.

7.9.5 Installation of door

Install the door and frame by following the instructions in section *7.9.4 Removing door, page 174* in reverse order.

Remember to hang the door horizontally when attaching it to the pulleys on both sides. Secure the wire to the door frame with the screws at the door's top corners.

Once the door and the door frame have been refitted, it may be necessary to adjust the door. For more information, see section *5.5.1 Checking and adjusting door, page 101*.

7.9.6 Modification when installing at high altitudes

For installations at high altitudes, the door may need to be modified to avoid problems with deformed glass windows due to pressure differences. See the Installation manual for instructions.

7.9.7 Automatic door opening

At the end of a completed program, the door opens automatically according to the standard setting. This setting may be changed by resetting the relevant DIP switch. For more information, see section *4.4.5.1 Machine Setup - DIP Switches, page 66*.

7.9.8 Opening the door in the event of a power failure

During a power failure, the chamber door fails to unlock or open automatically and the following steps must be followed:

1. Remove the upper cover panel on the soiled side. For more information, see section *7.4 Cover panels, page 144*.
2. Locate the nut used to adjust the door's locking force. For more information, see section *7.9.10 Adjustment of locking force, page 175*.
3. Measure and note the actual setting (spring length) for the locking force.
4. Loosen the nut completely to release the door.
5. Press the door down to access the chamber.

Reset the machine:

1. Open the door manually. For more information, see section *4.3.3.4 Digital outputs, page 40*.
2. Re-fit the nut and spring.
3. Adjust the spring to the same setting it was at previously.

7.9.9 Adjustment of crush protection

For more information, see section *2.1.1 Door pinch protection, page 12*.

7.9.10 Adjustment of locking force

For more information, see section *5.5.1.9 Adjustment - door locked, page 109*.

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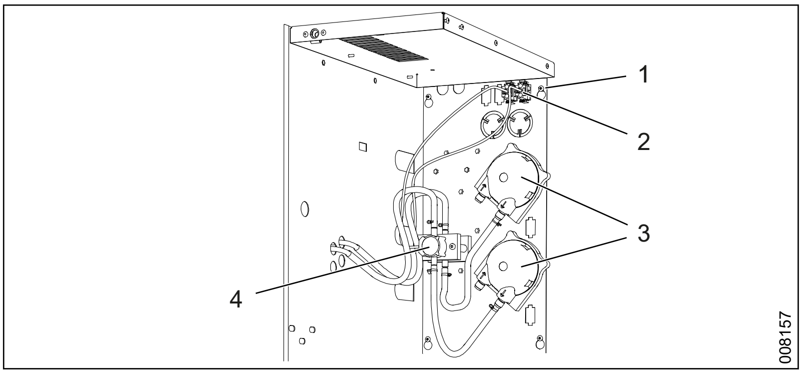
7.10 Dosing system

7.10.1 Overview



1 Screws, bracket
2 Switches
3 Dosing pumps
4 Flow sensor (option)

7.10.2 Replacing a dosing pump or flow sensor

1. Loosen the screws on the panel.
2. Run the hoses out through the panel and lift off the panel.
3.
   Dosing pump:
   3.1. Remove the dosing pump from the back side of the of the pump by pressing the plastic snaps on each side of the pump.
   3.2. Replace the faulty dosing pump with a new one.
   Flow sensor:
   3.3. Remove the faulty flow sensor.
   3.4. Replace the faulty flow sensor with a new one.
4. Refit the panel.
5. Fill the hoses with detergent by running the dosing pump in manual mode. For more information, see section *4.3.3.4 Digital outputs, page 40*.
6. Check the new hoses for leaks.
7. Perform a calibration. For more information, see section *6.7 Calibration of dose monitors, page 136*.

7.10.3 Installation and calibration

For installation and calibration of dosing amounts and flow meters, see sections *6.7 Calibration of dose monitors, page 136* and *7.10.3 Installation and calibration, page 176*.

Flow monitoring may only be turned off for non-process-critical agents and is done at the user's own risk.

Flow monitoring can be disconnected for non-process-critical agents, such as rinse and lubricants. To meet the requirements in ISO EN 15883, disconnection of the flow monitoring is not permitted for detergents.

The user has sole responsibility for creating hands-on routines to ensure that the dosing system works.

To turn off the flow monitoring, when possible on the machine, deactivate the corresponding DIP switches via the Web server. For more information, see section *4.4.5.1 Machine Setup - DIP Switches, page 66*.

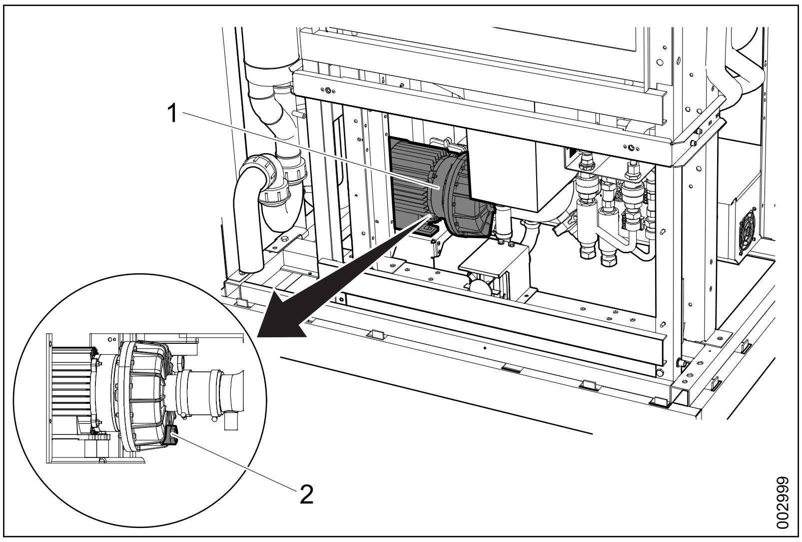
7.11 Circulation system

7.11.1 Overview

If a machine will not be used for more than 72 hours, the Complete drain program must be run and the circulation pump must be emptied.

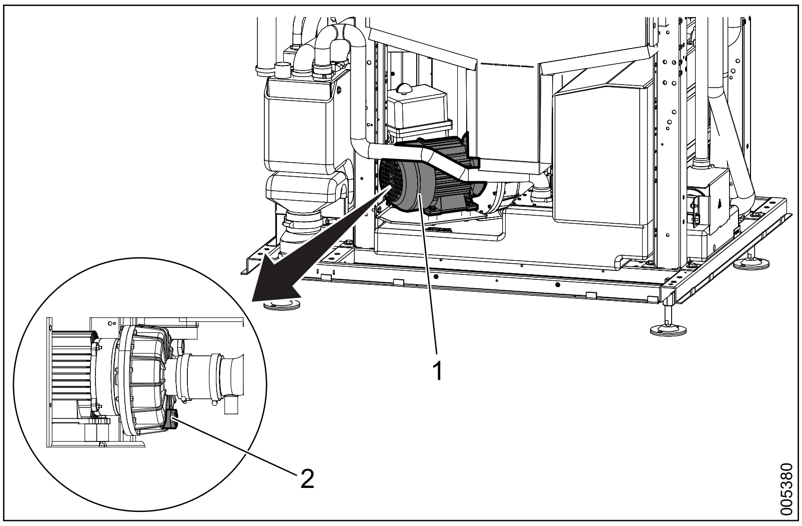
The circulation pump on S-8666 & S-8668 has a dead volume of <300 ml.

The circulation pump on S-8668T has a dead volume of <50 ml.



1 Circulation pump (S-8666 and S-8668

2 Drainage plug (S-8666 and S-8668)



1 Circulation pump (S-8668T)

2 Drainage plug (S-8668T)

If the machine has not been used for 72 hours, the **Complete drain** program must be run. Getinge Disinfection AB recommends running a program with an empty chamber before using the machine to clean items.

7.11.2 Checking the circulation pump

Check that there are no leaks from the pumps or hoses.

7.11.3 Draining the circulation pump

S-8666 and S-8668:

1. Ensure that the chamber is empty of water. If not, run the **Complete drain** program.
2. Ensure that there is a container that can collect the drainage water. Dead volume is about 300 ml.
3. Find the drainage plug (see the previous image).
4. Loosen the drainage plug and empty out the water.

S-8668T:

There are two ways of draining the circulation pump: by removing the drainage plug or by activating digital output for drain valve.

Remove the drainage plug

1. Ensure that the chamber is empty of water. If not, run the **Complete drain** program.
2. Ensure that there is a container that can collect the drainage water. Dead volume is about 50 ml.
3. Find the drainage plug (see the previous image).
4. Loosen the drainage plug and empty out the water.

Activate the digital output for the drain valve

1. Ensure that the chamber is empty of water. If not, run the **Complete drain** program.
2. Activate the digital output for the drain valve from the touchscreen (For more information, see section *4.3.3.4 Digital outputs, page 40*). Set “DO35 Drain valve open” to 1 and “DO41 Drain valve close” to 0.
3. Let the drain valve stay open for 3 minutes.
4. Set “DO35 Drain valve open” to 0 and “DO41 Drain valve close” to 1.

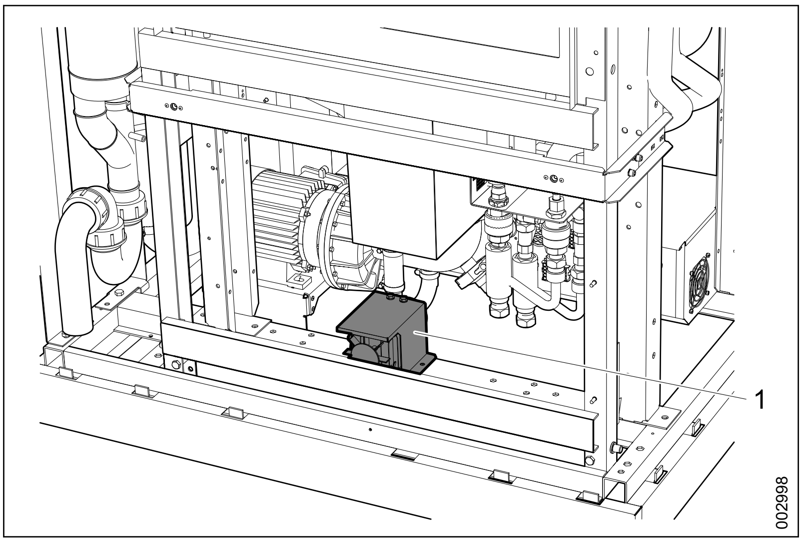
7.11.4 Replacing the circulation pump

1. Ensure that the chamber is empty of water. If not, run the **Complete drain** program.
2. Drain the pump. For more information, see section *7.11.3 Draining the circulation pump, page 178*.
3. Release the hose clips and remove the pump from its attachment. For S-8668T, access from both the machine’s side and from the clean side is required.
4. Install the new pump and reconnect the hoses.
5. After installing the pump, check the connections for leaks.

7.12 Drain system

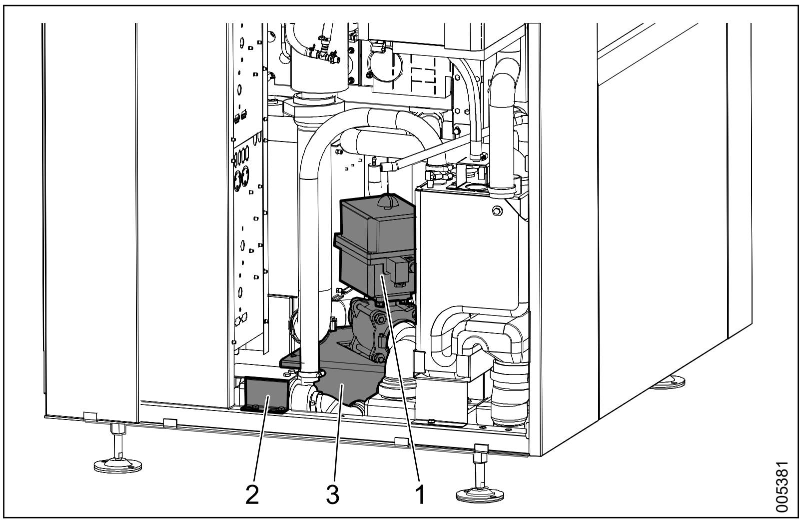
7.12.1 Overview

S-8666 and S-8668



1 Drain pump S-8666 and S-8668

S-8668T



1 Drain valve S-8668T
2 Drain pump S-8668T
3 Drain tank S-8668T

7.12.2 Checking the drain pump

Check that there are no leaks from the pumps or hoses.

7.12.3 Replacing the drain pump

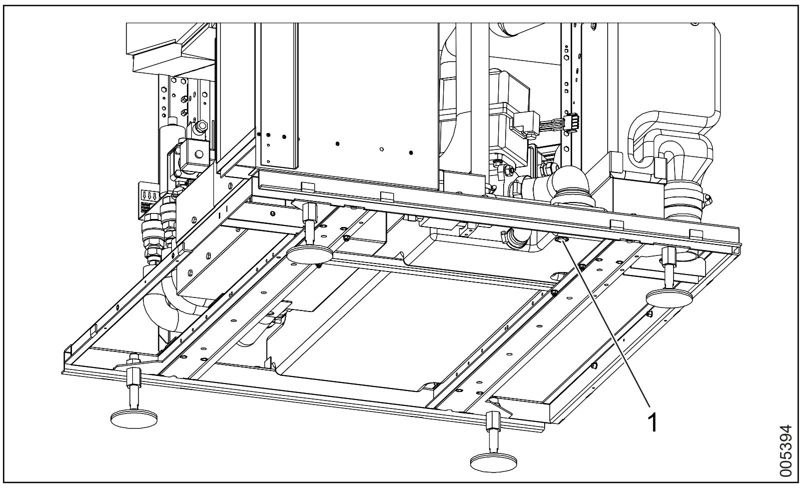
1. For S-8666/S-8668: ensure that the chamber is empty of water. For S-8668T: ensure that the drain tank is empty of water.
2. Release the hose clips and remove the pump from its attachment.
3. Install the new pump and reconnect the hoses.
4. After installing the pump, check the connections for leaks.

7.12.4 Replacing the drain valve (S-8668T)

1. Ensure that the chamber is empty of water.
2. Release the hose clips and remove the drain valve from its attachment.
3. Install the new drain valve and reconnect the hoses.
4. After installing the drain valve, check the connections for leaks.

7.12.5 Drain tank (S-8668T)

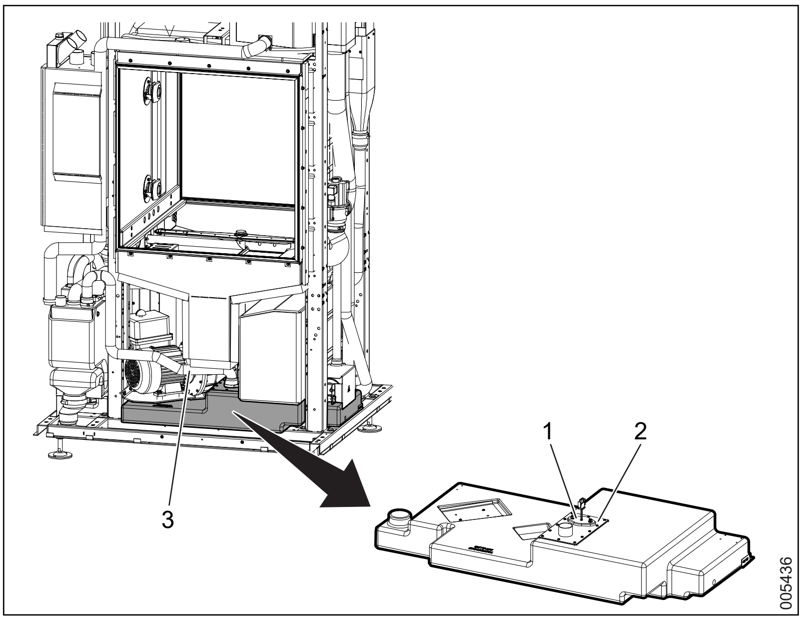
7.12.5.1 Draining drain tank



1 Drain plug

1. Locate the G1/4" drain plug and remove it.
2. Let the water out.
3. Screw the plug back into the tank.

7.12.5.2 Manual cleaning of drain tank



1 Level switch
2 Access panel for cleaning
3 Overflow hose

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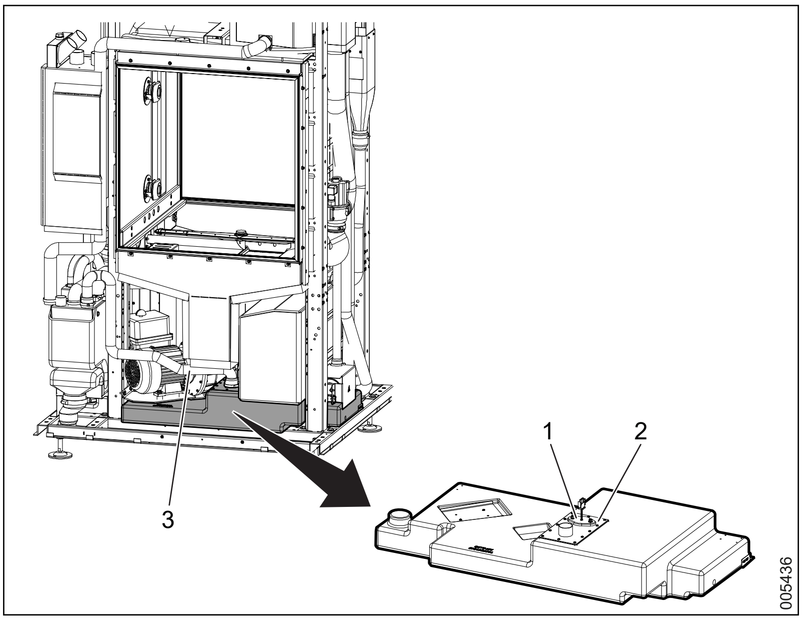
1. Release the overflow hose.
2. Release the access panel for cleaning where the level switch is mounted and remove it.
3. Clean it with a brush.

7.12.5.3 Replacing level switch



1 Level switch	3 Overflow hose
2 Access panel for cleaning	

1. Release the overflow hose.
2. Disconnect the level switch.
3. Unscrew the two nuts that hold the level switch in place.
4. Replace the level switch and mount everything back in place.

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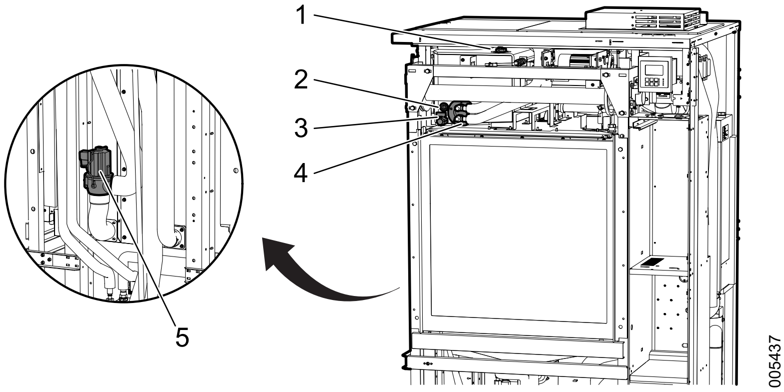
7.13 Booster tank (S-8666 and S-8668 option)

Drain the booster tank before starting service.

7.13.1 Overview



1 Level switch

2 emperature senso

3 Drain tap

4 Heating element

5 Automatic draining valve, with manual lever. (May be difficult to access in some configurations).

7.13.2 Drainage

The booster tank is primarily drained by running the **Complete drain** program from the touchscreen.

If the booster tank cannot be drained with the **Complete drain** program, drain it manually:

- by activating the digital output from the touchscreen. For more information, see section *4.3.3.4 Digital outputs, page 40*.

  or
- by using the manual lever on the automatic draining valve. For more information, see section *7.13.1 Overview, page 182*).

  or
- by using the drain tap on the booster tank.

To access the booster tank, remove the front cover panels on the soiled side. For more information, see section *7.4 Cover panels, page 144*).

Draining by manually opening draining valve

1. Find the draining valve, which is fitted to the left side of the machine as seen from the soiled side. For more information, see section *7.13.1 Overview, page 182*).
2. Pull the handle on top of the valve straight up.

Draining with drain tap

1. Locate the drain tap, which is accessible from the soiled side. For more information, see section *7.13.1 Overview, page 182*.
2. Ensure that the tap is closed before removing the plug.
3. Remove the plug.
4. Install the nipple and hose on the drain tap. Use a nipple with thread G1/4 outward and a matching hose.
5. Open the tap and drain the tank. The tank capacity is approximately 38 liters.

After draining (before using the machine

1. Check the function of the drain tap and reinstall the plug.
2. Seal with Teflon tape.
3. Refill the booster tank manually by activating the digital output. For more information, see section *4.3.3.4 Digital outputs, page 40*.
4. Check the level sensor's function in the booster tank when filling with water. For more information, see section *4.3.3.2 Digital inputs, page 38*.
5. Check the entire system for leakage after completing service.

7.13.3 Replacing a temperature sensor

Sensor: ***BT04***

Connector: ***UH03 -BT04-X1***

1. Remove the old temperature sensor by unscrewing the gland and pulling the sensor out.
2. Disconnect the old sensor from the connector.
3. Push in the new sensor and tighten the gland.
4. Connect the new sensor to the connector.

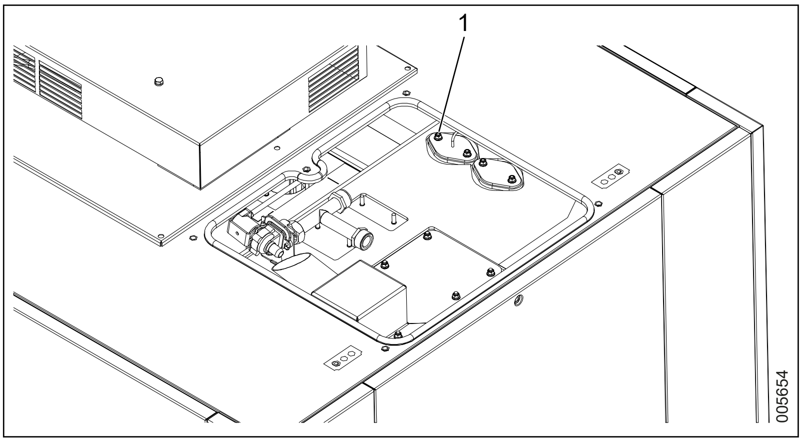
7.13.4 Replacing an element

The booster tank is heated with electricity or steam.

Electrical elements have a built-in overheating protection. If the overheating protection has tripped, the whole element must be replaced:

1. Remove the protective cover over the element.
2. Unscrew the nuts holding the element flange in place and remove it.
3. Disconnect the element from cabling/wires.
4. Remove the element.
5. Fit an O-ring to the element.
6. Fit the new element in the reverse order.

7.13.5 Replacing level switch



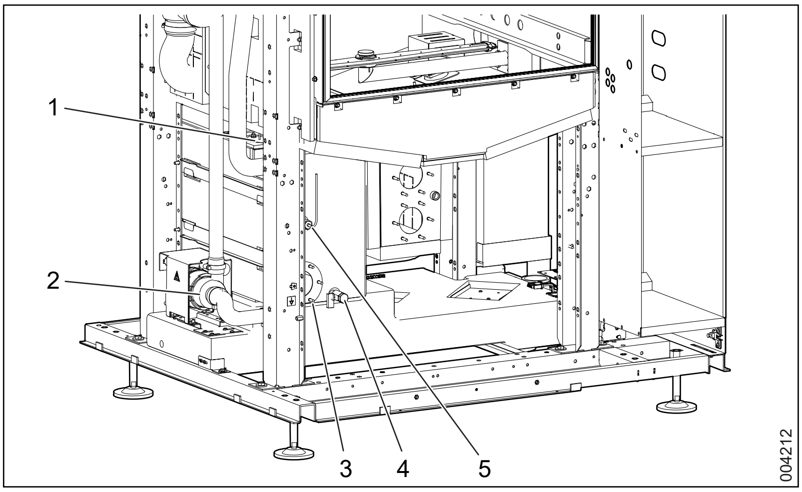
1 Level switch

1. Remove the top lid.
2. Disconnect the level switch.
3. Unscrew the two nuts that holds the level switch in place.
4. Replace the level switch and mount it back into place.

7.14 Booster tank (S-8668T)

Drain the booster tank before starting service.

7.14.1 Overview



1 Level switch
2 Booster pump
3 Heating element
4 Drain tap
5 Temperature sensor.

7.14.2 Drainage

The booster tank is primarily drained by running the **Complete drain** program from the touchscreen.

If the booster tank cannot be drained with the **Complete drain** program, drain manually:

- by activating the digital output from the touchscreen. For more information, see section *4.3.3.4 Digital outputs, page 40*.

  or
- by using the drain tap on the booster tank. For more information, see section *7.14.1 Overview, page 184*.

To access the booster tank, remove the front cover panels on the soiled side. For more information, see section *7.4 Cover panels, page 144*.

Draining with drain tap

1. Locate the drain tap, which is accessible from the soiled side. For more information, see section*7.13.1 Overview, page 182*.
2. Ensure that the tap is closed before removing the plug.
3. Remove the plug.
4. Install the nipple and hose on the drain tap. Use a nipple with thread G1/4 outward and a matching hose.
5. Open the tap and drain the tank. The tank capacity is approximately 30 liters.

After draining (before using the machine)

1. Check the function of the drain tap and reinstall the plug.
2. Use Teflon tape to seal.
3. Refill the booster tank manually by activating the digital output. For more information, see section *4.3.3.4 Digital outputs, page 40*.
4. Check the level sensor's function in the booster tank when filling with water. For more information, see section *4.3.3.2 Digital inputs, page 38*.
5. Check the entire system for leakage after completing service.

7.14.3 Replacing a temperature sensor

*Sensor: **BT04***

*Connector: **UH03 -BT04-X1***

1. Remove the old temperature sensor by unscrewing the gland and pulling the sensor out.
2. Disconnect the old sensor from the connector.
3. Push in the new sensor and tighten the gland.
4. Connect the new sensor to the connector.

7.14.4 Replacing an element

The booster tank is heated with electricity or steam.

Electrical elements have a built-in overheating protection. If the overheating protection has tripped, the whole element must be replaced.

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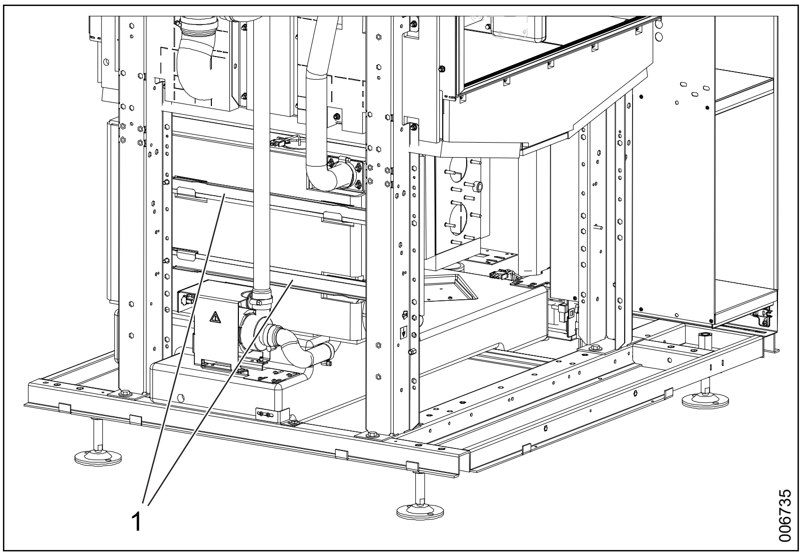
1. Remove the protective cover over the element.
2. Unscrew the nuts holding the element flange in place and remove the element flange.
3. Disconnect the element from cabling/wires.
4. Remove the element.
5. Fit an O-ring to the element.
6. Fit the new element in the reverse order.

7.14.5 Replacing booster pump

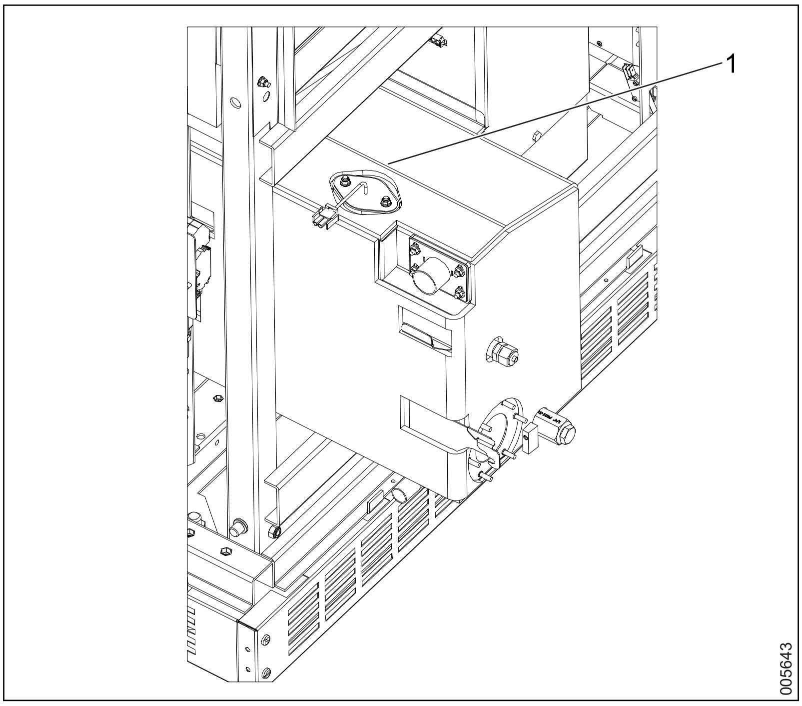
The booster pump can be removed from the soiled side of the machine or from the clean side of the machine. For easier access from the front, disconnect the booster tank and slide it outwards on the guide rails.



1 Guide rails

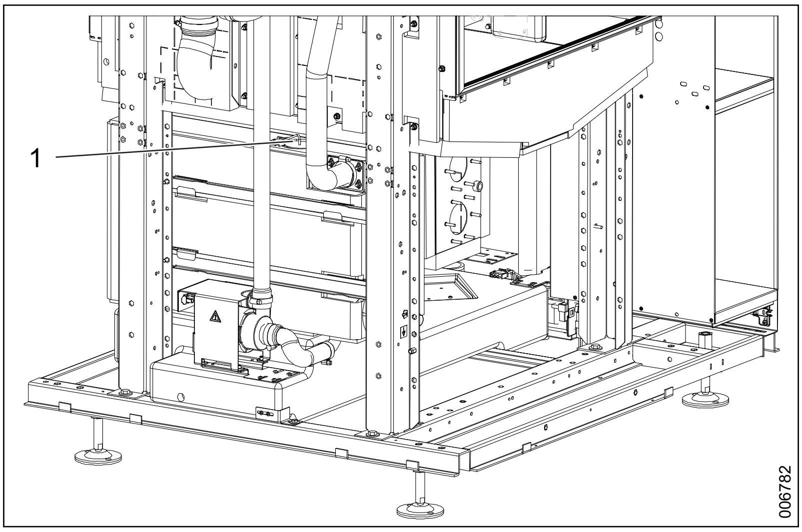
1. Ensure that the tank is empty of water.
2. Release the hose clamps and remove the pump from its attachment on the drain tank.
3. Install the new pump and reconnect the hoses.
4. After installing the pump, check the connections for leaks.

7.14.6 Replacing level switch



1 Booster tank

For access to the booster level switch from the front, disconnect the booster tank and slide it outwards on the guide rails.

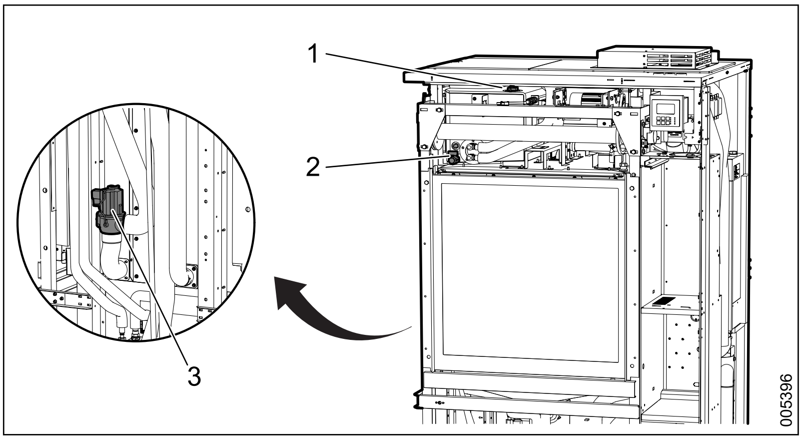


1 Level switch

The booster tank level switch can be removed from the side of the machine if it is equipped with a service sledge or placed with easy access to the side.

7.15 Process tank (S-8668T)

7.15.1 Overview



1 Level switches

2 Drain tap

3 Automatic draining valve, with manual lever. (May be difficult to access in some configurations).

7.15.2 Drainage

The process tank is primarily drained by running the **Complete drain** program from the touchscreen.

If the booster tank cannot be drained with the **Complete drain** program, drain it manually:

- by activating the digital outputs from the operator panel (see section *4.3.3.4 Digital outputs, page 40*).

  or
- by using the manual lever on the automatic draining valve (see section *7.15.1 Overview, page 188*).

  or
- by using the drain tap on the booster tank.

To access the process tank, remove the front cover panels on the soiled side (see section *7.4 Cover panels, page 144*).

Draining by manually opening draining valve

1. Find the draining valve, which is fitted to the left side of the machine as seen from the soiled side (see section *7.15.1 Overview, page 188*).
2. Pull the handle on top of the valve straight up.

Draining with drain tap

1. Locate the drain tap, which is accessible from the soiled side (see section *7.15.1 Overview, page 188*).
2. Ensure that the tap is closed before removing the plug.
3. Remove the plug.

4. Install the nipple and hose on the drain tap. Use a nipple with thread G1/4 outward and a matching hose.
5. Open the tap and drain the tank. The tank capacity is approximately 38 liters.

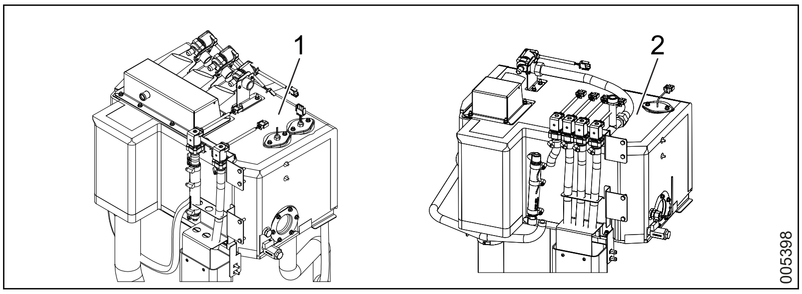
After draining (before using the machine)

1. Check the function of the drain tap and reinstall the plug.
2. Seal with Teflon tape.
3. Refill the process tank manually by activating the digital outputs (see section *4.3.3.4 Digital outputs, page 40*).
4. Check the level sensor's function in the booster tank when filling with water (see section *4.3.3.2 Digital inputs, page 38*).
5. Check the entire system for leakage after completing service.

7.16 Water valves

Perform necessary maintenance according to section *5.7 Water connections, page 113*.

7.16.1 Overview

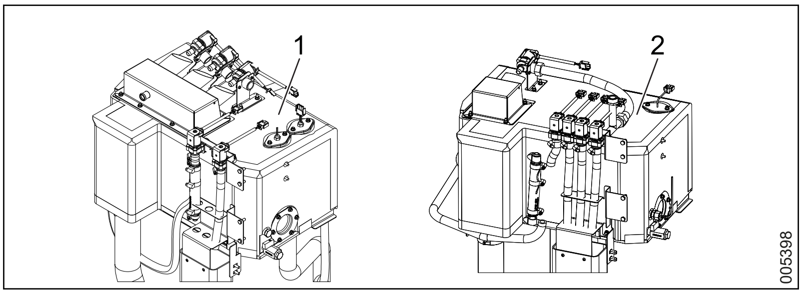


1 Water valves S-8668T

2 Water valves S-8666/S-8668

To access the water valves, remove the middle side cover plate and the top cover plate.

7.16.2 Replacing water valves



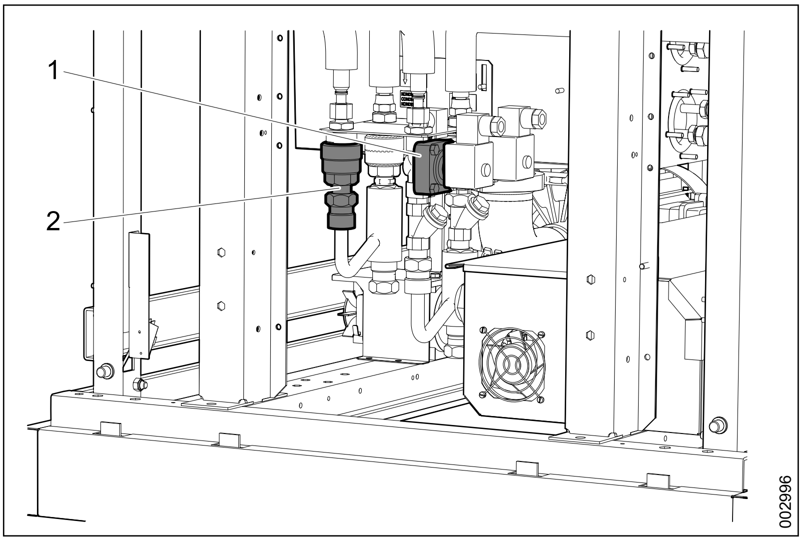
1 Water valves S-8668T

2 Water valves S-8666/S-8668

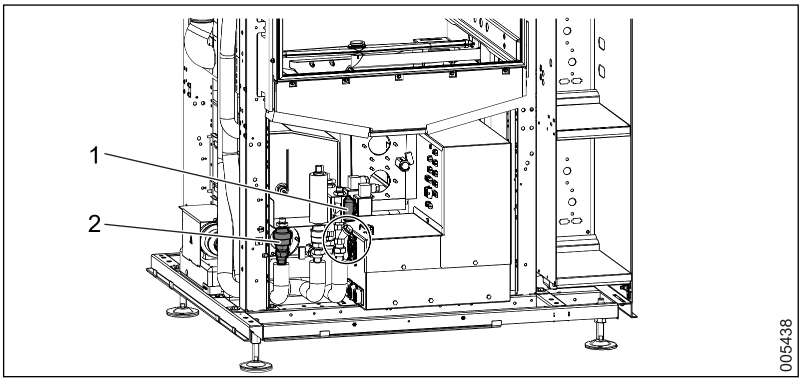
1. Locate the valve to be replaced.
2. Release the hose clamp and the pipe/hose.
3. Remove the old valve and replace it with the new one.
4. After securing the connections, check for leaks.

7.17 Steam system

7.17.1 Steam valves and condensate traps



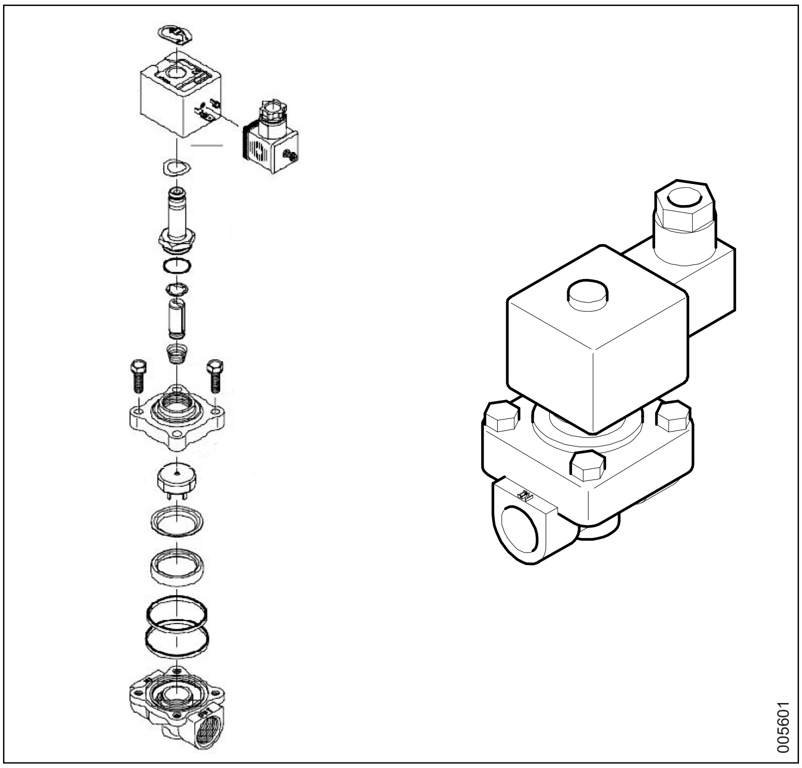
1 Steam valve (S-8666/S-8668)
2 Condensate trap (S-8666/S-8668)



1 Steam valve S-8668T
2 Condensate trap S-8668T

7.17.1.1 Repairing steam valve

1. Dismantle the steam valve and inspect it.



2. Check for deposits and clean if necessary.
3. Check that there is no leakage.

The steam valve is available as a spare part, but there is also a rebuild kit available for repairing steam valves.

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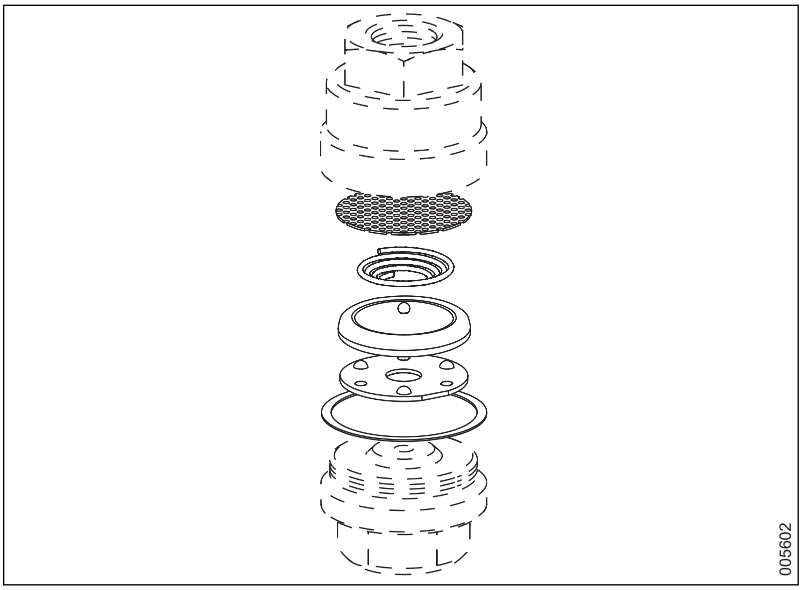
7.17.1.2 Repairing condensate trap

1. Dismantle the condensate trap and inspect it.

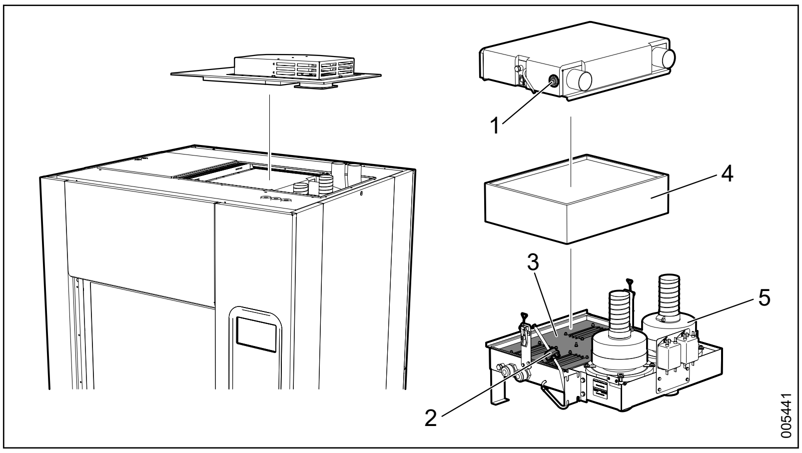


2. Check for deposits and clean if necessary.
3. Check that there is no leakage.

The condensate trap is available as a spare part.

7.18 Drying system

7.18.1 Overview



1 Temperature sensor	4 Filter
2 Differential pressure sensors	5 Drying fan
3 Dryer element	

7.18.2 Changing the filter

Replace the sterile filters according to instructions, when necessary, or if an alarm or warning of high differential pressure is triggered. For more information, see sections *5.1 Service table, page 94* and *5.11.2 Replacing a HEPA filter, page 123*.

7.18.3 Fan inspection and replacement

For the location of the dryer fans, see section *7.18.1 Overview, page 192*.

1. Check the fans' function. For more information, see section *5.11.5 Checking dryer fans, page 124*.
2. Replace the fan if it is damaged.

7.18.4 Checking and replacing sensors

7.18.4.1 Replacing a temperature sensor

*Sensor: **BT03***

*Connector: **UH03 -BT03-X1***

1. Remove the old sensor by pulling it out of the seal.
2. Disconnect the old sensor from the connector.
3. Push the new sensor in through the seal.
4. Connect the new sensor to the connector.

Ensure that the sensor does not penetrate too far into the chamber. After replacing the sensor in the chamber, the sensor must be calibrated according to section *6 CALIBRATION, page 129*.

7.18.4.2 Replacing a differential pressure sensor, HEPA filter

The differential pressure sensor is connected to the drying filter in the machine's upper space. For more information, see section *7.18.1 Overview, page 192*.

The differential pressure sensor is used to monitor:

- that both fans are running and working as intended.
- that the HEPA filter is properly mounted.
- that the HEPA filter is not blocked.
- that the HEPA filter is not damaged.

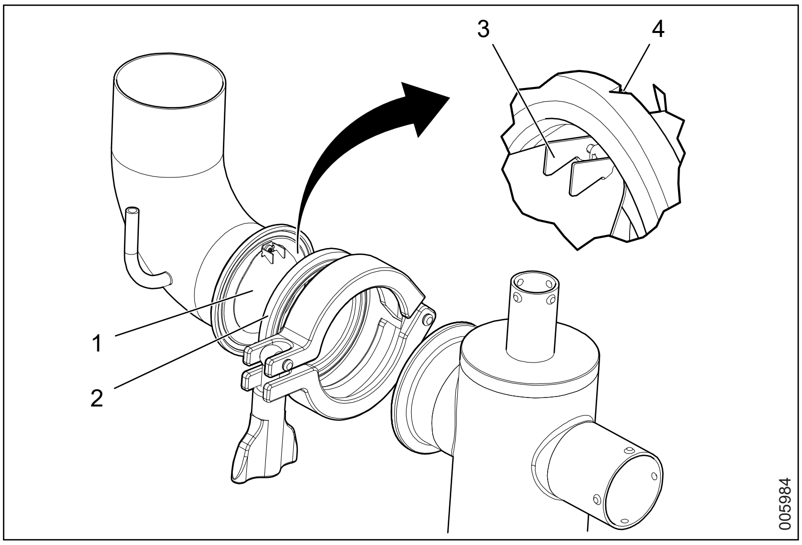
1. Remove the faulty differential pressure sensor and replace it with a new one.
2. Fit cables and hoses in the same way as with the old sensor.
3. Calibrate the new sensor. For more information, see section *6.5 Calibration of signals from differential pressure sensor, HEPA filter, page 134*.

7.18.4.3 Humidity sensor (option)

The humidity sensor is located adjacent to the heat exchanger. For more information, see section *5.11.1 Overview, page 122*.

Check, clean or replace the dryer sensors according to instructions. For more information, see sections *5.11 Drying system, page 122* and *5.11.4 Checking humidity sensor, page 123*.

7.18.5 Replacing check valves



1 Check valve
2 Gasket
3 Hinge
4 Notch on gasket

1. Locate the check valves. For more information, see section *5.11.1 Overview, page 122*. Replacing check valves is not possible from the soiled side.
2. Undo the wing nut and pull out the metal flap.
3. Remove the existing check valve and replace it with a new one. Make sure that the notch on the gasket is in line with the hinge when mounting the new check valve.
4. Check that the valve is correctly mounted and that the metal flap is not obstructed when opening/closing.

7.18.6 Replacing an element

The position of the dryer element is shown in the illustration in section *7.18.1 Overview, page 192*.

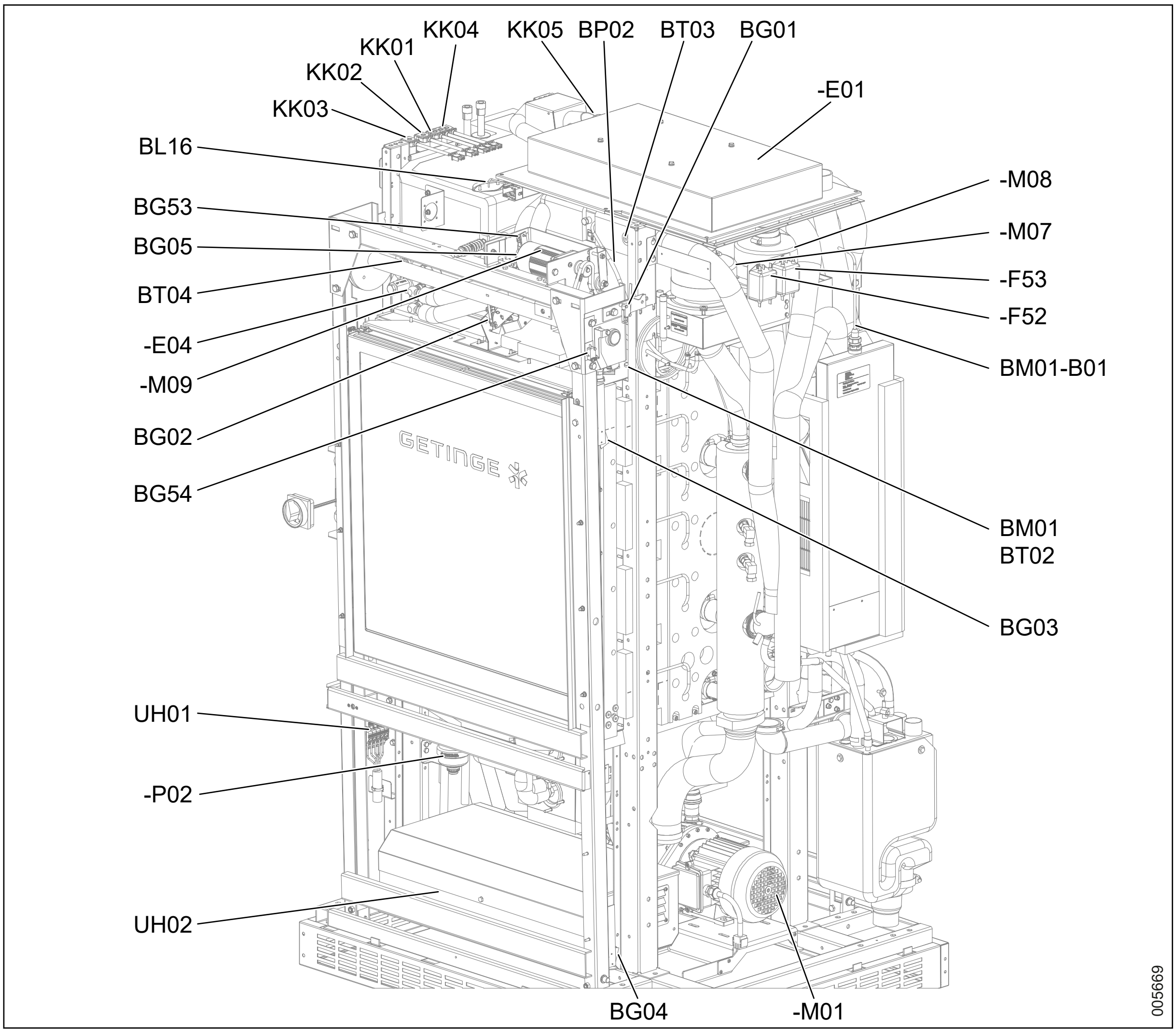
Access can be provided through the machine's cover in a way that is similar to what is described in section *5.11.2 Replacing a HEPA filter, page 123*.

The element is not accessible until the filter box has been removed.

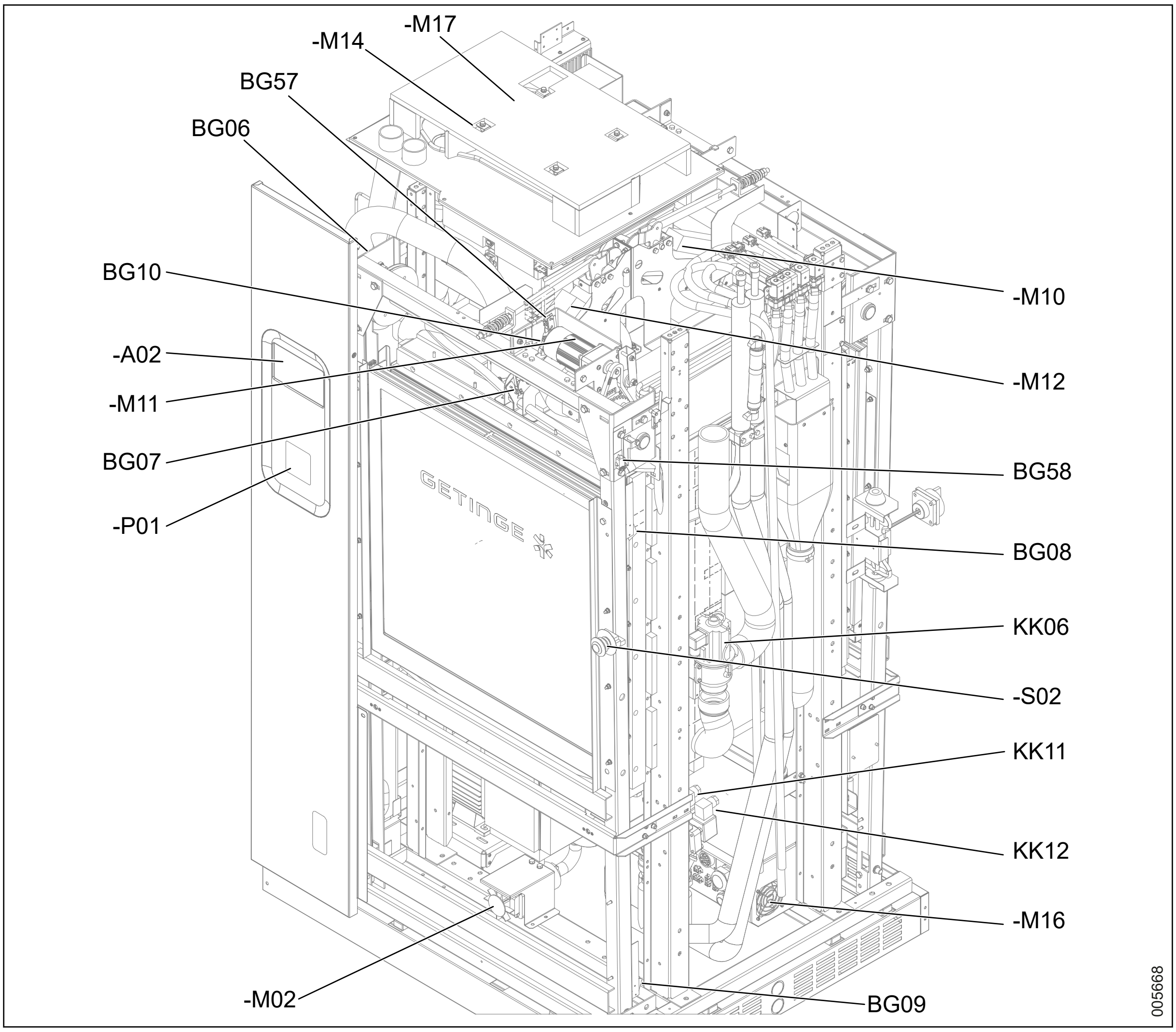
8 LIST OF COMPONENTS

8.1 S-8666 and S-8668

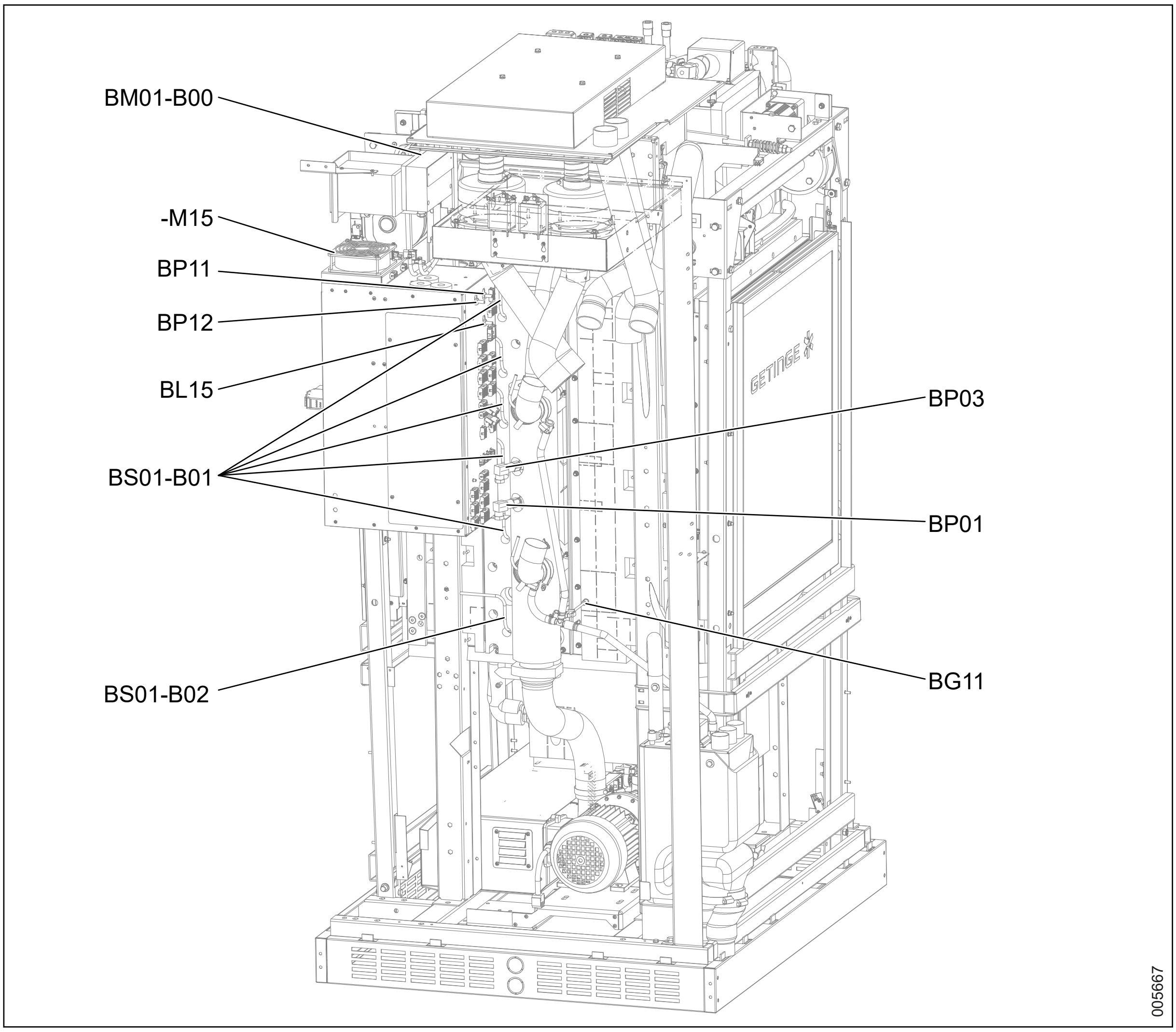
8.1.1 Components, illustration 1/7 (Overview)



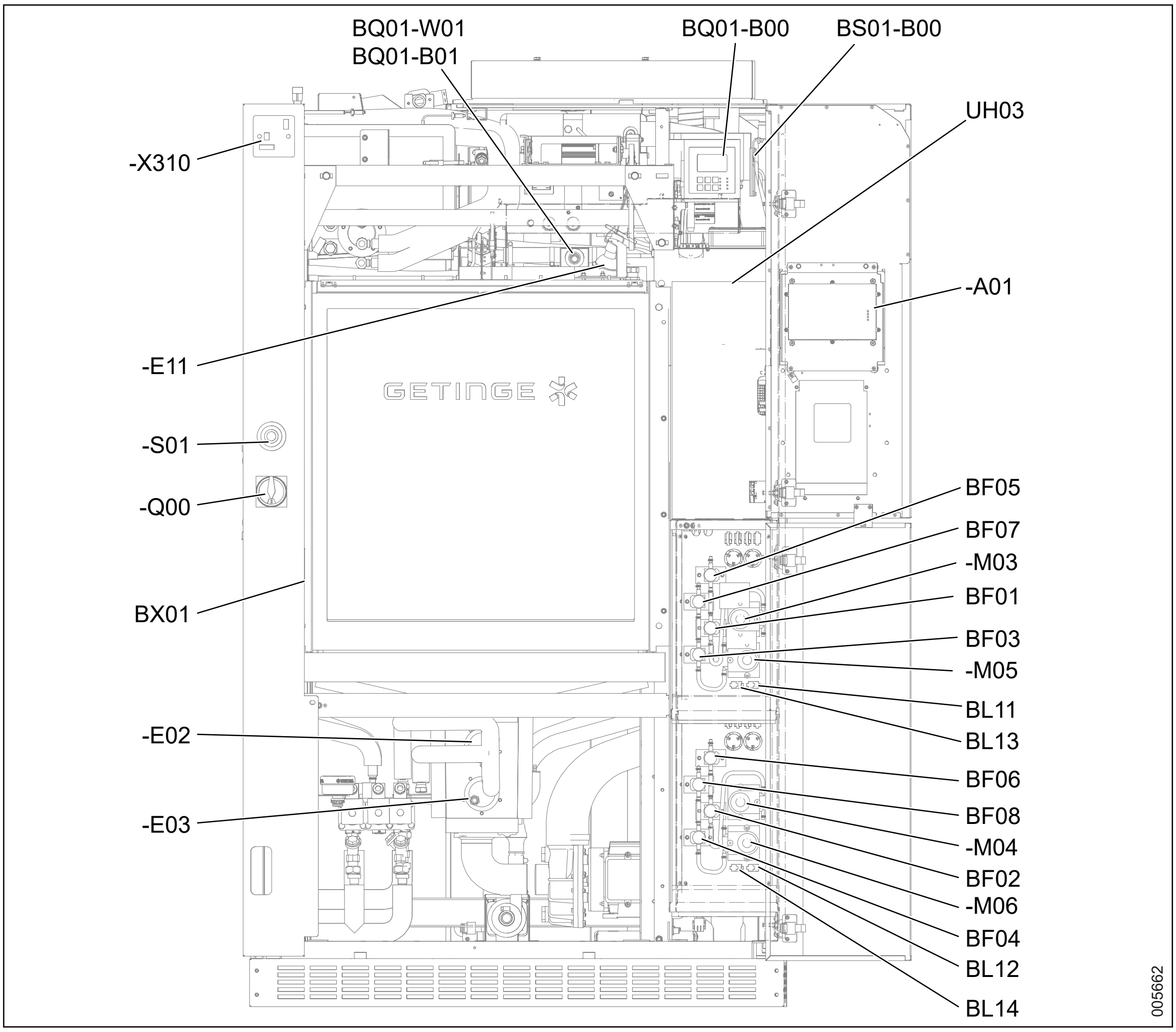
8.1.2 Components, illustration 2/7 (Overview)



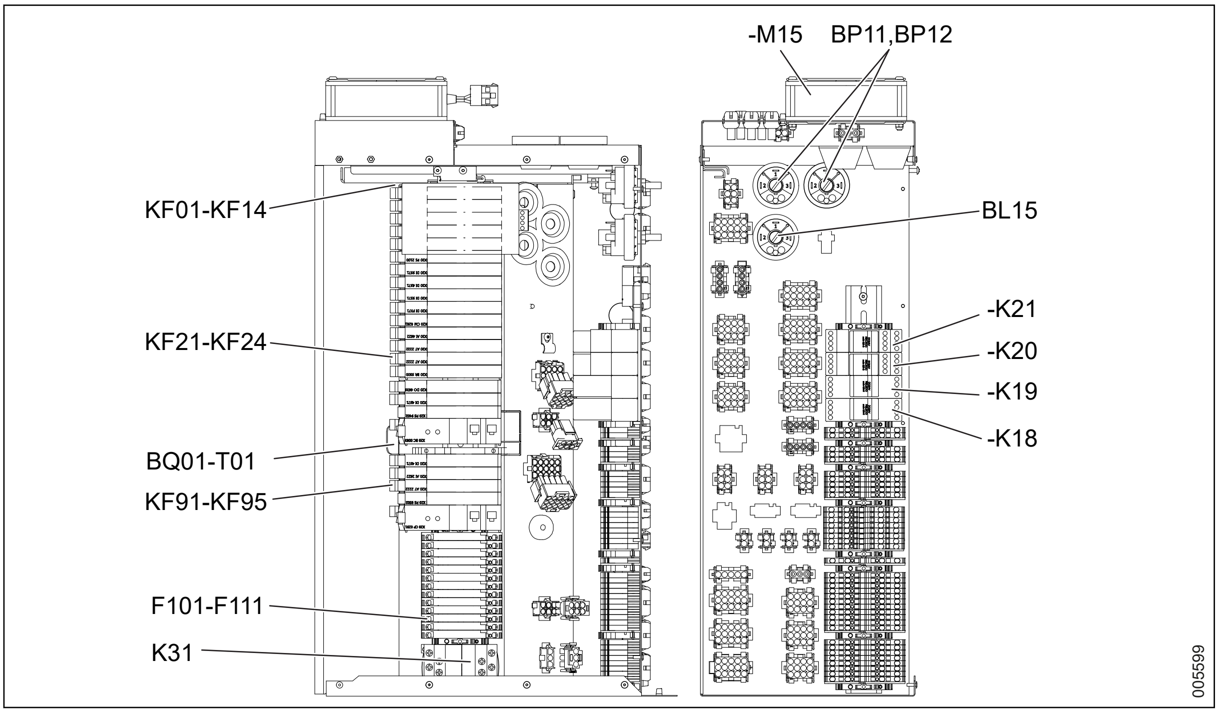
8.1.3 Components, illustration 3/7 (Overview)



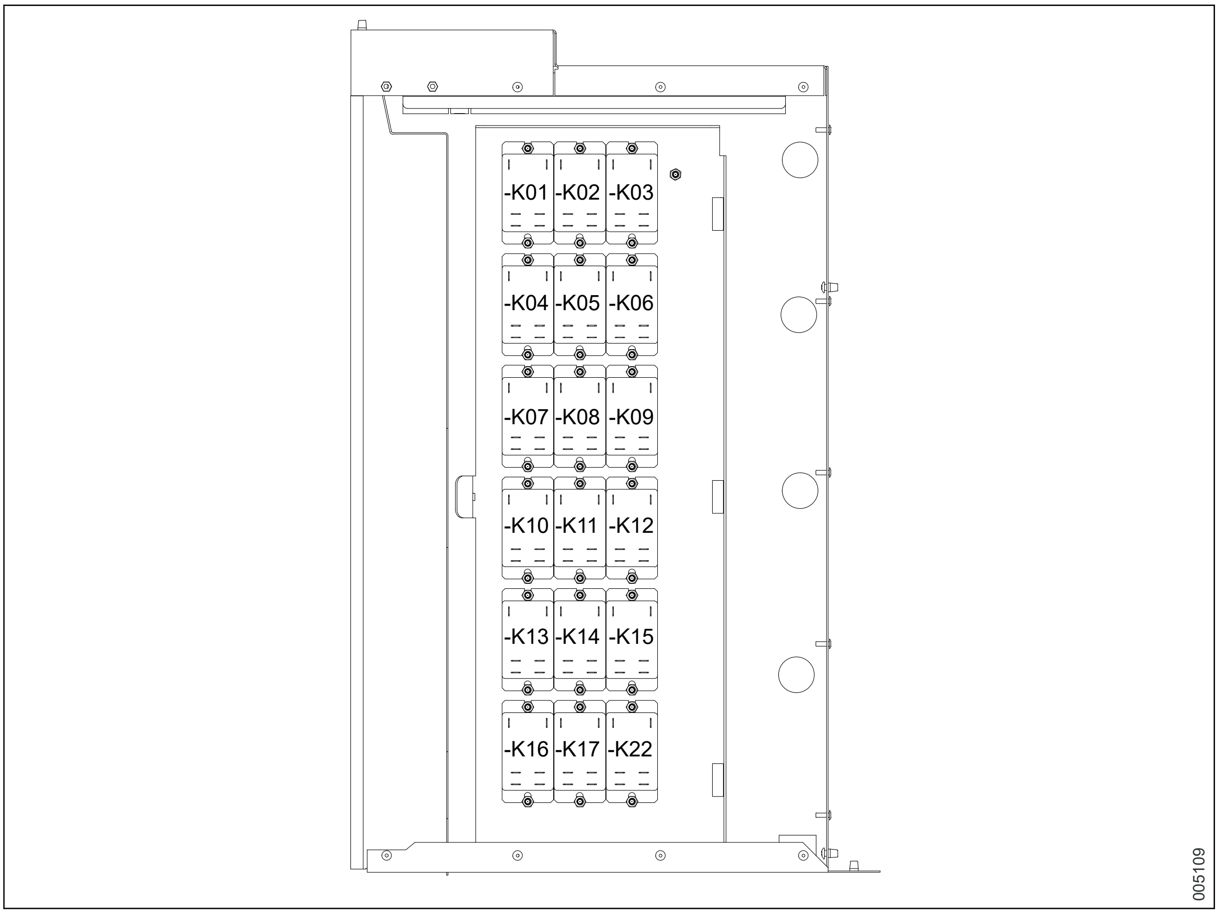
8.1.4 Components, illustration 4/7 (Overview)



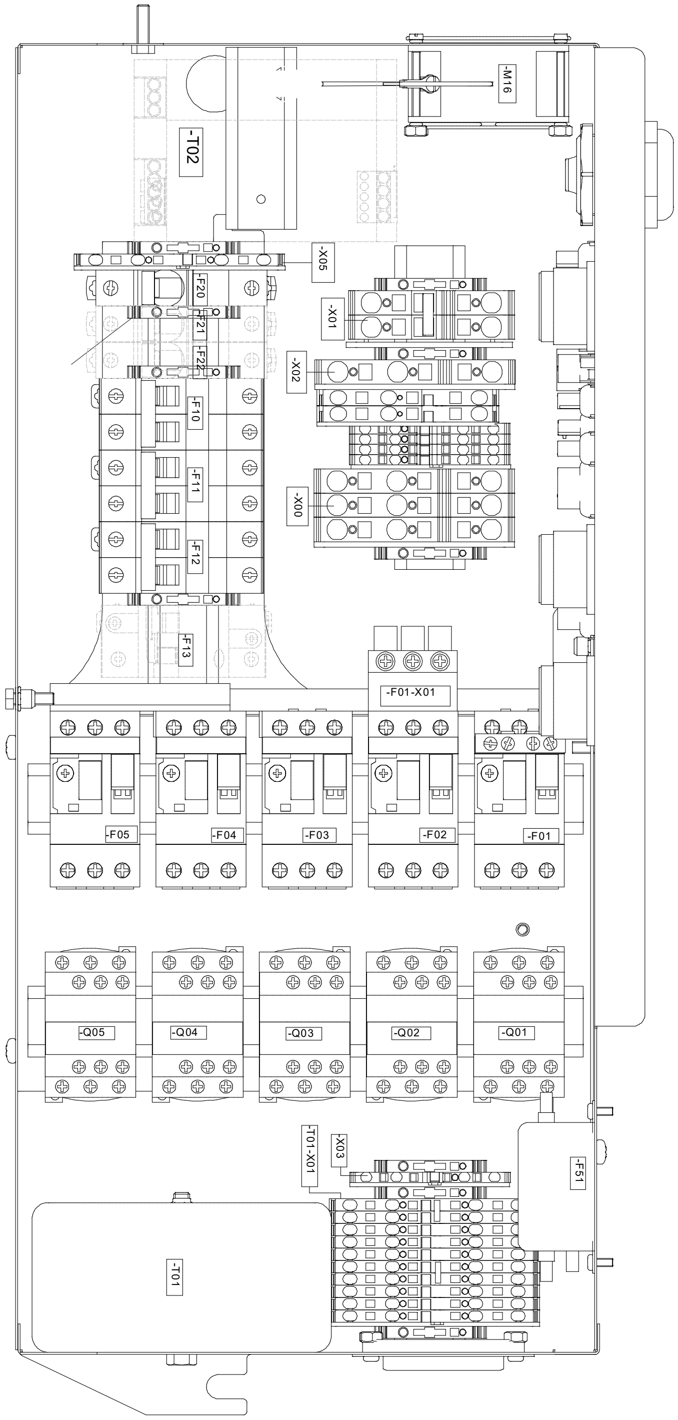
8.1.5 Components, illustration 5/7 (Control cabinet)



8.1.6 Components, illustration 6/7 (Control cabinet)



8.1.7 Components, illustration 7/7 (Power box)



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8.1.8 Components, table with explanations

POS	DESCRIPTION
-M01	Circulating pump
-M02	Drain pump
-M03	Dosing pump 1

POS	DESCRIPTION
-M04	Dosing pump 2
-M05	Dosing pump 3
-M06	Dosing pump 4
-M07	Drying fan 1
-M08	Drying fan 2
-F52	Noise filter drying fan 1
-F53	Noise filter drying fan 2
-M09	Door soiled side
-M11	Door clean side
-M10	Door locking soiled side
-M12	Door locking clean side
-M14	Cooling fan roof
-M15	Cooling fan control system
-M16	Cooling fan power cubicle
-M17	Cooling fan roof
-M18	Cooling fan roof
-E01	Dryer heater
-E02	Main heater el 1
-E03	Main heater el 2
-E04	Booster tank el heater
-E11	Lighting chamber
KK01	Cold water valve
KK02	Hot water valve
KK03	Treated water valve
KK04	Drain cooling valve
KK05	Booster valve
KK06	Fill from booster valve
KK11	Main heater steam valve
KK12	Booster tank steam valve
BT01	Chamber temperature
BT02	Chamber ind temperature
BT03	Drying air temperature
BT04	Booster tank temperature
BP01	Circulating water pressure
BP02	Drying air filter d pressure
BP03	Circulating water ind pressure
BP11	Water leakage switch 1
BP12	Water leakage switch 1
BL11	Chemical 1 level ok
BL12	Chemical 2 level ok
BL13	Chemical 3 level ok
BL14	Chemical 4 level ok
BL15	Chamber water level H
BL16	Booster tank level H

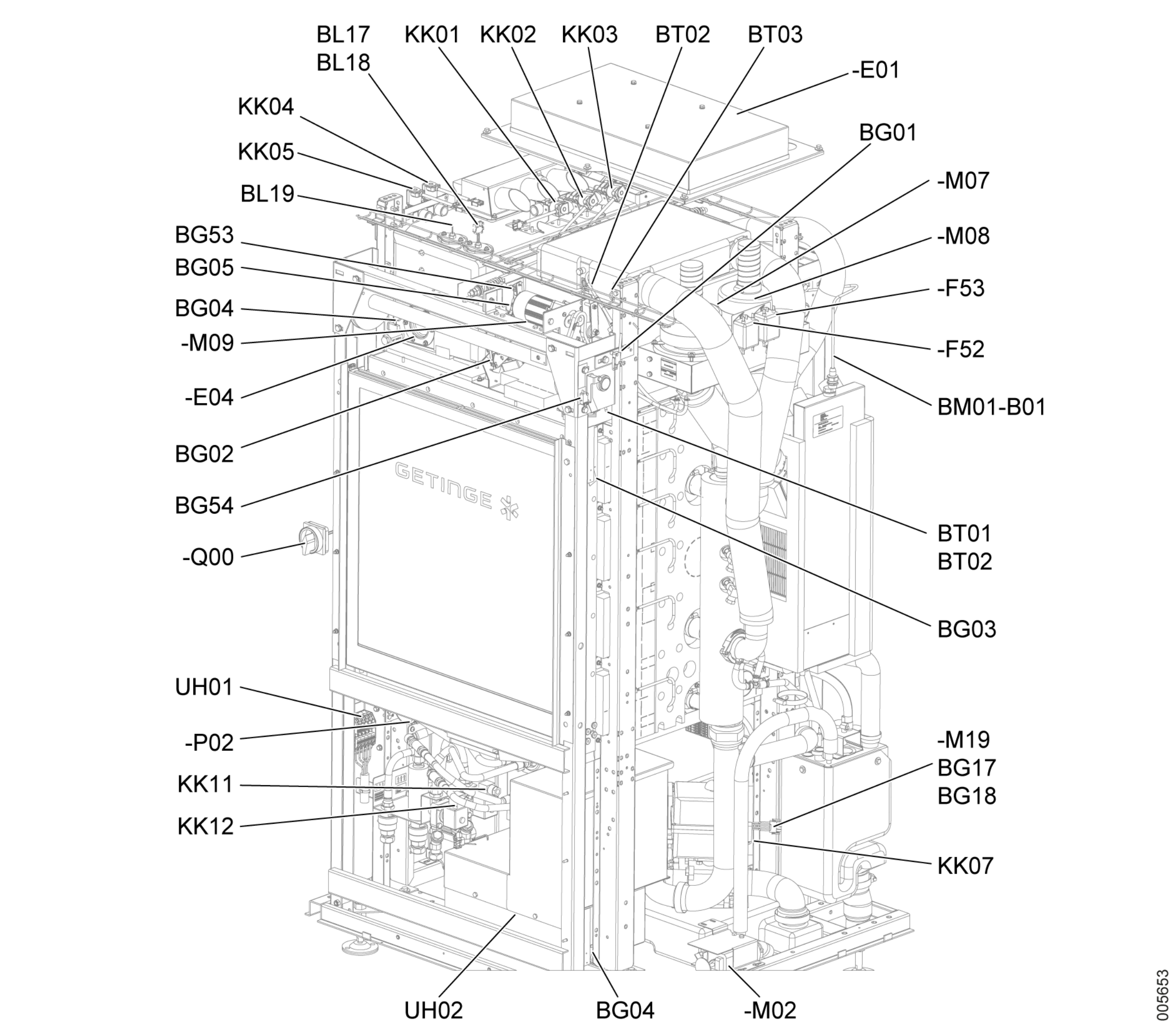
POS	DESCRIPTION
BF01	Chemical 1 flow pulses
BF02	Chemical 2 flow pulses
BF03	Chemical 3 flow pulses
BF04	Chemical 4 flow pulses
BF05	Chemical 1 ind flow pulses
BF06	Chemical 2 ind flow pulses
BF07	Chemical 3 ind flow pulses
BF08	Chemical 4 ind flow pulses
BQ01-B00	Rinse water conductivity transmitter
BQ01-B01	Rinse water conductivity sensor
BQ01-W01	Rinse water conductivity sensor cable
BG01	Door soiled side unlocked
BG02	Door soiled side locked
BG03	Door soiled side closed
BG04	Door soiled side opened
BG05	Door soiled side protection
BG53	Door soiled side closed hardware
BG54	Door soiled side opened hardware
BG06	Door clean side unlocked
BG07	Door clean side locked
BG08	Door clean side closed
BG09	Door clean side opened
BG10	Door clean side protection
BG57	Door clean side closed hardware
BG58	Door clean side opened hardware
BG11	Trolley in correct position
BG12	Recipe selection bit 1
BG13	Recipe selection bit 2
BG14	Recipe selection bit 3
BX01	Bar code scanner
BS01-B00	SWRSS
BS01-B01	SWRSS Antenna 1-5
BS01-B02	SWRSS Antenna 6
BM01-B00	Exhaust air humidity transmitter
BM01-B01	Exhaust air humidity sensor
-A01	OP panel soiled side
-A02	OP panel clean side
-P01	Printer
-P02	Buzzer
UH01	Connection box
-Q00	Main switch
UH02	Power distribution cubicle
-F01	Motor protector circulating pump
-Q01	Contactor circulating pump

POS	DESCRIPTION
-F02	Motor protector dryer heater
-Q02	Contactor dryer heater
-F03	Motor protector main heater el 1
-Q03	Contactor main heater el 1
-F04	Motor protector main heater el 2
-Q04	Contactor main heater el 2
-F05	Motor protector booster heater el
-Q05	Contactor booster heater el
-F10	Circuit breaker
-F11	Circuit breaker
-F12	Circuit breaker
-F13	Circuit breaker with earth fault
-F20	Circuit breaker
-F21	Circuit breaker
-F22	Circuit breaker
-T01	Transformer 230VAC
-T02	24 VDC internal supply
-T03	Transformer 230VAC
-F51	Noise filter
UH03	Control cubicle
KF01	Bus receiver X2X
KF02	Analogue input module RTD
KF03	Analogue input module RTD
KF04	Analogue input module gen 4xAI
KF05	Combi module
KF06	Digital input module 16xDI
KF07	Digital input module 12xDI
KF08	Digital input module 4xDI (count)
KF09	Digital input module 12xDI
KF10	Power supply module
KF11	Digital output module 16xDO
KF12	Digital output module 12xDO
KF13	Digital output module 6xDOr 5A
KF14	Digital output module 6xDOr 5A
KF15	Digital output module 6xDOr 5A
KF21	Bus controller powerlink
KF22	Supply module
KF23	Digital input module 4xDI
KF24	Digital output module 6xDOr 5A
KF31	ASI module
KF91	Compact plc
KF92	Supply module
KF93	Analogue input module RTD
KF94	Analogue input module gen 2xAI

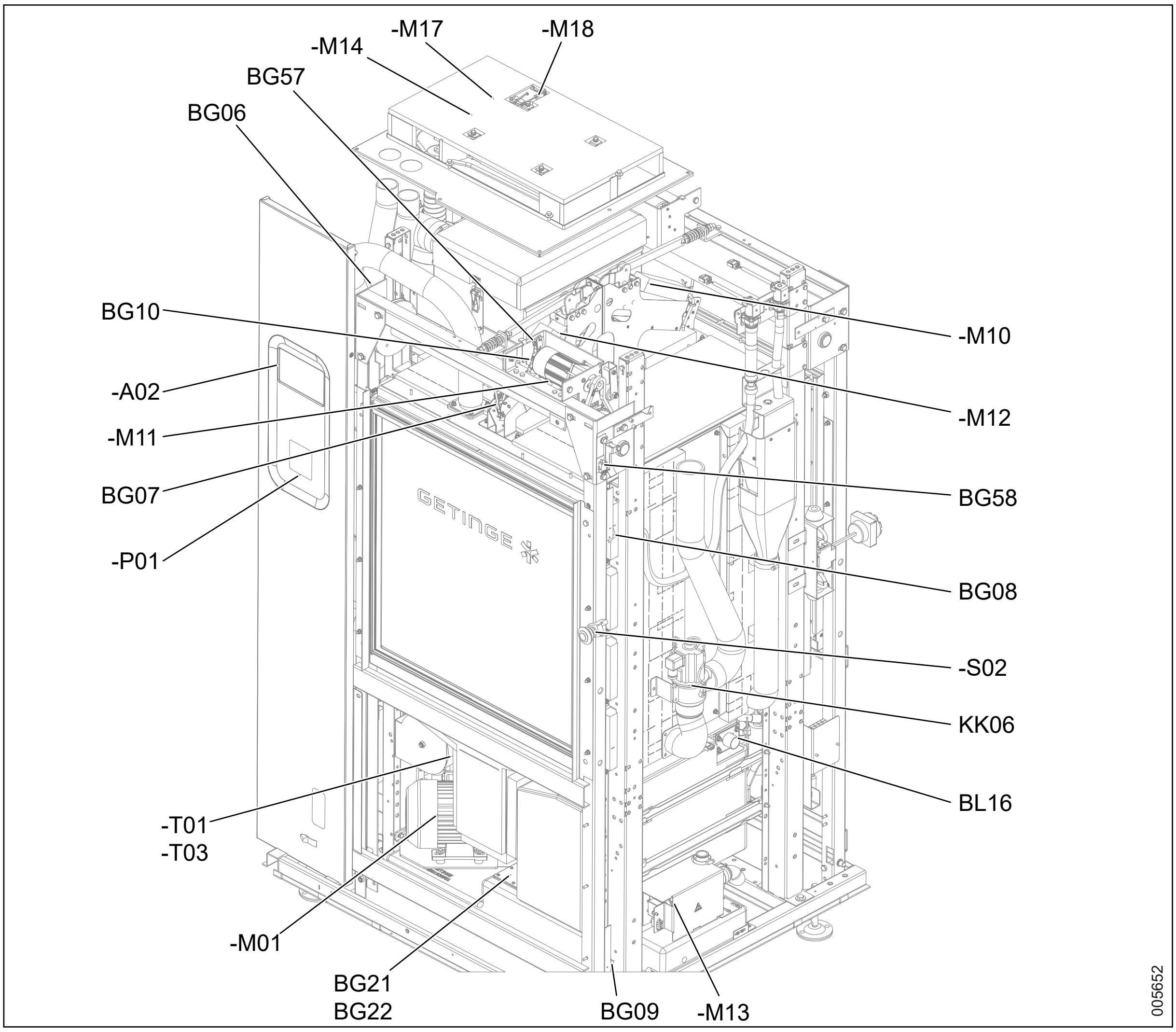
POS	DESCRIPTION
KF95	Digital input module 4xDI
-K01-K18	Digital output relay
-A11	Ethernet switch
BQ01-T01	Rinse water conductivity galvanic isolator
-K31	Contactor machine stop control
F101-F111	Fuse terminal
-S01	Machine stop soiled side Button
-S02	Machine stop clean side Button
-X310	UK Power socket

8.2 S-8668T

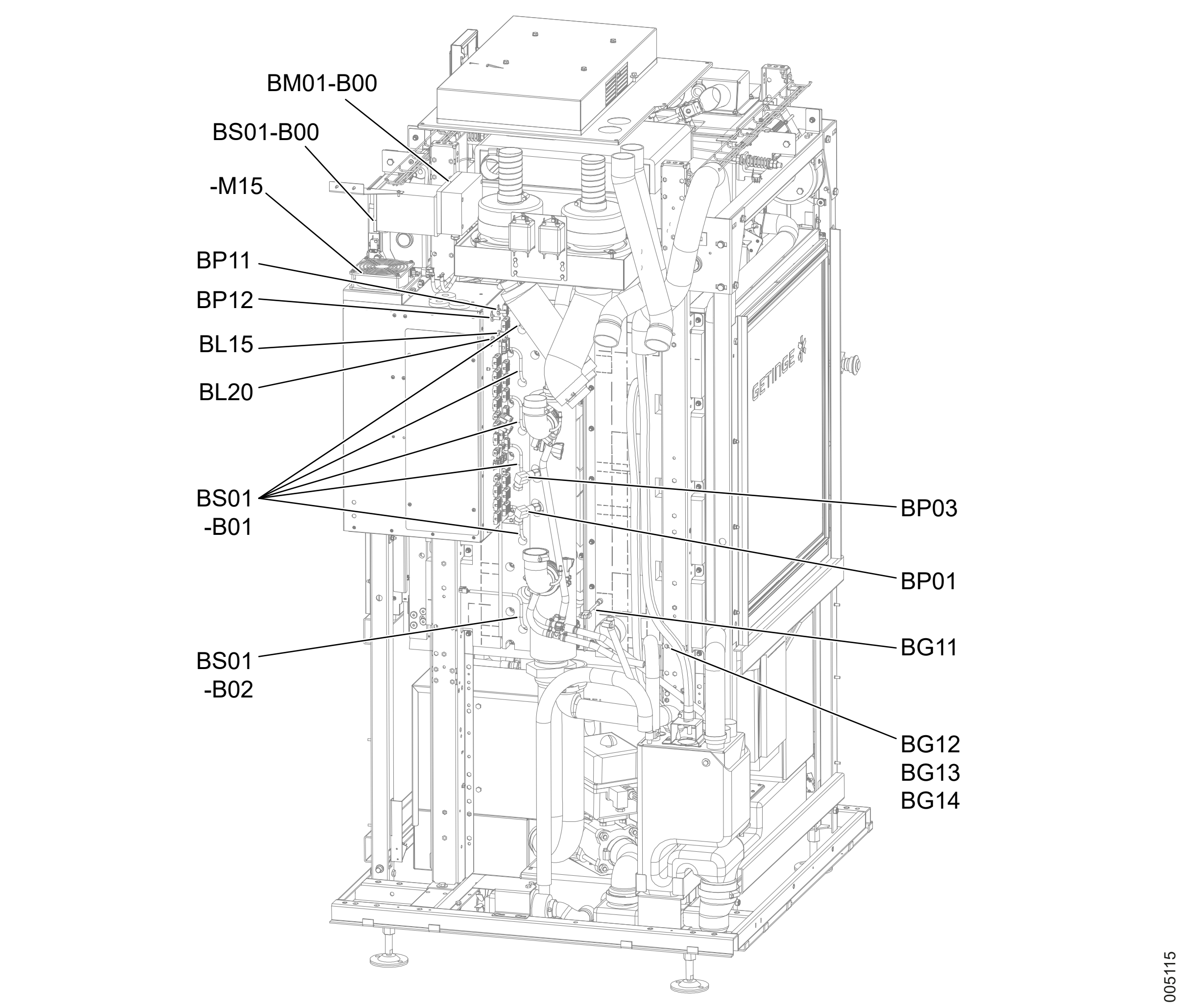
8.2.1 Components, illustration 1/9 (Overview)



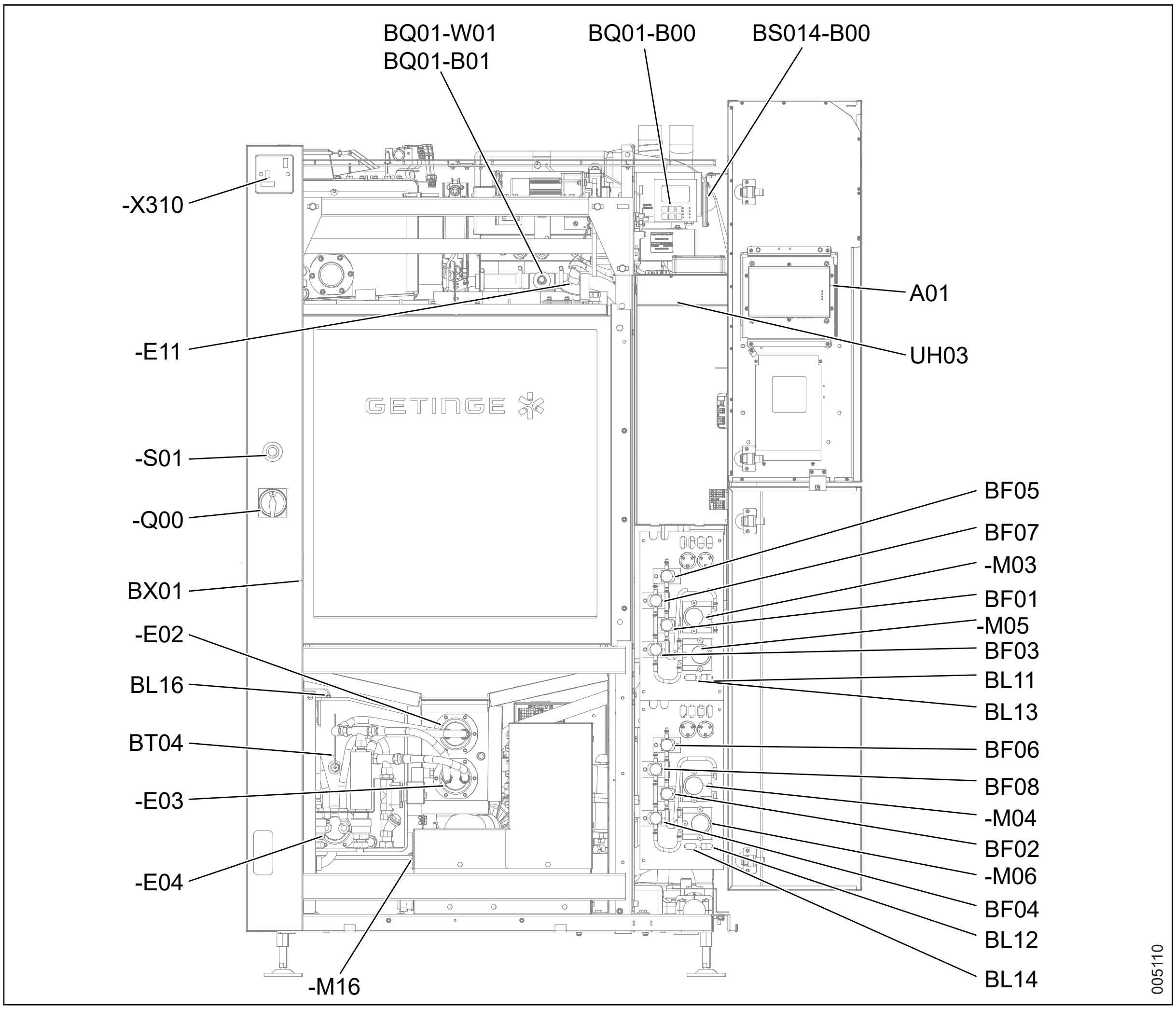
8.2.2 Components, illustration 2/9 (Overview)



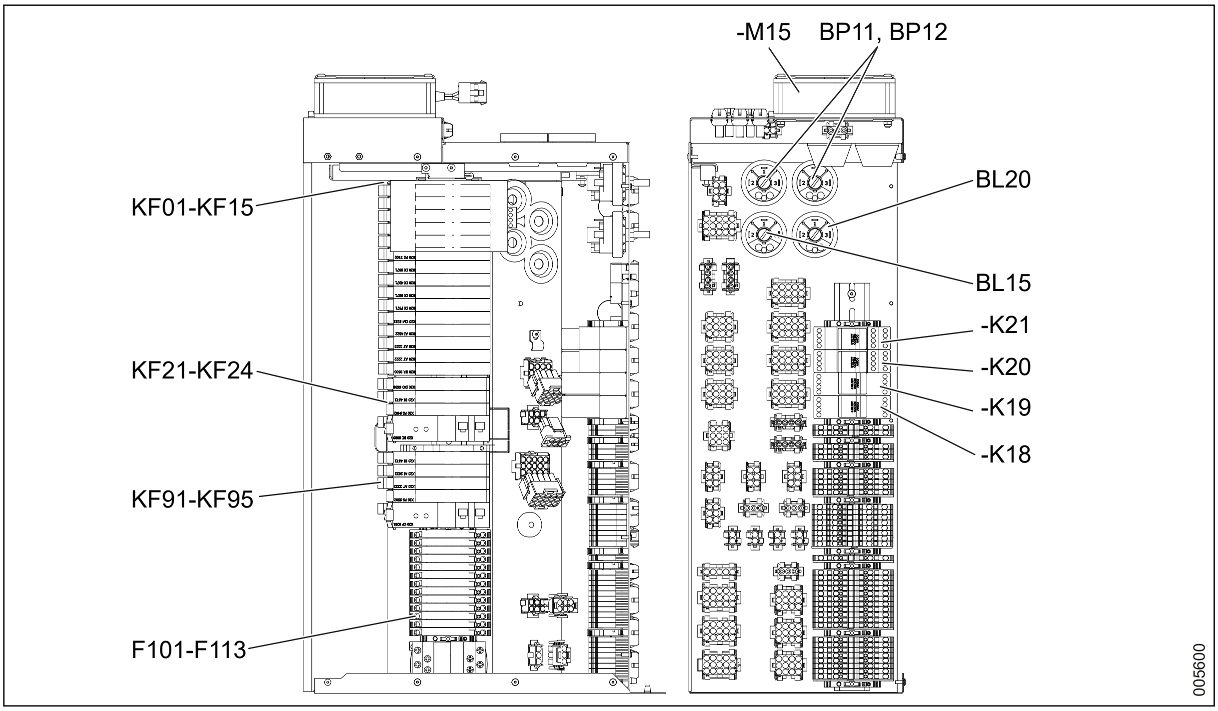
8.2.3 Components, illustration 3/9 (Overview)



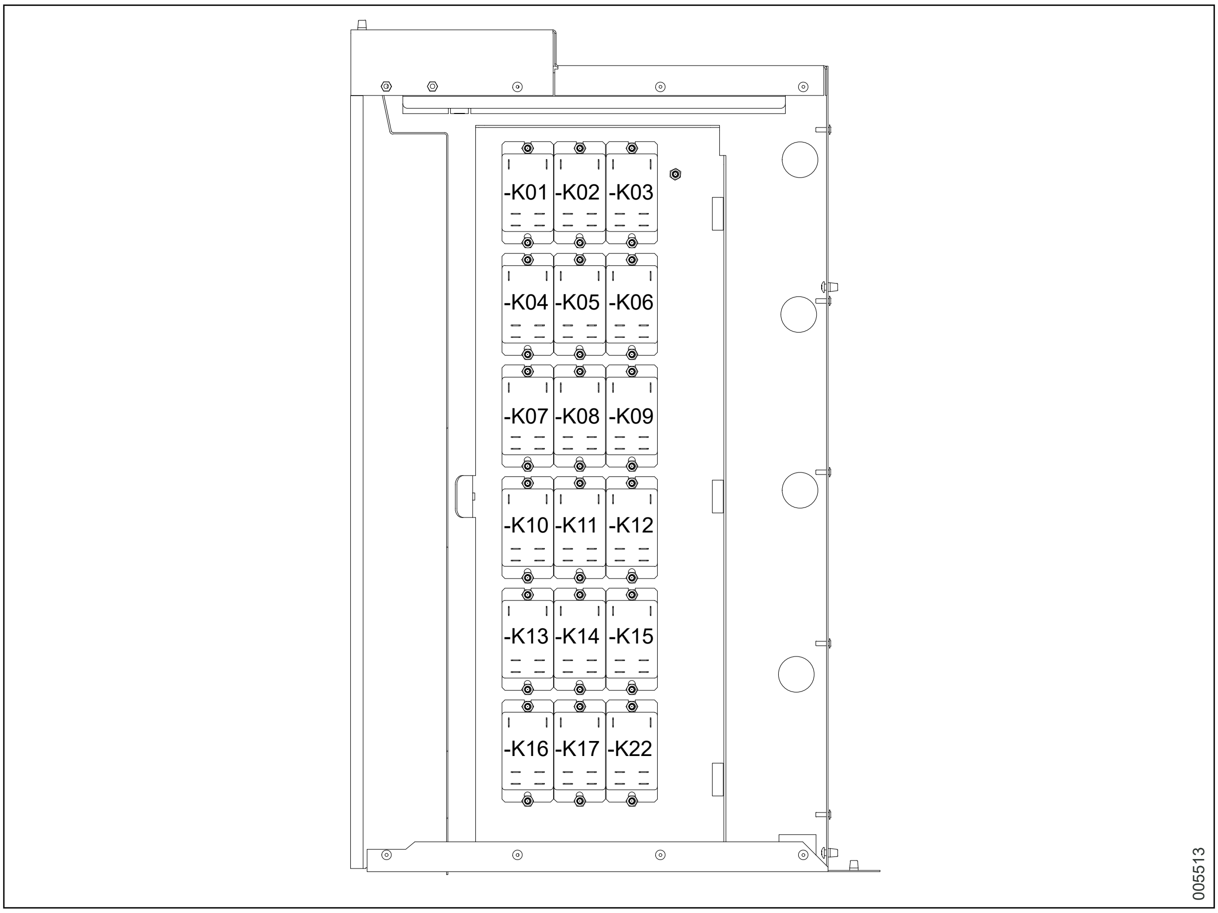
8.2.4 Components, illustration 4/9 (Overview)



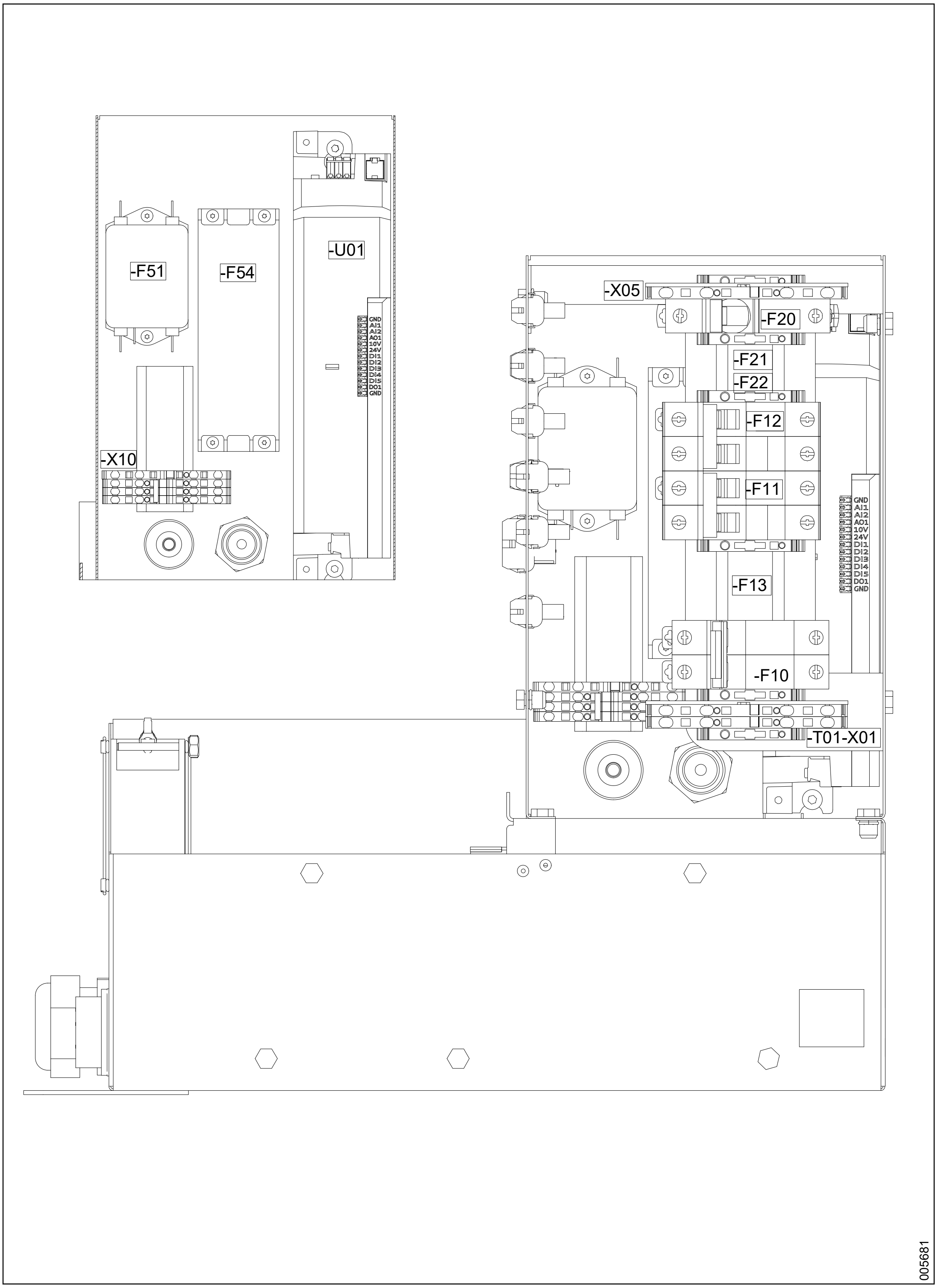
8.2.5 Components, illustration 5/9 (Control cabinet)



8.2.6 Components, illustration 6/9 (Control cabinet)



8.2.7 Components, illustration 7/9 (Power box)



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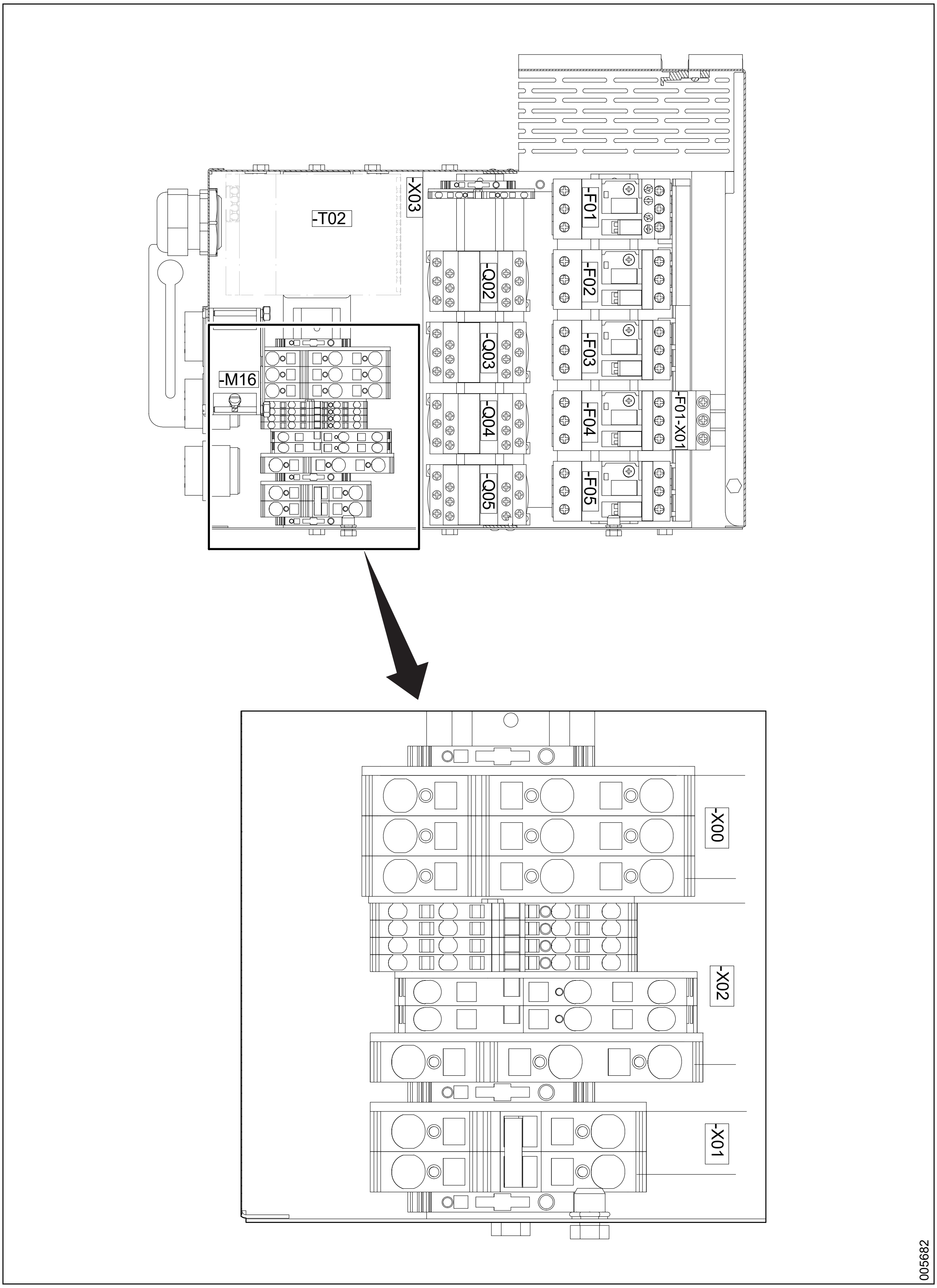
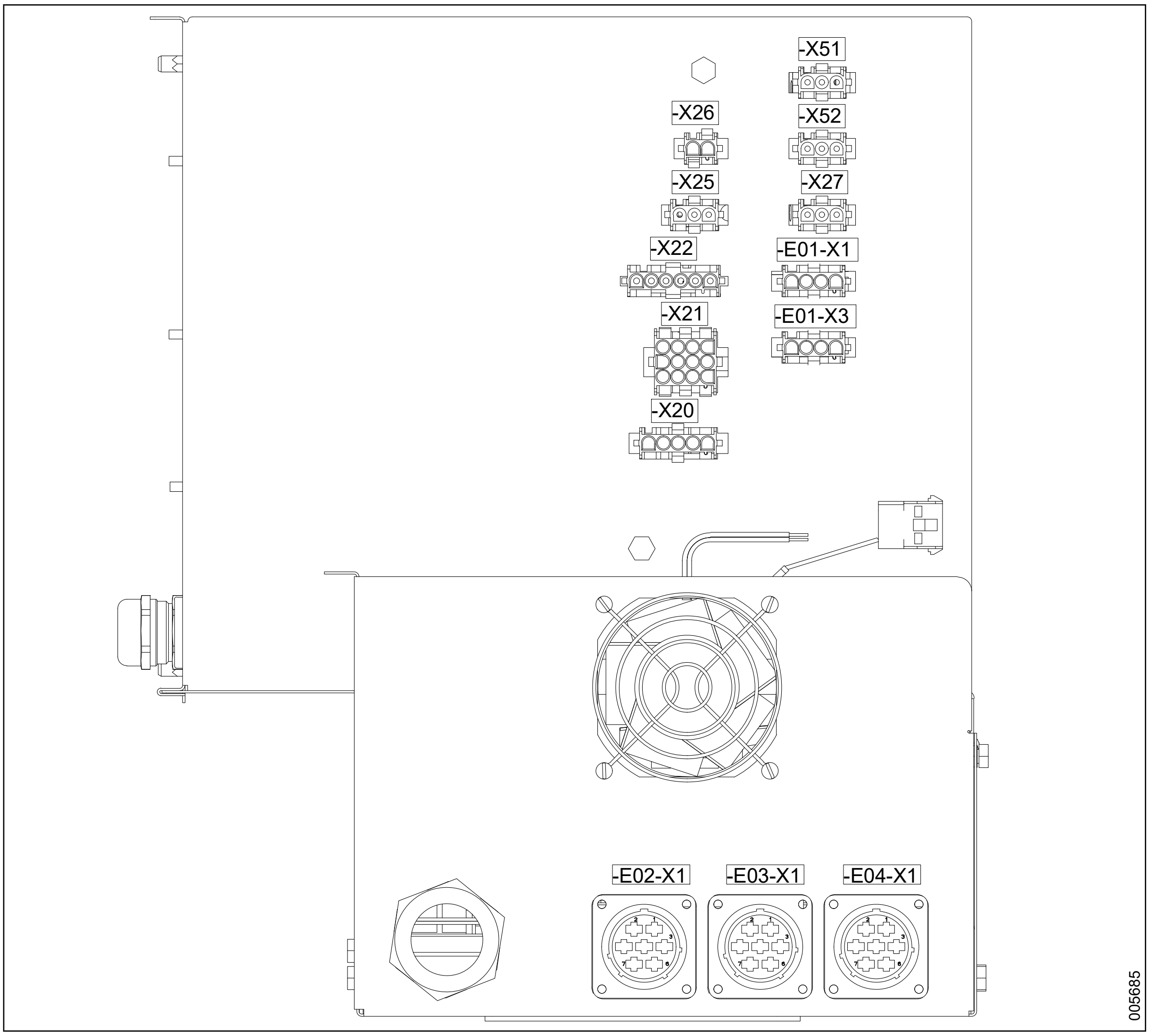
8.2.8 Components, illustration 8/9 (Power box)



8.2.9 Components, illustration 9/9 (Power box)



8.2.10 Components, table with explanations

POS	DESCRIPTION
-M01	Circulating pump
-M02	Drain pump
-M03	Dosing pump 1
-M04	Dosing pump 2
-M05	Dosing pump 3
-M06	Dosing pump 4
-M07	Drying fan 1
-M08	Drying fan 2
-F51	Noise filter
-F52	Noise filter drying fan 1
-F53	Noise filter drying fan 2
-M09	Door soiled side
-M11	Door clean side

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POS	DESCRIPTION
-M10	Door locking soiled side
-M12	Door locking clean side
-M14	Cooling fan top
-M15	Cooling fan control system
-M16	Cooling fan power cubicle
-M17	Cooling fan top
-M18	Cooling fan top
-M19	Draining valve
-E01	Dryer heater
-E02	Main heater el 1
-E03	Main heater el 2
-E04	Booster tank el heater
-E11	Lighting chamber
KF01	Bus receiver X2X
KF02	Analogue input module RTD
KF03	Analogue input module RTD
KF04	Analogue input module gen 4xAI
KF05	Combi module
KF06	Digital input module 16xDI
KF07	Digital input module 12xDI
KF08	Digital input module 4xDI (count)
KF09	Digital input module 12xDI
KF10	Power supply module
KF11	Digital output module 16xDO
KF12	Digital output module 12xDO
KF13	Digital output module 6xDOr 5A
KF14	Digital output module 6xDOr 5A
KF15	Digital output module 6xDOr 5A
KK01	Cold water valve
KK02	Hot water valve
KK03	Treated water valve
KK04	Drain cooling valve
KK05	Booster valve
KK06	Fill from booster valve
KK11	Main heater steam valve
KK12	Booster tank steam valve
BT01	Chamber temperature
BT02	Chamber ind temperature
BT03	Drying air temperature
BT04	Booster tank temperature
BP01	Circulating water pressure
BP02	Drying air filter d pressure
BP03	Circulating water ind pressure
BP11	Water leakage switch 1

POS	DESCRIPTION
BP12	Water leakage switch 1
BL11	Chemical 1 level ok
BL12	Chemical 2 level ok
BL13	Chemical 3 level ok
BL14	Chemical 4 level ok
BL15	Chamber water level H
BL16	Booster tank level H
BL 17	Process tank level 1
BK 18	Process tank level 2
BL 19	Process tank overflow
BL 20	Chamber water level L
BL 21	Drain tank level 1
BL22	Drain tank level 2
BF01	Chemical 1 flow pulses
BF02	Chemical 2 flow pulses
BF03	Chemical 3 flow pulses
BF04	Chemical 4 flow pulses
BF05	Chemical 1 ind flow pulses
BF06	Chemical 2 ind flow pulses
BF07	Chemical 3 ind flow pulses
BF08	Chemical 4 ind flow pulses
BQ01-B00	Rinse water conductivity transmitter
BQ01-B01	Rinse water conductivity sensor
BG01	Door soiled side unlocked
BG02	Door soiled side locked
BG03	Door soiled side closed
BG04	Door soiled side opened
BG05	Door soiled side protection
BG53	Door soiled side closed hardware
BG54	Door soiled side opened hardware
BG06	Door clean side unlocked
BG07	Door clean side locked
BG08	Door clean side closed
BG09	Door clean side opened
BG10	Door clean side protection
BG57	Door clean side closed hardware
BG58	Door clean side opened hardware
BG11	Trolley in correct position
BG12	Recipe selection bit 1
BG13	Recipe selection bit 2
BG14	Recipe selection bit 3
BX01	Barcode scanner
BS01-B00	SWRSS
BS01-B01	SWRSS Antenna 1-5

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POS	DESCRIPTION
BS01-B02	SWRSS Antenna 6
BS01-B10	SWRSS RFID tag trolley
BS01-B11	SWRSS RFID tag wing 1
BS01-B12	SWRSS RFID tag wing 2
BS01-B13	SWRSS RFID tag wing 3
BS01-B14	SWRSS RFID tag wing 4
BS01-B15	SWRSS RFID tag wing 5
BS01-B16	SWRSS RFID tag wing 6
BM01-B00	Exhaust air humidity transmitter
BM01-B01	Exhaust air humidity sensor
-A01	OP panel soiled side
-A02	OP panel clean side
-P01	Printer
-P02	Buzzer
-U01	Frequency converter
UH01	Connection box
-Q00	Main switch
UH02	Power distribution cubicle
-T01	Transformer 480/230 V
-T03	24 VDC internal supply
UH03	Control cubicle
UH04	Relay box
-A11	Ethernet switch
BQ01-T01	Rinse water conductivity galvanic isolator
-K31	Contactor machine stop control
-S01	Machine stop soiled side button
-S02	Machine stop clean side button
-X310	UK power socket

9 TROUBLESHOOTING

The following sections contain instructions for confirming and troubleshooting alarms and messages. For other functions, see section *4.3.10 View alarms and messages, page 53*.

Use the tables in the following sections when troubleshooting problems based on the alarms and messages displayed on the touchscreen.

Messages such as Warnings and Alarms are further divided into subgroups for Process and System, based on whether the warning or alarm is process or system-related.

The tables include the following information:

Name	Description
Error Code (only alarms)	Displays the numerical code associated with a specific alarm. The code is only visible during the alarm.
Name	The alarm, warning or information text displayed on the touchscreen.
General description	Displays a short description of the cause of the alarm, warning or information message.

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Name	Description
Detailed description (only process alarms)	Displays a more detailed description of the cause of the alarm.
Measure	Displays possible measures to solve the problem.

If the water does not empty out of the machine at the end of a program, a service technician is required.

9.1 System alerts

9.1.1 Handle alarms (login required)

When a serious problem occurs, a red alarm message appears on the touchscreen. The program is stopped and an alarm signal sounds.

1 **Mute Alarm**

Name	Description
Mute Alarm	Mutes the alarm signal.

Alarm during ongoing program

1. Tap **Mute Alarm** to turn off the alarm signal.
2. Note the displayed message, error code and phase. Tap **Guide** for more details.
3. Take action according to the Alarms and Warnings list. For more information, see sections *9.1.1.1 Process alarm, page 217* and *9.1.1.2 System Alarms, page 228*.
4. Login and tap **Confirm Alarm**.
5. Ensure that the doors are closed. If they are not closed, tap **Close Door.**

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6. Tap **End Program** and **Program Ends** is displayed while the machine is drained of water. When the program shut-down is completed, the door on the soiled side can be opened and the message **Goods not processed** is displayed on the touch screen.
7. Because the load is not processed, a new program has to be selected. Select a new program to process the load correctly. Tap **Open Door** in the **Goods not processed** view and the **Select Program** view appears.
8. Select an appropriate program and tap **Start Program**.

Alarm during standby

1. Follow steps 1-4 from Alarm during ongoing program.
2. Tap continue and the **Select Program** view appears.

9.1.1.1 Process alarm

Error code	Name	Description 1	Description 2	Proposed measure
11002	**Power failure**	Wash cycle interrupted due to power outage.	N/A	Acknowledge the alarm and start the cycle again.
11003	**I/O module fault**	The system has discovered a problem with the I/O module	N/A	1. Check the I/O module status. The LED diodes "R" or "S" must be green. 2. Check that all of I/O modules are correctly fitted. 3. Replace faulty I/O modules.
11004	**Process stop**	Wash cycle interrupted due to triggered process stop.	The process stop button is pressed or there is a breakdown in the control circuit.	1. Check that the process stop is reset. 2. If the machine does not have a process stop button, check the -S01-X1, -S02-X1, -K31-X1 and -K31-X2 jumpers.
11010	**SS door lock**	Door on the soiled side cannot be locked.	Switch door locked (BG02) is not activated within 10 seconds of when door begins to lock, or has been deactivated during an ongoing wash cycle.	1. Check if something is obstructing the door. 2. Check that switch door locked (BG02) is activated within 10 seconds when the door locks. 3. Check that the motor (-M10) starts. 4. Check the door adjustment. For more information, see section *7.9.2 Adjusting door switches, page 173*.

Error code	Name	Description 1	Description 2	Proposed measure
11011	**SS door unlock**	Door on the soiled side cannot be unlocked.	Switch door unlocked (BG01) is not activated within 10 seconds from when the door begins to unlock.	1. Check if something is obstructing the door. 2. Check that switch door unlocked (BG01) is activated within 10 seconds when the door is being locked. 3. Check that the motor (-M10) starts. 4. Check the door adjustment. For more information, see section *7.9.2 Adjusting door switches, page 173*.
11012	**SS door close**	Door on the soiled side cannot be closed.	Switch door closed (BG03) is not activated within 20 seconds from when the door begins to close, or becomes inactive during an ongoing wash cycle. This error code is also displayed if the door crush protection has been tripped.	1. Check if something is obstructing the door. 2. Check that the switch door closed (BG03) is activated within 20 seconds when the door closes. 3. Check that the motor (-M09) starts and the crush protection (BG53) is not activated. 4. Check the chain and wire. 5. Check door adjustment. For more information, see section *7.9.2 Adjusting door switches, page 173*.
11013	**SS door open**	Door on the soiled side cannot be opened.	Switch door open (BG04) is not activated within 30 seconds from when the door beings to open.	1. Check if something is obstructing the door. 2. Check that the switch door open (BG04) is activated within 30 seconds when the door opens. 3. Check that the motor (-M09) starts and that the cord slack switch (-BG54) is not activated. 4. Check the chain and wire. 5. Check the door adjustment. For more information, see section *7.9.2 Adjusting door switches, page 173*.

Error code	Name	Description 1	Description 2	Proposed measure
11015	**CS door lock**	Door on the clean side cannot be locked.	Switch door locked (BG07) is not activated within 10 seconds of when door begins to lock, or has been deactivated during an ongoing wash cycle.	1. Check if something is obstructing the door. 2. Check that switch door locked (BG07) is activated within 10 seconds when the door locks. 3. Check that the motor (-M12) starts. 4. Check the door adjustment. For more information, see section*7.9.2 Adjusting door switches, page 173*.
11016	**CS door unlock**	Door on the clean side cannot be unlocked.	Switch door unlocked (BG05) is not activated within 10 seconds from when the door begins to unlock.	1. Check if something is obstructing the door. 2. Check that switch door unlocked (BG05) is activated within 10 seconds when the door is being locked. 3. Check that the motor (-M12) starts. 4. Check the door adjustment. For more information, see section *7.9.2 Adjusting door switches, page 173*.
11017	**CS door close**	Door on the clean side cannot be closed.	Switch door closed (BG07) is not activated within 20 seconds from when the door begins to close, or becomes inactive during an ongoing wash cycle. This error code is also displayed if the door crush protection has been tripped.	1. Check if something is obstructing the door. 2. Check that the switch door closed (BG07) is activated within 20 seconds when the door closes. 3. Check that the motor (-M11) starts and the crush protection (-BG57) is not activated. 4. Check the chain and wire. 5. Check door adjustment. For more information, see section *7.9.2 Adjusting door switches, page 173*.

Error code	Name	Description 1	Description 2	Proposed measure
11018	**CS door open**	Door on the clean side cannot be opened.	Switch door open (BG08) is not activated within 30 seconds from when the door begins to open.	1. Check if something is obstructing the door. 2. Check that the switch door open (BG08) is activated within 30 seconds when the door opens. 3. Check that the motor (-M11) starts and that the cord slack switch (-BG58) is not activated. 4. Check the chain and wire. 5. Check the door adjustment. For more information, see section *7.9.2 Adjusting door switches, page 173*.
11020	**Slow fill chamber**	Filling the chamber with water takes too long.	The water level switch (-BL15) is not activated within 8 minutes from when filling starts or the draining valve has been open for 30 sec (only applies for S-8668T).	1. Check that the shut-off valves are open and that water reaches the machine. 2. Check that the water valves (-KK01, -KK02 or -KK03) open. 3. Check the level sensors.
11021	**Slow drain chamber**	Emptying the chamber takes too long.	The water level switch (-BL15) is still activated after the drainage pump has been running for 2 minutes.	1. Check that the drainage pump works. 2. Check that the water outlet is not blocked. 3. Check the level sensors.
11022	**Low level chamber**	The water level in the chamber is too low.	Water level switch (-BL15) is deactivated during wash cycle. For S-8668T BL15 or BL20.	1. Check that there are no leaks in the chamber or circulation system. 2. Check the level sensors.
11023	**Low circ. press**	Pressure in the circulation system is too low.	The pressure (BP01) is < 20 kPa (adjustable value)	1. Check that the wash cart is properly docked in the machine. 2. Check that the pump impeller rotates freely. 3. Check that the pump impeller rotates in the right direction (see the Installation manual). 4. Check that the detergent is not foaming. 5. Check the dosage amount. 6. Check the pressure sensor.
11024	**High circ. press**	Pressure in the circulation system is too high.	The pressure (BP01) is > 160 kPa (adjustable value)	1. Check the spray arms and spray tubes for blockage. 2. Check the pressure sensor.

Error code	Name	Description 1	Description 2	Proposed measure
11025	**Water leakage dryer**	Water has been detected in the drying system.	Leakage detectors (-BP11 and -BP12) have been activated during an ongoing wash cycle.	1. Check that the check valves in the drying system are undamaged and working as intended (no leaks). 2. Check the pressure sensor that warns of leaks.
11026	**Slow drain D. tank**	The draining of the drain tank is taking too long. Only applicable for S-8668T.	The lower drain tank level switch (-BL21) is still activated after the drain pump has been running for 3 minutes.	Check that the drain pump (-M02) runs.
11027	**Slow spray wing**	One or more of the spray arms are rotating too slowly.	If the problem is not solved within 15 minutes after the warning, an alarm is triggered.	1. Check that nothing is preventing the spray arms from rotating. 2. Check that each spray arm can rotate by hand. 3. Check that the clamps holding the trays in place on the wash cart are securely fastened. 4. Check that objects are not in contact with more than one tray on the wash cart. 5. Check the F107 (24 VDV) fuse. 6. Check that the detergent is not foaming inside the chamber. 7. Check that the spray arms are not damaged.
11028	**Circ. pump fail**	Circulation pump error	The circulation pump's load protection (-F01) is triggered.	Reset the load protection (-F01).
11029	**Drain valve fail**	The drain valve is not closed/ open. Only applicable for S-8668T.	The drain valve close/ open switches (-BG17/18) are not activated within 30 seconds, or are activated at the same time.	Check for drain valve (-M18).

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Error code	Name	Description 1	Description 2	Proposed measure
11030	**Low temp chamber**	Temperature in the chamber is too low	The chamber temperature (-BT01) has not increased by at least 5°C (adjustable) after heating for 5 minutes. Or the chamber temperature has been lower than the set temperature for a long time (longer than the set time).	**Electric heating:** 1. Check that both elements are working at full effect (measure all currents). If the internal overheating protection has tripped, the whole element must be replaced. 2. Check the heating contactors (-Q03, -Q04). 3. Check the calibration of the temperature sensor (-BT01). **Steam heating:** 1. Check the steam valve (-KK11). 2. Check that the ball valve is open and that the steam filter is clean. 3. Check the steam pressure (see the Installation manual). 4. Check condensate outlet 6. Check that there is no back pressure in the condensate outlet. 5. Check the calibration of the temperature sensor (-BT01).
11031	**High temp chamber**	Temperature in the chamber is too high.	The chamber temperature (-BT01) exceeds the set temperature by more than 10°C. Or 5°C above the maximum temperature set in the program.	1. Check water selection in the program. **Electric heating:** 2. Check that the heat contactor (-Q03, -Q04) has not failed closed. 3. Check the calibration of the temperature sensor (-BT01). **Steam heating:** 2. Check that the steam valve (-KK11) has not failed open. 3. Check the calibration of the temperature sensor (-BT01).
11032	**Temp deviation**	The temperature sensors in the chamber are registering different temperatures.	Temperatures from sensors for controlling and independent temperature monitoring (-BT01 and -BT02), deviate from each other by more than ±1°C for 5 seconds.	Check the calibration of the temperature sensors (-BT01, -BT02).

Error code	Name	Description 1	Description 2	Proposed measure
11033	**Disinfection fail**	The temperature during the disinfection phase is too low.	The process time exceeds the set disinfection time by more than double. Or the set disinfection time/temperature are too low to meet the program's A0 value.	**Electric heating:** 1. Check that both elements are working at full effect (measure all currents). If the internal overheating protection has tripped, the whole element must be replaced. 2. Check the heating contactors (-Q03, -Q04). 3. Check the calibration of the temperature sensor (-BT01). 4. Check the set temperature for time/temperature during the final rinse. **Steam heating:** 1. Check the steam valve (-KK11). 2. Check that the ball valve is open and that the steam filter is clean. 3. Check the steam pressure (see the Installation manual). 4. Check condensate outlet. 5. Check that there is no back pressure in the condensate outlet. 6. Check the calibration of the temperature sensor (-BT01).
11034	**Slow fill proc. tank**	The filling of the process tank is taking too long. Only applicable for S-8668T.	The process tank level switch (-BL17/18) is not activated within 8 minutes of filling.	1. Check that the shutoff valves are open and that water is reaching the machine. 2. Check that the water valves (-KK01, -KK02 or -KK03) are open. 3. Check the booster pump. 4. Check the water level switches.
11035	**Slow drain proc. tank**	The draining of the process tank is taking too long. Only applicable for S-8668T.	The process tank level switch (-BL17/18) is not deactivated within 20 seconds of filling of the chamber.	1. Check that the process tank valve (-KK06) opens. 2. Check the water level switches.
11036	**Proc. tank sensor fail**	Faulty process tank level sensor. Only applicable for S-8668T.	The process tank high level switch (-BL18) is activated, but not the low level switch (-BL17).	Check the water level switches.

Error code	Name	Description 1	Description 2	Proposed measure
11037	**High conductivity**	The conductivity in the rinse water is too high.	The conductivity (BQ01) is higher than the set value after the third rinse cycle (adjustable).	1. Check the water connections (that the right water is connected to each inlet). 2. Check that there are no detergent leaks. 3. Check that there are no deposits on the measuring cell. 4. Check that the chamber is empty after drainage.
11040	**Dosage pump 1**	The flow from the dosing pump is too low.	The measured dosing volume deviates >15% (adjustable value) from the set value.	1. Check that the dosing pump (-M03) works. 2. Check that there is detergent in the container. 3. Check the flow meter function and that there are no air pockets. Rinse the flow sensors with warm water (~40°C). 4. Check that the flow in the hoses is not blocked. 5. Check the dose monitor calibration in terms of both time and pulsing.
11041	**Dosage pump 2**	The flow from the dosing pump is too low.	The measured dosing volume deviates >15% (adjustable value) from the set value.	1. Check that the dosing pump (-M04) works. 2. Check that there is detergent in the container. 3. Check the flow meter function and that there are no air pockets. Rinse the flow sensors with warm water (~40°C). 4. Check that the flow in the hoses is not blocked. 5. Check the dose monitor calibration in terms of both time and pulsing.
11042	**Dosage pump 3**	The flow from the dosing pump is too low.	The measured dosing volume deviates >15% (adjustable value) from the set value.	1. Check that the dosing pump (-M05) works. 2. Check that there is detergent in the container. 3. Check the flow meter function and that there are not air pockets. Rinse the flow sensors with warm water (~40°C). 4. Check that the flow in the hoses is not blocked. 5. Check the dose monitor calibration in terms of both time and pulsing.

Error code	Name	Description 1	Description 2	Proposed measure
11043	**Dosage pump 4**	The flow from the dosing pump is too low.	The measured dosing volume deviates >15% (adjustable value) from the set value.	1. Check that the dosing pump (-M06) works. 2. Check that there is detergent in the container. 3. Check the flow meter function and that there are no air pockets. Rinse the flow sensors with warm water (~40°C). 4. Check that the flow in the hoses is not blocked. 5. Check the dose monitor calibration in terms of both time and pulsing.
11050	**Slow fill booster**	Filling the booster tank is too slow.	The water sensor (BL16) does not activate within 8 minutes from when filling begins.	1. Check that the shut-off valves are open and that water reaches the machine. 2. Check that the filling valve (-KK05) opens and that the filter is not clogged. 3. Check the level sensor.
11051	**Slow drain booster**	Emptying of the booster tank is too slow.	The water level sensor (BL16) is deactivated within 5 seconds (S-8668T: 20 seconds) from when drainage starts.	1. Check that the drainage valve (-KK06) opens. 2. Check the level sensors. 3. Check that the pump is running (only for S-8668T).
11052	**Low temp booster**	The temperature in the booster tank is too low.	The booster temperature (-BT04) has not increased by at least 5°C after heating for 5 minutes.	**Electric heating:** 1. Check that both elements are working at full effect (measure all currents). If the internal overheating protection has tripped, the whole element must be replaced. 2. Check the heating contactor (-Q05). 3. Check the calibration of the temperature sensor (-BT04). **Steam heating:** 1. Check the steam valve (-KK12). 2. Check that the ball valve is open and that the steam filter is clean. 3. Check the steam pressure (see the Installation manual). 4. Check the condensate outlet. 5. Check that there is no back pressure in the condensate outlet. 6. Check the calibration of the temperature sensor (-BT04).

Error code	Name	Description 1	Description 2	Proposed measure
11053	**High temp booster**	The temperature in the booster tank is too high.	The booster temperature (-BT04) exceeds the set temperature (the maximum temperature set in the program) by more than 5°C.	**Electric heating:** 1. Check that the heat contactor (-Q05) has not gotten stuck. 2. Check the calibration of the temperature sensor (-BT04). **Steam heating:** 1. Check that the steam valve (-KK12) has not failed open. 2. Check the water selection in the program. 3. Check the calibration of the temperature sensor (-BT04).
11060	**Low temp dryer**	The temperature in the drying system is too low.	The temperature (BT03) has not reached the set value -30°C after heating for 5 minutes.	1. Check that there is power on all phases up to the element (-E01). 2. Check that all 4 elements are working at full effect (measure current to all elements). NOTE overloading. 3. Check that there is no leakage in the drying system.
11061	**High temp dryer**	The temperature in the drying system is too high.	The temperature in the drying system is above 120°C.	1. Check that both fans are working (-M07, -M08). 2. Check that the heat contactor (-Q02) has not failed closed. 3. Check the calibration of the temperature sensor (BT03).
11062	**Low press dryer**	The pressure in the drying system is too low.	Pressure difference (-BP02) is < 100 Pa (adjustable).	Note! The fan is equipped with overheating protection that is automatically reset when the temperature has fallen and when the voltage has been turned off. If the fan does not have overheat protection: 1. Check that both fans are working (-M07, -M08). 2. Check the hoses for leaks. 3. Check that the filter is correctly fitted. 4. Check the pressure sensor (-BP02). 5. Check that there are no holes in the filter.
11063	**High press dryer**	The pressure in the drying system is too high.	Pressure difference (-BP02) is > 750 (adjustable).	1. Check if the filter is clogged. 2. Check the pressure sensor (-BP02).

Error code	Name	Description 1	Description 2	Proposed measure
11070	**S Disinfection fail**	See **11033.** The S prefix means that the alarm has been tripped by the Supervisor system.	The disinfection time is shorter than the set value. Or the process time exceeds the set value by more than double.	1. Check the calibration of the temperature sensor (BT02). 2. See alarm 11033.
11071	**S Temp deviation**	See **11032.** The S prefix means that the alarm has been tripped by the Supervisor system.	The chamber temperatures from the sensors for controlling and independent temperature monitoring (-BT01 and -BT02) deviate from each other by more than ±1°C for more than 20 seconds.	1. Check the calibration of the temperature sensors (-BT01, -BT02). 2. Check the temperature sensor.
11072	**S Circ. press**	The pump pressure is too high or too low (**11023, 11024**). The S prefix means that the alarm has been tripped by the Supervisor system.	The pressure (-BP03) is < 20 kPa (adjustable).	1. Check the calibration of the pressure sensor (-BP03). 2. See alarm 11023.
11073	**S H conductivity**	See **11037**. The S prefix means that the alarm has been tripped by the Supervisor system.	The conductivity (-BQ01) is greater than the set value.	1. Check the calibration of the conductivity meter (-BQ01). 2. See alarm 11037.
11074	**S Dosage**	The flow is too low from one of the dosing pumps (**11040, 11041, 11042, 11043**). The S prefix means that the alarm has been tripped by the Supervisor system.	The measured dosing volume from the Supervisor system deviates >15% (adjustable value) from the monitoring system's measured dosing volumes.	1. Check the calibration of the dose monitors (-BF05, -BF06, -BF07, -BF08). 2. See alarms 11040-11043.
11075	**Supervisor error**	The monitoring system is not getting a signal from the Supervisor system.	No communication between the monitoring system and Supervisor system.	Check the communication cable.

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Error code	Name	Description 1	Description 2	Proposed measure
11080	**AI fail-Chamb temp**	Error on sensor analogue in.	Analogue input is outside the permitted interval.	1. Check the sensor (for outage or short circuit). 2. Check the I/O module.
11081	**AI fail-Dryer temp**	Error on sensor analogue in.	Analogue input is outside the permitted interval.	1. Check the sensor (for outage or short circuit). 2. Check the I/O module.
11082	**AI fail-Chamb l temp**	Error on sensor analogue in.	Analogue input is outside the permitted interval.	1. Check the sensor (for outage or short circuit). 2. Check the I/O module.
11083	**AI fail-Boost temp**	Error on sensor analogue in.	Analogue input is outside the permitted interval.	1. Check the sensor (for outage or short circuit). 2. Check the I/O module.
11084	**AI fail-Circ press**	Error on sensor analogue in.	Analogue input is outside the permitted interval. The alarm may be caused by the monitoring system supervisor A1's sensor reacting to an error.	1. Check the sensor (for outage or short circuit). 2. Check the I/O module.
11085	**AI fail-Humidity**	Error on sensor analogue in.	Analogue input is outside the permitted interval.	1. Check the sensor (for outage or short circuit). 2. Check the I/O module.
11086	**AI fail-Conductivity**	Error on sensor analogue in.	Analogue input is outside the permitted interval. The alarm may be caused by the monitoring system supervisor A1's sensor reacting to an error.	1. Check the sensor (for outage or short circuit). 2. Check the I/O module.
11087	**AI fail-Dryer press**	Error on sensor analogue in.	Analogue input is outside the permitted interval.	1. Check the sensor (for outage or short circuit). 2. Check the I/O module.

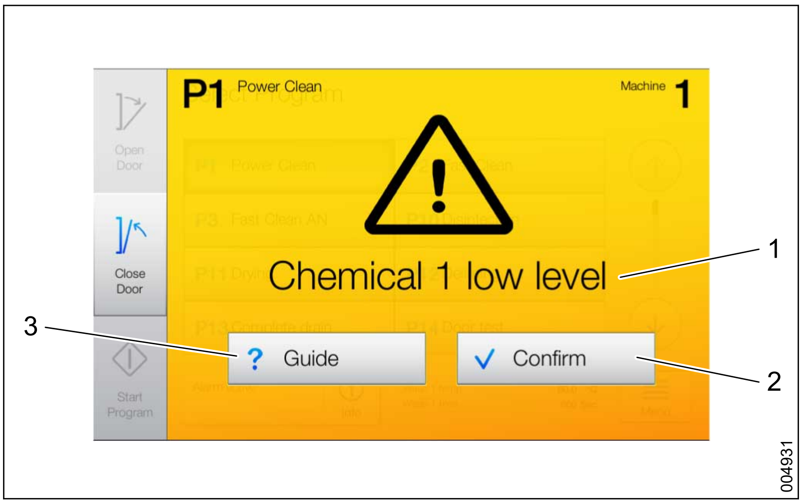
9.1.1.2 System Alarms

Error code	Name	Description	Measure
93000	**Config. File: Read Error**	The configuration file is corrupt or cannot be opened.	Restore the system from a backup, see section *7.1.2 Restore control system software from backup, page 141*.
93001	**System Time Monitor Err**	Excessive deviation between Real Time Clock and CPU clock detected. Suspected hardware failure.	Restart the machine. If the error remains, replace the operator panel.
93002	**Data Integrity Error**	A data corruption (software error) has been detected.	Restart the machine.
93003	**Internal System Error**	Internal system error.	Restart the machine. If the error remains, replace the operator panel.

Error code	Name	Description	Measure
93004	**Program Database Error**	The CRC did not match when the program file was read from the flash memory.	Restart the machine. If the error remains, restore the system from a backup, see section *7.1.2 Restore control system software from backup, page 141*.
93100	**Users File: Read Error**	Failed to read the user file at the start.	Import/restore the file from the Web server, see section *4.4.4 Access Management, page 64*.
93101	**Rights File: Read Error**	Failed to read the rights file at start.	Import/restore the file from the Web server, see section *4.4.4 Access Management, page 64*.
93102	**X20Config File: Read Err**	Failed to read the license file for I/O modules during the boot process.	Restart the machine. If the error remains, restore the system from a backup, see section *7.1.2 Restore control system software from backup, page 141*.
93103	**License File: Error**	Failed to read the license file for software addition during the boot process.	Restart the machine. If the error remains, restore the system from a backup, see section *7.1.2 Restore control system software from backup, page 141*.

9.1.2 Handle warnings

When a minor problem occurs, a yellow warning message appears on the touchscreen. The program continues.



1 Warning message
2 **Confirm**
3 **Guide**

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Name	Description
Warning message	Displays the warning message.
Confirm	Confirms the warning.
Guide	Displays detailed information about the warning and gives guidance to the operator.

A warning indicates a minor problem. The program is not stopped.

1. Note the displayed message, error code and phase. Tap **Guide** for more details.
2. Tap **Confirm** to return to the start screen.
3. Take action according to the Alarms and Warnings list. For more information, see sections *9.1.2.1 Process warnings, page 230* and *9.1.2.2 System warnings, page 231*.

9.1.2.1 Process warnings

Name	Description	Measure
Chemical 1 low level	The detergent is running out.	Replace the detergent container as soon as possible.
Chemical 2 low level	The detergent is running out.	Replace the detergent container as soon as possible.
Chemical 3 low level	The detergent is running out.	Replace the detergent container as soon as possible.
Chemical 4 low level	The detergent is running out.	Replace the detergent container as soon as possible.
Scanner read error	The machine's scanner was not able to read the barcode on the wash cart.	Pull out the wash cart and ensure that the barcode is clean and not worn. Wait 15 seconds. Push the wash cart back in. If the barcode cannot be read, use the touchscreen to select the correct program.
ID tag not detected	ID tag on wash cart cannot be found. The tag is used to indicate how many spray arms must be monitored for rotation.	Pull out the wash cart. Ensure that the ID tag is in place. Wait 15 seconds. Push the wash cart back in and start the program. If the problem persists, use another wash cart.
Slow spray wing	One or more of the spray arms are rotating too slowly. The warning is only active in the first pre rinse phase.	Open the door and check the wash cart. Ensure that the spray arms, nozzles or spray pipes are not blocked by objects. Close the door and continue the program. (The phase time will restart).
Chemical 1 empty	The detergent container is empty.	Replace the detergent container immediately.
Chemical 2 empty	The detergent container is empty.	Replace the detergent container immediately.
Chemical 3 empty	The detergent container is empty.	Replace the detergent container immediately.
Chemical 4 empty	The detergent container is empty.	Replace the detergent container immediately.

9.1.2.2 System warnings

Name	Cause	Action
License File: Error	The operator panel serial number and the serial number in the license file do not match. The license file has been copied/ reset from another operator panel.	Configure a new file via the Web server, see section *7.1.3 Restore serial number after replacing operator panel , page 143*.
Process Report Error	A process report could not be created. This is a subsequent error that occurs in conjunction with connection or writing errors.	Identify and correct the source of the error. • If there is a fault on the local printer: Print the latest process report manually, see section *4.3.8 Printouts, page 51*. • In case of a PDF generation error: The process report is saved automatically when the device is reconnected.
Netw. Printer: Conn Err	Failed to connect to the network printer.	Check connections and settings, see section *4.3.8.1 Print emergency print-outs (login required), page 51*. If the problem persists, contact your local IT administrator.
Fascia Printer: Conn Err	Unable to connect to the control panel printer.	Check connections and settings.
USB Storage: Conn Err	Failed to detect the presence of a USB memory stick.	Replace the USB memory stick.
Netw. Storage: Conn Err	Cannot connect to the network storage server.	Check connections and settings. If the problem persists, contact your local IT administrator.
USB Storage: Write Err	Unable to save to the USB memory stick.	Replace the USB memory stick. The report has been temporarily archived and is saved as soon as a new USB memory stick has been connected.
USB Storage: Mem Full	Not enough space on the USD memory stick.	Replace the USB memory stick. The report has been temporarily archived and is saved as soon as a new USB memory stick has been connected
USB Storage: Mem Full	Memory nearly full on the USB memory stick.	Replace the USB memory stick.
Netw. Storage: Write Err	Failed to send the report to the network storage server.	Check the connections and settings. If the problem persists, contact your local IT administrator. The report has been archived and a new attempt to send the report will be made as soon as the connection is reestablished.
Fascia Printer: Write Err	Failed to print on fascia printer.	• Replace the paper roll if needed. • Check that the printer cover is closed.

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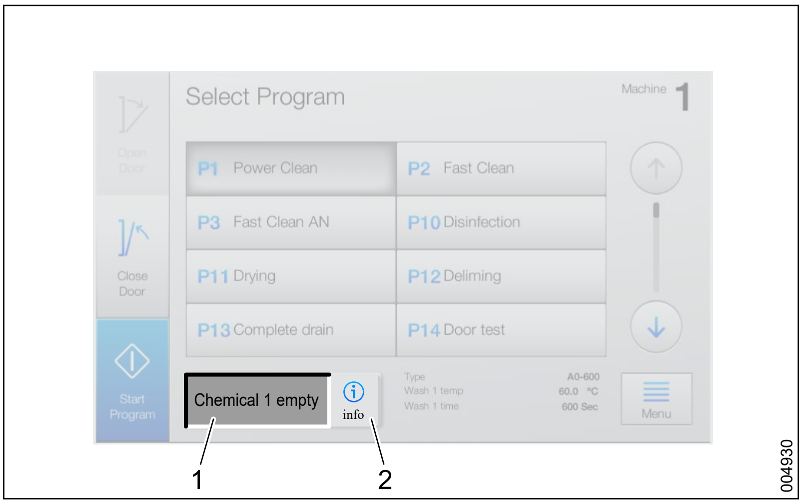
Name	Cause	Action
Fascia Printer: No Paper	Failed to print on fascia printer.	• Replace the paper roll if needed. • Check that the printer cover is closed.
Netw. Printer: Write Err	Failed to send report to network printer.	Check the connections and settings. If the problem persists, contact your local IT administrator. The report has been archived and a new attempt to send the report will be made as soon as the connection is reestablished.
Barcode Reader: Conn Err	Failed to connect to the barcode reader via USB.	Check the connections.
T-DOC:System Unavailable	An error occurred that affects T-DOC.	See the accompanying documentation for the system.
T-DOC:Connection Lost	An error occurred that affects T-DOC.	See the accompanying documentation for the system.
GO:System Unavailable	An error occurred that affects Getinge Online.	See the accompanying documentation for the system.
GO:Connection Lost	An error occurred that affects Getinge Online.	See the accompanying documentation for the system.
GO:Hostname Fail	An error occurred that affects Getinge Online.	See the accompanying documentation for the system.
GO:Proxy Login Fail	An error occurred that affects Getinge Online.	See the accompanying documentation for the system.
GO:Gateway Error	An error occurred that affects Getinge Online.	See the accompanying documentation for the system.
GO:Gateway Config	An error occurred that affects Getinge Online.	See the accompanying documentation for the system.
Netw. Printer: Report OK	-	No action needed.
Fascia Printer: Report OK	-	No action needed.
USB Storage: Report OK	-	No action needed.
USB Storage: Report OK	-	No action needed.

9.1.3 Handle information messages

Information text displays problems that be solved before starting a program. For more information, see section *9 TROUBLESHOOTING, page 215*.



1 Information text

2 **Info**

Name	Description
Information text	Displays information text.
Info	Button for detailed information.

1. Check information text.
2. Tap **Info** to get more detailed information.
3. Resolve the issue. For more information, see section *9.1.3.1 Information messages, page 233*.
4. Start the program. No confirmation is required.

9.1.3.1 Information messages

Displayed message/ text	Cause	Action
CS Door not locked	The door on the clean side is not closed and locked.	Close the clean side door.
Scanner read error	The machine's scanner was not able to read the barcode on the wash cart.	Pull out the wash cart and ensure that the barcode is clean and not worn. Wait 15 seconds. Push the wash cart back in. If the barcode cannot be read, use the touchscreen to select the correct program
ID tag not detected	ID tag on wash cart cannot be found. The tag indicates the number of spray arms to be monitored for rotation.	Pull out the wash cart. Ensure that the ID tag is in place. Wait 15 seconds. Push the wash cart back in and start the program. If the problem persists, use another wash cart.
Alarm active	An alarm (or warning) is still active.	For more information, go to **Menu** > **Alarms & Messages.**
No program selected.	No program selection made.	Select a program on the touchscreen.
Config error	Disinfection is not correctly configured.	Configure the program parameters for disinfection temperature and time to match the configured A_0 value.

Displayed message/ text	Cause	Action
Config error	Detergent dosing is not correctly configured.	Configure the program parameters. The dosing pump must be present on the machine and the dosage temperature must not be higher than the wash temperature.
Config error	The range for the water type is not correctly configured. Only applicable for S-8668T.	Configure the range for the water type: only 0-5 or 11-16 is valid.
Log not ready	A log-file is being prepared.	Wait until the task is completed.
Chemical 1 empty	The detergent container is empty.	Refill detergent or replace the container.
Chemical 2 empty	The detergent container is empty.	Refill detergent or replace the container.
Chemical 3 empty	The detergent container is empty.	Refill detergent or replace the container.
Chemical 4 empty	The detergent container is empty.	Refill detergent or replace the container.
Manual output active	One or more outputs are active.	Deactivate the outputs. For more information, see section *4.3.3.3 Analog outputs, page 39*

10 HANDLING OF WORN-OUT PRODUCTS

This section describes how to dispose of worn out products and parts.

10.1 WEEE directive

- This product and its accessories comply with the requirements in the WEEE directive.
- WEEE stands for “Waste Electrical & Electronic Equipment”.

10.2 Disposal of detergent or detergent containers

- Left-over detergent and used detergent containers must be disposed of at recycling stations in accordance with national laws and local regulations.
- If required, contact your detergent supplier for more information.

10.3 Disposal of the machine, accessories or spare parts

Before recycling, run a disinfection program to make sure that the machine is disinfected.

- Worn-out products must be disposed of at recycling stations in accordance with national laws and local regulations.
- If required, contact your reseller for more information.

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10.4 List of materials

Each type of material must be handled and recycled according to current national laws on waste handling. There is no mercury in the machine or its components.

Materials	**Approximate share of materials**					
	S-8666		**S-8668**		**S-8668T**	
	kg	**%**	**kg**	**%**	**kg**	**%**
Stainless steel	299.0	70	313.0	70	342.3	70
Steel	7.9	2	8.2	2	9.0	2
Cast iron	2.8	< 1	2,9	1	3.2	< 1
Copper	15.3	4	16.0	4	17.5	4
Plastic	69.8	16	73.1	16	80.0	16
Glass	18.4	4	19.3	4	21.1	4
Rubber/Silicone	0.3	< 1	0.3	< 1	0.4	< 1
Wiring *	5.3	1	5.6	1	6.1	1
Circuit boards *	2.5	< 1	2.7	< 1	2.9	< 1
Other components **	7.0	2	7.3	1	8.0	2
Total, all materials	428.3	100	448.4	100	490.5	100

* contains environmentally hazardous substances such as lead, cadmium, PVC and flame retardants and must be recycled as electronic scrap.

** consists of pumps, door motors and other parts that cannot be weighed separately.

Packing material

The machine is delivered on wooden pallets and the machine parts are covered with corrugated cardboard or wrapped in plastic.

11 LIST OF APPENDICES

Appendix 1 - S-8666/S-8668 - P&I Diagram - Electrically heated

Appendix 2 - S-8666/S-8668 - P&I Diagram - Steam heated

Appendix 3 - S-8666/S-8668 - P&I Diagram - Electrically/Steam heated

Appendix 4 - S-8668T - P&I Diagram - Electrically heated

Appendix 5 - S-8668T - P&I Diagram - Steam heated

Appendix 6 - S-8668T - P&I Diagram - Electrically/Steam heated

Appendix 7 - S-8666/S-8668/S-8668T - Installation drawing

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